QUANTITATIVE ANALYSIS

CPA

FOUNDATION LEVEL

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STUDY TEXT

Revised: September 2021

CONTENT

1. Basic mathematical techniques

Functions

- Functions, equations and graphs: Linear, quadratic, cubic, exponential and logarithmic
- Application of mathematical functions in solving business problems

Matrix algebra

- Types and operations (addition, subtraction, multiplication, transposition, and inversion)
- Application of matrices: statistical modelling, Markov analysis, input- output analysis and general applications

Calculus

- **Differentiation**
  - Rules of differentiation (general rule, chain, product, quotient)
  - Differentiation of exponential and logarithmic functions
  - Higher order derivatives: Turning points (maxima and minima)
  - Ordinary derivatives and their applications
  - Partial derivatives and their applications
- **Integration**
  - Rules of integration
  - Applications of integration to business problems

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2. Probability

Set theory

- Types of sets
- Set description: Enumeration and descriptive properties of sets
- Operations of sets: Union, intersection, complement and difference
- Venn diagram

Probability theory and distribution Probability theory

- Definitions: Event, outcome, experiment, sample space
- Types of events: Elementary, compound, dependent, independent, mutually exclusive, exhaustive, mutually inclusive
- Laws of probability: Additive and multiplicative rules - Baye's Theorem
- Probability trees
- Expected value, variance, standard deviation and coefficient of variation using frequency and probability

Probability distributions

- Discrete and continuous probability distributions (uniform, normal, binomial, poisson and exponential)
- Application of probability to business problems

3. Hypothesis testing and estimation

- Hypothesis tests on the mean (when population standard deviation is unknown)
- Hypothesis tests on proportions
- Hypothesis tests on the difference between means (independent samples)
- Hypothesis tests on the difference between means (matched pairs)

- Hypothesis tests on the difference between two proportions

4. Correlation and regression analysis

Correlation analysis

- Scatter diagrams
- Measures of correlation -product moment and rank correlation coefficients (Pearson and Spearman)

Regression analysis

- Assumptions of linear regression analysis
- Coefficient of determination, standard error of the estimate, standard error of the slope, t and F statistics
- Computer output of linear regression
- T-ratios and confidence interval of the coefficients
- Analysis of Variances (ANOVA)
- Simple and multiple linear regression analysis

5. Time series

- Definition of time series
- Components of time series (circular, seasonal, cyclical, irregular/ random, trend)
- Application of time series
- Methods of fitting trend: free hand, semi-averages, moving averages, least squares methods
- Models- additive and multiplicative models
- Measurement of seasonal variation using additive and multiplicative models
- Forecasting time series value using moving averages, ordinary least squares method and exponential smoothing
- Comparison and application of forecasts for different techniques

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6. Linear programming

- Definition of decision variables, objective function and constraints
- Assumptions of linear programming
- Solving linear programming using graphical method
- Solving linear programming using simplex method
- Sensitivity analysis and economic meaning of shadow prices in business situations
- Interpretation of computer assisted solutions
- Transportation and assignment problems

7. Decision theory

- Decision process
- Decision making environment - deterministic situation (certainty), analytical hierarchical approach (AHA), risk and uncertainty, stochastic situations (risk), situations of uncertainty
- Decision making under uncertainty - maximin, maximax, minimax regret, Hurwicz decision rule, Laplace decision rule
- Decision making under risk - expected monetary value, expected opportunity loss, minimising risk using coefficient of variation, expected value of perfect information
- Decision trees - sequential decision, expected value of sample information
- Limitations of expected monetary value criteria

8. Game theory

- Assumptions of game theory
- Zero sum games
- Pure strategy games (saddle point)
- Mixed strategy games (joint probability approach)
- Dominance, graphical reduction of a game
- Value of the game.
- Non zero sum games
- Limitations of game theory

9. Network planning and analysis

- Basic concepts - network, activity, event
- Activity sequencing and network diagram
- Critical path analysis (CPA)
- Float and its importance
- Crashing of activity/project completion time
- Project evaluation and review technique (PERT)
- Resource scheduling (levelling) and Gantt charts
- Limitations and advantages of CPA and PERT

10. Queuing theory

- Components/elements of a queue: arrival rate, service rate, departure, customer behaviour, service discipline,' finite and infinite queues, traffic intensity
- Elementary single server queuing systems
- Finite capacity queuing systems
- Multiple server queues

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11. Simulation

- Types of simulation
- Variables in a simulation model
- Construction of a simulation model
- Monte Carlo simulation
- Random numbers selection
- Simple queuing simulation: Single server, single channel "first come first served" (FCFS) model
- Application of simulation models

12. Current developments

- Role of advancement in information technology in solving quantitative analysis problems

13. Emerging issues and trends

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TOPIC 1

BASIC MATHEMATICAL TECHNIQUES

FUNCTIONS

Definitions

1. Variables

A variable is any quantity that assumes different values in a particular analysis.

Examples

i. Production costs
ii. Material costs
iii. Sales revenue

2. Constant

This is any quantity whose value remains unchanged in a particular analysis.

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Examples

- Fixed costs
- Rents
- Tuition fees

Note: In a given analysis there are two types of variables namely:

i. Independent variable/predictor variable
ii. Dependent / response variable

Independent variable is that which influences the value of the other variables in a particular analysis.

Dependent variable isthat whose value is influenced or changes when the value of other variables (independent) changes.

3. Functions

A function is a mathematical expression which describes a relationship between two or more variables in a particular analysis specifically one dependant variable and one or more independent variables.

Examples

If the price of the consumer product is Sh 40 per Kg, then the total sales revenue, S when Q units of the products are produced and sold is obtained as follows:

$S = 40q$

In this case S is the dependent variable, q the independent variable and 40 is a constant.
In terms of number of variables in a function, functions can be classified into the following categories:

i. Univariate function
ii. Bivariate function
iii. Multivariate function

A univariate function is that which involves two variables only, one dependent variable and one independent and is generally written as:
y = f (x) where y = dependent variable
x = independent variable
and f(x) = Function of x

Example of univariate function
The price of a house is dependent among other factors, on the size of the house. In functional form, this could be written as follows:
Price = f (size)
Where price is dependent variable
Size is independent variable

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A Bivariate function is that which involves three variables only, one dependent variable and two independent variables:

Example
A student's performance or grade in an examination could be dependent upon the following factors

i) IQ
ii) Time spent on studying in terms of Hours, H

In functional form, this is written as follows:
Grade = f (IQ,H)
Grade is dependent variables
IQ, H Are independent variables

Multivariable function is that function which involves four or more variables, one dependent variable and three or more independent variables.

Example
The price of a house depends on the following factors:

i) Size
ii) Location
iii) Security
iv) Nature of the house

In functional form this is written as follows;
Price = f (size, location, security, nature of the house)
Where price – is dependent variable
Size, location, security, nature of the house are independent variables.

Graph of a function

A graph is a visual method of illustrating the behaviour of a particular function. It is easy to see from a graph how as x changes, the value of f(x) is changing.
The graph is thus much easier to understand and interpret than a table of values. For example by looking at a graph we can tell whether f(x) is increasing or decreasing as x increases or decreases.
We can also tell whether the rate of change is slow or fast. Maximum and minimum values of the function can be seen at a glance. For particular values of x, it is easy to read the values of f(x) and vice versa i.e. graphs can be used for estimation purposes
Different functions create different shaped graphs and it is useful knowing the shapes of some of the most commonly encountered functions. Various types of equations such as linear, quadratic, trigonometric, exponential equations can be solved using graphical methods.

TYPES OF FUNCTIONS IN BUSINESS

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These include

1. Linear functions
2. Quadratic functions. Polynominals
3. Cubic functions
4. Exponential functions
5. Logarithmic functions
6. Hybrid functions

1. Linear functions

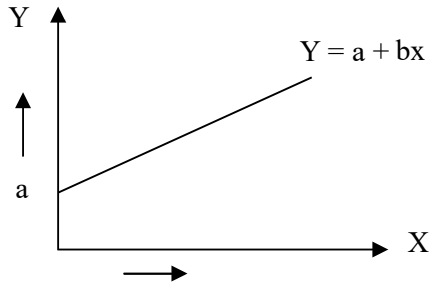
A linear function is a first degree polynomial function that takes the following general form.
y= a +bx
Where y is dependent variable
x is independent variable
a is y-intercept or the value of y when x = 0
b is the slope or gradient or the amount by which y changes in value when x changes by a unit

Properties/characteristics of linear functions

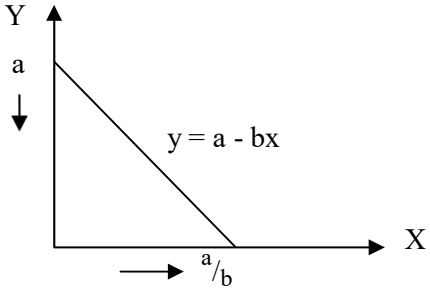
When plotted on an x-y coordinate system, the result is a straight line whose general direction is dependent on the slope, b of the function.

Specifically, if

a) Slope, b > 0 (+ve)



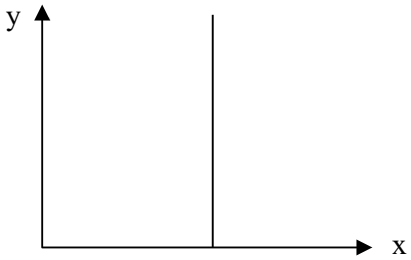
b) Slope, b < 0 (negative)



c) Slope, b = 0



d) Slope, b is undefined or b = ∞



2. A linear equation has only one root or solution

3. A linear function is completely specified if either

a) Two points or

b) One point and the slope of the function are given.

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ILLUSTRATIONS

Properties of linear functions or equations

1. Find the equation of the straight line which passes through the two point given as :
   When x = 1, y = 8
   x = -2, y = 4
2. Find the expression for the linear function which passes through the two points given as:
   (x,y) = (1,1)
   (x,y) = (-2,6)
3. Find the equation of the straight line with a slope of -5 which passes through the point (3,5)

SOLUTIONS

1. Let the linear equation be y = a +bx
   i) $8 = a + b$ $8 = a + b$ (i)
   ii) $4 = a + -2b$ $4 = a - 2b$ (ii)
   $4 = a = 2b$ $4 = 3b$ $b = 4/3$
   Substitute b in (i) $8 = a + \frac{4}{3}$
   $a = \frac{8}{1}$ $\frac{y}{3} = \frac{24-4}{3} = \frac{20}{3}$
   Hence the equation of the straight line is:
   $y = \frac{20}{3}$ $\frac{4}{3}x$
   $3y = 20 + 4x$
2. Let the linear equation be y = a + bx
   Let the linear equation be y = a+bx
   1 = a+b. .....(i)

   $\frac{-6=a-2b}{-5=3b}$ ∴ b = $5/3$

   $1 = a$ $5/3$

   $a = \frac{1}{1} + \frac{5}{3} = \frac{3+5}{3} = \frac{8}{3}$
   ∴ The equation will be

   $y = \frac{8}{3}$ $\frac{5x}{3}$

   <u>3y = 8 – 5x</u>

3. Let the equation be y = a+bx
   b= - 5, x = 3, y=5
   5 = a – 5x
   5 = a – 15

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$a = 5 + 15 = 20$ Hence the equation will be $y = 20 – 5x$.

EQUATIONS

An equation, in a mathematical context, is generally understood to mean a mathematical statement that asserts the equality of two expressions. In modern notation, this is written by placing the expressions on either side of an equals sign (=), for example + 3 = 5 asserts that *x*+3 is equal to 5. The = symbol was invented by Robert Recorde (1510–1558), who considered that nothing could be more equal than parallel straight lines with the same length.

Centuries ago, the word "equation" frequently meant what we now usually call "correction" or "adjustment". This meaning is still occasionally found, especially in names which were originally given long ago. The "equation of time", for example, is a correction that must be applied to the reading of a sundial in order to obtain mean time, as would be shown by a clock.

Equations often express relationships between given quantities, the *knowns*, and quantities yet to be determined, the *unknowns*. By convention, unknowns are denoted by letters at the end of the alphabet, *x*, *y*, *z*, *w*, …, while knowns are denoted by letters at the beginning, *a*, *b*, *c*, *d*, … . The process of expressing the unknowns in terms of the knowns is called solving the equation. In an equation with a single unknown, a value of that unknown for which the equation is true is called a *solution* or *root* or zero of the equation. In a set of simultaneous equations, or *system of equations*, multiple equations are given with multiple unknowns. A *solution to the system* is an assignment of values to all the unknowns so that all of the equations are true.

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Equations are classified into two main groups linear equations and non linear equations. Examples of linear equations are

$x + 13 = 15$

$7x + 6 = 0$

Non linear equations includes unknowns having higher degrees transcendentalfuntions etc.

$5x^2 + 3x + 7 = 0$ (quadratic equation)

$2x^3 + 4x^2 + 3x + 8 = 0$ (cubic equation)

The solution of equations or the values of the variables for which the equations hold is called the roots of the equation or the solution set.

Solution of Linear Equation

Supposing M, N, and P are expressions that may or may not involve variables, then the following constitute some rules which will be useful in the solution of linear equations

Rule 1: Additional rule

If $M = N$ then $M + P = N + P$

Rule 2: Subtraction rule

If $M = N$, Then $M – P = N – P$

Rule 3: multiplication rule

If $M = N$ and $P \neq O$ then $M \times P = N \times P$

Rule 4: Division rule

If P x M = N and P ≠ O
And N/P = Q Q being a rational number then
M = N/P

Example

i. Solve 3x + 4 = - 8

ii. Solve $\frac{y}{3} = -4$

Solutions

i. $3x + 4 = -8$

$3x + 4 - 4 = -8 - 4$ (by subtraction rule)

$3x = -12$ (simplifying)

$\frac{3x}{3} = -\frac{12}{3}$ (by division rule)

$x = -4$ (simplifying)

ii. $3 \times \frac{y}{3} = -4 \times 3$

$y = -12$ (simplifying)

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Solution of quadratic equations

Suppose that we have an equation given as follows

$ax^2 + bx + c = 0$

Where a, b and c are constants, and a≠ 0. Such an equation is referred to as the general quadratic equation in x. if b = 0, then we have

$ax^2 + c = 0$

which is a pure quadratic equation

There are 4 general methods for solving quadratic equations; solution by **factorization**, solution by **completing the square,** solution by the **quadratic formula** and solution using graphical method.

1. Solution by Factorization

The following are the general steps commonly used in solving quadratic equations by factorization

(i) Set the given quadratic equation to zero

(ii) Transform it into the product of two linear factors

In step (ii) to find the factors find the a×c and then factors of a×c which add up to b. If these factors are p and q, replace bx by px + qx then complete the factorization.

(iii) Set each of the two linear factors equal to zero (using a null factor law).

(iv) Find the roots of the resulting two linear equations

Example

Solve the following equation by factorization

ii. $6x^2 = 18x$

iii. $15x^2 + 16x = 15$

Solutions

i. $6x^2 = 18x$

$6x^2 - 18x = 0$ .....(step 1)

$6x(x - 3) = 0$ .....(step 2)

$6x = 0$ .....(step 3)

and $x - 3 = 0$

$\therefore x = 0$ or $x = 3$ .....(by step 4)

ii. $15x^2 + 16x = 15$

$15x^2 + 16x - 15 = 0$.....(step 1)

$(5x - 3)(3x + 5) = 0$.....(step 2)

$(5x - 3) = 0\}$ Step 3

$\{3x + 5 = 0\}$

$\therefore x = -\frac{5}{3}$ or $+\frac{3}{5}$ .....(step 4)

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2.Solution by Completing the Square

The process of completing the square involves the construction of a perfect square from the members of the equation which contains the variable of the equation.

Consider the equation – $9x^2 - bx = 0$

The method of completing the square will involve the following steps

i. Make the coefficient of x^2 unity by dividing by a whenever it is not one.

ii. Add the square of ½ the coefficient of x to both sides of the equal sign. Theleft hand side is now a perfect square

iii. Factorize the perfect square on the left hand side.

iv. Find the square root of both sides

v. Solve for x

Example

Solve by completing the square.

i. $3x^2 = 9x$

ii. $2x^2 + 3x + 1 = 0$

Solutions

i. $3x^2 = 9x$ or

$(3x^2 - 9x = 0)$

$x^2 - 3x = 0$.....(Step 1)

$x^2 - 3x + \left(-\frac{3}{2}\right)^2 = \left(-\frac{3}{2}\right)^2$.....(Step 2)

$\left(x - \frac{3}{2}\right)^2 = \frac{9}{4}$.....(Step 3)

$x - 3 = \pm\sqrt{\frac{9}{4}}$ ..... (Step 4)

$\therefore \quad x = \left(\frac{3}{2} \pm \frac{3}{2}\right)$

$= \frac{3+3}{2}$ or $\frac{3}{2} - \frac{3}{2} =$

(= 3 or 0)

ii. $2x^2 + 3x + 1 = 0$ or $(2x^2 + 3x = -1)$

$x^2 + \frac{3x}{2} = -\frac{1}{2}$..... (Step 1)

$X^2 + \frac{3x}{2} + \left(\frac{3}{4}\right)^2 = \left(\frac{3}{4}\right)^2 - \frac{1}{2}$ …… (Step 2)

$\left(x + \frac{3}{4}\right)^2 = \frac{1}{16}$..... (Step 3)

$x + \frac{3}{4} = \pm\sqrt{\frac{1}{16}}$

$x = \frac{3}{4} \pm \frac{1}{4}$

$-\frac{3}{4} + \frac{1}{4}$ or $-\frac{3}{4} - \frac{1}{4}$

$x = -\frac{1}{2}$ or $x = -1$

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3. Solution by Quadratic Formula

Consider the general quadratic equation

$ax^2 + bx + c = 0$ where $a \neq 0$

The roots of the equation are obtained by the following formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example

Solve for x by formula

$5x^2 + 2x - 3 = 0$

Solution

$a = 5, b = 2, c = -3$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(5)(-3)}}{2(5)}$$

$$x = \frac{3}{5} \text{ or } -1$$

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4. Graphical Method .

Given the general equation $ax^2 + bx + c = 0$, draw the graph of $y = ax^2 + bx + c$. The x – intercepts give the solution to the equation $ax^2 + bx + c = 0$.

Example

Solve for x in $x^2 - 5x + 6 = 0$

Using graphical approach use $2 \leq x \leq 5$

Inequalities

An inequality or inequation is an expression involving an inequality sign (i.e. >, <, ≤, ≥, i.e. greater than, less than, less than or equal to, greater than or equal to) The following are some examples of inequations in variable x.

$3x + 3 > 5$

$x^2 - 2x - 12 < 0$

The first is an example of linear inequation and the second is an example of a quadratic inequation.

Solutions of inequations

The solutions sets of inequations frequently contain many elements. In a number of cases they contain infinite elements.

Example

Solve the following inequation

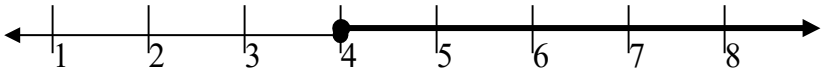
$x - 2 > 2$; $x \subset w$ (where x is a subset of w)

Solution

$x - 2 > 2$ so $x - 2 + 2 > 2 + 2$

Thus, $x>4$

The solution set is infinite, being all the elements in w greater than 4. This can be illustrated using the following number line.



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Example

Solve

$3x - 7 < -13$;

Solution

$3x - 7 < -13$

$\Rightarrow 3x - 7 + 7 < -13 + 7$

$\Rightarrow 3x < -6$

$\frac{3x}{3} < \frac{-6}{3}$

$x < -2$

This answer can be illustrated on the number line as shown below;-



Linear inequation in two variables: relations

An expression of the form

$$y \geq 2x - 1$$

is technically called a relation. It corresponds to a function, but different from it in that, corresponding to each value of the independent variable x, there is more than one value of the dependent variable y

Relations can be successfully presented graphically and are of major importance in linear programming.

Linear simultaneous equations:

Two or more equations will form a system of linear simultaneous equations if such equations be linear in the same two or more variables.

For instance, the following systems of the two equations is simultaneous in the two variables x and y.

$$2x + 6y = 23$$
$$4x + 7y = 10$$

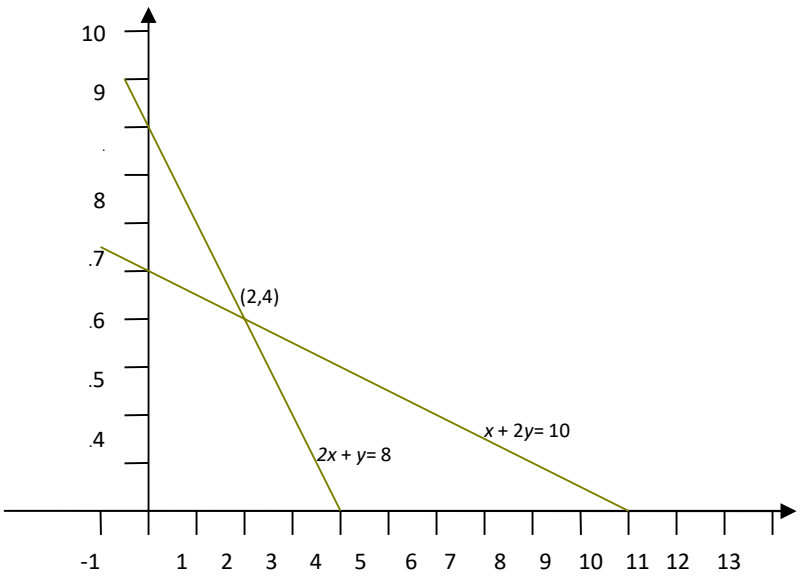
The solution of a system of linear simultaneous equations is a set of values of the variables which simultaneously satisfy all the equations of the system.

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Solution techniques

a) The graphical technique

The graphical technique of solving a system of linear equations consists of drawing the graphs of the equations of the system on the same rectangular coordinate system. The coordinates of the point of intersection of the lines from equations of the system would then be the solution.



Example

The above figure illustrates:

Solution by graphical method of two equations

$2x + y = 8$
$x + 2y = 10$

The system has a unique solution (2, 4) represented by the point of intersection of the two lines.

b) The elimination technique

This method requires that each variable be eliminated in turn by making the absolute value of its coefficients equal in the equations of the system and then adding or subtracting the equations. Making the absolute values of the coefficients equal necessitates the multiplication of each equation by an appropriate numerical factor.

Consider the system of two equations (i) and (ii) below

$2x - 3y = 8$ .....(i).
$3x + 4y = -5$.....(ii).

Step 1

Multiply (i) by 3
$6x - 9y = 24$.....(iii).
Multiply (ii) By 2
$6x + 8y = -10$ .....(iv).
Subtract iii from iv.
$17y = -34$ .....(v).
$\therefore \quad y = -2$

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Step 2

Multiply (i) by 4
$8x - 12y = 32$.....(vi)

Multiply (ii) by 3
$9x + 12y = -15$ .....(vii)
Add vi to vii
$17x = 17$.....(viii)
$\therefore \quad x = 1$
Thus $x = 1, y = -2$ i.e. {1,-2}

c) The substitution technique

To illustrate this technique, consider the system of two equations (i). and (ii) reproduced below
$2x - 3y = 8$ …….. (i).
$3x + 4y = -5$ …… (ii).

The solution of this system can be obtained by

a) Solving one of the equations for one variable in terms of the other variable;

b) Substituting this value into the other equation(s) thereby obtaining an equation with one unknown only
c) Solving this equation for its single variable finally
d) Substituting this value into any one of the two original equations so as to obtain the value of the second variable

Step 1

Solve equation (i) for variable x in terms of y

$2x - 3y = 8$

$x = 4 + 3/2\ y$ (iii)

Step 2

Substitute this value of x into equation (ii). And obtain an equation in y only

$3x + 4y = -5$

$3\ (4 + 3/2\ y) + 4y = -5$

$8\ ½\ y = -\ 17$ ……. (iv)

Step 3

Solve the equation (iv). For y

$8½y = -17$

$y = -2$

Step 4

Substitute this value of y into equation (i) or (iii) and obtain the value of x

$2x - 3y = 8$

$2x - 3(-2) = 8$

$x = 1$

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Example

Solve the following by substitution method

$2x + y = 8$

$3x - 2y = -2$

Solution

Solve the first equation for y

$y = 8 - 2x$

Substitute this value of y into the second equation and solve for x

$3x - 2y = -2$

$3x - 2\ (8-2x) = -2$

$x = 2$

Substitute this value of x into either the first or the second original equation and solve for y

$2x + y = 8$

$(2)\ (2) + y = 8$

$y = 4$

d) Using matrix algebra (either Cramers rule or Matrix inverse method)

This method will be discussed later under matrices.

APPLICATIONS OF LINEAR FUNCTIONS IN BUSINESS

Application areas are:

1. Computations of salaries / wages and commissions
2. Fixed asset accounting
3. Demand, supply and market equilibrium analysis
4. Cost-volume-profit (C-V-P analysis)and break even analysis

1. Computations of salaries / wages and commissions

ILLUSTRATION

A salesman's daily wages is composed of a fixed amount and a variable component which is dependent on the number of ice cream units sold. He finds that when he sells 10 units on a given day, he earns Shs 600 whereas when he doubles his sales, his earnings increase by only Ksh 100.

Determine

i) Fixed daily earnings
ii) Level of commission per unit sold and hence
iii) What are the salesman's earnings if he sells 30 units?
iv) On a given day the salesman is determined to earn Kshs 3500. Suppose on the previous day he had guaranteed order to achieve his target earnings, how many units must he sell over 20 units to achieve his target?

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SOLUTION

i) Daily earnings = Fixed daily earning + variable earnings

$E = a+bx$

Where E = daily earnings (total)

a = Fixed daily earnings

b = level of commission or earning per unit ice cream

x = Number of ice cream units sold

bx = variable earnings

When E=600, x = 10

$600 = a + 10b.$ ..... (i)

$\underline{700 = a + 20b.}$ ..... (ii)

$-100 = -10b$

$\underline{b = 10}$

$600 = a + 10(10)$

$a = 600 - 100 = 500$

Therefore $E = 500 + 10x$

Fixed daily earnings = 500

ii) Level of commission per unit sold is Sh. 10

iii) When x = 30

E = 500 + 10x

= 500 + 10 (30) = Sh 800

iv) Let x be the number of ice cream needed.

E = 500 + 10x

3500 = 500 + 10x

10x = 3000

x = 300

He must sell 300 – 20 = 280 units to meet his target.

2. Supply /Demand Relationship

Suppose a certain commodity has linear demand and supply functions going through the following points.

1) When P = Shs 7,500 q = 1,000 Units
   and when P = Shs 4,625 q = 750 Units

2) When p = Shs 2525 q = 100 Units
   and when P = Shs 1525 q = 200 Units

a) Obtain the linear function that go through the points given in 1 and 2 above and clearly explain which the supply|function and which the demand function. Assume this is a normal commodity.

b) Explain what is meant by market equilibrium and obtain the same for the above. Indicate your results on a graphical sketch.

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SOLUTION

a) Linear dd or ss function is

P = a +bq

Function 1

7500 = a + 100 b.....(i)

4625 = a + 750b.....(ii)

Function 2

2522 = a + 100b.....(i)

1525 = a + 200b.....(ii)

Solution

Function 1

7500 = a + 1000 b

$$\underline{4625 = a + 750b}$$

$$2875 = 250\,b$$

$\therefore b = \frac{2875}{250} = 115$

a = 7500 – 1000 (11.5)

= 7500 – 11500

= - 4000

Hence P = - 4000 + 11.5q.....This is a supply function due to positive slope.

Function 2

P = a + bq

2525 = a + 100b

1525 = a + 200b

1000 = - 100b

$$b = \frac{1000}{-100} = -10$$

Substitute in equation

a = 2525 – 100b

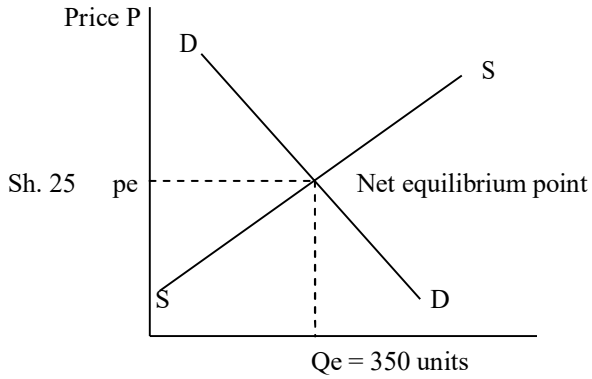
= 2525 – 100 (-10)

= 2525 + 1000 = 3525

Hence P = 3525 – 10q ..... Demand function due to a negative slope.

b) The market equilibrium for a commodity is the point at which Qs = Qd so that the equilibrium supply and demand prices are also equal.

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Hence P = - 4000 + 11.5 q.....Supply function due = +ve slope.

There at equilibrium, -400 + 11.5q = 3525 – 10q

- 4000 – 3525 = 10q – 11.5q

$$\therefore q = \frac{-7525}{-21.5} = \underline{\underline{350 \text{ Units}}}$$

Using supply Function substitute

P=-4000+11.5(350)

= -4000+4025

P=Sh.25

3. Accounting for fixed assets – Straight Line Depreciation Method

ILLUSTRATION

Sakuz transporters depreciate its fleet of trucks using a straight line method. The current accounting year is coming to an end and external auditors are examining the books of accounts. However they cannot get complete records concerning a truck which was acquired 3 years ago. Its current book value is Kshs 1,800,000 while its purchase cost was Shs. 4,200,000. This type of truck is usually disposed off after 5 years.

a) Determine the linear function y = a + bt which relates the book value y and time in years t. Interpret a and b.
b) What is the book value at the end of the 2nd year of the truck?
c) Determine the disposal value of the truck stating any assumptions you may make.

SOLUTION

a)

t	**Value (sh)**
3 years	1.8 Million
-	4.2 Million

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V = a + bt

1.8 = a+3b ..... (i)

4.2 = a + 0b ..... (ii)

∴ a = Sh 4.2 Million..... Purchase (historical) cost

Substitute a in equation (i)

1.8 = 4.2 + 3b $b = \frac{-2.4}{3} = -0.8$

3b = 1.8 – 4.2

3b = -2.4 b = (sh 800,000) ..... annual depreciation rate

∴ V = 4.2 – 0.8t

b) What is the book value at the end of 2nd year of the truck?

V = 4.2 – 0.8t

= 4.2 – 0.8 (2)

= 4.2 – 1.6

= Shs 2.6 Million

c) Disposal value of the truck

V = 4.2 – 0.8t

= 4.2 – 0.8(5)

= 0.2 Million shillings or shs.200,000

d) Find the time at which the trucks book is shs. 2 million.

$V = 4.2 - 0.8t$

$2 = 4.2 - 0.8t$

$0.8t = 4.2 - 2$

$0.8t = 2.2$

$t = \frac{2.2}{0.8} = 2.75$ years $\quad = 2$ years, 9 months

4. Cost – Volume – Profit Analysis (C-V-P) / Profit Planning

Profit = f (Prices, costs, volume/output)

Problem

How management should manipulate the profit determining factors so as to maximise this profit?

For linear certainity C-V-P model or analysis, we make the following assumptions:

A model is a representation of a reality or some aspects of reality e.g. a toy car, a diagram or a map, a graph, an equation

Assumptions are like requirements or conditions.

They include:

1. The revenue, cost and profit functions are linear with respect to the level of activity or output or volume.
2. Units selling price is constant e.g. no discounts, the market is perfect.
3. Unit variable cost is constant
4. Fixed cost does not change
5. All cost can be classified as either fixed or variable i.e. there is no semi variable costs.
6. The only factor which influences revenue, costs and profits is the level of activity (output/volume).
7. All output is sold (in relevant period)
8. There are no demand nor other constraints or restrictions.
9. All factors under consideration (prices, demand, costs) are known with certainty.
10. The firm produces a single product (single product C-V-P)
11. There are no taxes.

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Equation form of the model – Sales in Physical units, x

Let x represents sales in units

R – Represents sales revenue in shillings

P – Represents selling price

The equation relating them i.e. X, R, P

Note 1 is $R = Px$

v = Unit variable cost

V = Total variable cost

f = Fixed cost

C = Total cost = V + f

Note 2 C = Vx + f

Let π represent Profit = R – C

Note 3

π = Px – (Vx + f)

π = Px – Vx – f

π = (P-V)x – f

Generally, P> V

P – V = Unit Contribution margin (Cm)

∴ π = Cmx – f

Quadratic Function (QF)

This is the second degree polynomial

General form

$y = a + b_1x + b_2x^2$ where $b_2 \neq 0$

Properties / Characteristic of Quadratic Function

i) It can cross the x-axis a maximum of two times i.e. it has two roots or two solutions.
ii) It has a single turning point
iii) It is fully defined once any three points which lie on the curve are provided.

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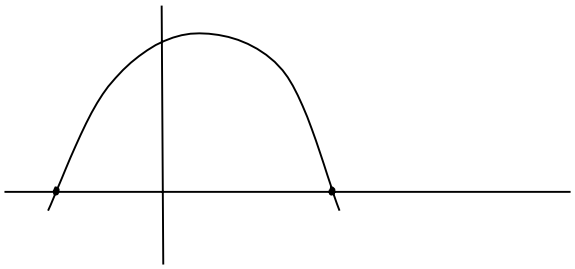
For property No (i)

Recall the quadratic formula

Then $x = \frac{-b \pm \sqrt{b^2-4aC}}{2a}$

Quadratic Function sketches

1)

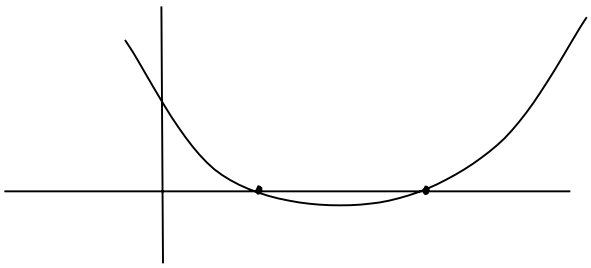


For (1) and (2)

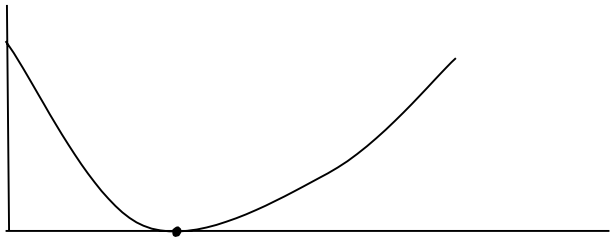
- Two real roots (solutions)

- (1) and (2) $b^2 > 4ac$

2)



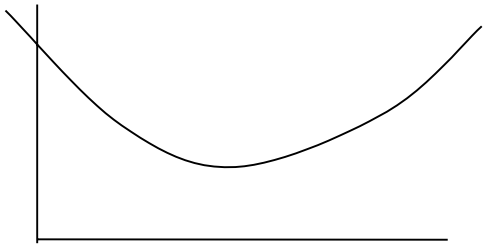
3)



For (3)

- $b^2 = 4ac$
- Two identical or coincidental real roots

4)



For (4)

- $b2 < 4ac$
- Two imaginary or complex Roots

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ILLUSTRATION

A revenue function is quadratic in nature. When x = 5, R = 50 whereas when x = 4, R = 48. Determine the revenue function.

SOLUTION

$R = a + b_1x + b_2x^2$

X	5	4
R	50	48

When x = 0, R = 0

$\therefore 0 = a + 0 + 0$

$a = 0$

$\underline{R = b_1x + b_2x^2}$

Equations

$50 = 5b_1 + 25b_2$

$48 = 4b_1 + 16b_2$

Matrix Format

$$\begin{bmatrix} 5 & 25 \\ 4 & 16 \end{bmatrix} \times \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 50 \\ 48 \end{bmatrix}$$

$$A \quad x \quad X \quad = B$$

$$x = \frac{A}{B} \quad = A^{-1}B$$

Crammer's rule

$$b_1 = \frac{\begin{vmatrix} 50 & 25 \\ 48 & 16 \end{vmatrix}}{\begin{vmatrix} 5 & 25 \\ 4 & 16 \end{vmatrix}} = \frac{(50 \times 16) - (48 \times 25)}{(5 \times 16) - (4 \times 25)} = \frac{-400}{-20} = \underline{\underline{20}}$$

$$b_2 = \frac{\begin{vmatrix} 5 & 50 \\ 4 & 16 \end{vmatrix}}{\begin{vmatrix} 5 & 25 \\ 4 & 16 \end{vmatrix}} = \frac{(5 \times 48) - (4 \times 50)}{-20} = \frac{40}{-20} = \underline{\underline{-2}}$$

Hence revenue function is $R(x) = 20x - 2x^2$

Cubic Function (CF)

This is the 3rd degree polynominal and the general structure is:

$y = a + b_1x + b_2x^2 + b_3x^3$ and $b_3 \neq 0$

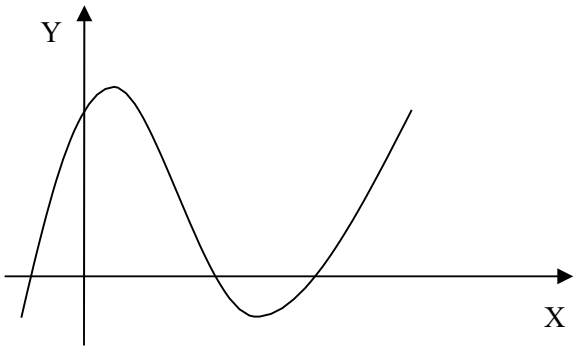
Properties of Cubic Function

1. It can cross the x – axis a maximum of 3 times i.e. it has 3 roots or 3 solutions.
2. It has either 2 turning points (one a maximum and the other a minimum) or point of inflexion.
3. It is completely described once any 4 points which lie on the curve are provided.

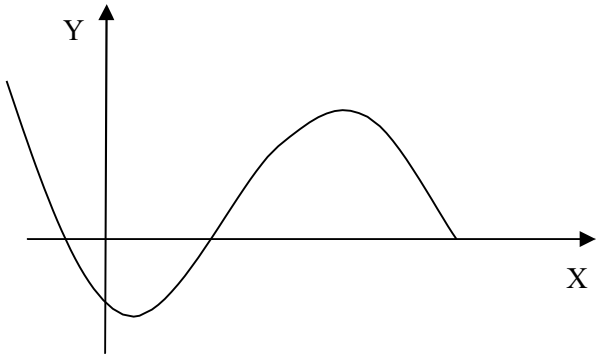
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Cubic Sketches

1)

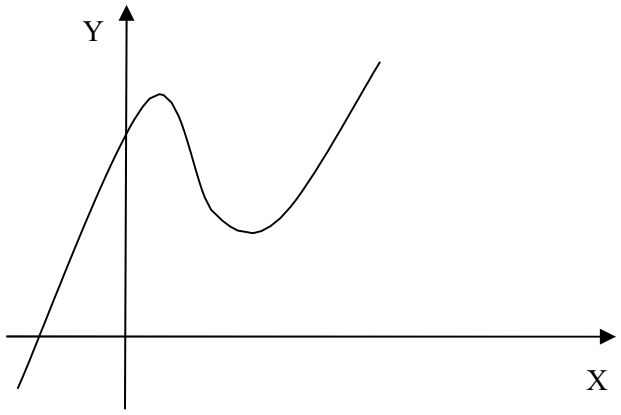


2)



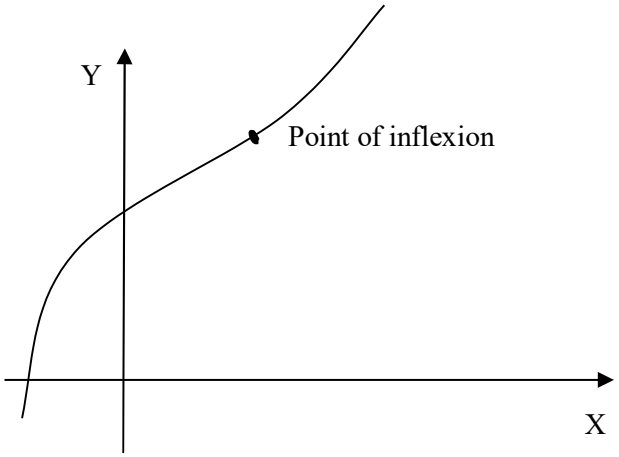
For (1) and (2)
They have 3 roots(solutions)

3)



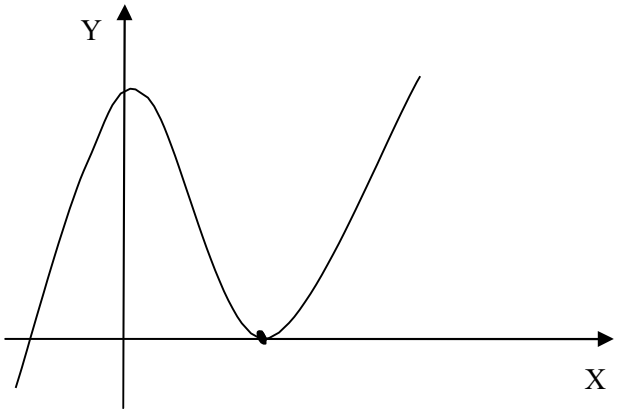
For (3) and (4)
- 1 Real
- 2 Imaginary (complex root)

4)



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5)



For (5)
- 3 real roots, 2 of which are identical or coincidental

ILLUSTRATION

A Management Accountant is studying the relationship between the number of units of output in a year and the total cost incurred for a given product. From the records of the firm, the following data was extracted;

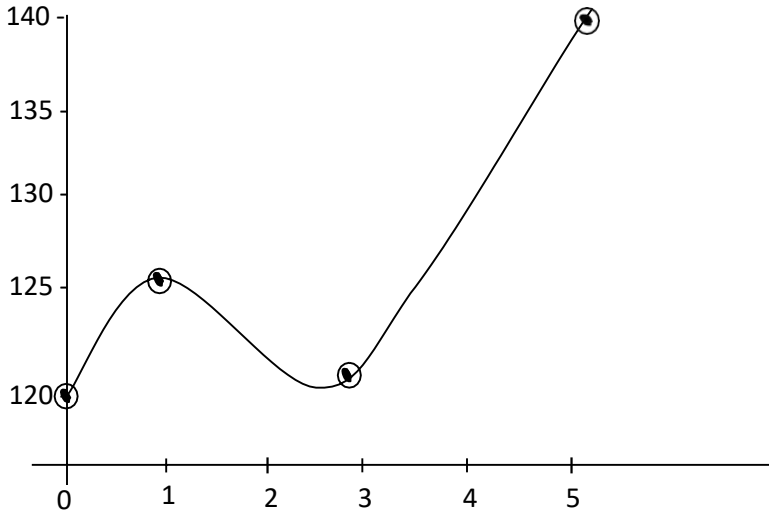
Output, Q	Total Cost, C
0	120
1	124
3	120
5	140

a) Determine the firm's fixed cost
b) Plot the above data on a graph and hence recommend the best functional form within the given range.
c) Without prejudice to your answer in (b) above, fit a 3rd degree polynomial to the data above i.e. of the form;
$C = a + b_1Q + b_2Q^2 + b_2Q^3$ and hence estimate the total cost if level of output equals eleven units.

SOLUTION

a) Fixed cost = TC when Q = 0
and hence f = Sh 120

b) Graphical Sketch



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Comment
The best functional form is cubic since it has 2 turning points.

c) Equations
$C = a + b_1Q + b_2Q^2 + b_3Q^3$
a = Fixed cost = Sh 120
1) $124 = 120a + b_1 + b_2 + b_3 \Rightarrow b_1 + b_2 + b_3 = 4\ldots$ .....(i)
2) $120 = 120 + 3b_1 + 9b_2 + 27b_3 \Rightarrow 3b_1 + 9b_2 + 27b_3 = 0$
$\Rightarrow b_1 + 3b_2 + 9b_3 = 0$.....(ii)
3) $140 = 120 + 5b_1 + 25b_2 + 125b_3 \Rightarrow 5b_1 + 25b_2 + 125b_3 = 20$
$\Rightarrow b_1 + 5b_2 + 25b_3 = 4\ldots$.....(iii)

Solving the 3 equations can be done using different methods. One is; subtract equation (i) from each of the other two equations to give;-

(ii) – (i) $2b_2 + 8b_3 = -4$ ..... (iv)

$4b_2 + 24b_3 = 0$.....(v)

Multiply (iv) by 2 and subtract from (v) this gives

$4b_3 + 24b_3 = 0$

$\underline{4b_2 + 16b_3 = -8}$

$8b_3 = 8$

$b_3 = 1$

Now $2b_2 + 8(1) = -4$

$2b_2 = -12$

$b_2 = -6$

but $b_1 + b_2 + b_3 = 4$

$b_1 - 6 + 1 = 4$

$b_1 - 5 = 4$

$b_1 = 9$

$\therefore b_1 = 9, b_2 = -6, b_3 = 1$

The equation is $C = 120 + 9Q - 6Q^2 + Q^3$

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Solution

$C = 120 + 9Q - 6Q^2 + Q^3$

For $Q = 11$

$C = 120 + 9(11) = 6(11)^2 + 11^3$

$= \underline{\text{Sh. 824}}$

The Multivariate Function

This is a function which has more than one independent variable e.g.

Sales = f (Its price, prices of substitutes and compliments, incomes)

ILLUSTRATION

The following information relates to Mulamba, a dealer in standard wooden tables:

Mulamba realized profits of Sh.12,000 from 7 tables, Sh.12,400 from 9 tables and Sh.11,300 from 4 tables sold respectively.

Mulamba has approached you for assistance in forecasting future profits. The profit function is believed to be quadratic in nature.

Required:

(i) Derive the profit function.

(ii) The profit maximizing output and the maximum profit.

Solution

(i) Profit function = $P=ax^2+bx+c$

$1200=a(7)^2 + b(7) + c \rightarrow 12000 = 49a+7b+c$

$12400 = a(9)^2 + b(9) + c \rightarrow 12400 =81a + 9b+ c$

$11300 = a(4)^2 + b(4) + c \rightarrow 11300 = 16a + 4b +c$

Solve the 3 equations simultaneously

$a= -\frac{20}{3}$ $b = \frac{920}{3}$ $c=10,180$

$\therefore P = 10180 + \frac{920S}{3} \frac{20S^2}{3}$ is the required profit function.

ii) Profit maximizing output

at maximum profit $\frac{dp}{dS} = 0$

FOC $\frac{920}{3} \frac{40S}{3} = 0 \quad x = 23$

$\frac{40S}{3} = \frac{920}{2}$

x = 23

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Maximum profit

$P = 10180 + \frac{920x}{3} - \frac{20x^2}{3}$

$= 10,180 + \frac{920(23)}{3} - \frac{20(23)^2}{3}$

= Sh13,706.67

The Exponential Function

An exponential function has at a least one term for the independent variable as part of an exponent or power e.g. $y = 10^{2x}$

=> 10 is called the base at the function

-> 2x is the exponent or power

Important classes of exponential functions in business are those which have naturally occurring constant, e as their base i.e.

$y = ae^{kx}$

Where a, e and k are constants and specifically, e is a specific constant associated with continuous growth or continuous decay.

$e = \text{Lim } (1 + \frac{1}{x})^x$ as $x \rightarrow \infty$

Lim means Limit and means approaches or tends to.

Approximations of e

x = 1 $e\ (1 + \frac{1}{1})^{1} = 2$

x = 2 $e\ (1 + \frac{1}{2})^{2} = 2.25$

x = 10 $e\ (1 + \frac{1}{10})^{10} = 2.5937$

x = 100 $e\ (1 + \frac{1}{100})^{100} = 2.7048$

x = 1000 $e\ (1 + \frac{1}{1000})^{1000} = 2.7169$

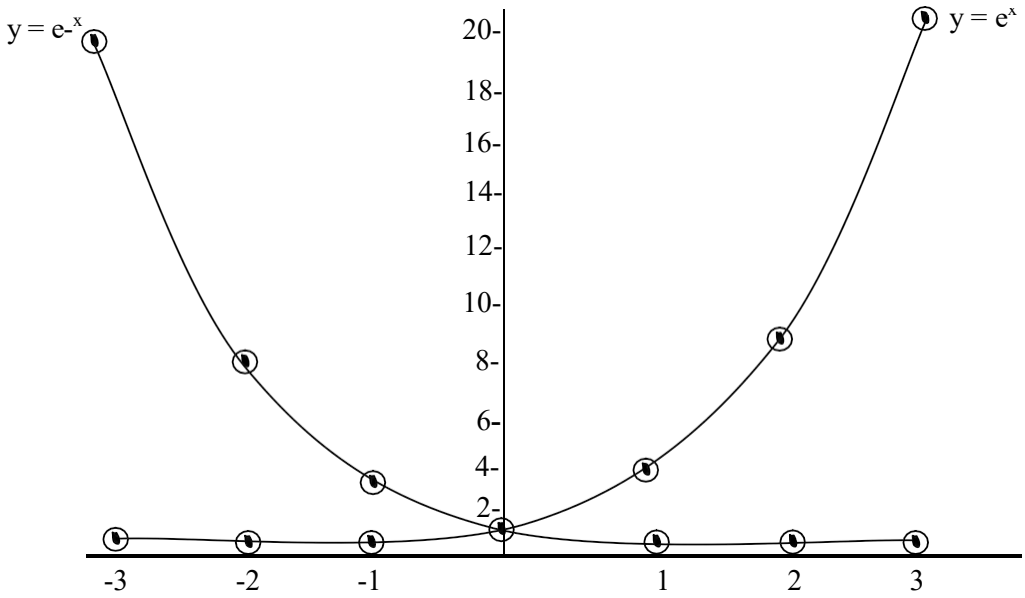
ILLUSTRATION

Sketch the following 2 functions on the same graph

(1) $y = e^{S}$ (2) $y = e^{-S}$

X	-3	-2	-1	0	1	2	3
e^{x}	0.05	0.14	0.37	1	2.72	7.39	20.09
e^{-x}	20.09	7.39	2.72	1	0.37	0.14	0.05

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Notes

1. For most application, exponential functions have either time or space as the independent variable e.g. population level= f (time)
   Population level = f (area of distance covered)
2. Equal changes in the independent variable for an exponential function results in constant % change in the value of the dependent variables; this constant % change is the coefficient of the independent variable.

For $y = ae^{kx}$

=> a is the value of y when x = 0 $\quad a > 0$

e.g Initial population, initial (purchase cost of an asset. .....)

=> k is the constant % change per unit of x if k is positive then, it is a growth function but if k is negative, it is a decay function.

e.g. $y = 10e^{0.2S}$

Initial value of y is 10

Growth rate is 20% per unit of x e.g. per annum

$y = 25e^{-0.05S}$

Initial value of y is 25

Rate of decay/decrease is 5% per unit of x ..... e.g. per m^3

Logarithmic Functions

A logarithmic is a power which a base must be raised in order to give a certain number i.e. a logarithmic is an exponent.

e.g. $2^3 = 8$

This is equivalent to $\log_2 8 = 3$ i.e 3 is the log to base 2 of the number 8

Equivalent Exponential and Logarithmic Forms

Exp. Form	**Log Form**
$10^2 = 100$	$\log_{10} 100 = 2$
$5^3 = 125$	$\log_5 125 = 3$
$4^3 = 64$	$\log_4 64 = 3$

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Although logarithms can be taken to any base, the most commonly used bases are base 10 and base 2.

Further, base 10 logarithms are denoted "log" while base e logarithms are denoted "ln" –> e^x (also known as natural logarithms e.g. log 100 = 2. And ln 100 = 4.605 $\quad e^{4.605} = 100$

Properties of Logarithms

1. Log uv = Log u + Log v
   e.g Log 100 x 1000 = Log 100 + Log 1000
   = 2 + 3 = 5
   = Log 100000
2. Log $\frac{u}{v}$ = Log u – Log v
   e.g. Log $\frac{1000}{100}$
   = log 1000 – log 100
   = 3 – 2 = 1 = log 10
3. Log u^n = n log u
   e.g. Log10^2 = 2log10
   = 2 x 1 = 2

$= \log 100$

4. $\text{Log}_b b = 1$ Since $b^1 = b$

5. $\text{Log}_b 1 = 0$ Since $b^0 = 1$

6. $\text{Log}_b b^x = x$ $\text{Log}_b{}^b = x$
   $=$ since $\text{Log}_b b = 1$
   $\therefore \log_b b^x = x$

Applications of exponential and logarithmic functions

1. Growth processes
   i) Population growth is exponential
   ii) Spread of a contagious disease
   iii) Growth in value of certain assets e.g. land
   iv) Rate of inflation
2. Decay process
   i) Asset depreciation e.g. computers and electronics generally.
   ii) Decrease in purchasing power of the shilling.
   iii) Decline in the rate of incidence of certain disease such as polio as medical research and technology advances.
   iv) Decrease in the value of a share in the stock exchange as negative sentiments concerning it spread etc.

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APPLICATION PROBLEMS IN COST, REVENUE AND PROFIT

PROBLEM 1

Super Toys Ltd. (STL) manufactures and sells toys. "Super car" is one of their popular models. The marketing department has estimated the demand function for the model to be linear. If the price was fixed at Sh. 570, the daily sales of the model would be 400 toys, whereas if the price was increased to Sh. 820, the daily sales would drop to 200 toys

Data from the production department indicate that the incremental cost of producing q toys of the model is given by the equation;

$C(q) = 2q - 570$

and that the daily fixed cost is Sh. 1,100.

Required:

(i) The revenue functions if q toys are sold.
(ii) The total cost function.
(iii) The daily break-even number of toys

(iv) The point elasticity of demand when the demand is 110 toys. Interpret the economic meaning of your result.

SOLUTION

(a) (i) Demand slope $= \frac{570 - 820}{400 - 200} = -1.25$

Equation of demand

$\frac{P - 570}{q - 400} = -1.25$

$P = -1.25 (q - 400) + 570 = -1.25q + 1070$

$\therefore$ Revenue, $R = (1070 - 1.25q)\, q = 1070q - 1.25q^2$

(ii) Total cost, $TC = \int (2q - 570)\, dq$

$= q^2 - 570q + C$

C = fixed cost = 1,100

$TC = q^2 - 570q + 1,100$

(iii) Profit, $\pi = 1070q - 1.25q^2 - q^2 + 570q - 1100$

$= -2.25q^2 + 1640q - 1100$

At B.E.P, profit = 0 $\quad -2.25q^2 + 1640q - 1100 = 0$

$q = \frac{-1640 \pm \sqrt{1640^2 - 4(-2.25)(-1100)}}{2(-2.25)}$

$q = 0.67$ or $q = 728$

(iv) $P = 10\ 70 - 1.25q$

$\frac{dp}{dq} = -1.25 \Rightarrow \frac{dq}{dp} = \frac{1}{-1.25} = -0.8$

When $q = 110$, $p = 932.5$

Point of elasticity, $E = \frac{p}{q} \times \frac{dp}{dp}$

$= \frac{932.5}{110} \times -0.8$

$= -6.78$

Demand is elastic

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PROBLEM 2

Puda Development Company (PDC) is a small real estate developer operating in the Eastlands Valley. It has seven permanent employees whose monthly salaries are given below:

Employee	Monthly salary (Sh)
Managing Director	100,000
Manager, Development	60,000
Manager, Marketing	45,000
Project Manager	55,000
Finance Manager	40,000
Office Manager	30,000
Receptionist	20,000

PDC leases a building for Sh. 20,000 per month. The cost of suppliers, utilities and leased equipment runs for another Sh. 30,000 per month. PDC builds only one style house in the valley. Land for each house costs. Sh. 550,000 and lumber, supplies and others run for another Sh. 280,000 per house. Total labour costs amount to Sh. 200,000 per house. The one sales representative of PDC is paid a commission of Sh. 20,000 on the sale of each house. The selling price of the house is Sh. 1,150,000.

Required:

i) Identify all the costs and deduce the marginal revenue and marginal cost for each house.
ii) Determine the monthly cost function; C(x), revenue function; R(x) and the profit function; P(x)
iii) Determine the break-even point for monthly sales of the houses.
iv) Determine the monthly profit if 12 houses per month are build and sold.

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SOLUTION

(i) Salaries (Sh '000):

100 + 60 + 45 + 55 + 40 + 30 + 20
= 350
Office lease and supply costs= 20 + 30= 50
Fixed cost= 350,000 + 50,000= 400,000

- Land, Material, labour and sales commission per house is the variable or marginal cost for the house. It is given as:
  = 550,000 + 280,000 + 200,000 + 20,000
  = 1,050,000
- The selling price of Sh. 1,150,000 is the marginal revenue per house.

(ii) Total cost function;

TC = VC + FC = 1,050,000x + 400,000
= 1,050,000 + 400,000 = 1,450,000

TR = 1,150,000 (x)
= 1,150,000x

Profit = TR – TC
= 1,150,000x – 1,050,000x – 400,000
= 100,000x – 400,000

(iii) Break even in number of houses;

At BEP TR = TC … substituting
⇒ 1,150,000x = 1,050,000x + 400,000
⇒ 100,000x - 400,000
⇒ x = 4 houses

(iv) The profit if 12 houses are built and sold is computed as equal to

= 100,000 x (12) – 400,000
= 1,200,000 – 400,000
= Sh. 800,000.

PROBLEM 3

The following information relates to M. Mutuma, a dealer in standard wooden tables:

M. Mutuma realized profits of Sh.12,000 from 7 tables, Sh.12,400 from 9 tables and Sh.11,300 from 4 tables sold respectively.

M. Mutuma has approached you for assistance in forecasting future profits. The profit function is believed to be quadratic in nature.

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Required:

(i) Derive the profit function.
(ii) The profit maximizing output and the maximum profit.

SOLUTION

a (i) Profit function = $P=ax^2+bx+c$

$1200=a(7)^2 + b(7) + c \rightarrow 12000 = 49a+7b+c$
$12400 = a(9)^2 + b(9) + c \rightarrow 12400 =81a + 9b+ c$
$11300 = a(4)^2 + b(4) + c \rightarrow 11300 = 16a + 4b +c$

Solve the 3 equations simultaneously

$a= -\frac{20}{3}$ $b = \frac{920}{3}$ $c=10,180$

$\therefore P = 10180 + \frac{920S}{3} \frac{20S^2}{3}$ is the required profit function.

ii) Profit maximizing output at maximum profit $\frac{dp}{dS} = 0$

FOC $\frac{920}{3} \frac{40S}{3} = 0$ $x = 23$

$\frac{40S}{3} = \frac{920}{2}$

x = 23

Maximum profit

$$P = 10180 + \frac{920x}{3} - \frac{20x^2}{3}$$

$$= 10180 + \frac{920(23)}{3} - \frac{20(23)^2}{3}$$

= Sh13,706.67

PROBLEM 4

The data below relate to products A and B, manufactured by Mauzo Limited.

$Q_1 = 2(P_2 \quad P_1) + 4$ is the demand function for product A

$Q_2 = \frac{1}{4}P_1 \quad \frac{5}{2}P_2 + 52$ is the demand function for product B

Q_1 is the quantity of product A

Q_2 is the quantity of product B

P_1 is the selling price of product A

p_2 is the selling price per unit of product B

The variable costs per unit are sh. 9 and sh. 12 for products A and B respectively.

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Required:

i) The total revenue function of Manzo Limited.
ii) The total cost function of Mauzo Limited
iii) The total profit function of Mauzo Limited.
iv) The profit maximizing prices and quantities of products A and B

SOLUTION

(i) $R_1 = P_1q_1 = P_1(2P_2-2P_1+4)$

$R_1=2P_1P_2-2P^2_1+4P_1$

$R_2=P_2q_2=P_2 \quad \frac{P_1}{4} - \frac{5}{2}P_2+52$

$$R_2 = \frac{1}{4}P_1P_2 \quad \frac{5}{2}P_2^2 + 52P_2$$

Total revenue function, $R=R_1+R_2$

$$R = \frac{9}{4}P_1P_2 \quad 2P_1^2 + 4P_1 \quad \frac{5}{2}P_2^2 + 52P_2$$

(ii) $C_1=9q_1=9(2P_2-2P_1+4)$

$C_1=18P_2-18P_1+36$

$C_2= 12q_2 = 12 \quad \frac{P_1}{4} - \frac{5}{2}P_2+52 \quad = 3P_1 – 30P_2 + 624$

Total Cost Function, $C=C_1+C_2$

$C=660-12P_2-15P_1$

(iii) Total profit function, $\Pi=R-C$

$$\pi = \frac{9}{4}P_1P_2 \quad 2P_1^2 + 4P_1 \quad \frac{5}{2}P_2^2 + 52P_2 \quad 660 + 12P_2 + 15P_1$$

$$\pi = \frac{9}{4}P_1P_2 \quad 2P_1^2 + 19P_1 \quad \frac{5}{2}P_2^2 + 64P_2 \quad 660$$

(iv) At maximum profit $\frac{dh}{dP_1} = 0$ and $\frac{dh}{dP_2} = 0$

F.O.C: $\frac{dh}{dP_1} = {}^9/_4 P_2 - 4P_1 + 19 = 0$ ..... (i)

$$\frac{dh}{dP_2} = \frac{9}{4}P_1 \quad 5P_2 + 64 = 0$$

$4P_1-\frac{80}{9}P^2+\frac{1024}{9}=0$

$-4P_1+{}^9/_4 P_2+19=0$

$-\frac{239}{36}P_2+\frac{1195}{9}=0$

P_2=Sh.20

P_1 =Sh.16

S.O.C: $\frac{d^2h}{dP_1^2}$ = - 4 (-ve) hence maximum

$\frac{d^2h}{dP_2^2}$ = - 5 (-ve) hence maximum

Profit is maximized when P_1 =sh.16, P_2 = Sh.20, $q_1 = 2(20-16) + 4 = 12$ units and $q_2 = 16-5x20+52=6$ units.

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MATRICES

A matrix is a rectangular array of items or numbers. These items or numbers are arranged in rows and columns to represent some information.

The position of an element in one matrix is very important as will be seen later; therefore an element is located by the number of the row and column which it occupies.

The size of a matrix is defined by the number of its rows (m) and column (n).

For example $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & & i \end{pmatrix}$

are (2 x 2) and (3 x 3) matrices since A has 2 rows and 2 columns and B has 3 rows and 3 columns. A matrix A with three rows and four columns is given by one of:

$$A=\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

or

$$A = (a_{ij}) \quad i = 1, 2, 3$$
$$j = 1, 2, 3, 4$$ *where i represents the row number whereas j represents the column number*

Types of matrices

Equal Matrices

Two matrices A and B are said to be equal, that is

$$A = B \quad \text{or} \quad (a_{ij}) = (b_{ij})$$

If and only if they are identical if they both have the same number of rows and columns and the elements in the corresponding locations in the two matrices should be the same, that is, $a_{ij} = b_{ij}$ for all i. And j.

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Example

The following matrices are equal $\begin{pmatrix} 3 & 4 & 0 \\ 2 & 2 & 3 \\ 5 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 4 & 0 \\ 2 & 2 & 3 \\ 5 & 1 & 1 \end{pmatrix}$

Column Matrix or column vector

A column matrix, also referred to as column vector is a matrix consisting of a single column.

For example $x = \begin{pmatrix} x_1 \\ x_2 \\ . \\ . \\ . \\ x_n \end{pmatrix}$

Row matrix or row vector

It is a matrix with a single row

For example $y = (y_1, y_2, y_3 ..... y_n)$

Transpose of a Matrix

The transpose of an mxn matrix A is the nxm matrix A^T obtained by interchanging the rows and columns of A.

A = a_{ij}

The transpose of A i.e. A^T is given by

$$A^T = [a_{ij}]^T = [a_{ji}]$$

mxn nxm

Example

Find the transposes of the following matrices

$$A = \begin{pmatrix} 1 & 5 & 7 \\ 2 & 1 & 4 \\ 0 & 9 & 3 \end{pmatrix}$$

$$B = (b_1, b_2, b_3, b_4)$$

$$C = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

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Solution

i. $A^T = \begin{pmatrix} 1 & 5 & 7 \\ 2 & 1 & 4 \\ 0 & 9 & 3 \end{pmatrix}^T = \begin{pmatrix} 1 & 2 & 0 \\ 5 & 1 & 9 \\ 7 & 4 & 3 \end{pmatrix}$

ii. $B^T = (b_1, b_2, b_3, b_4)^T = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$

iii. $C^T = \begin{pmatrix} x_1 \\ (x_2) \\ x_3 \end{pmatrix}^T = (x_1, x_2, x_3)$

Square Matrix

A matrix A is said to be square when it has the same number of rows as columnse.g.

$$A = \begin{matrix} 2 & 5 \\ 3 & 7 \end{matrix}$$ is a square matrix of order 2

B = n × n is a square matrix of the order n

Diagonal matrices

It is a square matrix with zeros everywhere in the matrix except on the principal diagonal

e.g.

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix}, \quad B = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

An identity or unity matrix

It is a diagonal matrix in which each of the diagonal elements is a positive one (1)

e.g.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2×2 unit matrix 3×3 unit matrix

A null or zero matrix

A null or zero matrix is a matrix whose elements are all equal to zero.

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Sub matrix

The sub matrix of the matrix A is another matrix obtained from A by deleting selected row(s) and/or column(s) of the matrix A.

$$\text{e.g, if } A = \begin{pmatrix} 7 & 9 & 8 \\ 2 & 3 & 6 \\ 1 & 5 & 0 \end{pmatrix}$$

$$\text{then } A_1 = \begin{pmatrix} 2 & 3 & 6 \\ 1 & 5 & 0 \end{pmatrix} \quad \text{and } A_2 = \begin{pmatrix} 7 & 9 \\ 1 & 5 \end{pmatrix}$$

are both sub matrices of A

OPERATION ON MATRICES

Matrix addition and subtraction

We can add any number of matrices (or subtract one matrix from another) if they have the same sizes. Addition is carried out by adding together corresponding elements in the matrices. Similarly subtraction is carried out by subtracting the corresponding elements of two matrices as shown in the following example

Example: Given A and B, calculate A + B and A – B

$$A = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & -13 & -6 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 12 & 4 & -7 & 3 \\ 0 & -4 & 10 & -4 \\ 7 & -3 & 7 & 9 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & -13 & -6 & 0 \end{pmatrix} + \begin{pmatrix} 12 & 4 & -7 & 3 \\ 0 & -4 & 10 & -4 \\ 7 & -3 & 7 & 9 \end{pmatrix} = \begin{pmatrix} 18 & 3 & 3 & 8 \\ 3 & 0 & 12 & -9 \\ -2 & -16 & 1 & 9 \end{pmatrix}$$

$$A - B = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & -13 & -6 & 0 \end{pmatrix} - \begin{pmatrix} 12 & 4 & -7 & 3 \\ 0 & -4 & 10 & -4 \\ 7 & -3 & 7 & 9 \end{pmatrix} = \begin{pmatrix} -6 & -5 & 17 & 2 \\ 3 & 8 & -8 & -1 \\ -16 & -10 & -13 & -9 \end{pmatrix}$$

If it is assumed that A, B, C are of the same order, the following properties are fulfilled:

a) Commutative law: $A + B = B + A$

b) Associative law: $(A + B) + C = A + (B + C) = A + B + C$

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Multiplying a matrix by a number

In this case each element of the matrix is multiplied by that number

Example

$$\text{If } A = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & 13 & -6 & 0 \end{pmatrix}$$

$$\text{then } (10)A = \begin{pmatrix} 60 & -10 & 100 & 50 \\ 30 & 40 & 20 & -50 \\ -90 & 130 & -60 & 0 \end{pmatrix}$$

Matrix Multiplication

a) Multiplication of two vectors

Let row vector A represent the selling price in shillings of one unit of commodity P, Q and R respectively and let column vector B represent the number of units of commodities P, Q, R sold respectively. Then the vector product $A \times B$ will be equal to the total sales value

i. e. $A \times B =$ Total sales value

$$\text{Let } A = \begin{pmatrix} 4 & 5 & 6 \end{pmatrix} \text{ and } B = \begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix}$$

$$\text{then } \begin{pmatrix} 4 & 5 & 6 \end{pmatrix}\begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix} = 400 + 1{,}000 + 1{,}800 = \text{Shs } 3{,}200$$

Rules of multiplication

i) The row vector must have the same number of elements as the column vector
ii) The first vector is a row vector and the second is a column vector
iii) The corresponding elements in each vector are multiplied together and the results obtained are added. This addition is always a single number

Going back to the example given before

$$A \times B = \begin{pmatrix} 4 & 5 & 6 \end{pmatrix}\begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix} = 4 \times 100 + 5 \times 200 + 6 \times 300 = \text{Shs}3{,}200, \text{ a single number}$$

b) Multiplication of two matrices

Rules

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i) Multiplication is only possible if the first matrix has the same number of columns as the rows of the second matrix. That is if A is the order a×b, then B has to be of the order b×c. If the A×B = D, then D must be of the order a×c.
ii) The general method of multiplication is that the elements in row m of the first matrix are multiplied by the corresponding elements column n of the second matrix and the products obtained are then added giving a single number.

We can express this rule as follows

$$\text{Let } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \text{and } b = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{pmatrix}$$

$$\text{Then } A \times B = D = \begin{pmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \end{pmatrix}$$

A = 2 x 2 matrix B = 2 x 3 matrix D = 2 x 3 matrix

Where

$d_{11} = a_{11} \times b_{11} + a_{12} \times b_{21}$

$d_{12} = a_{11} \times b_{12} + a_{12} \times b_{22}$

Example I

$$\begin{pmatrix} 6 & 1 \\ 2 & 3 \end{pmatrix} \times \begin{pmatrix} 3 & 0 & 2 \\ 4 & 5 & 8 \end{pmatrix} = \begin{pmatrix} 6\times3+1\times4 & 6\times0+1\times5 & 6\times2+1\times8 \\ 2\times3+3\times4 & 2\times0+3\times5 & 2\times2+3\times8 \end{pmatrix}$$

$$= \begin{pmatrix} 22 & 5 & 20 \\ 18 & 15 & 28 \end{pmatrix}$$

Example II

Matrix X gives the details of component parts used in the make up of two products P_1 and P_2 matrix Y gives details of products made on each day of the week as follows:

Matrix X

Parts

Products	A	B	C
P_1	3	4	2
P_2	2	5	3

Matrix Y

Product

	P_1	P_2
Mon	1	2
Tues	2	3
Wed	3	2
Thur	2	2
Fri	1	1

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Use matrix multiplication to find the number of component parts used on each day of the week.

Solution:

After careful consideration, it will be easy to decide that the correct order of multiplication is Y×X (Note the order of multiplication). This multiplication is compatible and also it gives the desired answer.

$$Y \times X = \begin{pmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{pmatrix} \times \begin{pmatrix} 3 & 4 & 2 \\ 2 & 5 & 3 \end{pmatrix} = \begin{pmatrix} 1\times3+2\times2 & 1\times4+2\times5 & 1\times2+2\times5 \\ 2\times3+3\times2 & 2\times4+3\times5 & 2\times2+3\times3 \\ 3\times3+2\times2 & 3\times4+2\times5 & 3\times2+2\times3 \\ 2\times3+2\times2 & 2\times4+2\times5 & 2\times2+2\times3 \\ 1\times3+1\times2 & 1\times4+1\times5 & 1\times2+1\times3 \end{pmatrix}$$

5 x 2 matrix 2 x 3 matrix = 5 x 3 matrix

	A	B	C
Mon	7	14	8
Tues	12	23	13
Wed	13	22	12
Thur	10	18	10
Fri	5	9	5

Interpretation

On Monday, number of component parts A used is 7, B is 14 and C is 8. in the same way, the number of component parts used for other days can be interpreted.

The determinant of a square matrix

The determinant of a square matrix A det (A) or |A| is a number associated to that matrix. If the determinant of a matrix is equal to zero, the matrix is called singular matrix otherwise it is called non-singular matrix. The determinant of a non square matrix is not defined.

i) Determinant of a 2 × 2 matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ = ad – bc

$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ = ad – bc

ii) Determinant of a 3 × 3 matrix

$$A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & & i \end{vmatrix} = a \begin{vmatrix} e & f \\ & i \end{vmatrix} \quad b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & \end{vmatrix}$$

a(ei – fh) – b(di – gf) + c(dh – eg)

Simplified

iii) Determinant of a 4 × 4 matrix

$$A = \begin{pmatrix} a & b & c & d \\ e & f & g & \\ i & j & k & l \\ m & k & o & p \end{pmatrix}$$

$$|A| = a \begin{vmatrix} f & g & \\ j & k & l \\ k & o & p \end{vmatrix} \quad b \begin{vmatrix} e & g & \\ i & k & l \\ m & o & p \end{vmatrix} + c \begin{vmatrix} e & f & \\ i & j & l \\ m & k & p \end{vmatrix} \quad d \begin{vmatrix} e & f & g \\ i & j & k \\ m & k & o \end{vmatrix}$$

Simplify 3 ×3 determinants as in ii and then evaluate the 4 x 4 determinants.

Inverse of a matrix

If for an n x n square matrix A, there is another n x n square matrix B such that their product is the identity of the order n x n, In, that is A × B = B×A = I, then B is said to be the inverse of A. Inverse is generally written as A^{-1}

Hence $AA^{-1} = I$

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Note: Only non singular matrices have an inverse and therefore the inverse of a singular matrix is undefined.

General method for finding inverse of a matrix

In order to introduce the rule to calculate the determinant as well as the inverse of a matrix, we should introduce the concept of minor and cofactor.

The minor of an element

Given a matrix A = (aij), the minor of an element aij in row i and column j (call it mij), is the value of the determinant formed by deleting row i and column j in matrix A.

Example

$$\text{Let matrix } A = \begin{bmatrix} 4 & 2 & 3 \\ 5 & 6 & 1 \\ 2 & 3 & 0 \end{bmatrix}$$

The minors of A are,

$$m_{11} = \begin{vmatrix} 6 & 1 \\ 3 & 0 \end{vmatrix} = 6\times0 - 3\times1 = -3$$

$$m_{12} = \begin{vmatrix} 5 & 1 \\ 2 & 0 \end{vmatrix} = 5\times0 - 1\times2 = -2$$

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Similarly

$$m_{13} = \begin{vmatrix} 5 & 6 \\ 2 & 3 \end{vmatrix} = 15 - 12 = 3 \qquad m_{21} = \begin{vmatrix} 2 & 3 \\ 3 & 0 \end{vmatrix} = 0 - 9 = -9$$

$$m_{22} = \begin{vmatrix} 4 & 3 \\ 2 & 0 \end{vmatrix} = 0 - 6 = -6 \qquad m_{23} = \begin{vmatrix} 4 & 2 \\ 2 & 3 \end{vmatrix} = 12 - 4 = 8$$

$$m_{31} = \begin{vmatrix} 2 & 3 \\ 6 & 1 \end{vmatrix} = 2 - 18 = -16 \qquad m_{32} = \begin{vmatrix} 4 & 3 \\ 5 & 1 \end{vmatrix} = 4 - 15 = -11 \qquad m_{33} = \begin{vmatrix} 4 & 2 \\ 5 & 6 \end{vmatrix} = 24 - 10 = 14$$

The cofactor of an element

The cofactor of any element a_{ij} (known as c_{ij}) is the signed minor associated with that element.
The sign is not changed if (i+j) is even and it is changed if (i+j)is odd. Thus the sign alternated whether vertically or horizontally, beginning with a plus in the upper left hand corner.

i.e. 3 x 3 signed matrix will have signs $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$

Hence the cofactor of element a_{11} is $m_{11} = -3$, cofactor of a_{12} is $-m_{12} = +2$ the cofactor of element a_{13} is $+m_{13} = 3$ and so on.

Matrix of cofactors of A $= \begin{pmatrix} -3 & 2 & 3 \\ 9 & -6 & -8 \\ -16 & 11 & 14 \end{pmatrix}$

in general for a matrix M $= \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$

Cofactor of a is written as A, cofactor of b is written as B and so on.
Hence matrix of cofactors of M is written as

$$= \begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}$$

The determinant of a n×n matrix

The determinant of a n×n matrix can be calculated by adding the products of the element in any row (or column) multiplied by their cofactors. If we use the symbol for determinant.

Then $= aA + bB + cC$

or

$= dD + eE + fF$ e.t.c

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Note: Usually for calculation purposes we take $= aA + bB + cC$

Hence in the example under discussion

$= (4 \times -3) + (2 \times 2) + (3 \times 3) = 1$

The adjoint of a matrix

The ad joint of matrix $\begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}$ is written as

$\begin{pmatrix} A & D & G \\ B & E & H \\ C & F & I \end{pmatrix}$

i.e. change rows into columns and columns into rows (transpose of the matrix of cofactors)

The inverse of the matrix $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$

is written as $\frac{1}{determinant}$ x (adjoint of the matrix)

i.e. $A^{-1} = \frac{1}{\Delta} \times \begin{pmatrix} A & D & G \\ B & E & H \\ C & F & I \end{pmatrix}$

Where = aA + bB + cC

Hence inverse of $\begin{pmatrix} 4 & 2 & 3 \\ 5 & 6 & 1 \\ 2 & 3 & 0 \end{pmatrix}$

is found as follows

= (4 × –3) + (2 × 2) + (3 (3) = 1

A = -3 B = 2 C = 3

D = 9 E = -6 F = -8

G = -16 H = 11 I = 14

$$A^{-1} = \frac{1}{1}\begin{pmatrix} 3 & 9 & 16 \\ 2 & 6 & 11 \\ 3 & 8 & 14 \end{pmatrix}$$

(Note: Check if A (A-1 = A^{-1}× A = 1)

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Solution of simultaneous equations

In order to determine the solutions of simultaneous equations, we may use either of the following 2 methods

i) Matrix inverse method
ii) Cramers rule

The cofactor method

This method requires that we obtain

a) The minors and cofactors
b) The adjoint of the matrix
c) The inverse of the matrix
d) Premultiply the original by the inverse on both sides of the matrix equation

Example

Solve the following

$4x_1 + x_2 – 5x_3 = 8$

$-2x_1 + 3x_2 + x_3 = 12$

$3x_1 – x_2 + 4x_3 = 5$

Solution

a) From the system above, we have

$$\begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \\ 5 \end{pmatrix}$$

A X b

We need to determine the **minors** and the **cofactors**for the above matrix

Definition

A **minor** is a determinant of a sub matrix obtained when other elements are as shown below.
A cofactor is the product of (-1) i + j and a minor where

i = Ith row i = 1, 2, 3 …….

j = Jth row j = 1, 2, 3 …….

Cofactor of 4 $(a_{11}) = (-1)^{1+1}$ $\begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} = 13$

Cofactor of -2 $(a_{21}) = (-1)^{2+1}$ $\begin{vmatrix} 1 & -5 \\ -1 & 4 \end{vmatrix} = 1$

Cofactor of 3 $(a_{31}) = (-1)^{3+1}$ $\begin{vmatrix} 1 & -5 \\ 3 & 1 \end{vmatrix} = 16$

Cofactor of 1 $(a_{12}) = (-1)^{1+2}$ $\begin{vmatrix} -2 & 1 \\ 3 & 4 \end{vmatrix} = 11$

Cofactor of 3 $(a_{22}) = (-1)^{2+2}$ $\begin{vmatrix} 4 & -5 \\ 3 & 4 \end{vmatrix} = 31$

Cofactor of -1 $(a_{23}) = (-1)^{2+3}$ $\begin{vmatrix} 4 & 5 \\ -2 & 1 \end{vmatrix} = 6$

Cofactor of -5 $(a_{13}) = (-1)^{1+3}$ $\begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} = -7$

Cofactor of +1 $(a_{23}) = (-1)^{2+3}$ $\begin{vmatrix} 4 & 1 \\ 3 & -1 \end{vmatrix} = 7$

Cofactor of 4 $(a_{33}) = (-1)^{3+3}$ $\begin{vmatrix} 4 & 1 \\ -2 & 3 \end{vmatrix} = 14$

The matrix of C of cofactors is

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$$\begin{pmatrix} 13 & 11 & -7 \\ 1 & 31 & 7 \\ 16 & 6 & 14 \end{pmatrix}$$

$C^T = \begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix}$ = Adjoin of the original matrix of coefficients

The original matrix of coefficients

$$= \begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix}$$

Therefore determinant is

$$\begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix} = \begin{matrix} 4 & 1 & -5 & 4 & 1 \\ -2 & 3 & 1 & -2 & 3 \\ 3 & -1 & 4 & 3 & -1 \end{matrix}$$

$= (48 + 3 - 10) - (-45 - 4 - 8)$

$= 41 + 57$

$= 98$

The inverse of the matrix of coefficients, will be

$$= \frac{1}{98} \begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix}$$

By multiplying the inverse on both sides of the equation we have,

$$\frac{1}{98} \begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix} \begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= \frac{1}{98} \begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix} \begin{pmatrix} 8 \\ 12 \\ 5 \end{pmatrix}$$

$$= \frac{1}{98} \begin{pmatrix} 98 & 0 & 0 \\ 0 & 98 & 0 \\ 0 & 0 & 98 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{98} \begin{pmatrix} 196 \\ 490 \\ 98 \end{pmatrix}$$

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$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$$

$\therefore X_1 = 2, X_2 = 5, X_3 = 1$

Cramers Rule in Solving Simultaneous Equations

Consider the following system of two linear simultaneous equations in two variables.

$a_{11} x_1 + a_{12} x_2 = b_1$.....(i)

$a_{21} x_1 + a_{22} x_2 = b_2$.....(ii)

after solving the equations you obtain

$$x_1 = \frac{b_1 a_{22} - b_2 a_{12}}{a_{11} a_{22} - a_{12} a_{21}} = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

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and

$$x_2 = \frac{a_{11} b_2 - a_{21} b_1}{a_{11} a_{22} - a_{12} a_{21}} = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

Solutions of x_1 and x_2 obtained this way are said to have been derived using Cramers rule, practice this method over and over to internalize it. It is advisable for exam situation since it is shorter.

Example

Solve the following systems of linear simultaneous equations by Cramers' rule:

i) $2x_1 - 5x_2 = 7$

$x_1 + 6x_2 = 9$

ii) $x_1 + 2x_2 + 4x_3 = 4$

$2x_1 + x_3 = 3$

$3x_2 + x_3 = 2$

Solutions

i. $2x_1 - 5x_2 = 7$

$x_1 + 6x_2 = 9$

can be expressed in matrix form as

$$\begin{pmatrix} 2 & -5 \\ 1 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7 \\ 9 \end{pmatrix}$$

and applying cramers' rule

$$x_1 = \frac{\begin{vmatrix} 7 & -5 \\ 9 & 6 \end{vmatrix}}{\begin{vmatrix} 2 & -5 \\ 1 & 6 \end{vmatrix}} = \frac{87}{17} = 5\frac{2}{17}$$

$$x_2 = \frac{\begin{vmatrix} 2 & 7 \\ 1 & 9 \end{vmatrix}}{\begin{vmatrix} 2 & -5 \\ 1 & 6 \end{vmatrix}} = \frac{11}{17}$$

(ii) can be expressed in matrix form as

$$\begin{pmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$

and by Cramers' rule

$$x_1 = \frac{\begin{vmatrix} 4 & 2 & 4 \\ 3 & 0 & 1 \\ 2 & 3 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{vmatrix}} = \frac{22}{17}$$

$$x_3 = \frac{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 3 \\ 0 & 3 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{vmatrix}} = \frac{7}{17}$$

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$$x_2 = \frac{\begin{vmatrix} 1 & 4 & 4 \\ 2 & 3 & 1 \\ 0 & 2 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{vmatrix}} = \frac{9}{17}$$

Solving simultaneous Equations using matrix algebra

i. Solve the equations

$2x + 3y = 13$

$3x + 2y = 12$

in matrix format these equations can be written as

$$\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 13 \\ 12 \end{pmatrix}$$

pre multiply both sides by the inverse of the matrix

$$\Delta = \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5$$

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and inverse of the matrix is

$$-\frac{1}{5}\begin{pmatrix} 2 & -3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} -\frac{2}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{pmatrix}$$

Pre multiplication by inverse gives

$$\begin{pmatrix} -\frac{2}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{pmatrix} \begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} -\frac{2}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{pmatrix} \begin{pmatrix} 13 \\ 12 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Therefore x = 2 y = 3

ii. Solve the equations

$4x + 2y + 3z = 4$

$5x + 6y + 1z = 2$

$2x + 3y = -1$

Solution:

Writing these equations in matrix format, we get

$$\underset{A}{\begin{pmatrix}4 & 2 & 3\\ 5 & 6 & 1\\ 2 & 3 & 0\end{pmatrix}} \times \underset{B\bar{X}}{\begin{pmatrix}x\\ y\\ z\end{pmatrix}} = \underset{\bar{b}}{\begin{pmatrix}4\\ 2\\ -1\end{pmatrix}}$$

Pre-multiply both sides by the inverse

the inverse of A as found before is A^{-1} $= \begin{pmatrix}-3 & 9 & -16\\ 2 & -6 & 11\\ 3 & -8 & 14\end{pmatrix}$

$$\begin{pmatrix}-3 & 9 & -16\\ 2 & -6 & 11\\ 3 & -8 & 14\end{pmatrix}\begin{pmatrix}4 & 3 & 2\\ 5 & 6 & 1\\ 2 & 3 & 0\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix} = \begin{pmatrix}-3 & 9 & -16\\ 2 & -6 & 11\\ 3 & -8 & 14\end{pmatrix}\begin{pmatrix}4\\ 2\\ -1\end{pmatrix} = \begin{pmatrix}22\\ -15\\ -18\end{pmatrix}$$

hence x = 22 y = -15 z = -18

(Note: under examination conditions it may be advisable to check the solution by substituting the value of x, y, z into any of the three original equations)

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DIFFERENTIATION AND INTEGRATION

Introduction

Calculus is concerned with the mathematical analysis of change or movement. There are two basic operations in calculus.

1. **Differentiation**
2. **Integration**

These two basic operations are inverse to one another like addition and subtraction or multiplication and division.

Importance of calculus in business management

1. Often we must be involved in optimisation i.e. maximum revenues, profits and minimise costs, losses, waste
   For optimisation, we apply differential calculus.
2. Calculus is also used in marginal analysis e.g. to obtain a Total cost (TC) function from marginal cost (MC) function.

Total Revenue (TR) function from Marginal Revenue (MR) function

For marginal analysis, we use indefinite integration.

3. Certain problems require that they are solved by finding the area under a curve e.g. Total $\pi$for a number of days if profit is dependent or is a function of time.
   To find the area under a curve, we apply definite integration.

Differentiation and integration

Differentiation deals with the determination of the rates of change of business activities or simply the process of finding the derivative of a function.

Integration deals with the summation or totality of items produced over a given period of time or simply the reverse of differentiation

The derivative and differentiation

The process of obtaining the derivative of a function or slope or gradient function is referred to as derivation or differentiation.

The derivative is denoted by $\frac{dy}{dx}$ or f′(x) and is given by dividing the change in y variable by the change in x variable.

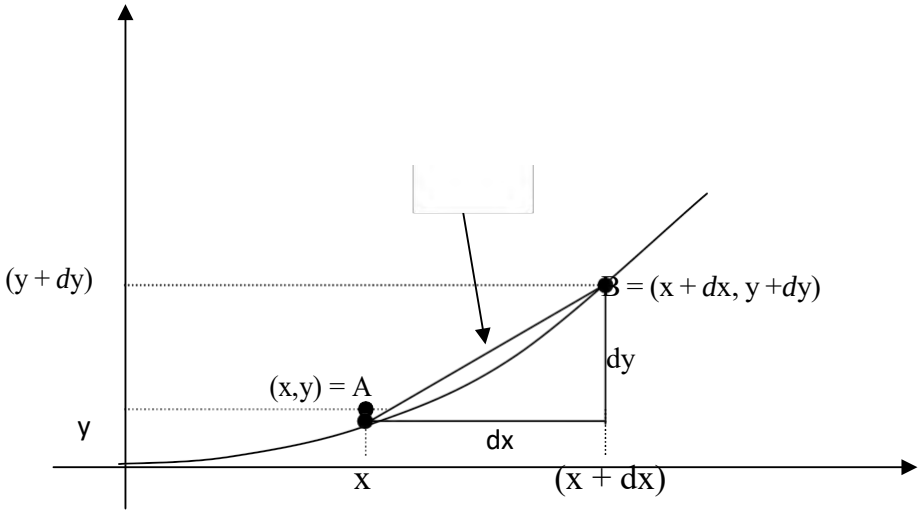
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The derivative or slope or gradient of a line AB connecting points (x,y) and (x+dx, y + dy) is given by

$$\frac{\Delta y}{\Delta x} = \frac{\text{Change in y}}{\text{Change in x}} = \frac{(y+dy)-y}{(x+dx)-x} = \frac{dy}{dx}$$

Where dy is a small change in y and dx is a small change in x variables.

Illustration



Rules of Differentiation

1. The constant function rule

If given a function y = k where k is a constant then $\frac{dy}{dx} = 0$

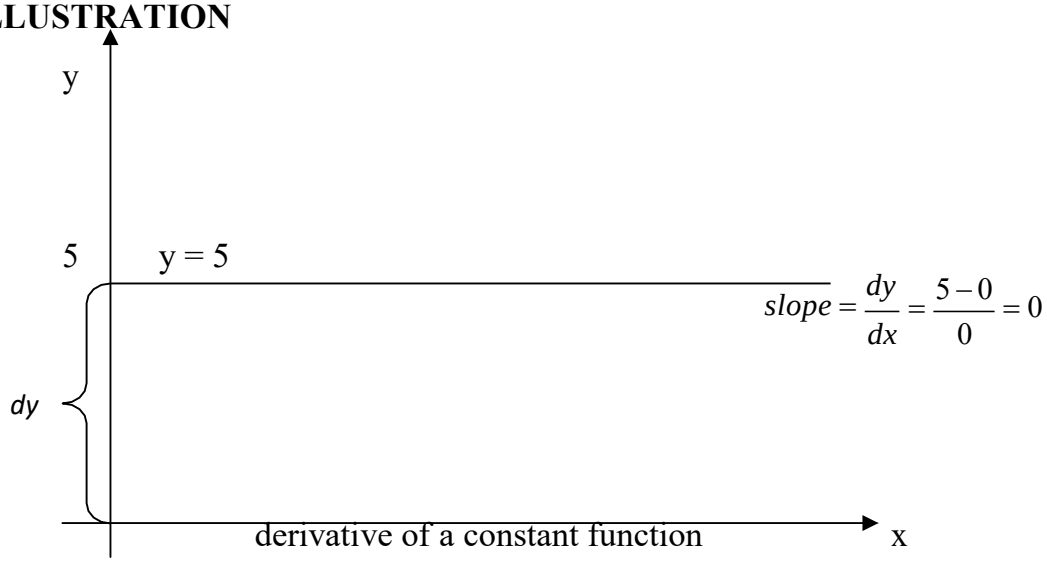
Example

Find the derivative of (i) y = 5

Solution

i. y = 5 $\quad \frac{dy}{dx} = 0$

ILLUSTRATION



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2. Power function rule

Given a function $y = x^r$

$$Then \quad \frac{dy}{dx} = rx^{r-1}$$

Example

Find $\frac{dy}{dx}$ for;

(i). $y = x^7$
(ii). $y = x^2$
(iii). $y = x^{-3}$
(iv). $y = x$

Solution

i. $y = x^7$

$\frac{dy}{dx} = 7x^{7-1} = 7x^6$

ii. $y = x^2$

$\frac{dy}{dx} = 2\ x^{(2\ -\ 1)}$

iii. $y = x^{-3}$

$\frac{dy}{dx} = -3x^{-3-1} = -3x^{-4}$

iv. $y = x$

$\frac{dy}{dx} = 1x^{1-1} = 1.x^0 = 1$ (since $x^0=1$)

3. Power function multiplied by a constant

If given $y = Ax^r$, then $\frac{dy}{dx} = rAx^{r-1}$

4. The sum rule

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The derivative of the sum of two or more functions equals the sum of the derivatives of the functions.

For instance

If $H(x) = h(x) + g(x)$

Then $\frac{dy}{dx}$ or $H'(x) = h'(x) + g'(x)$

5. The difference rule

The derivative of the difference of two or more functions equals the difference of the derivatives of the functions

If $H(x) = h(x) - g(x)$

Then $H'(x) = h'(x) - g'(x)$

Examples

Find the derivatives of

i. $y = 3x^2 + 5x + 7$

ii. $y = 4x^2 - 2x^b$

Solution

i. $y = 3x^2 + 5x + 7$

$$\frac{dy}{dx} = \frac{d\left(3x^2\right)}{dx} + \frac{d\left(5x\right)}{dx} + \frac{d\left(7\right)}{dx}$$

$$= 6x + 5 + 0$$

$$= 6x + 5$$

ii. $y = 4x^2 - 2x^b$

$$\frac{dy}{dx} = \frac{d\left(4x^2\right)}{dx} - \frac{d\left(2x^b\right)}{dx}$$

$$= 8x - 2bx^{b-1}$$

6. The product rule – both factors are functions

The derivative of the product of two functions equals the derivative of the first function multiplied by the second function PLUS the derivative of the second function multiplied by the first function.

given that $H\left(x\right) = h\left(x\right).g\left(x\right)$

Then $H'\left(x\right) = h'\left(x\right).g\left(x\right) + h\left(x\right).g'\left(x\right)$

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Example

Find $\frac{dy}{dx}$ for

i. $y = x^2(x)$

ii. $y = (x^2 + 3)\,(2x^3 + x^2 - 3)$

SOLUTION

i. $y = x^2(x)$

$$\frac{dy}{dx} = x.\frac{d\left(x^2\right)}{dx} + x^2.\frac{d\left(x\right)}{dx}$$

$$= x.2x + x^2.1$$

$$= 2x^2 + x^2$$

$$= 3x^2$$

Note that $y = x^2(x) = x^3$. Directly differentiating this we get $3x^2$.

ii. $y = (x^2 + 3)\,(2x^3 + x^2 - 3)$

$$\frac{dy}{dx} = \frac{d\left(x^2+3\right)}{dx}.\left(2x^3+x^2-3\right)+\left(x^2+3\right).\frac{d\left(2x^3+x^2-3\right)}{dx}$$

$$2x.\left(2x^3+x^2-3\right)+\left(x^2+3\right).\left(6x^2+2x\right)$$

$$10x^4+4x^3+18x^2$$

7. Quotient Rule

The derivative of the quotient of two functions equals the derivative of the numerator times the denominator MINUS the derivative of the denominator times the numerator, all which are divided by the square of the denominator

If given H (x) = $\frac{h(x)}{g(x)}$

then $H'(x) = \frac{h'(x).g(x)-h(x).g'(x)}{\left[g(x)\right]^2}$

For example

Find $\frac{dy}{dx}$ for

i. $\frac{x}{3+x^2}$

ii. $\frac{x}{3x+7}$

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Solutions

i. $\frac{x}{3+x^2}$

$$\frac{dy}{dx} = \frac{\frac{d(x)}{dx}.\left(3+x^2\right)-\frac{d\left(3+x^2\right)}{dx}.x}{\left(3+x^2\right)^2}$$

$$=\frac{(3+S^2)-(2S)S}{(3+S^2)^2}$$

$$=\frac{3+S^2-2S^2}{(3+S^2)^2}=\frac{3+S^2}{(3+S^2)^2}$$

ii. $y = \frac{x^3}{(3x+7)}$

$$\frac{dy}{dx} = \frac{\left(3x^2\right)(3x+7)-(3)\left(x^3\right)}{3x+7)} = \frac{6x^3+21x^2}{(3x+7)}$$

dx (2 2

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ILLUSTRATION

A farmer of a large farm of poultry announced that egg production per month follows the equation;

$$w = \frac{3m^3 - m^2}{m^2 + 10}$$

Where w – Total no of eggs produced per month

m – Amount in kilograms of layers mash feed.

Required

Determine the rate of change of w with respect to m (i.e. the rate at which the number of eggs per month increase or decrease depending on the rate at which the kilos of layers marsh are increased).

SOLUTION

Let $u = 3m^3 - m^2$

$\therefore \quad \frac{du}{dm} = 9m^2 - 2m$

Let $v = m^2 + 10$

$\therefore \quad \frac{dv}{dm} = 2m$

$$\therefore \quad \frac{dw}{dm} = \frac{(m^2+10)(9m^2-2m) - (3m^3 - m^2)2m}{(m^2+10)^2}$$

$$= \frac{9m^2+90^2-2m^3 - 20m^3 - 6m^4+2m^3}{(m^2+10)^2}$$

$$= \frac{3m^4 + 90m^2 - 20m}{\left(m^2 + 10\right)^2}$$

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8. Chain Rule

This rule is generally applied in the determination of the derivatives of composite functions, which can be defined as a function in which anotherfunction can be considered to have taken the place of the independent variable. The composite function is also referred to as a function of a function.

It is normally of the form $y = (2x^2 + 3)^3$. If we let $u = (2x^2 + 3)$, then $y = u^3$.

In order to differentiate such an equation we use the formula

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Solution

$y = (2x^2 + 3)^3$

Let $u = 2x^2 + 3$

$\therefore \quad \frac{du}{dx} = 4x$

Let $y = u^3$

$\therefore \quad \frac{dy}{du} = 3u^2$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 3u^2 \times 4x = 12xu^2$

$= 12x(2x^2 + 3)^2$

Example

Consider the function

$y = (x^2 + 16x + 5)^2$

which can be decomposed into

$y = u^2$ and $u = x^2 + 16x + 5$. in this case y is a function of $(x^2 + 16x + 5)$

Hence $y = f(u)$ and $u = g(x)$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= (2u)(2x + 16)$

$= 2(x^2 + 16x + 5)(2x + 16)$

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9. The derivative of a function raised to power r; the composite function rule.

The derivative of a function raised to power r equals to the power r times the function which is raised by power (r-1), all of which is multiplied by the derivative of the function

If $y = [g(x)]^r$

Then $\frac{dy}{dx} = r[g(x)]^{r-1} \cdot g'(x)$

For example

Find $\frac{dy}{dx}$ given $y = (3x^2 + 4x)^5$

Solution

$\frac{dy}{dx} = 5(3x^2 + 4x)^4 \cdot (6x + 4)$

Differentiation of an implicit function

An Implicit function is one of the $y = x^2\ y + 3x^2 + 50$. it is a function in which the dependent variable (y) appears also on the right hand side.

To differentiate the above equation we use the differentiation method for a product, quotient or function of a function.

Solution

$y = x^2 y + 3x^2 + 50$

$$\frac{dy}{dx} = \frac{d\left(x^2 y\right)}{dx} + \frac{d\left(3x^2\right)}{dx} + \frac{d\left(50\right)}{dx}$$

$$\frac{dy}{dx} = \left[y(2x) + x^2 \frac{dy}{dx} \right] + 6x + 0$$

$$0 = 2xy + x^2 \frac{dy}{dx} - \frac{dy}{dx} + 6x$$

$0 = 2xy + (x^2 \quad 1)\frac{dy}{dS} + 6x$

$$(x^2 \quad 1)\frac{dy}{dx} = 2xy + 6x$$

$$\frac{dy}{dx} = \frac{(2xy + 6x)}{(x^2 \quad 1)} = \frac{2xy \ + 6x}{1 \quad x^2}$$

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Partial derivatives

These derivatives are used when we want to investigate the effect of one independent variable on the dependent variable.

For example, the revenues of a farmer may depend on two variables namely; the amount of fertilizer applied and also the type of the natural soil.

Let = $30x^2y + y^2 + 50x + 60y$

Where = annual revenue in £ '000'

x = type of soil

y = amount of fertilizer applied

Required;-

Determine the rate of change of the with respect to x and y

Solution

= $30x^2y + y^2 + 50x + 60y$

Differentiating with respect to x keeping y constant we have

d = 60xy + 50

dx

Differentiating with respect to y keeping x constant we have

$\frac{d}{dy} = 30x^2 + 2y + 60$

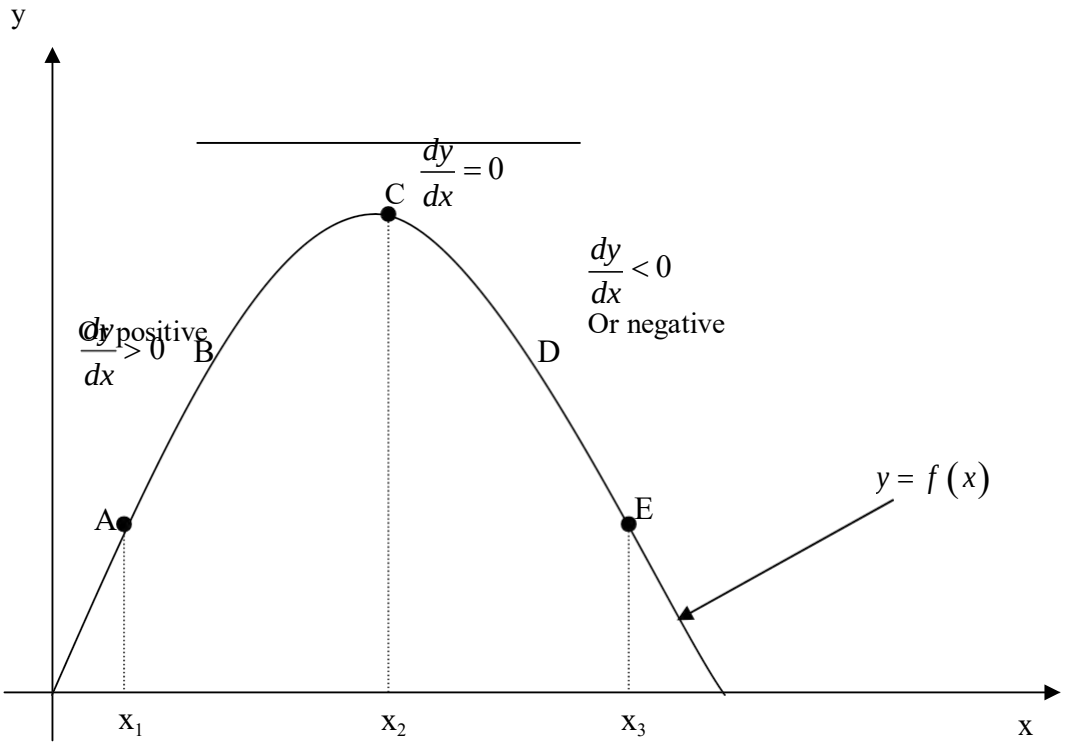
Maxima, minima and points of inflexion

a) Test for relative maximum

Consider the following function of x whose graph is represented by the figure below

$y = f(x)$

$\frac{dy}{dx} = f'(x)$



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Relative maximum point

The graph of the function slopes upwards to the right between points A and C and hence has a positive slope between these two points. The function has a negative slope between points C and E. At point C, the slope of the function is Zero.

Between points X_1 and X_2 $\frac{dy}{dx} > 0$ Where $X_1 \leq X < X_2$

and between X_2 and X_3 $\frac{dy}{dx} < 0$ Where $X_2 < X \leq X_3$.

Thus the first test of the maximum points require that the first derivative of a function equals zero or

$$\frac{dy}{dx} = f'(x) = 0$$

The second test of a maximum point requires that the second derivative of a function is negative or

$$\frac{d^2y}{dx^2} = f''(x) < 0$$

It should be noted that maximum, minimum or points of inflexion are also called critical points.

Example

Determine the critical value for the following functions and find out the critical value that constitutes a maximum

$y = x^3 - 12x^2 + 36x + 8$

Solution

$y = x^3 - 12x^2 + 36x + 8$

then $\frac{dy}{dx} = 3x^2 - 24x + 36 + 0$

The critical values for the function are obtained by equating the first derivative of the function to zero, that is:

$\frac{dy}{dx} = 0$ or $3x^2 - 24x + 36 = 0$

Hence $(x-2)(x-6) = 0$

And $x = 2$ or 6

The critical values for x are x = 2 or 6 and critical values for the function are y = 40 or 8

To ascertain whether these critical values of x will give rise to a maximum, we apply the second derivative test that is

$\frac{d^2y}{d^2x} < 0$

$\frac{dy}{dx} = 3x^2 - 24x + 36$ and

$\frac{d^2y}{d^2x} = 6x - 24$

a) When x = 2

Then $\frac{d^2y}{d^2x} = -12 < 0$

b) When x = 6

Then $\frac{d^2y}{d^2x} = +12 > 0$

Hence a maximum occurs when x = 2, since this value of x satisfies the second condition. X = 6 does not give rise to a local maximum i.e. it is a local minimum.

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b) Tests for relative minimum

There are two tests for a relative minimum point

i. The first derivative, that is

$$\frac{dy}{dx} = f'(x) = 0$$

ii. The second derivative, that is

$$\frac{d^2y}{dx^2} = f''(x) > 0$$

Example

For the function

$h(x) = 1/3\ x^3 + x^2 - 35x + 10$

Determine the critical values and find out whether these critical values are maxima or minima. Determine the extreme values of the function

Solution

i. Critical values

$h(x) = 1/3\ x^3 + x^2 - 35x + 10$ and

$h'(x) = x^2 + 2x - 35$

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by first text,

then $h'(x) = x^2 + 2x - 35 = 0$

or $(x-5)(x+7) = 0$

Hence $x = 5$ or $x = -7$

ii. The determinant of the maximum and the minimum points requires that we test the value $x = 5$ and -7 by the second text

$h'(x) = 2x + 2$

a) When $x = -7, h''(x) = -12 < 0$

b) When $x = 5, h''(x) = 12 > 0$

There $x = -7$ gives a maximum point and $x = 5$ gives a minimum point.

iii. Extreme values of the function

$h(x) = 1/3\ x^3 + x^2 - 35x + 10$

when $x = -7$, $h(x) = 189\ 2/3$

when $x = 5$, $h(x) = -98\ 1/3$

The extreme values of the function are $h(x) = 189\ 2/3$ which is a relative maximum and $h(x) = -98\ 1/3$, a relative minimum

c) Points of inflexion

Given the following two graphs, points of inflexion can be determined at points P and Q as follows:

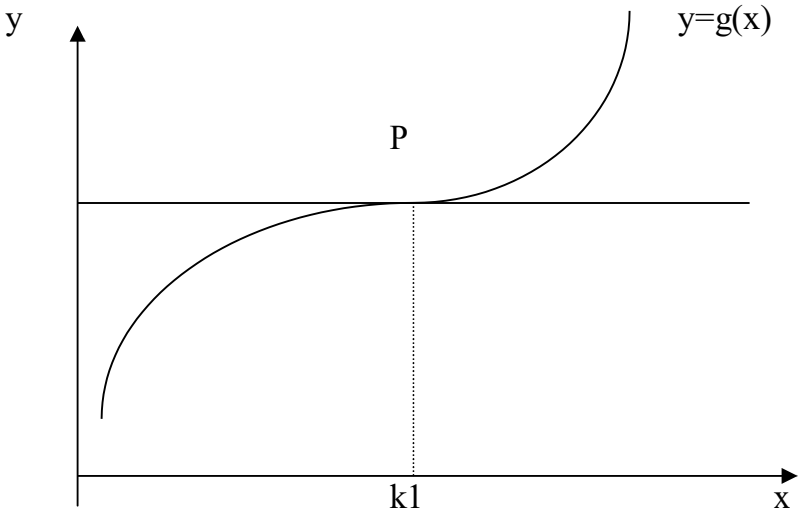
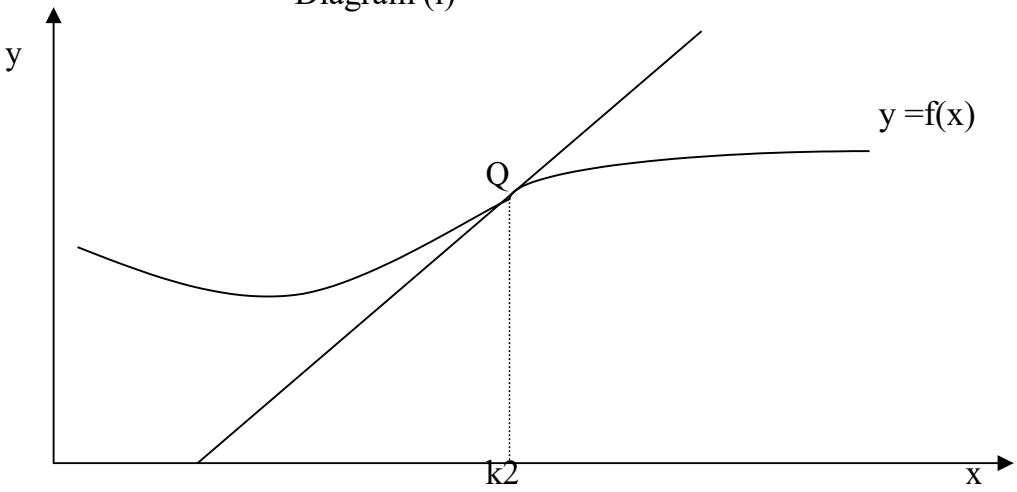


Diagram (i)



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The points of inflexion will occur at point P when

$g''(x) = 0$ at $x = k_1$

$g''(x) < 0$ at $x < k_1$

$g''(x) > 0$ at $x > k_1$

and at point Q when

$f''(x) = 0$ at $x = k_2$

$f''(x) > 0$ at $x < k_2$

$f''(x) < 0$ at $x > k_2$

Example

Find the points of inflexion on the curve of the function

$y = x^3$

Solution

The only possible inflexion points will occur where

$$\frac{d^2y}{dx^2} = 0$$

From the function given

$$\frac{dy}{dx} = 3x^2 \quad \text{and} \quad \frac{d^2y}{dx^2} = 6x$$

Equating the second derivative to zero, we have

$6x = 0$ or $x = 0$

We test whether the point at which $x = 0$ is an inflexion point as follows

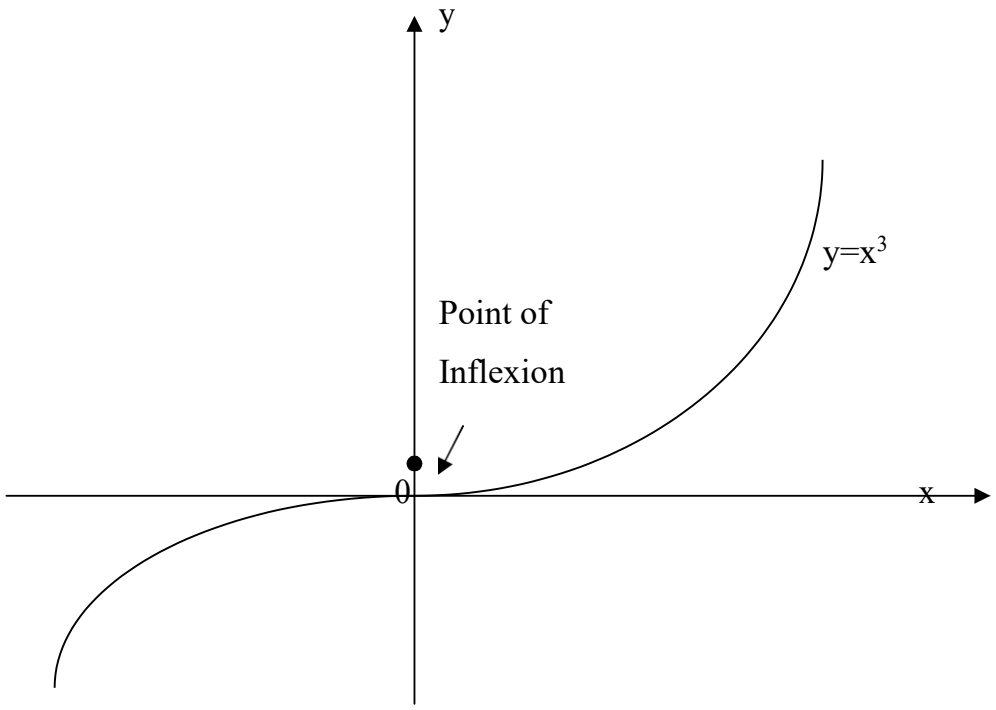
When x is slightly less than 0, $\frac{d^2y}{dx^2} < 0$ which means a downward concavity

When x is slightly larger than 0, $\frac{d^2y}{dx^2} > 0$ which means an upward concavity

Therefore we have a point of inflexion at point $x = 0$ because the concavity of the curve changes as we pass from the left to the right of $x = 0$

ILLUSTRATION

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Example

The weekly revenue Sh. R of a small company is given by

$R = 14 + 81x - \frac{x^3}{12}$ Where x is the number of units produced.

Required

i) Determine the number of units that maximize the revenue
ii) Determine the maximum revenue
iii) Determine the price per unit that will maximize revenue

Solution

i. To find maximum or minimum value we use differential calculus as follows

$$R = 14 + 18x - \frac{x^3}{12}$$

$$\frac{dR}{dx} = 81 - \frac{1}{12}.3x^2$$

$$\frac{d^2R}{dx^2} = 0 - \frac{1}{12}.3.2x = -\frac{x}{2}$$

$$put \quad \frac{dR}{dx} = 0 \quad i.e. \quad 81 - \frac{1}{4}x^2 = 0$$

which gives $x = 18$ or $x = -18$

$$\frac{d^2R}{dx^2} = -\frac{x}{2}$$

thus when $x = 18$; $\frac{d^2R}{dx^2} = -9$ *which is negative – indicating a* maximum *value*

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Therefore at x = 18, the value of R is a maximum. Similarly at x = -18, the value of R is a minimum. Therefore, the number of units that maximize the revenue = 18 units

ii. The maximum revenue is given by

$$R = 14 + 81 + 18 - \frac{(18)^3}{12}$$

= Shs. 986

ii. The price per unit to maximize the revenue is

$\frac{986}{18}$ = 54.78 or Shs.54.78

INTEGRATION

It is the reversal of differentiation
An integral can either be indefinite (when it has no numerical value) or definite (have specific numerical values)
It is represented by the sign $\int f(x)dx$.

Rules of integration

i. The integral of a constant

adx = ax +c where c = constant

Example

Find the following

a) 23dx

b) ^{2}dx. (where is a variable independent of x, thus it is treated as a constant).

Solution

a) 23dx = 23x + c

b) ^{2}dx. = ^{2} x + c

ii. The integral of x raised to the power n

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + c$$

Example

Find the following integrals

a) x^2dx

b) $x^{-5/2}$ dx

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Solution

i) $\int x^2 dx = \frac{1}{3}x^3 + c$

ii) $\int x^{-\frac{5}{2}} dx = -\frac{2}{3}\ \bar{x}^{\frac{3}{2}} + c$

iii). Integral of a constant times a function

$$\int af\ (x)dx = a\int f\ (x)\ dx$$

Example

Determine the following integrals

i. ax^3dx

ii. x^5dx

Solution

a) $\int ax^3dx = a\int fx^3dx$

$= \frac{a}{4}x^4 + c$

b) $\int 20x^5dx = 20\int fx^5dx$

$= \frac{10}{3}x^6 + c$

iv) Integral of sum of two or more functions

$\{f(x) + g(x)\}\,dx = f(x)dx + g(x)\,dx$

$\{f(x) + g(x) + h(x)\}dx = f(x)dx + g(x)dx + h(x)dx$

Example

Find the following

i. $(4x^2 + ½\,x^{-3})\,dx$

ii. $(x^{3/4} + 3/7\,x^{-½} + x^5)$

Solution

i) $\int\left(4x^2 + \frac{1}{2}x^{-3}\right)dx = \int 4x^2dx + \int \frac{1}{2}x^{-3}dx$

$= \frac{4}{3}x^3 - \frac{1}{4}x^2 + c$

ii) $\int\left(x^{\frac{3}{4}} + \frac{3}{7}x^{-\frac{1}{2}} + x^5\right)dx = \int x^{\frac{3}{4}}dx + \int \frac{3}{7}x^{-\frac{1}{2}}dx + \int x^5dx$

$= \frac{4}{7}x^{\frac{7}{4}} + \frac{6}{7}x^{\frac{1}{2}} + \frac{1}{6}x^6 + c$

v) Integral of a difference

$\{f(x) - g(x)\}\,dx = f(x)dx - g(x)\,dx$

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Definite integration

Definite integrals involve integration between specified limits, say a and b

The integral $\int_a^b f(x)dx$ Is a definite integral in which the limits of integration are a and b

The integrals is evaluated as follows

1. Compute the indefinite integral f(x)dx. Supposing it is F(x) + c
2. Attach the limits of integration
3. Substitute b(the upper limit) and then substitute a (the lower limit) for x.
4. Take the difference and the result is the numerical value for the definite integral.

Applying these steps to the definite integral

$$\int_a^b f(x)dx = \left[F(x) + c\right]_a^b$$

$$= \left\{\left[F(b) + c\right] - \left[F(a) + c\right]\right\}$$

$$= F(b) - F(a)$$

Example

Evaluate

i. $\int_1^3 (3x^2 + 3)dx$

ii. $\int_0^5 (x + 15)dx$

Solution

a. $\int_1^3 (3x^2 + 3)dx = [(x^3 + 3x + c)]$

$= (27 + 9 + c) - (1 + 3 + c)$

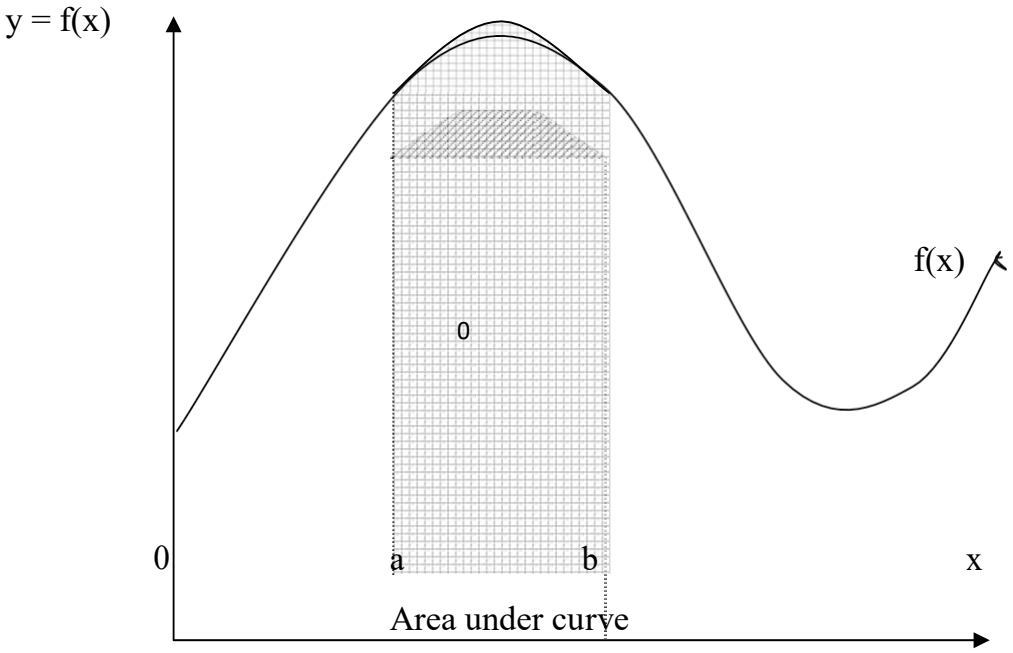
$= 32$

b. $\int_0^5 (x + 15)dx = [(\frac{1}{2}x^2 + 15x + c)]_0^5$

$= (12\frac{1}{2} + 75 + c) - (0 + 0 + c)$

$= 87\frac{1}{2}$

The numerical value of the definite integral $\int_a^b f(x)dx$ can be interpreted as the area bounded by the function f(x), the horizontal axis, and x=a and x=b see the figure below:



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Therefore $\int_a^b f(x)dx$ = A or area under the curve

Example

You are given the following marginal revenue function

$MR = a + a_1q$

Find the corresponding total revenue function

Solution

Total revenue $= \int MR.dq = \int (a + a_1q)dq$

$$= aq - \frac{1}{2} a_1 q^2 + c$$

Example

A firm has the following marginal cost function

$$MC = a - a_1 q + a_2 q^2$$

Find its total cost function.

Solution

The total cost C is given by

$C = MC.dq$

$= (a + a_1 q + a_2 q^2).dq$

$$= aq + \frac{a_1}{2} q^2 + \frac{a_2}{3} q^3 + c$$

Note: **Exams focus**: Note the difference between marginal function and total function. You differentiate total function to attain marginal function, this is common in exams,

total profit = total revenue – total cost.

Example

Your company manufactures large scale units. It has been shown that the marginal variable cost, which is the gradient of the total cost curve, is (92 – 2x) Shs. thousands, where x is the number of units of output per annum. The fixed costs are Shs. 800,000 per annum. It has also been shown that the marginal revenue which is the gradient of the total revenue is (112 – 2x) Shs. thousands.

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Required;-

i) Establish by integration the equation of the total cost curve
ii) Establish by integration the equation of the total revenue curve
iii) Establish the break even situation for your company
iv) Determine the number of units of output that would
   a) Maximize the total revenue and
   b) Maximize the total costs, together with the maximum total revenue and total costs

Solution

i. First find the indefinite integral limit points of the marginal cost as the first step to obtaining the total cost curve

   Thus $(92 – 2x)\,dx = 92x – x^2 + c$

   Where c is constant

   Since the total costs are the sum of variable costs and fixed costs, the constant term in the integral represents the fixed costs, thus if Tc are the total costs then,

   $Tc = 92x – x^2 + 800$

   or $Tc = 800 + 92x - x^2$

ii. As in the above case, the first step in determining the total revenue is to form the indefinite integral of the marginal revenue

Thus $(112 - 2x)\,dx = 112x - x^2 + c$

Where c is a constant

The total revenue is zero if no items are sold, thus the constant is zero and if Tr represents the total revenue, then

$$Tr = 112x - x^2$$

iii. At break even the total revenue is equal to the total costs

Thus $112x - x^2 = 800 + 92x - x^2$

$20x = 800$

$x = 40$ units per annum

iv. At maximum total revenue $\frac{dR}{dS} = 0$

a) $Tr = 112x - x^2$

$$\frac{d(Tr)}{dx} = 112 \quad 2x$$

Equating this to 0 we have

$112 - 2x = 0$

$x = 56$

Testing for critical point we have

$$\frac{d^2(Tr)}{dx^2} = -2$$

at the maximum point

$\frac{d^2(Tr)}{dS^2} < 0$ that is $112 - 2x = 0$

$x = 56$ units per annum

Since $\frac{d^2(Tr)}{dx^2} = -2$ this confirms the maximum

The maximum total revenue is Shs. (112 x 56 – 56 x 56) x 1000

= Shs. 3,136,000

ii. $Tc = 800 + 92x - x^2$

$$\frac{d(Tc)}{dx} = 92 - 2x$$

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$$\frac{d^2(Tc)}{dx^2} = \quad 2$$

At this maximum point

$$\frac{d(Tc)}{dx} = 0$$

92 – 2x = 0

92 = 2x

x = 46 units per annum

since

$\frac{d^2(Tc)}{dS^2} =$ 2this confirms the maximum

the maximum costs are Shs. (800 + 92 x 46 - 46 x 46) x 1000

= Shs. 2,916,000

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PRACTICE EXERCISES

QUESTION 1

Demand function for a firm is given by

$$P = 12 - 0.4Q$$

P is the price of the product, Q is the quantity demanded, and the total cost (C) is given by

$$C = 5 + 4Q + 0.6Q^2$$

At what price and quantity will the firm have maximum profit? If the firm aims at maximizing sales, what price should it charge?

Solution:

Let profit = z

Profit z = PQ – C

= (12 – 0.4Q) Q – (5 + 4Q + 0.6Q^2)

= 12Q – 0.4Q^2 – 5 – 4Q – 0.6Q^2

= 8Q – Q^2 – 5

For maximum profit, the differentiation of z with respect to Q equals zero.

$\frac{dz}{dQ} = 8 - 2Q = 0 \Rightarrow 2Q = 8 \Rightarrow$ Q = 4

So P = 12 – 0.4Q and for Q =4

= 12 – 1.6

= 10.4

$\frac{d^2z}{dQ^2} = -2Q < 0$ Profit is maximized.

Profit is maximised at a price of 10.4 and when quantity = 4

To maximize sales then,

$$\frac{d(PQ)}{dQ} = \frac{d(12Q - 0.4Q^2)}{dQ} = 0$$

$= 12 - 0.8Q = 0$

$\Rightarrow Q = \frac{12}{0.8} = 15$ and since $\frac{d^2(PQ)}{dQ^2} = -0.8 < 0$ then sales is maximized

So $P = 12 - 0.4 \times 15$

$= 6$

QUESTION 2

a) Two CPA students were discussing the relationship between average cost and total cost. One student said that since average cost is obtained by dividing the cost function by the number of units Q, it follows that the derivative of the average cost is the same as marginal cost, since the derivative of Q is 1.

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Required:
Comment on this analysis.

b) Gatheru and Kabiru Certified Public Accountants have recently started to give business advise to their clients. Acting as consultants, they have estimated the demand curve of a clients firm to be;

$AR=200-8Q$

Where AR is average revenue in millions of shillings and Q is the output in units.

Investigation of the client firm's cost profile shows that marginal cost (MC) is given by:

$MC=Q^2-28Q+211$(In million shillings)

Further investigations have shown that the firm's cost when not producing output is sh.10 million.

Required:

i) The equation of total cost
ii) The equation of total revenue
iii) An expression for profit.
iv) The level of output that maximizes profit
v) The equation of marginal revenue.

Solution:

a) Taking the following to mean:

TC – Total cost

AC – Average cost

MC – Marginal cost

Q – Number of units

Then $AC = \frac{TC}{Q}$

And $MC = \frac{d(TC)}{dQ}$

These are the relationships that link TC, AC, and MC.

To comment on the CPA students analysis,

The derivative of AC is as follows,

$$\frac{d(AC)}{dQ} = \frac{d(TC/Q)}{dQ} = \frac{\frac{d(TC)}{dQ} \times Q - TC}{Q^2} = \frac{1}{Q}\frac{d(TC)}{dQ} - \frac{TC}{Q^2}$$

Since $\frac{d(AC)}{dQ} \neq \frac{d(TC)}{dQ} = MC$ then the students comment is wrong in getting marginal cost. The student is right though in saying that $AC = \frac{TC}{Q}$.

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b)

i) Total cost function can be obtained from expression of marginal cost (MC) since,

$\frac{d(TC)}{dQ} = (MC)$ Then

$dTC = (MC)dQ$

Integrating both sides gives:

$TC = \int (MC)dQ$

Given $MC = Q^2 - 28Q + 211$

then $TC = \int (Q^2 - 28Q + 211)dQ$

$= \frac{Q^3}{3} - \frac{28Q^2}{2} + 211Q + A$

A – is a constant of integration.

Given that when Q = 0, TC = Sh 10 million

then $A = 10$

So the total cost function is as follows:

$$TC = \frac{Q^3}{3} - 14Q^2 + 211Q + 10$$

ii) The total revenue (TR) function can be obtained from Average revenue (AR) function as follows,

$$AR = \frac{TR}{Q} \quad So \quad TR = Q \times AR$$

$$= Q \times (200 - 8Q)$$
$$= 200Q - 8Q^2$$

iii) Profit equal to TR – TC. Since TR and TC expression have been obtained from (i) and (ii), then profit P is as follows,

$$P = TR - TC$$
$$= 200Q - 8Q^2 - (\frac{Q^3}{3} - 14Q^2 + 211Q + 10)$$
$$= 11Q + 6Q^2 - \frac{Q^3}{3} - 10$$

iv) The level of output that maximizes profit is got by equating the derivative of profit P with respect to Q to zero, as follows

$$\frac{d(P)}{dQ} = \frac{d(11Q + 6Q^2 - Q^3/3 - 10)}{dQ} = 0$$
$$= 11 + 12Q - Q^2 = 0$$

The solution to this quadratic equation is as follows:

$$Q = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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Where a, b, c are the coefficients of the equation as follows:

a = - 1, b = 12, and c = 11.

$$So \quad Q = \frac{-12 \pm \sqrt{12^2 - 4 \times (-1) \times 11}}{2 \times (-1)}$$

$$So\ Q = \frac{-12 + 10}{-2} = 1 \text{ or } Q = \frac{-12 - 1}{-2} = 11$$

Since two points of maximum profit exist, then the Q that gives more profit is the one to be used.

At Q = 11,

$$P = 11 \times 11 + 6 \times 11^2 - \frac{11^3}{3} - 10$$

= 151.333 million

At Q = 1,

$$P = -11 \times 1 + 6 \times 1^2 - \frac{1^3}{3} - 10$$

= 15.333 million

So the level that maximizes profit is Q = 11.

v) Marginal revenue (MR) can be obtained from Total revenue (TR) as follows:

$$\frac{d(TR)}{dQ} = MR$$

$TR = 200Q - 8Q^2$

So $$\frac{d(TR)}{dQ} = \frac{d(200Q - 8Q^2)}{dQ}$$

$= 200 - 16Q$

QUESTION 3

XYZ Company Limited invests in a particular project and it has been estimated that after X months of running, the cumulative profit (Sh.'000') from the project is given by the function $10x - x^2 - 5$, where x represents time in months. The project can run for eleven months at most.

Required:

i) Determine the initial cost of the project.

ii) Calculate the break-even time in months for the project.

iii) Determine the best time to end the project.

iv) Determine the total profit within the break-even points.

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Solution:

i) The initial cost of the project is determined when the time is zero. That is when the project is started. Given Profit $P = 10x - x^2 - 5$, then the initial cost of the project is when x = 0.

$Profit = 10 \times 0 - 0^2 - 5 = -5$

The initial cost is sh. 5000.

ii) Equating the profit function to zero and solving the function for the time determines break-even time in months for the project.

$P = 10x - x^2 - 5 = 0$

Since this is a quadratic equation, the solution is as follows,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Given a = - 1

b = 10

c = - 5

Then

$$x = \frac{-10 + \sqrt{10^2 - 4 \times (-1) \times (-5)}}{2 \times (-1)} \quad \text{or} \quad \frac{-10 - \sqrt{10^2 - 4 \times (-1) * (-5)}}{2 \times (-1)}$$

$= \frac{-10+8.94}{-2}$ or $\frac{-10-8.94}{-2}$

$= 0.527$ or 9.472 months

Break-even time is 0.527 and 9.472 months.

iii) The best time to end the project is when profit is at maximum. This is determined by differentiating the profit function with respect to time and equating to zero as follows:

$\frac{dP}{dx} = 10 - 2x = 0$

$\Rightarrow x = 5$

The best time to end the project is after 5 months.

iv) To obtain the total profit within the break-even points, the profit function is integrated within those break-even points as follows,

$\text{Profit} = \int_{0.527}^{9.472} P\,dx \int_{0.527}^{9.472} (10x \quad x^2)\,dx$

$= \left[\frac{10x^2}{2} - \frac{x^3}{3} - 5x \right]_{0.527}^{9.472}$

$= \left[\left(5 \times 9.472^2 \right) - \frac{(9.472)^3}{3} - \left(5 \times 9.472 \right) \right] - \left[\left(5 \times 0.527^2 \right) - \frac{(0.527)^3}{3} - \left(5 \times 0.527 \right) \right]$

$= 117.96 - (-1.30) = 119.26 \times 1000$

$=$ Sh. 119,260

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QUESTION 4

a) The number of shoppers queuing at any given time in a certain supermarket in downtown Nairobi can be approximately represented by the equation:
$y = x^3 - 14x^2 + 50x$ over the range $0 \le x \le 8.5$, where y is the number queuing and x is the time in hours after the store opens at 9.00a.m. (So that, for example 10.30a.m. is x=1.5, and 5.30p.m. - when the store closes is x= 8.5).

Required:

i) The management wants to know when they should deploy more cashiers and the number queuing at that time.

ii) Determine the number of man-hours spent per day by shoppers queuing.

b) An electronics firm carries out a small-scale test launch of a new low-priced pocket calculator. It estimates from this test that if it went into full-scale production it would sell between 1,000 and

2,500 calculators per month, and that its monthly revenue in thousands of shillings over this range of sales could be represented by the equation:

$R = -x^2 + 5x$

Where: x is the monthly output in thousands of calculators (it is assumed that it sells its entire output).

From experience of calculator production, the firm estimates its marginal cost in thousands of shillings could be represented by the equation:

$MC = x^2 - x + 2$

and that its fixed costs will be Sh.500 per month.

Required:

i) Determine the average cost and revenue equations for this firm.

ii) Determine the profit-maximizing output, the price that should be charged to maximize profit, and how much each calculator will then cost to make.

Solution:

a)

i) To obtain the time when there are a maximum number of people queuing, the derivative of the equation is equated to zero.

$$\frac{dy}{dx} = \frac{d(x^3 - 14x^2 + 50x)}{dx} = 0$$

$$= 3x^2 - 28x + 50 = 0$$

This is a quadratic equation with the following solutions,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Given a = 3

b = - 28

c = 50

Then

$$x = \frac{28 + \sqrt{28^2 - 4 \times (50) \times (3)}}{2 \times (3)} \quad \text{or} \quad \frac{28 - \sqrt{28^2 - 4 \times (50) \times (3)}}{2 \times (3)}$$

$$= \frac{28 + 13.56}{6} \text{ or } \frac{28 - 13.56}{6}$$

= 6.92 or 2.41

The number of shoppers queuing at these particular times is as follows,

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$y = 6.92^3 - 14 \times (6.92)^2 + 50 \text{ x } 6.92$

$= 6.96$

or $y = 2.41^3 - 14 \times (2.41)^2 + 50 \times 2.41$

$= 53.18$

The management should deploy more cashiers after 2.41 hours, that is at 11.25 am. The number of people queuing at this particular time is 53.

ii) The number of man hours spent is equal to Y × x. To get the man-hours spent per day, the function is integrated within the limits $0 \le x \le 8.5$

$$\int_0^{8.5} (Yx)\partial x = \int_0^{8.5} (x^4 - 14x^3 + 50x^2)dx$$

$$\left[\frac{x^5}{5} - \frac{14x^4}{4} + \frac{50x^3}{3}\right]_0^{8.5}$$

$= 839.3 - 0 = 839.3$ man-hours

b)

i) Average cost AC is given by the following expression.

$AC = \frac{TC}{x}$ where TC – Total cost

$TC = \int (MC)dx$ and given $MC = x^2 - x + 2$

Then $TC = \int (x^2 - x + 2)dx$

$= \frac{x^3}{3} - \frac{x^2}{2} + 2x + A$ where A is a constant of integration.

Given the information that when x = 0; TC = 500, then A is determined as follows,

$$500 = \frac{0^3}{3} - \frac{0^2}{2} + 2 \times 0 + A$$

$\Rightarrow A = 500$

So the TC function is as follows,

$$\frac{x^3}{3} - \frac{x^2}{2} + 2x + 500$$

The AC function then is

$$\frac{TC}{x} = \frac{x^2}{3} - \frac{x}{2} + \frac{500}{x}$$

Average revenue AR is given by the following expression

$AR = \frac{R}{x}$ Given $R = -x^2 + 5x$ then

$AR = -x + 5$

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ii) Profit maximizing output is obtained by equating the differential of profit to zero and solving for the x values as follows:

Profit P = R – TC

$$= -x^2 + 5x - \left(\frac{x^3}{3} - \frac{x^2}{2} + 2x + 500\right)$$

$$= -\frac{x^2}{2} - \frac{x^3}{3} + 3x - 500$$

$$\frac{dP}{dx} = 0 = -1x - x^2 + 3 \qquad x^2 + x - 3 = 0$$

Since this is a quadratic equation, the solutions are obtained as follows:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Given a = - 1

b = - 1

c = 3

Then

$$x = \frac{1 + \sqrt{1^2 - 4\times(-1)\times(3)}}{2\times(-1)} \quad \text{or} \quad \frac{1 - \sqrt{1^2 - 4\times(-1)\times(3)}}{2\times(-1)}$$

= - 2.3 or 1.3

So x = 1300 calculators.

Price to charge = AR = - 1.3 + 5 = sh. 3700

Cost per calculator = AC = $\frac{1.3^2}{3} - \frac{1.3}{2} + \frac{500}{1.3}$ = Sh. 384.5

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QUESTION 5

a) Explain the following terms as used in calculus:
   i) Turning point.
   ii) Second order derivative condition.
   iii) Partial derivative.
   iv) Mixed partial derivative.
   v) Saddle point.

b) Drumstick Chicken Wings Ltd supplies chicken wings for Kuku Inn with the following demand and cost functions for a given week:

P = 100 – 0.01 x - Price

TC = 50x + 30,000 - Total cost

Where:

x – number of chicken wings supplied.

Required:

i) Total revenue for Drumstick Chicken Wings Ltd.

ii) Determine the number of chicken wings that maximize weekly profit.

iii) What is the difference in profit if the Drumstick Chicken Wings Ltd. objective is to maximize revenue rather than profit?

Solution:

a)

i) Turning point is the point where a curve changes direction. It can either be local minimum, local maximum or point of inflexion.

ii) Second order derivative condition states that if the first derivative equals zero and the second derivative is defined then the given point is a relative minimum if the second derivative is greater than zero, or maximum if the second derivative is less than zero or a point of inflexion of the second derivative is equal to zero.

iii) Partial derivative is the derivative of a multivariate function (function of more than one variable). It is usually with respect to each of the independent variables.

iv) Mixed or cross partial derivative is obtained by first getting the derivative of multivariate function with respect to one variable then the second derivative with respect to the second variable.

v) A saddle point is a stationery point that is neither a maximum nor a minimum. Here the difference between the product of pure second partial derivative (second derivative of a function with respect to one variable) and square of mixed partial derivative is less than zero.

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b)

i) Total revenue $= Px$

$= (100-0.01x)\times x$

$= 100x-0.01x^2$

ii) Profit $=$ Revenue-cost

$= 100x-0.01x^2-50x-30000$

$= 50x-0.01x^2-30000$

$$\frac{d(Profit)}{dx} = 50 - 0.02x = 0 \Rightarrow x = \frac{50}{0.02} = 2500 \text{ Chicken wings}$$

iii) To maximize revenue

$$\frac{d(Revenue)}{dx} = 100 - 0.02x = 0 \Rightarrow x = 5000$$

So profit when revenue is maximized is

Profit$=50\times5000-0.01\times(5000)^2-30000=-30,000$

Maximum profit $=50\times2500-0.01\times(2500)^2-30000$

$125000-0.01\times6250000-30000=32,500$

So the difference in profit is 32,500-(-30000)=62,500

QUESTION 6

a) Given the following input – output matrix and demand vector of shoes S, rubber R and glue G industries, determine the production vector.

Input – output matrix

$$\begin{array}{c} \\ S \\ R \\ G \end{array}\begin{array}{c} \begin{array}{ccc} S & R & G \end{array} \\ \begin{pmatrix} 0.3 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ 0.2 & 0.3 & 0.4 \end{pmatrix} \end{array}$$

Demand vector $\begin{pmatrix} 40 \\ 50 \\ 60 \end{pmatrix}$

b) If in (a) above the demand of industries changes as follows:

S decreases by 10 units

R increases by 5 units

C increases by 10 units.

What should be the production levels?

Solution:

a) The Leontief open model is

Mx+d=x

So rearranging the equation

(I-M)x=d

$x=(I-M)^{-1}d$

Where M-matrix of technical coefficients

x-required production

d-the external demand

I-Identity matrix

Given that $M=\begin{pmatrix} 0.3 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ 0.2 & 0.3 & 0.4 \end{pmatrix}$

$d=\begin{pmatrix} 40 \\ 50 \\ 60 \end{pmatrix}$ Then

$$(I-M)=\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0.3 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ 0.2 & 0.3 & 0.4 \end{pmatrix}=\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ 0.2 & 0.3 & 0.4 \end{pmatrix}$$

$$(I-M)^{-1}=\frac{1}{\text{Determinant}(I-M)}\times \text{Adjoint}(I-M)$$

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Determinant (I-M) $= |1 \quad M| = 0.7(0.36 \quad 0.06 + 0.2(\quad 0.06 \quad 0.04) \quad 0.1\,(0.03 + 0.12)$

$= 0.21 - 0.02 - 0.015 = -0.175$

Ad joint (I-M)=Transpose of the co-factors of (I-M)

Co-factors of (I-M)$= \begin{pmatrix} 0.3 & 0.1 & 0.15 \\ 0.15 & 0.4 & 0.25 \\ 0.1 & 0.15 & 0.4 \end{pmatrix}$

So adjoint (I-M)$= \begin{pmatrix} 0.3 & 0.15 & 0.1 \\ 0.1 & 0.4 & 0.15 \\ 0.15 & 0.25 & 0.4 \end{pmatrix} = A$

$$X = (I \quad M)^{-1}d = \frac{1}{0.1_{75}} x \begin{pmatrix} 0.3 & 0.15 & 0.1 \\ 0.1 & 0.4 & 0.15 \\ 0.15 & 0.25 & 0.4 \end{pmatrix} x \begin{pmatrix} 40 \\ 50 \\ 60 \end{pmatrix} = \begin{pmatrix} 145.7 \\ 188.6 \\ 242.9 \end{pmatrix}$$

b) If $d = \begin{pmatrix} 30 \\ 55 \\ 70 \end{pmatrix}$ the production vector will be

$$X = (I \quad M)^{-1}d = \frac{1}{0.1_{75}} x \begin{pmatrix} 0.3 & 0.15 & 0.1 \\ 0.1 & 0.4 & 0.15 \\ 0.15 & 0.25 & 0.4 \end{pmatrix} x \begin{pmatrix} 40 \\ 50 \\ 60 \end{pmatrix} = \begin{pmatrix} 138.57 \\ 202.86 \\ 264.29 \end{pmatrix}$$

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TOPIC 2

PROBABILITY THEORY

SET THEORY

A Set is a collection of distinct items or objects e.g. members, letters, people, houses etc.
The items or objects in a set are called **members or elements** of the set.
Any set is denoted using a capital letter while the elements are denoted using small letters.
The members or elements of the set are enclosed within the curly brackets and separated using comas, e.g. a set of vowels can be written as follows; A = {a, e, i, o, u}
If element x is a member of set A it is denoted as follows
x ∈ A (x belongs to set A)
If X is not an element of A it is denoted as
x A (x doesn't belong to set A)

We may consider all the ocean in the world to be a set with the objects being whales, sea plants, sharks, octopus etc, similarly all the fresh water lakes in Africa can form a set. Supposing A to be a set

A = {4, 6, 8, 13}

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The objects in the set, that is, the integers 4, 6, 8 and 13 are referred to as the members or elements of the set. The elements of a set can be listed in any order. For example,

A = {4, 6, 8, 13} = {8, 4, 13, 6}

Sets are always precisely defined. Each element occurs once and only once in a set.

The notation ∈ is used to indicate membership of a set. represents non membership. However, in order to represent the fact that one set is a subject of another set, we use the notation ⊂. A set "S" is a subset of another set "T" if every element in "S" is a member of "T"

Example

If A = {4, 6, 8, 13} then

i) 4 ∈ {4, 6, 8, 13} or 4 ∈ A; 16 A

ii) {4, 8} ⊂ A; {5, 7} ⊄ A; A ⊂ A

Methods of set representation

Capital letters are normally used to represent sets. However, there are two different methods for representing members of a set:

i. The descriptive method and
ii. The enumerative method

The descriptive method involves the description of members of the set in such a way that one can determine the elements of the set without difficulty.
The enumerative method requires that one writes out all the members of the set within the curly brackets.
For example, the set of numbers 0, 1, 2, 3, 4, 5, 6 and 7 can be represented as follows

P = {0, 1, 2, 3, 4, 5, 6, 7} , enumerative method

P = {X/x = 0, 1, 2…7} descriptive method

Or

P = {x/0 ≤ x ≤7} *where x is an integer.*

Application of set Theory

i) It is used in capturing statistical data.
ii) It is used in solving counting problems
iii) It shows the logical relationship between two or more sets.
iv) It creates a basis for probability theory
v) It is a research tool that can be used in data capturing.

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TYPES OF SETS

Subset – This is a portion of a set where the elements of that set belongs to another bigger set.
Universal set (U) – This is a set containing all the elements under consideration e.g. a set of all the students in college, a set of alphabetical letters, a set of all the months in the source of the year.
Finite set – This is a set containing countable elements e.g. a set of weekdays a set of students in sec iv etc.
Null/Empty /void set ( ) – A set without elements, e.g. a set of married bachelors.
Infinite sets – This is a set containing countless elements e.g. a set of counting numbers.

Sets concepts and Operations

Concepts;

1. Overlapping sets

These are two or more sets with some common elements.
Eg: A{1,2,3,4,5,6}
B{2,4,6,8,10} Overlapping set.

2. Sets equality

Two or more sets are said to be equal if and only if they have the same elements but not necessarily the same order of elements.
Eg: A- {a, b, c, d}

C = {b,c, a, d,}

A = C

3. **Disjoint sets**

These are two or more sets without common elements

Eg: A- {a, b, c, d}

C = {1,2, 3, 4,}

Set operation;

1) **Sets intersection (n)**

This operation represents a set containing the common elements in two or more sets.

If A = {1 2 3 4 5 6}

B = {2, 4, 6, 8, 10}

Then AnB = {2 4 6}

If set C = {11, 12, 13,14}

Then AnC =( )

2) **Set Union**

This operation represents a collection of all the elements in two or more sets without repetition if the sets are overlapping.

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If A = {1 2 3 4 5 6} n (A) = 6

B = { 2, 4, 6, 8, 10} n (B) = 5

AUB = {1, 2, 3, 4, 5, 6, 8, 10} n(AUB) = 8

3) **Set difference (-)**

Given two sets A & B which are overlapping, the difference between A & B is a set of elements that are in set A but not in set B.

Similarly B difference A is a set of elements in B but not in A.

If A = {1, 2, 3, 4, 5, 6}

B= {2, 4, 6, 8, 10}

Then A – B = {1, 3, 5}

B – A = {8, 10}

4) **Compliment (C)**

Compliment of a set is a set of elements that are not in the original set but they are part of the universal set, e.g.

If A = {1, 2, 3, 4, 5, 6}

Then compliment of A = A^c = A1 = {7, 8, 9, 10 ..... $\propto$ }

NB//

Set theory begins with a fundamental binary relation between an object *o* and a set *A*. If *o* is a **member** (or **element**) of *A*, write $o \in A$. Since sets are objects, the membership relation can relate sets as well.

A derived binary relation between two sets is the subset relation, also called **set inclusion**. If all the members of set *A* are also members of set *B*, then *A* is a **subset** of *B*, denoted $A \subseteq B$. For example, {1, 2} is a subset of {1,2,3} , but {1,4} is not. From this definition, it is clear that a set is a subset of itself; for cases where one wishes to rule out this, the term **proper subset** is defined. *A* is called a **proper subset** of *B* if and only if *A* is a subset of *B*, but *B* is **not** a subset of *A*.

Just as arithmetic features binary operations on numbers, set theory features binary operations on sets. The:

- **Union** of the sets *A* and *B*, denoted $A \cup B$, is the set of all objects that are a member of *A*, or *B*, or both. The union of {1, 2, 3} and {2, 3, 4} is the set {1, 2, 3, 4} .
- **Intersection** of the sets *A* and *B*, denoted $A \cap B$, is the set of all objects that are members of both *A* and *B*. The intersection of {1, 2, 3} and {2, 3, 4} is the set {2, 3} .
- **Set difference** of *U* and *A*, denoted $U \setminus A$, is the set of all members of *U* that are not members of *A*. The set difference {1,2,3} \ {2,3,4} is {1} , while, conversely, the set difference {2,3,4} \ {1,2,3} is {4} . When *A* is a subset of *U*, the set difference $U \setminus A$ is also called the **complement** of *A* in *U*. In this case, if the choice of *U* is clear from the context, the notation A^c is sometimes used instead of $U \setminus A$, particularly if *U* is a universal set as in the study of Venn diagrams.
- **Symmetric difference** of sets *A* and *B*, denoted $A \circ B$ or $A \quad B$, is the set of all objects that are a member of exactly one of *A* and *B* (elements which are in one of the sets, but not in both). For instance, for the sets {1,2,3} and {2,3,4} , the symmetric difference set is {1,4} . It is the set difference of the union and the intersection, $(A \cup B) \setminus (A \cap B)$ or $(A \setminus B) \cup (B \setminus A)$.
- **Cartesian product** of *A* and *B*, denoted $A \times B$, is the set whose members are all possible ordered pairs (a,b) where *a* is a member of *A* and *b* is a member of *B*. The cartesian product of {1, 2} and {red, white} is {(1, red), (1, white), (2, red), (2, white)}.
- **Power set** of a set *A* is the set whose members are all possible subsets of *A*. For example, the power set of {1, 2} is { {}, {1}, {2}, {1,2} } .

Some basic sets of central importance are the empty set (the unique set containing no elements), the set of natural numbers, and the set of real numbers.

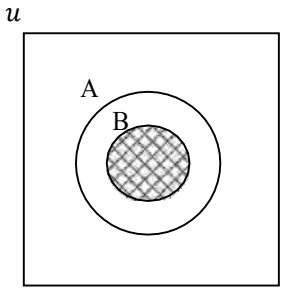
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VENN DIAGRAMS

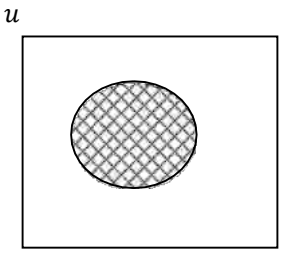
This is a pictorial representation of sets and their relationships.

They involve the use of loops enclosed within a square or a rectangle. The loop represent a specific set while the square / rectangle represents the universal set from where the set was drawn.

If set B is a subset of A then the venn diagram of subset B is (BCA).



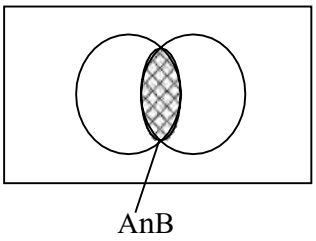
Set A

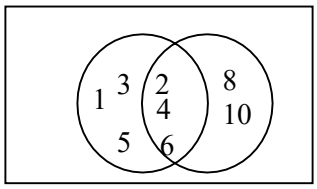


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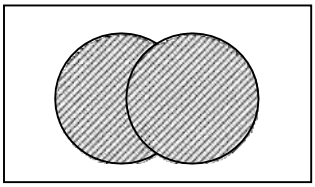
Intersection of set A & B (AnB) (overlapping sets)
IF A = {1, 2, 3, 4 ,5, 6}
B= {2, 4, 6, 8, 10}

Then;

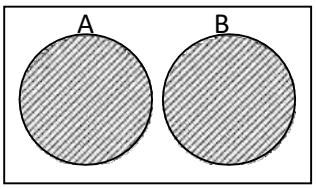




AUB (A union B) (Overlapping sets)

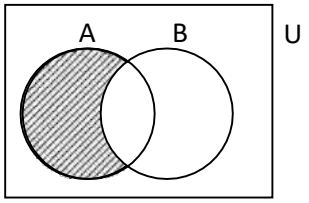


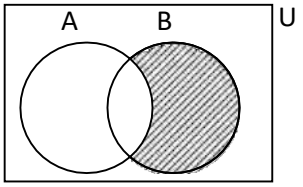
AUB (Disjoint Sets)



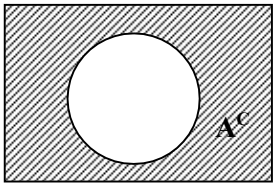
A – B (over lapping sets) i.e A difference B

B – A

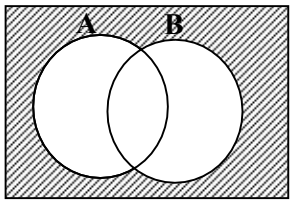




Complement of A (A^c)

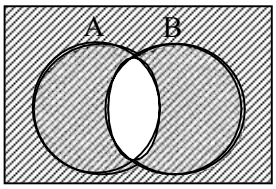


$(A \cup B)^C$

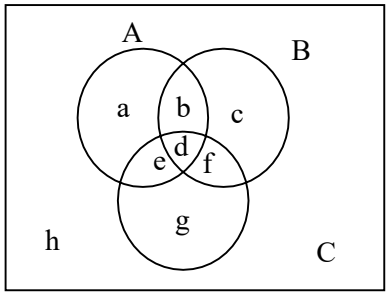


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$(A \cap B)^C$



Venn diagram for sets A, B, & C (overlapping set)



Observation

The venn diagram has 8 sectors i.e: a, b, c, d, e, f, g, & h.

The small letters represents number of elements in each sector.

Sector	Interpretation
a, c and g	– Number of elements in set A only, B only and C only.
b, e and f	– Number of elements at the intersection of A and B only, A and C only, B and C only respectively. eb = AnB-C; e = AnC-B; f = BnC – A.
d	– Number of common elements in all the three sets i.e. AnBnC
h	– Number of elements outside the three sets i.e. (AUBUC)C
b+d	– AnB (A and B)
d + e	– AnC (A and C)
d + f	– BnC
a + b + c	– A or B only. (AuB only) (AUB – C) Same as c + f + g and a + e + g
a + b + d + e	– A
c+ b + d + f	– B
a + b+c +d+e +f +g + h	– U (Universal set)
a+b+c + d + e + f	– A or B (AUB)

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SOLVING PROBLEMS USING VENN DIAGRAMS

ILLUSTRATION

a) A quick survey of 1,000 children in a refugee camp produced the following results:

320 children were fed on beans

200 children were fed on rice.

450 children were fed on potatoes.

150 children were fed on beans and potatoes.

70 children were fed on beans and rice.

100 children were fed on rice and potatoes.

300 children were fed on none of the three types of food.

Required:

(i) Present the above information in the form of a Venn diagram.

(ii) The number of children who were fed on all the three types of food.

(iii) The number of children who were fed on exactly one of the three types of food.

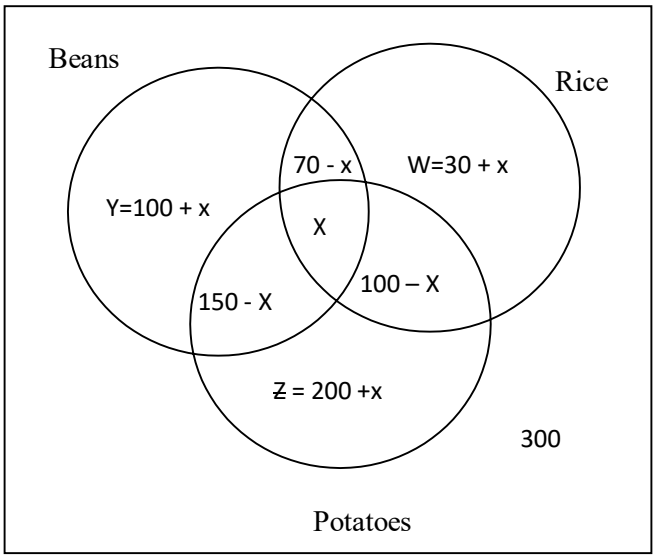
(iv) The number of children who were fed on at least two types of food.

Solution

i.

VENN DIAGRAM

$$C = 1000$$



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$Y + 70 – X + X + 150 – X = 320$

$Y – X = 320 – 220$

$Y = 100 + X$

$W + 70 – X + X + 100 – X = 200$

$W = 30 + X$

$W = 200 – 170 + X$

$Z + 150 – X + X + 100 – X = 450$

$Z = 450 – 250+X$

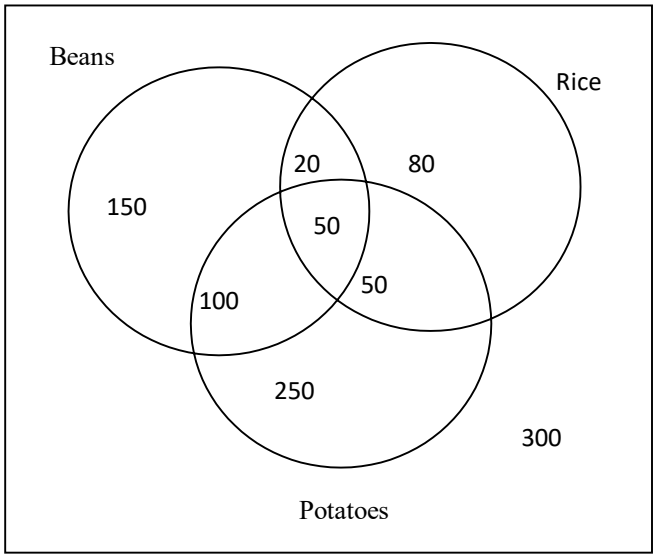
$Z= 200 + X$

$100 + X + 70 – X + X + 150 – X + 30 + X + 200 + X + 300 = 1,000$

$X = 1,000 – 950$

$X = 50$

CORRECT VENN DIAGRAM



ii) The number of children who fed on all the three types of food = 50

iii) The number of children who fed on exactly one of the three types of food

150 +50 + 250 = 480

iv) The number of children who fed on at least two types of food

100 + 20 + 50 + 50 = 220

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PROBABILITY THEORY AND DISTRIBUTION

Probability (or **likelihood**) is a measure or estimation of how likely it is that something will happen or that a statement is true. Probabilities are given as values between 0 (0% chance or *will not happen*) and 1 (100% chance or *will happen*).The higher the degree of probability, the more likely the event is to happen, or, in a longer series of samples, the greater the number of times such event is expected to happen.

These concepts have been given an axiomatic mathematical derivation in probability theory, which is used widely in such areas of study as mathematics, statistics, finance, gambling, science, artificial intelligence/machine learning and philosophy to, for example, draw inferences about the expected frequency of events. Probability theory is also used to describe the underlying mechanics and regularities of complex systems.

The word *Probability* derives from the Latin word *probabilitas*, which can also mean *probity*, a measure of the authority of a witness in a legal case in Europe, and often correlated with the witness's nobility. In a sense, this differs much from the modern meaning of *probability*, which, in

contrast, is a measure of the weight of empirical evidence, and is arrived at from inductive reasoning and statistical inference.

When dealing with experiments that are random and well-defined in a purely theoretical setting (like tossing a fair coin), probabilities describe the statistical number of outcomes considered divided by the number of all outcomes (tossing a fair coin twice will yield HH with probability 1/4, because the four outcomes HH, HT, TH and TT are possible). When it comes to practical application, however, the word *probability* does not have a singular direct definition. In fact, there are two major categories of **probability interpretations**, whose adherents possess conflicting views about the fundamental nature of probability:

1. Objectivists assign numbers to describe some objective or physical state of affairs. The most popular version of objective probability is frequentist probability, which claims that the probability of a random event denotes the *relative frequency of occurrence* of an experiment's outcome, when repeating the experiment. This interpretation considers probability to be the relative frequency "in the long run" of outcomes.A modification of this is propensity probability, which interprets probability as the tendency of some experiment to yield a certain outcome, even if it is performed only once.
2. Subjectivists assign numbers per subjective probability, i.e., as a degree of belief.The most popular version of subjective probability is Bayesian probability, which includes expert knowledge as well as experimental data to produce probabilities. The expert knowledge is represented by some (subjective) prior probability distribution. The data is incorporated in a likelihood function. The product of the prior and the likelihood, normalized, results in a posterior probability distribution that incorporates all the information known to date. Starting from arbitrary, subjective probabilities for a group of agents, some Bayesiansclaim that all agents will eventually have sufficiently similar assessments of probabilities, given enough evidence.

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Basic terms in probability

i) Probability experiment

This is any process that yields some outcomes e.g. taking an examination or tossing a coin.
A probability experiment can be theoretical or experimental.

ii) Sample space (s)

This is a collection of all the possible outcomes in a probability experiment e.g. in throwing a fair dice the sample space S = {1,2,3,4,5,6}
In an examination the sample space is S = {Pass, fail}

iii) Event

An event is a collection of either one or some of the outcomes in a probability experiement.
If the event has only one outcome it is known as elementary event.
If the event has more than one outcome it is known as compound event.
In case of a dice S = {1,2,3,4,5,6}

P (1) = $\frac{1}{6}$ - Elementary event.

P (Even faces) = P (2,4,6) = $\frac{3}{6}$ = ½ - Compound event.

iv) **Marginal probability**
This is the probability of either elementary or compound event

v) **Joint probability**
This is the resultant probability when two or more marginal probabilities, are combined through the use of probability laws.

vi) **Collectively exhaustive events.**
These are events or outcomes whereby one of them must occur on a single trial of a probability experiment. Eventually it is possible to list down all the outcomes of the collectively exhaustive events e.g. in tossing a fair coin, the events of head and tail are collectively exhaustive.

Probability Approaches

1. Theoretical / Prior Probability
2. Experimental / Empirical
3. Exhaustive / personalistic

There are three approaches to probability as described above.

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Classical / Theoretical/ prior probability

This is the probability based on the known physical situation and hence no need carrying out a probability experiment.

In this case, all events are equally likely implying that they have same chances of occurrence and the same value of probability.

Any probability based on either throwing a dice, throwing a coin, drawing a playing card is classified as theoretical probability.

This approach to probability doesn't have business application but mainly for learning purposes.

Experimental / empirical probability

This is the probability based on either probability experiment or past records or data.

In this case, the events are not equally-likely and therefore the probability of different events will differ.

Probability of a given event under this context is a ratio between the favourable outcomes to the total number of outcomes.

Most of the probabilities on a business environment are based on this approach.

Subjective / Personalistic approach

This approach doesn't have either theoretical or experimental background mainly based on either personal judgment or experience.

Therefore the probability of the same event in this approach will differ from one person to another. This approach is applicable in managerial decision making where there are no data and no time for experimentation.

Relationship of events

There are four relationships of events as described below.

1. **Mutually exclusive events**

These are two or more events which cannot happen simultaneously on a single trial of probability experiment. Therefore, the occurrence of one event excludes the occurrence of the other e.g. in taking an examination, one can either pass or fail but cannot pass and fail at the sometime therefore events of passing and failing are mutually exclusive.

2. **Mutually non-exclusive events.**

These are two or more events which can occur at the same time on a single trial or a probability experiment hence the occurrences of one cannot prevent the occurrence of the other e.g. the events of raining or sunshine can occur simultaneously and hence mutually non-exclusive.

3. **Independent events**

These are two or more events where the occurrence or non-occurrence of one does not affect the occurrence or non-occurrence of the other. These events are said that they have nothing to do with each other.

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4. **Dependent events**

These are events where the occurrence or non-occurrence of one affects the occurrence or non-occurrence of the other.

The events have a conditional relationship e.g. the event of passing an exam is dictated by a number of other events like teaching, reading, revising etc.

Probability Conjunctions

These are connecting terms in probability namely “AND and “OR” ‘AND’ implies happening at the same time of two or more events but not necessarily multiplication.

Multiplication of probability events is mainly used when the events are either independent or dependent.

The conjunction “AND is similar to intersection symbolin set theory.

The conjunction OR implies either one event happens or both happen. The conjunction represents the union of a probability values through addition of the marginal probabilities.

RULES OF PROBABILITY

(a) Addition Rule – This rule is used to calculate the probability of two or more mutually exclusive events. In such circumstances the probability of the separate events must be added.

Let A and B be two events then the addition law states that P(A or B) = P (A) + P(B) - P (A and B

If A and B are mutually exclusive events then P (A and B) = 0
hence the law becomes P (A or B) = P (A) + P (B)

Example
What is the probability of throwing a 3 or a 6 with a throw of a dice?

Solution
P (throwing a 3 or a 6) = $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

(b) Multiplication rule
This is used when there is a string of independent events for which individual probability is known and it is required to know the overall probability.
Let A and B be any two events the multiplication law states that P(A and B) = P (A) x P (B/A)
If A and B are independent events then P(B/A) = P (B)
Hence the rule becomes P (A and B) = P(A) x P (B)

Example
What is the probability of a 3 and a 6 with two throws of a dice?

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Solution
P(throwing a 3) and P(6)
= P(3) and P(6) = $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$

Note: In probability 'and' is replaced by 'x' – multiplication.

P(x) and P(y) ≠ P(x and y) note that these two are different. The first implies P(x) happening and P(y), but if the order of which happened first is unimportant then we have p(x and y).

In the example above:
P (3) and P(6) = $\frac{1}{36}$

but
P (3 and 6) = P(3 followed by 6) or P(6 followed by 3)
= [P(3) P(6)] or [P(6) P(3)]
= $\frac{1}{36} + \frac{1}{36} = \frac{1}{18}$

(c) Conditional probability
This is the probability associated with combinations of events but given that some prior result has already been achieved with one of them.

Its expressed in the form of

P(x|y) = Probability of x given that y has already occurred.

$P(x|y) = \frac{P(xy)}{P(y)}$ → conditional probability formula.

Example:
In a competitive examination 30 candidates are to be selected in all 600 candidates who appear in a written test, and 100 will be called for the interview.

(i) What is the probability that a person will be called for the interview?

(ii) Determine the probability of a person getting selected if he has been called for the interview?

(iii) Probability that person is called for the interview and is selected?

Solution:
Let event A be that the person is called for the interview and event B that he is selected.

(i) $\therefore P(A) = \frac{100}{600} = \frac{1}{6}$

(ii) $P(B|A) = \frac{30}{100} = \frac{3}{10}$

(iii) $P(AB) = P(A) \times P(B|A)$

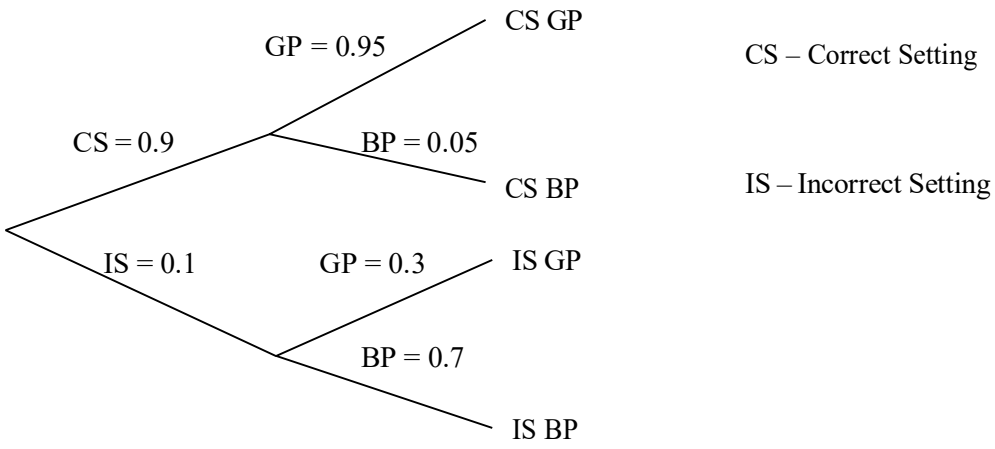
$= \frac{1}{6} \times \frac{3}{10} = \frac{3}{60} = \frac{1}{20}$

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Example:
From past experience a machine is known to be set up correctly on 90% of occasions. If the machine is set up correctly then 95% of good parts are expected but if the machine is not set up correctly then the probability of a good part is only 30%.

On a particular day the machine is set up and the first component produced and found to be good. What is the probability that the machine is set up correctly.

Solution
This is displayed in the form of a probability tree or diagram as follows:



P(CSGP) = 0.9 × 0.95 = 0.855
P(CSBP) = 0.9 × 0.05 = 0.045
P(ISGP) = 0.1 × 0.3 = 0.03
P(ISBP) = 0.1 × 0.7 = 0.07
1.00

- Probability of getting a good part (GP) = P(CSGP) or P(ISGP)
= P(CSGP) + P(ISGP)
= 0.855 + 0.03 = 0.885

Note: Good parts may be produced when the machine is correctly set up and also when its incorrectly setup. In 1000 trials, 855 occasions when its correctly setup and good parts produced (CSGP) and 30 occasions when its incorrectly setup and good parts produced (ISGP).

- Probability that the machine is correctly set up after getting a good part.

$$= \frac{\text{Number of favourable outcomes}}{\text{Total possible outcomes}} = \frac{P(CSGP)}{P(GP)} = \frac{0.855}{0.885} = 0.966$$

Or

$$= P(CS|GP) = \frac{P(CSGP)}{P(GP)} = \frac{0.855}{0.885} = 0.966$$

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Example
In a class of 100 students, 36 are male and are studying accounting, 9 are male but not studying accounting, 42 are female and studying accounting, 13 are female and are not studying accounting. Use these data to deduce probabilities concerning a student drawn at random.

Solution

	Accounting A	**Not accounting $\bar{A}$**	**Total**
Male M	36	9	45
Female F	42	13	55
Total	78	22	100

$P(M) = \frac{45}{100} = 0.45$
$P(F) = \frac{55}{100} = 0.55$
$P(A) = \frac{78}{100} = 0.78$
$P(\bar{A}) = \frac{22}{100} = 0.22$
$P(M \text{ and } A) = P(A \text{ and } M) = \frac{36}{100} = 0.36$
$P(M \text{ and } \bar{A}) = 0.09$
$P(F \text{ and } \bar{A}) = 0.13$

These probabilities can be express differently as;

P(M) = P(M and A) or P(M and $\bar{A}$)
= 0.36 + 0.09 = 0.45

P(F) = P(F and A) or P(F and $\bar{A}$)
= 0.42 + 0.13 = 0.55

P(A) = P(A and M) + P(A and F) = 0.36 + 0.42 = 0.78

$P(\bar{A})$ = P($\bar{A}$ and M) + P($\bar{A}$ and F) = 0.09 + 0.13 = 0.22

Now calculate the probability that a student is studying accounting given that he is male.

This is a conditional probability given as P(A|M)

$$P(A|M) = \frac{P(A \text{ and } M)}{P(M)} = \frac{0.36}{0.45} = 0.80$$

From the formula above we get that,

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P(A and M) = P(M) P(A|M) ..... (i)
Note that P(A|M) ≠ P(M|A)

Since $P(M|A) = \frac{P(A \text{ and } M)}{P(A)}$ this is known as the Bayes' rule.

BAYES' RULE/THEOREM

In probability theory and statistics, **Bayes' theorem** (alternatively **Bayes' law**) is a theorem with two distinct interpretations. In the *Bayesian interpretation*, it expresses how a subjective degree of belief should rationally change to account for evidence. In the *frequentist interpretation*, it relates inverse representations of the probabilities concerning two events. In the Bayesian interpretation, Bayes' theorem is fundamental to Bayesian statistics, and has applications in fields including science, engineering, economics (particularlymicroeconomics), game theory, medicine and law. The application of Bayes' theorem to update beliefs is called Bayesian inference.

Bayes' theorem is named after Thomas Bayes (1701–1761), who first suggested using the theorem to update beliefs. His work was significantly edited and updated by Richard Price before it was posthumously read at the Royal Society. The ideas gained limited exposure until they were independently rediscovered and further developed by Laplace, who first published the modern formulation in his 1812 *Théorie analytique des probabilités*. Until the second half of the 20th

century, the Bayesian interpretation was largely rejected by the mathematics community as unscientific However, it is now widely accepted. This may have been due to the development of computing, which enabled the successful application of Bayesianism to many complex problems.

Sir Harold Jeffreys wrote that Bayes' theorem "is to the theory of probability what Pythagoras's theorem is to geometry".

Introductory example

Suppose someone told you they had a nice conversation with someone on the train. Not knowing anything else about this conversation, the probability that they were speaking to a woman is 50%. Now suppose they also told you that this person had long hair. It is now more likely she was speaking to a woman, since most long-haired people are women. Bayes' theorem can be used to calculate the probability that the person is a woman.

To see how this is done, let

W represent the event that the conversation was held with a woman, and
L denote the event that the conversation was held with a long-haired person.

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It can be assumed that women constitute half the population for this example. So, not knowing anything else, the probability that W occurs is

$$P(W) = 0.5.$$

Suppose it is also known that 75% of women have long hair, which we denote as

$$P(L|W) = 0.75$$

(read: the probability of event L given event W is 0.75).
Likewise, suppose it is known that 30% of men have long hair, or

$$P(L|M) = 0.30,$$

where M is the complementary event of W, i.e., the event that the conversation was held with a man (assuming that every human is either a man or a woman).

Our goal is to calculate the probability that the conversation was held with a woman, given the fact that the person had long hair, or, in our notation, $P(W|L)$. Using the formula for Bayes' theorem, we have:

$$P(W|L) = \frac{P(L|W)P(W)}{P(L)} = \frac{P(L|W)P(W)}{P(L|W)P(W) + P(L|M)P(M)}$$

where we have used the law of total probability. The numeric answer can be obtained by substituting the above values into this formula. This yields

$$P(W|L) = \frac{0.75 \cdot 0.50}{0.75 \cdot 0.50 + 0.30 \cdot 0.50} \approx \mathbf{0.714},$$

i.e., the probability that the conversation was held with a woman, given that the person had long hair, is about 71%.

Statement and interpretation

Mathematically, Bayes' theorem gives the relationship between the probabilities of A and B, $P(A)$ and $P(B)$, and the conditional probabilities of A given B and B given A ($P(A|B)akd\ P\ (B|A)$). In its most common form, it is:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}.$$

Or

$$P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}$$

It's used frequently in decision making where information is given in form of condition probabilities and the reverse of these probabilities must be found.

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Example

Analysis of questionnaire complete by holiday makers showed that 0.75 classified their holiday as good at Malindi. The probability of hot weather in the resort is 0.6. If the probability of regarding holiday as good given hot weather is 0.9, what is the probability that there was hot weather if a holiday maker considers his holiday good?

Solution

$$P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}$$

Let H = hot weather

G = Good

P(G) = 0.75

P(H) = 0.6 and P(G|H) = 0.9 (Probability of regard holiday as good given hot weather)

Now the question requires us to get

P(H|G) = Probability of (there was) hot weather given that the holiday has been rated as good).

$$= \frac{P(H)P(G|H)}{P(G)} = \frac{(0.6)(0.9)}{0.75}$$

$$= 0.72.$$

ILLUSTRATION

A machine comprises of 3 transformers A, B and C. The machine may operate if at least 2 transformers are working. The probability of each transformer working are given as shown below;

P(A) = 0.6, P(B) = 0.5, P(C) = 0.7

A mechanical engineer went to inspect the working conditions of these transformers. Find the probabilities of having the following outcomes

i) Only one transformer operating
ii) Two transformers are operating
iii) All three transformers are operating
iv) None is operating
v) At least 2 are operating
vi) At most 2 are operating

Solution

P(A) =0.6 P($\bar{A}$) = 0.4 P(B) = 0.5 P($\bar{B}$) = 0.5
P(C) = 0.7 P($\bar{C}$) = 0.3

P(only one transformer is operating) is given by the following possibilities

	1st	2nd	3rd	
P	(A	B	C)	= 0.6 x 0.5 x 0.3 = 0.09
P	($\bar{A}$	B	C)	= 0.4 x 0.5 x 0.3 = 0.06
P	($\bar{A}$	B	C)	= 0.4 x 0.5 x 0.7 = 0.14

∴ P(Only one transformer working)
= 0.09 + 0.06 + 0.14 = 0.29

i. P(only two transformers are operating) is given by the following possibilities.

	1st	2nd	3rd	
P	(A	B	C)	= 0.6 x 0.5 x 0.3 = 0.09
P	(A	B	C)	= 0.6 x 0.5 x 0.7 = 0.21
P	($\bar{A}$	B	C)	= 0.4 x 0.5 x 0.7 = 0.14

∴ P(Only two transformers are operating)
= 0.09 + 0.21 + 0.14 = 0.44

ii. P(all the three transformers are operating).
= P(A) x P(B) x P(C)
= 0.6 x 0.5 x 0.7
= 0.21

iii. P(none of the transformers is operating).
= P($\bar{A}$) x P($\bar{B}$) x P($\bar{C}$)
= 0.4 x 0.5 x 0.3

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= 0.06

iv. P(at least 2 working).
= P(exactly 2 working) + P(all three working)
= 0.44 + 0.21
= 0.65

v. P(at most 2 working).
= P(Zero working) + P(one working) + P(two working)
= 0.06 + 0.29 + 0.44
= 0.79

Probability Trees

This is a diagrammatic presentation of a probability experiment which is repeated severally and the events are either independent or dependent.

Probability tree cannot be used where the events are mutually exclusive or non-exclusive.

ILLUSTRATION

An accountant has a file with ten account receivables. The file has four accounts out of the ten being overdue. The accountant selected three accounts from the file randomly each at a time.

Required

Probability tree of the possible outcomes

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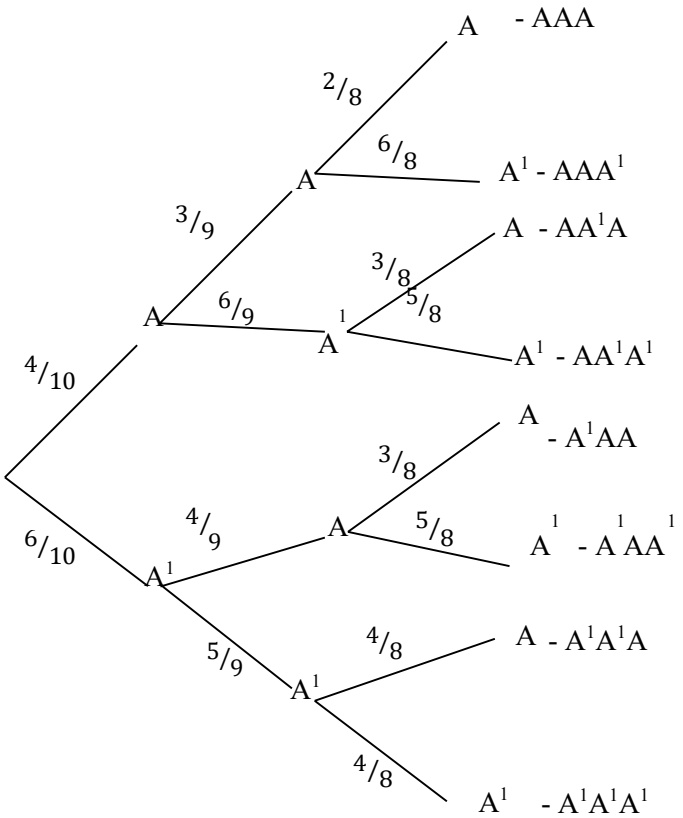
SOLUTION

Let A – Overdue account

A^1 – Not overdue account

Experiment of sampling $AA^1A^1 = \frac{4}{10} \times \frac{6}{9} \times \frac{5}{8} = \frac{1}{6}$ $A^1A^1A = \frac{6}{10} \times \frac{5}{9} \times \frac{4}{8}$

$$A^1AA^1 = \frac{6}{10} \times \frac{4}{9} \times \frac{5}{8} = \frac{1}{6}$$



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PROBABILITY DISTRIBUTIONS

PROBABILITY DISTRIBUTIONS

A **probability distribution** is either a probability formula or a probability table representing a frequency distribution. There are two categories namely:

1) Discrete probability distributions
2)Continuous discrete distributions

The use of either category above depends on the random variable (RV) being considered. A random variable is a variable whose values depend on the outcome of an experiment. It associates a single numerical value with each possible outcome of the experiment. If the numerical values are distinct (whole numbers) the random variable is known a discrete random variable (DRV). RVs that represent counts are usually discrete.

ILLUSTRATIONS OFDISCRETE RANDOM VARIABLE (DRV)

Experiment	Outcome	Random variable	Range of values
Tossing a coin twice	Number of heads	X= number of heads	0,1,2
Giving a test with 10 multiple choice question	Number of correct answers	X=number of correct answer	0,1,2,3……10
Inspection of a machine	Defective or non-defective	X = 0 if defective 1 if non defective	0,1
Consumers response to how they like a product	Good, average, poor	X = 0 if defective 2 if average 1 if poor	1,2,3
Inspecting 600 items	Number of acceptable items	X = number of acceptable	0,1,2……..600
Sending out 5000 sales letters	Number of people responding	X= number of people responding	0,1,2……. 5000

A continuous random variable (CRV) is a RV that has unlimited set of values.

Examples of CRV

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Experiment	Outcome	Random variable	Range of values
Building a house	% completed after 4 months	X= % of house complete	$0 \leq x \leq 100$
Testing lifetime of a light bulb (hrs)	Length of time the bulb last up to 800 hrs	X=time the bulb burns	$0 \leq x \leq 800$

Probability distribution of a DRV

If the probability of each x value of a DRV 'X' is known the arrangement of the value and their probabilities is called a probability distribution. A probability distribution can either be in from of a table or formula.

Example of a tabular discrete probability distribution

X	0.1	1	2	3
P(x)	0.1	0.4	0.3	0.2

NB:

$0 \leq P(x) \leq 1$

$\sum P(x) = 1$

Mean and the standard deviation of DRV

Mean (expected value) (μ)

The mean is the expected value if the experiment is performed a large number of times or indefinitely. The mean of a random variable denoted by either E(x) or μis given by:

$$E(x) = \mu = [\sum X\, P(X)]$$

Standard deviation (σ)

$$\sigma = \sqrt{\sum[x^2p(x)] \quad \mu^2} = E(x^2) \quad (E(x))^2$$

σ measures how much a probability distribution is spread around the mean of a DRV.

ILLUSTRATION

At a given stock market, shares of a certain company are selling at sh.10 a share. An investor plans to buy the shares and hold the stock for a year. If x is the price of stock after a year, the probability distribution is as shown below:

X	10	11	12	13	14
P(x)	0.35	0.25	0.2	0.15	0.05

Required;

a) The expected price of the stock after a year.

b) The standard deviation of the price of the stock over the 1 year period.

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SOLUTION

a. $E(x) = \sum[xp(x)] = 10\,(0.35) + 11(0.25) + 12(0.2) + 13\,(0.15) + 14(0.05) = \text{Sh. } 11.30$

b. $$\sigma = \sqrt{\sum[x^2p(x)] \quad \mu^2} = \sqrt{10^2(0.35) + 11^2(0.25) + 12^2(0.2) + 13^2(0.15) + 14^2(0.05) \quad 11.3^2}$$

$= \text{Sh } 1.23$

EXAMPLES OF DISCRETE PROBABILITY DISTRIBUTIONS

1) Binomial probability distribution
2) Poisson probability distribution
3) Hypergeometric distribution

1) Binomial probability distribution (BPD)

This is a probability distribution used to compute probability of specific discrete number of outcomes whereby the experiment follows a Bernoulli process. The process has the following characteristics or assumptions:

i. Each trial has only 2 possible outcomes called success or failure;(success of failure could also represent yes or no, head or tail, pass or fail , good or bad etc).
ii. The probability remains the same from one trial to the next.
iii. Trials are statistically independent.

iv. The number of trials is a positive integer (∩)

The BPD is used to find the probability of a specific number of successes out of n trials of a Bernoulli process. A common example of a Bernoulli process is tossing a coin.
For a Binomial experiment, the mathematical model representing the BPD of obtaining x successes in n trials is:

$P(x) = \binom{k}{x} P^S q^{n-S}$

x = 0,1, 2…..n

Where P(x) is probability of x successes, n is the total number of trials, p is the probability of success, q is the probability of failure= l-p,$\binom{n}{S}$is the num6er of ways of obtaining x successes in n trials

NB: P + q = 1
n and p are called the parameters of the binomial distribution.
Mean and standard deviation of BPD
Mean (μ) or E(x) = np
Standard deviation (σ) = $\sqrt{npq}$

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ILLUSTRATION
If a coin is tossed 5 times, find the probability of getting 4 heads if the experiment follows a binomial distribution.

SOLUTION
n = 5, p =0.5, q = 0.5, x = 4 therefore P(x = 4) = $\frac{n!}{S!(n-S)!} p^I q^{n-1} = \frac{5!}{4!(5-4)!} 0.5^I x\ 0.5^{(5-0)} = 0.15625$

ILLUSTRATION
After the analysis of accounts receivable, accounts either end up as paid or bad debts .A credit control accountant at Kargo Ltd. has established that in a financial year, 20% of the accounts receivable end up being bad debts. At the beginning of the current financial year the accountant had a hundred accounts receivable.

Required:
(i) The probability that exactly 10 of these accounts will eventually be bad debts.
(ii) State one assumption made in solving (i) above.
(iii) The expected number and the standard deviation of accounts receivable that will turn out to be bad debts.
(iv) The probability that at most 30 of the accounts receivable to be bad debts.

SOLUTION

(i). Accounts receivable are binomially distributed

P (bad debts) = 0.2 $\Rightarrow$ p

P (paid) = 0.8 $\Rightarrow$ q

n = 100

$P\ (x) = kC_S P^S q^{n-S}$P

P (x = 10) = 100 C_{10} $(0.2)^{10}$ $(0.8)^{90}$

= 0.00336

(ii). Assumption made in solving (i) above

- Accounts receivable are binomially distributed. Since the sample size of 100 is large, accounts receivables are also normally distributed.

(iii). μ = np = 100 x 0.2

= 20

Standard deviation $\overline{\text{ð}} = \sqrt{kPq}$

$= \sqrt{100 \text{ x } 0.2 \text{ x } 0.8}$

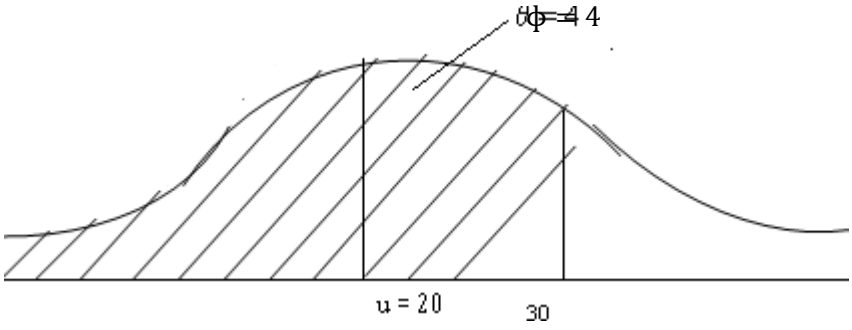
= 4

(iv). At most 30 of the accounts receivable are bad debts

P (x ≤ 30)

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Use binomial approximation to normal distribution



$Z = \frac{S-u}{\text{ð}} = \frac{30-20}{4} = 2.5$

Area = 0.4938

P (x ≤ 30) = 0.5 + 0.4938

= 0.9938

2) Poisson probability distribution (PPD)

Named after a French mathematician called S Poisson (1837). It is a discrete probability distribution that is used when the sample size n is not precisely known. The model of the distribution takes the form:

P(x) = $P\ (x) = \frac{m^x e^{-m}}{S!}$ where x = 0,1,2,…..

e … Constant = 2.7183

m= mean

Assumptions

i. The variable is discrete
ii. The event can only be either a success or a failure
iii. The outcomes are statistically independent
iv. The number of trials 'n' is finite and large but not precisely known.
v. Probability of success 'p' is so small that probability of failure 'q' is almost equal to unity. P ≤0.1
vi. The probability of an occurrence is the same.

Characteristics

i. Like binomial distribution, Poisson distribution is a discrete probability distribution where the random variable assumes a countable infinite number of values 0, I ,2, ...∞
ii. The main parameter of the Poisson distribution is the mean = m
iii. Mean and variance are the same i.e $\mu = \sigma^2 = m$
iv. As an approximation to binomial, Poisson distribution can be viewed as a limiting
v. form of binomial distribution when:
    a . N (number of trials) is indefinitely large i.e. n → ∞
    b. P - the constant probability of success for each trial is infinitesimally i.e. P 0

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In practice, the Poisson distribution may be used in place of the binomial when
n ≥ 20 and p ≥ 0.10

Uses or Importance of Poisson distribution

The distribution describes the following situations.

i. Number of customers arriving independently at a service facility like hospital or bank per unit of time "say an hour".
ii. Number of telephone calls arriving at a telephone switch board per unit time.
iii. Number of accidents on a particular road per day.
iv. Hospital emergencies per day
v. Number of goals in a football match
vi. Dimensional errors in engineering drawing

ILLUSTRATION

A random variable X follows a Poisson distribution with a mean of 6. Calculate:

1. P(x= 0)
2. P(x>2)

SOLUTION

$P(x= 0)= \frac{6^{0}e^{-6}}{0!} = \frac{1 \text{ S } 0.00248}{1} = 0.00248$

P (x>2) = 1- [p(x=0) + P (x=1) + P(x = 2) = 1 – [0.00248 + 6(0.00248) + 18(0.00248)] = 0.938

3) Hyper geometric distribution

Is a distribution applicable when sampling is done without replacement from a finite population and the basic condition of independence of Bernoulli trial fails and so does the use of binomial distribution, In this case the probability of success changes from trial to trial, because the sampling is without replacement. Under these circumstances, the hypergeometric distribution is applicable.

In general, suppose that we are sampling from a finite population of size N, and the elements in this population can be divided into two groups, say defective and non-defective items.

Defective and non-defective can be replaced by success and failure. Suppose there are D defective items in the population, then the number of non- defectives would be N-D. Let x be the number of defectives in a random sample of n elements. Then the x defectives in the sample must come from the D defectives in the population and (n - x) non-defectives must come from (N - D) elements in the population.

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The probability distribution in this case is hypergeometric which is expressed as follows:

$P(X= x) = \frac{[D_cS][(N-D)c\ (n-S)]}{N_cn}$ for x = 0,1,2 ……n

Where N is the population size, n is ~ sample size, D is the number of defectives in the population and x is the number of defectives in the sample.

Mean and standard deviation of hypergeometric distribution

Mean = np, where $p = \frac{D}{N}$ (population proportion of defective)

Standard deviation = $\sqrt{kpq}\ [\frac{N-n}{N-1}]$ where $[\frac{N-n}{N-1}]$ is called a finite correction factor.

NB: when n is very small and N is large, this factor is close to 1 and the hypergeometric distribution is an approximation to the binomial distribution.

ILLUSTRATION

Past experience indicates that in a box of 25 bulbs, five bulbs are defective. If a random of 5 bulbs are examined, what is the probability of having:

i) No defective items
ii) Less than 2 detectives

SOLUTION

i) Probability of no defective, P (X=0) = $\frac{(5_c0)S\ \{(25-5)c\ (5-0)\}}{25_c5}$ = 0. 292

ii) Probability of less than 2 defective

P(X<2) = P(X = 0) + P(X=0) + P(X=1) = 0.748

CONTINUOUS PROBABILITY DISTRIBUTIONS

Are associated with continuous random variables (CRV). A CRV takes continuous values

-Any variable relating to measurement is an example of a CRV

A CRV is normally expressed as a grouped frequency distribution. The frequency distribution can provide a relative frequency distribution. If the relative frequency is plotted, it smooth curve which describes the overall shape of the distribution. The curve is called **probability density curve**. The total area under the curve is normally I which is similar probability for CRV, probability cannot be assigned to a single value (since values arc continuous) but it can only be assigned to an interval, say' a' to 'b'

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i.e P(x) = P (a≤ x ≤b) = $\int_a^b f(x)dx$

NB: for a CRV, probability of a single discrete value=0

Any CRV is described using a function called probability density function (pdf). For a function to quality to be a pdf, the following conditions must be met:

$f(x) \geq 0$ for all x values

$\int_{-\infty}^{\infty} f(x)dx$= 1(total area under the curve

Expected value E(X) and the variance V(X) or a continuous random variable

The expected or mean value of a continuous random variable X with a pdf f(x) is:

μ_S = E(X) = $\int_{-\infty}^{\infty} f(x)dx$

The variance of the pdf is determined as: V(X) = $E(X^2) = [E(X)]^2$

Examples of continuous random distributions

Commonly used are:

1. The Normal distribution
2. The Exponential distribution
3. The Uniform distribution

1) The Normal Distribution

The Normal distribution is also known as Gaussian or Laplace.

A continuous random variable X that has a normal distribution is called normal random variable.

The normal random variable is said to have a normal distribution with parameters μ (mean) and σ^2 (variance) if it has the following density function.

$$y = f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x \quad x}{ð}\right)^2}$$

Where $\bar{x} - \mu$ = mean

σ = S = Standard deviation.

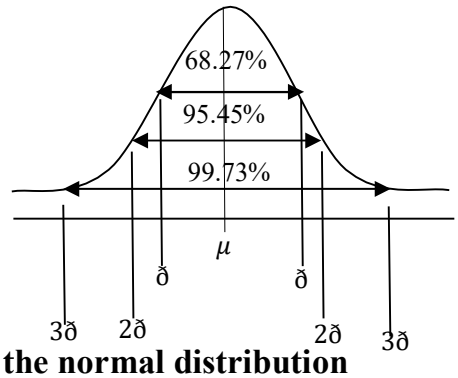
The above function is denoted as follows: $x \sim N(\mu, ð^2)$. Read as: the random variable X follows a normal distribution with mean μ and standard deviation σ

Where: σ is the standard deviation of the given normal distribution, π= constant = 3.1416, e = constant =2.7183 and μ is the mean of the random variable X.

Properties of the normal distribution

a) The normal curve is symmetrical about the mean i.e it is bell shaped.
b) The mean=median=mode
c) Height of the normal curve is maximum at the mean value
d) The curve is asymptote to the x axis i.e. it continues to approach but never touches the x axis.
e) The first and third quartiles are equidistant from the median.
f) 67.27% observations are within $\pm 1\sigma$ from the mean 95.45% are within ± 2ð from the mean and 99.7% are within $\pm 3\sigma$

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Importance of the normal distribution

The normal distribution is of importance in Quantitative analysis for several reasons:

i) Frequency distributions of many variables such as height, weight dimensions, and temperature often have the normal curve.
ii) The normal distribution is useful in approximating other distributions under certain limiting conditions like binomial and Poisson.
iii) Has a wide application in hypothesis testing and test of significance.
iv) Has extensive use in sampling theory where large samples are assumed to follow normal distribution.
v) It is useful in statistical quality control where the control limits are set by using the distribution.

Normal distribution in probability estimation

The calculation of probability for a normal random variable X requires the use of specialized tables known as z-score/ standard normal table. The use of the table requires that all given x values of the normal random variable X be standardized or transformed. The x values are standardized by converting them into new values called z-scores using a transformation formula where: Z-Score = $\frac{S-u}{\sigma}$

The standardized values give a standard normal distribution with a mean of 0 and a standard deviation of 1

ILLUSTRATION

A normal curve has a mean of 20 and a standard deviation of 10. Find the probability that an observed x value is:

i. Between 15 and 40. **ii.** Less than 15 **iii.** More than 40

SOLUTION

i) Required: P (15≤ x ≤ 40)

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Let x1 = 15 and x2 = 40, μ = 20 and σ = 10

Transform x1 and x2 to z – score 1 (Z_1) and z – score 2 (Z_2)

$Z_1 = \frac{15-20}{10} = -0.5$ $Z_2 = \frac{40-20}{10} = 2$

Hence P (-0.5≤ z ≥) = P (z ≤ 2) =0.1915 + 0.4772 = 0.6687

ii)P (z ≤ -0.5) =0.5 – 0.1915 = 0.3085

iii.P (z ≤ 2) =0.5 - 0.4772 = 0.0228

ILLUSTRATION

An electric utility company has found out that the weekly number of occurrences of lightning striking the transformers is a Poisson distribution with mean 0.4.

Required:

i) The probability that no transformer will be struck in a week.
ii) The probability that at most two transformers will be struck in a week.

SOLUTION

Poisson distribution is expressed by the following:

$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$ Where x - event transformer being struck.

e-natural logarithm ≈ 2.718

λ -mean=0.4

i) The probability x = 0, $P(0) = \frac{4^0 e^{-0.4}}{0!} = e^{-0.4} = 0.6703$

ii) $P(x \le 2) = P(0) + P(1) + P(2)$

$= 0.6703 + \frac{4e^{-0.4}}{1!} + \frac{4^2 e^{-0.4}}{2!} = 0.6703 + 0.2681 + 0.0536 = 0.9921$

Relationship between the Binomial Poisson and Normal distributions

The three distributions are very closely related to each other. When n is large and the probability of 'p' of occurrence of an event is close to zero so that np remains a finite constant, then the Binomial distribution tends to Poisson distribution.

Similarly, when n is very large i.e n → ∞ and neither p nor q is very small, then the Normal tends to Binomial distribution

2)The exponential distribution (negative exponential distribution)

This is a continuous distribution which is widely used in the analysis of queuing problems, as a probability model for service time or inter-arrival times i.e. the time span which lapses between the two successive arrivals. If ~ represents the rate of service, i.e. the average number of customers served per unit of time, then the probability density function is given by

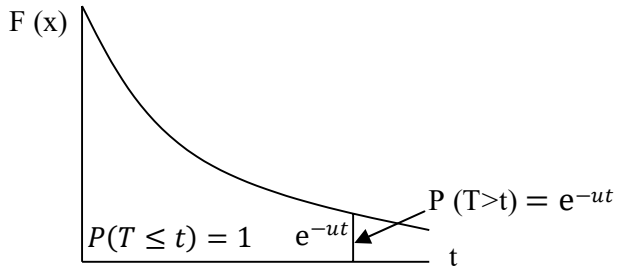
$f(t) = \mu e^{-ut}$ $0 < t < \infty$

Where; T is a random variable (e.g. arrival time or service time)

μ is the parameter of the distribution whose value performance of the service.

e =2.7183 (the base of natural logarithms)

The distribution is skewed to the right and takes the following shape:



The widely used form of the distribution in estimating probability is its cumulative form expressed as:

$P(T \le t) = e^{-ut}$ e.g. the probability that $T \le 2 = 1 - e^{-2u}$ for unknown μ

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$P(T \leq t) = 1 \quad e^{-ut}$

Hence P (T>t) = e^{-ut}

However, the probability can be obtained by integrating the p.d.f between the given limits.
The expected value E(t) and variance are given by:
E(t) = $1/\mu$ and variance = $1/\mu^2$

ILLUSTRATION
Suppose that a fuse has a life length which may be considered a s a continuous random variable with an exponential distribution. The manufacturing process yields an expected life length of 100hrs
Required:
The probability that 200 hrs will pass without the fuse becoming dead.

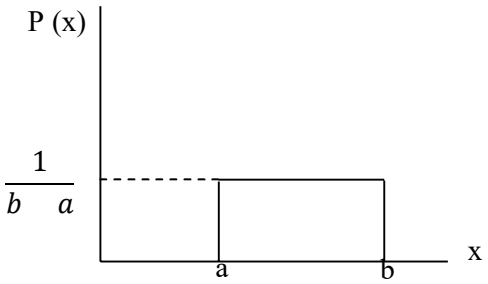
SOLUTION
E(t) = 100, hence μ = 1/100 = 0.01
Therefore P (T>200) = $e^{-ut} = e^{-0.01\ S\ (200)} = 0.135$

ILLUSTRATION
At Hilton hotel in the city, it takes 10 minutes to receive the order after placing. If the service exponentially distributed, find the probability that the customer waiting
(i) More than 10 minutes (ii) 10 minutes or less (iii) 3 minutes or less

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SOLUTION
μ= 1/10 = 0.1 per minute
i) P (T>10) = $e^{-ut} = e^{-0.01} = e^{-1}) = 0.368$
ii) P (T≤10) = 1 e^{-ut}= 1 – 0.368 = 0.632
iii) P (T≤3) = 1 $e^{-0.1\ S\ 3}$= 1 – 0.741 = 0.259

3) The Uniform distribution
This is a continuous distribution which is rectangular in nature and bounded by two points, say a and b in such a way that no value is more likely than the other. The range of a and b contains the possible outcomes. The area within the interval (a, b) is considered to be 1 and the height of the rectangle is assumed to be equal to l/(b - a) as shown below.



The area under the rectangle between any two intermediate point's c and d is given by:

$$P(c < x < d) = \frac{d}{b} \quad \frac{c}{a}$$

The mean and the standard deviation of the uniform distribution in the interval a and b are given by:
Mean = (a + b) /2 and standard deviation = $\sqrt{(a+b)^2 /12}$

ILLUSTRATION
The average daily procurement of fresh milk by a milk producer is 40000 litres and the minimum is 25000 litres per day. Assuming a Uniform distribution, find out the maximum milk procurement in a day and what percentage of days the procurement will exceed 35000

SOLUTION
Mean = (a + b) /2
Mean = 40000 and a = 25000
Therefore b = 80000 – 25000 = 55000
Hence, the minimum daily procurement would be 55000 litres.
The percentage of days that procurement will exceed 35000 litres is:

P (35000 <x< 55000) = $\frac{55000 - 35000}{55000 - 25000} = \frac{20000}{30000} = 0.67 = 67\%$

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Thus 67% of the days, the procurement of milk is beyond 35000 litres.

MARKOV ANALYSIS

This is a stochastic or probabilistic system whereby the state of a given phenomenon in future can be predicted from the current state using a matrix of transition probabilities. In other words, it is a quantitative technique that combines the use of probabilities and matrices in the prediction of future behaviour of some variable by using the current behaviour of that variable.

This analysis is used to analyze decision problems in which the occurrence of a specified event depends on the occurrence of a previous event.

Areas of application
The Markov processes or chains are frequently applied as follows:-

1. **Brand Switching**

By using the transitional probabilities we can be able to express the manner in which consumers switch their tastes from one product to another.

2. Insurance industry

Markov analysis may be used to study the claims made by the insured persons and also decide the level of premiums to be paid in future.

3. Movement of urban population

By formulating a transition matrix for the current population in the urban areas, one can be able to determine what the population will be in say 5 years.

4. Movement of customers from one bank to another.

It is a fact that customers tend to look for efficient banks. Therefore at a certain time when a given bank installs such machinery as computers it will tend to attract a number of customers who will move from certain banks to efficient ones.

5. Finance - to predict share prices in the stock exchange

6. Human resource management-to analyze shifting of personnel within the organization's units e.g. branches, departments, divisions etc.
7. Accounting - to estimate the provision for bad debts.
8. To analyze equipment replacement and failure problems
9. Introduction of new products into the market

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BASIC TERMS IN MARKOV CHAINS

a) Probability Vector

This is a row matrix whose elements are non-negative and also they add up to 1 e.g. u = 0.2, 0.1, 0.2, 0.5)

Example

State the ones which are probability vectors

Consider u =(¾ , 0, - ¼ , 1/2) Not because – ¼ is negative

v = (¾ , ½ , 0, ¼) Not because the sum of the elements is greater than 1.

w= (¼ , ¼ , 0, ½) Adds up to 1, each element is non negative.

Therefore it's a probability vector

Stochastic matrix

A matrix whose row elements are all non negative and also add up to 1

Example (i) $M = \begin{pmatrix} 0.1 & 0.2 & 0.3 & 0.4 \\ 0.0 & 0.7 & 0.1 & 0.2 \\ 0.5 & 0.1 & 0.1 & 0.3 \\ 0.3 & 0.4 & 0.2 & 0.1 \end{pmatrix}$

Example ii) = Consider the following matrices

$$A = \begin{pmatrix} \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{3}{4} & \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \qquad B = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \qquad C = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & 0 \end{pmatrix}$$

A is not stochastic matrix because the element in the 2nd row and 3rd column is negative.
B is not Stochastic matrix because the elements in the second row do not add up to 1
C is stochastic matrix because each element is non negative and they add up to 1 in each row.

Regular stochastic matrix

A matrix P is said to be regular stochastic matrix if all the elements in P^m are all positive, where m is a power, m = 1, 2, 3 e.t.c

Let $A = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ Where A is a Stochastic Matrix

$$A^2 = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \times \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \times \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{8} & \frac{5}{8} \end{pmatrix}$$

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Since the elements in A^2 and A^3 are all positive then A is regular Stochastic matrix.

State - any identified possible condition of a process or a system e.g. a machine can be in one of two states at any point in time i.e. either functioning correctly or not.
Markov process - a stochastic process where the future state depend on the current state.
State probability - probability of an event occurring at a point in time.
Vector of state probabilities - row matrix of all state probabilities for a given system or process.
Transition probability - conditional probability that will be in a future state given the current or existing state (it's the probability of moving from one state to another).
Matrix of transition probabilities- matrix containing all transition probabilities for a certain process or system,
Equilibrium condition - a condition that exist when the State probabilities for a future period are the same as the state probabilities for a previous state.
Absorbing state - a state when entered cannot be left. It has a transition probability of unity to itself and zero to all other states. In business, absorbing states include the payment of a bill, termination of employment, completion of a contract, a sale of a capital asset etc.
Steady state - refers to long- run state of the system. Provided the assumptions of Markov process persist, the system finally reaches an equilibrium called steady state. At equilibrium, (equilibrium state vector) x (transition matrix) =equilibrium state.
Recurrent state - refers to a state that can be left and re-entered many times.
Closed state - a state which once left cannot be re-entered.

Markov analysis assumptions

1. The probability of movement from one state to another over time can be determined and remains constant over the period under consideration.
2. The current state of the system depends only upon the immediately preceding state of the system and not on any prior state.
3. All the states of the system are known and can be listed down.
4. The various states-are mutually exclusive and collectively exhaustive i.e at any given time, a subject of analysis belongs to one and only one state.
5. No new states can join the system and none of the states in the system can leave.
6. the number and composition of possible states do not change

Forecasting using Markov process

Forecasting is possible once we have the initial states and the transition matrix.

Transition matrix, T = from

	To	
S_1	$S_1 \ldots\ldots\ldots\ldots\ldots\ldots\ldots S_n$	
S_2	$P_{11} \ldots\ldots\ldots\ldots\ldots\ldots\ldots P1_n$	
:	$P_{21} P2k$	
:	:	:
:	:	:
S_m	:	:
	$P_{m1} P_{mn}$	

Notes

a. P_{11} is the conditional probability of the system being in state j in future if the current state is i.

b. $P_{11} + P_{12} + \ldots\ldots\ldots\ldots\ldots + P1n = 1$
$P_{21} + P_{22} + \ldots\ldots\ldots\ldots\ldots P2n = 1$ exhaustive property
etc

c. T is a square matrix

d. T is obtained empirically i.e. through observations, data collection and analysis

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MARKOV ANALYSIS METHODOLOGY

The Markov process involves 4 major steps one has to go through in order to predict the future status of a given variable.

Step 1: determine and list all the possible states of the given system.

Step 2: determine the current or initial probability for each of the different states of the system. Such probabilities arc called market shares. They are normally symbolized by a row vector, V(o) that gives the probabilities at period zero (now) of the said variable

Step 3: formulate the transition probability matrix, T.

Step 4: predict the future behaviour of the system. The position of the system at any given period will be computed by multiplying the preceding period's market shares by the matrix of transition probabilities, T. For instance, the position of a variable at different time periods will be obtained by using the model below:

Position at period 0 (now) = V(0)
Position at period 1, V(1) = V(0).T
Position at period 2, (2) = V(1).T = V(0).T.T = V(0).T^2
Position at period 3, (3) = V(2).T = V(0).T^2.T = V(0).T^3
Position at period 2, (3) = V(0).T. T = V(0).T^2
Position at period n, V(n) = V(n -1).T = V(0).T^n
Therefore, the general Markov analysis model for prediction of what to expect in any given period, n in the future is V(n) = V(n -1).T = V(0).T^n

STEADY STATE (EQUILIBRIUM) CONDITION

After a number of transitions, the probabilities are expected to stabilize or come to an equilibrium position as the rate of change decreases with time. This implies that a time will come when the current variable values (probabilities) equal to the succeeding period probabilities throughout. Thus, a steady state exists if state probabilities do not change for a large number of periods.
If we suppose that these final fixed equilibrium market shares are p and q then:

$[Pq]$ T = $[Pq]$ for all the periods after the equilibrium period.

NB: p + q = 1

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ILLUSTRATION

Two TV stations S_1 and S_2 compete for viewers. Of those who view S1 on a given day; 40% view S2the next day. In the case of those who view S_2 on a given day, 30% switch over to S_1 the next day. Suppose yesterday; of the total viewers 60% view S_1 and the rest S_2;. Determine the percentage of viewers for each station:

a) Today
b) Tomorrow
c) At equilibrium/steady state or in the long run.

SOLUTION

Transition matrix m, T = from $\begin{matrix} & S_1 & S_2 \\ S_1 & 1 & 0 \\ S_2 & 0 & 1 \end{matrix}$

$\begin{matrix} S_1 & S_2 \end{matrix}$
Initial state vector = (0.6 0.4) (Yesterday)

a) Today's market shares (% of viewers) = $(0.6 \quad 0.4) \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} = (0.48 \quad 0.52)$

$S_1 = 48\%$ $S_2 = 52\%$

$\begin{matrix} S_1 & S_2 \end{matrix}$

b) Initial state vector = (0.48 0.52)
Tomorrow's market shares = $(0.48 \quad 0.52) \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} = (0.444 \quad 0.556)$

$S_1 = 44.4\%$ $S_2 = 55.6\%$

c) Provided the assumptions of the Markov process hold, the system finally reaches equilibrium (steady state, long-term or long run status). At equilibrium, the following hold: (Equilibrium state vector) (T) = (Equilibrium state vector)

Let p = Long-term % of viewers (market share) for s1
q = long-term % of viewers (market share) for s2
p + q = 1
q = 1-p

In the long run, (pq) $\begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$ = (p q)

$0.6 + 0.3p = p$
$0.4 + 0.7p = q$ drop one of the equation arbitrarily

Hence $0.6 + 0.3p = p$
$0.3q = p - 0.6p$
$0.3q = 0.4$..... (i)

However, $p + q = 1$ implying that $p = 1 - q$ .....(ii)

Substitute (ii) in (i) to get: $0.3q = 0.4 (1-q)$

$0.3q = 0.4 - 0.4q$
$0.3q + 0.4q = 0.4$
$\frac{0.7q}{0.7} = \frac{0.4}{0.7} \Rightarrow q = 0.5714 = 57.12\%$
$p = 1-0.5712 = 42.82\%$

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ABSORBING /TRAPPING IN MARKOV PROCESS

This is a state that has a zero probability of being left once entered. A common business application is when receivable (debtors) and their ageing. In this case, there are two absorbing states or "debt declared al bad debt",

The computation process for an absorbing state requires an initial change of the transition probability matrix to a referred as the canonical form:

$$T = \begin{pmatrix} I & 0 \\ R & Q \end{pmatrix}$$

I- An identity matrix defining the probability of staying within an absorbing state once it is entered.

O- A null matrix indicating the probabilities of going from an absorbing state to a non-absorbing state.

R- The probabilities of going from a non-absorbing state to another an absorbing state.

Q- A matrix showing the probabilities of going from one non absorbing state to another non absorbing

The analysis of determining how much will eventually; end up in absorbing state requires the use the fundamental matrix (F), derived from the canonical form:

F=(1-Q)$^{-1}$(inverse of matrix 1- Q)

Probability of absorption of the non-absorbing states, we employ the following relationship
Probability of absorption = FR = (I – Q)$^{-1}$R

ILLUSTRATION

An accountant has analysed a firm's sh. 100,000 accounts receivable and determined the following

State		Amount (Sh)
State 1 (S_1) Amount paid in full	=	45,000
State 2 (S_2) bad debt	=	15,000
State 3 (S_3) current debts	=	25,000
State 4 (S_4) over debts	=	15,000

The historical data have been collected and the following matrix of transition probabilities specified:

$$T = \begin{array}{c} \\ S_1 \\ S_2 \\ S_3 \\ S_4 \end{array} \begin{array}{c} \begin{array}{cccc} S_1 & S_2 & S_3 & S_4 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0.5 & 0.2 & 0.1 & 0.2 \\ 0.4 & 0.4 & 0.1 & 0.1 \end{bmatrix} \end{array}$$

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Required:

a) The proportions of absorption
b) The amount of money that will eventually end up as paid in full or bad debts.

SOLUTION

a) Probability of absorption = FR

$$F = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.1 & 0.2 \\ 0.1 & 0.1 \end{bmatrix} \right\}^{-1} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}^{-1} = \begin{bmatrix} 1.139 & 0.253 \\ 0.127 & 1.139 \end{bmatrix}$$

$$FR = \left\{ \begin{bmatrix} 1.139 & 0.253 \\ 0.127 & 1.139 \end{bmatrix} \begin{bmatrix} 0.5 & 0.2 \\ 0.4 & 0.4 \end{bmatrix} \right\} = \begin{bmatrix} 0.671 & 0.329 \\ 0.519 & 0.481 \end{bmatrix}$$

Amount that will end up as paid in full or bad debt=

$$(25{,}000\ 15{,}000) \begin{bmatrix} 0.671 & 0.329 \\ 0.519 & 0.481 \end{bmatrix} = 24560\ 15440$$

Conclusion: Sh24,560 will eventually be paid while sh. 15440 will eventually be bad debt.

PRACTICE EXERCISES

QUESTION 1

A problem is given to three managers A, B, C whose chances of solving are ½, ⅓, ¼ respectively. What is the probability that the problem will be solved?

Solution:

The product of the probabilities of each manager solving a problem gives probability of solving a problem. (Since one manager solving a problem is independent of the others)

P (solving)= 1- P (not solving)

$$= 1 - \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4}\right) = 1 \quad \frac{1}{4} = \frac{3}{4}$$

QUESTION 2

Three groups of children contain respectively 3 girls and 1 boy; 2 girls and 2 boys; 1girl and 3 boys. One child is selected at random from each group, show that the chance that the three selected, consist of 1 girl and 2 boys is 13/32.

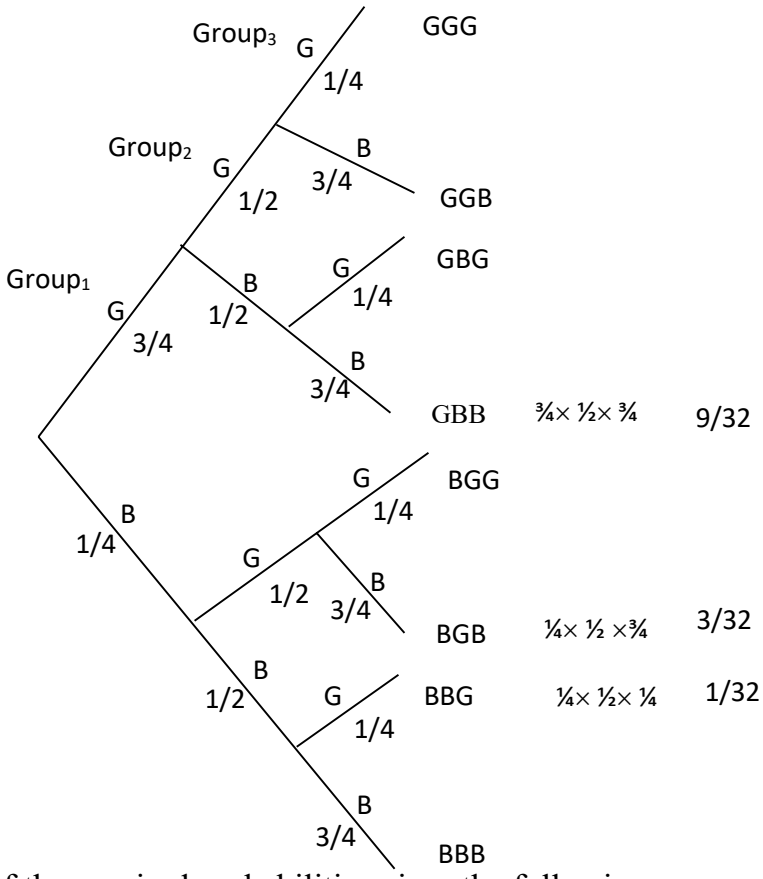
Solution:

The best way to solve this is by use of a probability tree as follows:

Let G be the event of a girl being chosen

And B be the event of a boy being chosen

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GBB	¾× ½× ¾	9/32
BGB	¼× ½ ×¾	3/32
BBG	¼× ½× ¼	1/32

Sum of the required probabilities gives the following.

P (GBB) + P(BGB) + P(BBG)

$\frac{3}{4}x\frac{1}{2}x\frac{3}{4}+\frac{1}{4}x\frac{1}{2}x\frac{3}{4}+\frac{1}{4}x\frac{1}{2}x\frac{1}{4}$

$P = {}^{9}\!/_{32} + {}^{3}\!/_{32} + {}^{1}\!/_{32} = {}^{13}\!/_{32}$

QUESTION 3

The following table gives a bi-variate frequency distribution of 50 managers according to their age and salary (in rupees).

Salary in rupees

Age in years	**1000-1500**	**1500-2000**	**2000-2500**	**2500-3000**	**Total**
20-30	2	3	-	-	5
30-40	5	4	2	1	12
40-50	-	2	10	3	15
50-60	-	1	8	9	18
Total	7	10	20	13	50

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If a manager is chosen at random from the above distribution, find the chance that; (i) he is in the age group of 30-40 and earns more than Rs.1500, (ii) his earnings are in the range of Rs.2000-2500 and is less than 50 years old.

Solution:

i) Let A be the age group 30-40
B be the earnings more than 1500
Then $P(B/A) = \frac{P(AB)}{P(A)} = \frac{{}^{7}\!/_{50}}{{}^{12}\!/_{50}} = {}^{7}\!/_{12}$ Then the probability of B given A
Where: P (AB) - Probability of A and B occurring.
P (A) - Probability of A occurring.

ii) Let A be the age group below 50 years
B be the earnings varying between 2000-2500
Then $P(B/A) = \frac{P(AB)}{P(A)} = \frac{{}^{12}\!/_{50}}{{}^{20}\!/_{50}} = {}^{12}\!/_{20}$

QUESTION 4

Computer analysis of satellite data has correctly forecast locations of economic oil deposits 80% of the time. The last 24 oil wells drilled produced only 8 wells that were economic. The latest analysis indicates economic quantities at a particular location. What is the probability that the well will produce economic quantities of oil?

Solution:

Let A be the event drilling a well and B be the computer analysis showing an economic well. Then $P(AB) = P(A)P(B/A)$. But then since computer analysis and drilling of economic well are independent then $P(B/A) = P(B)$. So that $P(AB) = P(A)P(B) = {}^{8}/_{24} \times 0.8 = 0.267$

QUESTION 5

A firm recently submitted a bid for a turnkey project for a 500 MW power plant. If its main competitor submits a bid, the chances of bid being awarded to the firm is 0.3. If the main competitor doesn't bid, there is a ¾ chance of the firm getting the contract. There is a 0.50 chance that the main competitor will bid.

i) What is the probability of the firm getting the contract?

ii) What is the probability that the competitor's bid given that the firm's bid is awarded?

Solution:

Let G-firms bid awarded

H-competitor submitting a bid

i) $P(G) = P(GH) + P(G\overline{H})$

$= P(H) \times P(G/H) + P(\overline{H}) \times P(G/\overline{H})$

$= 0.5 \times 0.3 + 0.5 \times {}^{3}/_{4} = 0.525$

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Where $\overline{H}$ - event that competitor does not submit a bid

ii) $P(H/G) = \dfrac{P(GH)}{P(G)} = \dfrac{0.3}{0.525} = 0.571$

QUESTION 6

a) Define probability as used in Quantitative Techniques.

b) What is Bayes Theorem? Explain how Bayes Theorem can be utilized practically.

c) KK accounting firm has noticed that of the companies it audits, 85% show no inventory shortages, 10% show small inventory shortages and 5% show large inventory shortages. KK firm has devised a new accounting test for which it believes the following probabilities hold:

P (Company will pass test/no shortage)	=	0.90
P (Company will pass test/small shortage)	=	0.50
P (Company will pass test/large shortage)	=	0.20

Required:

i) Determine the probability if a company being audited fails this test has large or small inventory shortage.

ii) If a company being audited passes this test, what is the probability of no inventory shortage?

Solution:

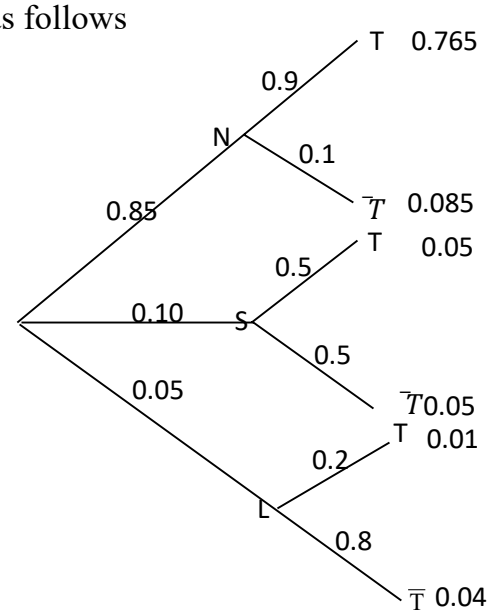
a) Probability is a measure of the likelihood of obtaining a particular outcome from an experiment. Given an experiment has n trials with no influence to each other, then having m outcomes of an event A, the probability of event A $P(A)=\frac{m}{n}$. It is between 0 and 1.

b) Bayes' theorem is as follows

$P(A/B)=\frac{P(AB)}{P(B)}$ Which is probability of occurrence of event A given that event B has occurred is given by the probability of occurrence of both events divided by the probability of occurrence of event B.

Bayes' theorem can be used to revise subjective probabilities made from beliefs. This is so when more information is added to what already exists.

c) The probability tree is as follows



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Let event T- Passing of test
N-No shortage
S-Small shortage
L-Large shortage

i) Probability of failing test=0.085+0.05+0.04=0.175
Probability of having large or small inventory shortage given the failed test $=\frac{0.05+0.04}{0.175}=\frac{0.09}{0.175}=0.514$

ii) The probability of passing test=0.765+0.05+0.01=0.825
Probability of no inventory shortage given the failed test $=\frac{0.765}{0.825}=0.927$

QUESTION 7

a) Define the following terms as used in Markovian analysis:
   i) Transition matrix.
   ii) Initial Probability vector
   iii) Equilibrium
   iv) Absorbing state

b) A company employs four classes of machine operators (A,B,C,D): all new employees are hired as class D and, through a system of promotion, may work up to a higher class. Currently, there are 200 class D, 150 class C, 90 class B and 60 class A employees. The company has signed an agreement with the union specifying that 20 percent of all employees in each class be promoted, one class in each year. Statistics show that each year 25 percent of the class D employees are separated from the company by reason such as retirement, resignation and death. Similarly 15 percent of class C, 10 percent of class B and 5 percent of class A employees are also separated. For each employee lost, the company hires a new class D employee.

   Required:
   i) The transition matrix.
   ii) The number of employees in each class two years after the agreement with the union.
   iii) The equilibrium state in number of employees.

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Solution:

a)
   i) Transition matrix is that which contains the probabilities of moving from any one state to another.
   ii) Initial probability vector is the vector that contains the current state before transition.
   iii) Equilibrium state is the state that a system settles on in the long run.
   iv) Absorbing state is one in which cannot be left once entered. It has a transition probability of unity to itself and of zero to other states.

b)
   i) To make the transition matrix, we can make a loss probability table and retention probability table first.
   Loss Probability table

	To A	**To B**	**To C**	**To D**	**Total Loss**
From A	0	0	0	0.05	0.05
From B	0.2	0	0	0.1	0.3
From C	0	0.2	0	0.15	0.35
From D	0	0	0.2	0	0.2

Retention Probability table

	Retention= (1 – Total loss)		
A	1 – 0.05	=	0.95
B	1 – 0.3	=	0.7
C	1 – 0.35	=	0.65
D	1 – 0.2	=	0.8

So the transition matrix will be as follows.

		To			
		A	**B**	**C**	**D**
From	**A**	0.95	0	0	0.05
	B	0.2	0.7	0	0.1
	C	0	0.2	0.65	0.15
	D	0	0	0.2	0.8

NOTE:

1) D only looses to C, since whatever it looses due to separation is immediately replenished. i.e. 25% loss is immediately returned by more employment. So the loss is zero.
2) The retentions are placed on the diagonal of transition matrix before the loss probabilities.
3) Notice that summation in rows of transition matrix is equal to one.
4) The matrix can be interchanged to be.

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		From			
		A	B	C	D
To	A	0.95	0.2	0	0
	B	0	0.7	0.2	0
	C	0	0	0.65	0.2

In this case the initial vector is post multiplied as follows⇒Transition matrix× Initial vector

ii) To get the initial probability vector after two periods, just multiply the initial vector with the transition matrix twice.
Initial vector

$$(A \quad B \quad C \quad D) = (60 \quad 90 \quad 150 \quad 200)$$

The first period

$$(60 \quad 90 \quad 150 \quad 200)\times\begin{pmatrix} 0.95 & 0 & 0 & 0.05 \\ 0.2 & 0.7 & 0 & 0.1 \\ 0 & 0.2 & 0.65 & 0.15 \\ 0 & 0 & 0.2 & 0.8 \end{pmatrix}=(75 \quad 93 \quad 137.5 \quad 194.5)$$

Calculations

- $60\times0.95+90\times0.2+150\times0+200\times0=75$
- $60\times0+90\times0.7+150\times0.2+200\times0=93$
- $60\times0+90\times0+150\times0.65+200\times0.2=137.5$
- $60\times0.05+90\times0.1+150\times0.15+200\times0.8=194.5$

The second period.

$$(75 \quad 93 \quad 137.5 \quad 194.5)\times\begin{pmatrix} 0.95 & 0 & 0 & 0.05 \\ 0.2 & 0.7 & 0 & 0.1 \\ 0 & 0.2 & 0.65 & 0.15 \\ 0 & 0 & 0.2 & 0.8 \end{pmatrix}=(89.85 \quad 92.6 \quad 128.273 \quad 189.275)$$

Calculations

- $75\times0.95+93\times0.2+137.5\times0+194.5\times0=89.85$
- $75\times0+93\times0.7+137.5\times0.2+194.5\times0=92.6$
- $75\times0+93\times0+137.5\times0.65+194.5\times0.2=128.273$
- $75\times0.05+93\times0.1+137.5\times0.15+194.5\times0.8=189.275$

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Approximately in A ≈ 89, B ≈ 92, C ≈ 128, D ≈ 189

iii) The equilibrium or steady state is determined from the following matrix and equation as follows:

$$(A \quad B \quad C \quad D)\times\begin{pmatrix} 0.95 & 0 & 0 & 0.05 \\ 0.2 & 0.7 & 0 & 0.1 \\ 0 & 0.2 & 0.65 & 0.15 \\ 0 & 0 & 0.2 & 0.8 \end{pmatrix}=(A \quad B \quad C \quad D) \qquad (1)$$

$$\text{and } A + B + C + D = 1 \qquad (2)$$

From the matrix multiplication (1), the following expressions are determined in terms of A.

$0.95\ A + 0.2\ B = A \Rightarrow 0.05\ A = 0.2\ B \quad \Rightarrow B = 0.25\ A$ (3)

$0.7\ B + 0.2\ C = B \Rightarrow 0.075\ A = 0.2\ C \Rightarrow C = 0.375A$ (4)

$0.65\ C + 0.2\ D = C \Rightarrow 0.13125\ A = 0.2\ D \Rightarrow D = 0.65625A$ (5)

$0.05\ A + 0.1\ B + 0.15\ C + 0.8\ D = D$ (6)

From equation (2), (3), (4) and (5),

$A + 0.25\ A + 0.375\ A + 0.65625\ A = 1$

$\Rightarrow 2.28125\ A = 1$

So $A = 0.4384$

$B = 0.25 \times A = 0.25 \times 0.4384 = 0.1096$

$C = 0.375 \times A = 0.375 \times 0.4384 = 0.1644$

$D = 0.65625 \times A = 0.65625 \times 0.4384 = 0.2877$

The number of employees in each class at equilibrium is obtained as follows:

$A = 0.4384 \times 500$, $B = 0.1096 \times 500$, $C = 0.1644 \times 500$, $D = 0.2877 \times 500$

$500 = 200 + 150 + 90 + 60$ (the initial state)

So at equilibrium $(A \quad B \quad C \quad D) \approx (219 \quad 55 \quad 82 \quad 144)$

QUESTION 8

a) Differentiate between overlapping sets and equal sets as used in set theory

b) Clean Wash Limited conducted a market survey to investigate customers' loyalty to the company's three brands of soap namely; Powerfoam, Ngarisha and Nguvu Zaidi.

The following results were obtained from the survey:

- 22 percent of the customers were loyal to the Powerfoam brand.
- 18 percent of the customers were loyal to the Ngarisha brand
- 16 percent of the customers were loyal to the Nguvu Zaidi brand
- 10 percent of the customers were loyal to both the Powerfoam and the Nguvu Zaidi brands.
- 7 percent of the customers were loyal to both the Powerfoam and the Nguvu Zaidi brands.
- 6 percent of the customers were loyal to both the Ngarisha and the Nguvu Zaidi brands.
- 4 percent of the customers were loyal to all the three brands of soap.

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Required.

The percentage of customers that were loyal to at least one of the three brands of soap.

Solution:

(a) Difference between over lapping sets and equal sets as used in set theory.

Overlapping sets are those which have some elements in common.

For example, the set of positive multiples of 2 would be {2, 4, 6, 8, 10, 12, 14, ...}

the set of positive multiples of 3 would be {3, 6, 9, 12, 15, ...}

Their overlap (**intersection**) is the set of all positive multiples of 6 ie {6, 12, 18, ...}

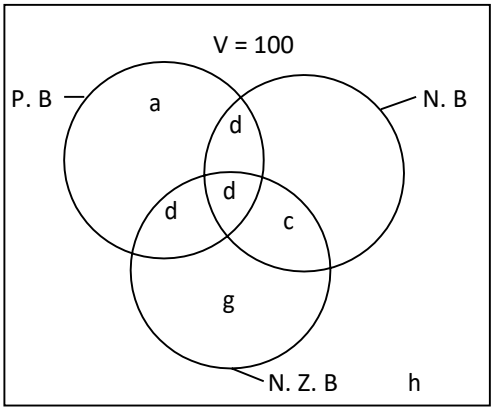
Equal sets are sets that have the same elements e.g. C = (7,8,9,1,0), D = (1,0,7,9,8)

b) Clean wash limited brands Power foam, Ngarisha and Nguvu zaidi

Required

Percentage of customers that were loyal to at least one of the three brands of soap.

Workings



a + b + d + d = 22

b + c + d + e = 18

d + e + d+ f + g = 16

b + d = 10

d + d = 7

4 + c	=	6
c = 6 – 4	=	2
4 + f	=	7
f = 7 – 4	=	3
b + 4	=	10
b = 10 – 4	=	6
d + e + f + g	=	16
4 + 2 + 3 + g	=	16
g = 16 – 9	=	7
b + c + d + e	=	18
b + c + 4 + 2	=	18
c = 18 – 12	=	6
a + b + d + f	=	22
a + 6 + 4 + 3	=	22
a = 22 – 13	=	9

∴

a + b + c + d + e f + g + h = 100

9 + 6 + 6 + 4 + 2 + 3 + 7 + h = 100

h = 100 – 37 = 63

Customers who were loyal to at least one of the three brands

a=9 c= 6 g=7

9+6 +6 +3 + 4+2 + 7 = 63%

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QUESTION 9

a) News Agency Limited deals in the distribution of three types of magazines namely; Newline, Informer and Update. The company recently conducted a market survey to determine the magazine preferences of 100 households in a certain town. The following results were obtained from the survey.

- 48 households read the Newsline magazine.
- 18 household read the Informer magazine.
- 26 households read the update magazine.
- 8 households read the Newsline and the Update magazines.
- 8 households read the Newsline and the Informer magazines.
- 3 households read the Update and the Informer magazines.
- 3 households read the three magazines.

Required:

(i) Represent the above information using a Venn diagram.

(ii) The number of households that read the Newsline magazine but did not read the Informer magazine.

(iii) The number of households that read the Update magazine and the Informer magazines but did not read the Newsline magazine.

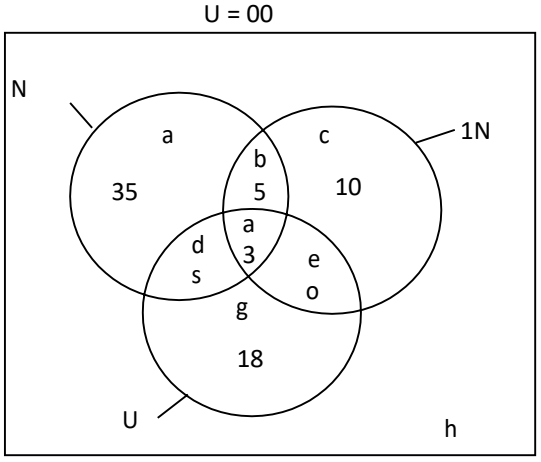
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The number of households that read none of the magazines

Solution:

a) Represent the information using a vein diagra

Workings



a + b + c + d+ e + f + g + h = 100

a + b + d + f= 48

b + c+ d+ e = 18

a + e + f + g = 26

f + d = 8

∴ if d = 3

d + e = 3

e = 3 – 3 = 0

b+ d = 8
b = 8 – 3
b = 5

f+ d = 8
f = 8 – 3
f = 5

b + c + d + e = 18
5 + c + 3 + 0 = 18
c = 18 – 8
c = 10

a + b + d + f = 48
a + 5 + 3 + 5 = 48
a = 48 – 13
a = 3

d + e + f + g = 26
3 + 0 + 5 + g = 26
g = 26 – 8
g = 18
a + b + c + d+ e + f + g + h = 100
35 + 5 + 12 + 3 + 0 + 5 + 18 + h = 100
h = 100 – 76
h = 24

ii) No. of households that read the Newsline magazine but did not read the informer magazine.
= 35 + 5 = 40

iii) No. of households that read the update magazine and the informer magazine but did not read the Newsline magazine =0

No. of households that read none of the magazine = 24

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TOPIC 3

HYPOTHESIS TESTING AND ESTIMATION

Meaning Hypothesis Testing

A **statistical hypothesis** is an assumption about a population parameter. This assumption may or may not be true. **Hypothesis testing** refers to the formal procedures used by statisticians to accept or reject statistical hypotheses.

Statistical Hypotheses

The best way to determine whether a statistical hypothesis is true would be to examine the entire population. Since that is often impractical, researchers typically examine a random sample from the population. If sample data are not consistent with the statistical hypothesis, the hypothesis is rejected.

There are two types of statistical hypotheses.

- **Null hypothesis**. The null hypothesis, denoted by H_0, is usually the hypothesis that sample observations result purely from chance.
- **Alternative hypothesis**. The alternative hypothesis, denoted by H_1 or H_a, is the hypothesis that sample observations are influenced by some non-random cause.

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For example, suppose we wanted to determine whether a coin was fair and balanced. A null hypothesis might be that half the flips would result in Heads and half, in Tails. The alternative hypothesis might be that the number of Heads and Tails would be very different. Symbolically, these hypotheses would be expressed as

$$H_0: P = 0.5$$
$$H_a: P \neq 0.5$$

Suppose we flipped the coin 50 times, resulting in 40 Heads and 10 Tails. Given this result, we would be inclined to reject the null hypothesis. We would conclude, based on the evidence, that the coin was probably not fair and balanced.

Hypothesis Tests

Statisticians follow a formal process to determine whether to reject a null hypothesis, based on sample data. This process, called **hypothesis testing**, consists of four steps.

- State the hypotheses. This involves stating the null and alternative hypotheses. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false.

- Formulate an analysis plan. The analysis plan describes how to use sample data to evaluate the null hypothesis. The evaluation often focuses around a single test statistic.
- Analyze sample data. Find the value of the test statistic (mean score, proportion, t statistic, z-score, etc.) described in the analysis plan.
- Interpret results. Apply the decision rule described in the analysis plan. If the value of the test statistic is unlikely, based on the null hypothesis, reject the null hypothesis.

Decision Errors

Two types of errors can result from a hypothesis test.

- **Type I error**. A Type I error occurs when the researcher rejects a null hypothesis when it is true. The probability of committing a Type I error is called the **significance level**. This probability is also called **alpha**, and is often denoted by α.
- **Type II error**. A Type II error occurs when the researcher fails to reject a null hypothesis that is false. The probability of committing a Type II error is called **Beta**, and is often denoted by β. The probability of *not* committing a Type II error is called the **Power** of the test.

Decision Rules

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The analysis plan includes decision rules for rejecting the null hypothesis. In practice, statisticians describe these decision rules in two ways - with reference to a P-value or with reference to a region of acceptance.

- P-value. The strength of evidence in support of a null hypothesis is measured by the **P-value**. Suppose the test statistic is equal to *S*. The P-value is the probability of observing a test statistic as extreme as *S*, assuming the null hypotheis is true. If the P-value is less than the significance level, we reject the null hypothesis.
- Region of acceptance. The **region of acceptance** is a range of values. If the test statistic falls within the region of acceptance, the null hypothesis is not rejected. The region of acceptance is defined so that the chance of making a Type I error is equal to the significance level.

The set of values outside the region of acceptance is called the **region of rejection**. If the test statistic falls within the region of rejection, the null hypothesis is rejected. In such cases, we say that the hypothesis has been rejected at the α level of significance.

These approaches are equivalent. Some statistics texts use the P-value approach; others use the region of acceptance approach. In subsequent lessons, this tutorial will present examples that illustrate each approach.

One-Tailed and Two-Tailed Tests

A test of a statistical hypothesis, where the region of rejection is on only one side of the sampling distribution, is called a **one-tailed test**. For example, suppose the null hypothesis states that the mean is less than or equal to 10. The alternative hypothesis would be that the mean is greater than

10. The region of rejection would consist of a range of numbers located on the right side of sampling distribution; that is, a set of numbers greater than 10.

A test of a statistical hypothesis, where the region of rejection is on both sides of the sampling distribution, is called a **two-tailed test**. For example, suppose the null hypothesis states that the mean is equal to 10. The alternative hypothesis would be that the mean is less than 10 or greater than 10. The region of rejection would consist of a range of numbers located on both sides of sampling distribution; that is, the region of rejection would consist partly of numbers that were less than 10 and partly of numbers that were greater than 10.

How to Test Hypotheses

This lesson describes a general procedure that can be used to test statistical hypotheses.

How to Conduct Hypothesis Tests

All hypothesis tests are conducted the same way. The researcher states a hypothesis to be tested, formulates an analysis plan, analyzes sample data according to the plan, and accepts or rejects the null hypothesis, based on results of the analysis.

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- **State the hypotheses.** Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false; and vice versa.

- **Formulate an analysis plan.** The analysis plan describes how to use sample data to accept or reject the null hypothesis. It should specify the following elements.

    - Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.

    - Test method. Typically, the test method involves a test statistic and a sampling distribution. Computed from sample data, the test statistic might be a mean score, proportion, difference between means, difference between proportions, z-score, t statistic, chi-square, etc. Given a test statistic and its sampling distribution, a researcher can assess probabilities associated with the test statistic. If the test statistic probability is less than the significance level, the null hypothesis is rejected.

- **Analyze sample data.** Using sample data, perform computations called for in the analysis plan.

    - Test statistic. When the null hypothesis involves a mean or proportion, use either of the following equations to compute the test statistic.

Test statistic = (Statistic - Parameter) / (Standard deviation of statistic)
Test statistic = (Statistic - Parameter) / (Standard error of statistic)

where *Parameter* is the value appearing in the null hypothesis, and *Statistic* is the point estimate of *Parameter*. As part of the analysis, you may need to compute the standard deviation or standard

error of the statistic. Previously, we presented common formulas for the standard deviation and standard error.

When the parameter in the null hypothesis involves categorical data, you may use a chi-square statistic as the test statistic. Instructions for computing a chi-square test statistic are presented in the lesson on the chi-square goodness of fit test.

    - P-value. The P-value is the probability of observing a sample statistic as extreme as the test statistic, assuming the null hypotheis is true.

- **Interpret the results.** If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Applications of the General Hypothesis Testing Procedure

The next few lessons show how to apply the general hypothesis testing procedure to different kinds of statistical problems.

- Proportions
- Difference between proportions
- Proportions from small samples
- Regression slope
- Means
- Difference between means
- Difference between matched pairs
- Goodness of fit
- Homogeneity
- Independence

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At this point, don't worry if the general procedure for testing hypotheses seems a little bit unclear. The procedure will be clearer after you read through a few of the examples presented in subsequent lessons.

Test Your Understanding

Problem 1

In hypothesis testing, which of the following statements is always true?

I. The P-value is greater than the significance level.
II. The P-value is computed from the significance level.
III. The P-value is the parameter in the null hypothesis.
IV. The P-value is a test statistic.
V. The P-value is a probability.

(A) I only
(B) II only
(C) III only
(D) IV only
(E) V only

Solution

The correct answer is (E). The P-value is the probability of observing a sample statistic as extreme as the test statistic. It can be greater than the significance level, but it can also be smaller than the significance level. It is not computed from the significance level, it is not the parameter in the null hypothesis, and it is not a test statistic.

CHI-SQUARE GOODNESS OF FIT TEST

This lesson explains how to conduct a **chi-square goodness of fit test**. The test is applied when you have one categorical variable from a single population. It is used to determine whether sample data are consistent with a hypothesized distribution.

For example, suppose a company printed baseball cards. It claimed that 30% of its cards were rookies; 60%, veterans; and 10%, All-Stars. We could gather a random sample of baseball cards and use a chi-square goodness of fit test to see whether our sample distribution differed significantly from the distribution claimed by the company. The sample problem at the end of the lesson considers this example.

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When to Use the Chi-Square Goodness of Fit Test

The chi-square goodness of fit test is appropriate when the following conditions are met:

- The sampling method is simple random sampling.
- The variable under study is categorical.
- The expected value of the number of sample observations in each level of the variable is at least 5.

This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Every hypothesis test requires the analyst to state a null hypothesis (H_0) and an alternative hypothesis (H_a). The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false; and vice versa.

For a chi-square goodness of fit test, the hypotheses take the following form.

- H_0: The data are consistent with a specified distribution.
- H_a: The data are *not* consistent with a specified distribution.

Typically, the null hypothesis (H_0) specifies the proportion of observations at each level of the categorical variable. The alternative hypothesis (H_a) is that *at least* one of the specified proportions is not true.

Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. The plan should specify the following elements.

- Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.
- Test method. Use the chi-square goodness of fit test to determine whether observed sample frequencies differ significantly from expected frequencies specified in the null hypothesis. The chi-square goodness of fit test is described in the next section, and demonstrated in the sample problem at the end of this lesson.

Analyze Sample Data

Using sample data, find the degrees of freedom, expected frequency counts, test statistic, and the P-value associated with the test statistic.

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- Degrees of freedom. The degrees of freedom (DF) is equal to the number of levels (k) of the categorical variable minus 1: DF = k - 1 .
- Expected frequency counts. The expected frequency counts at each level of the categorical variable are equal to the sample size times the hypothesized proportion from the null hypothesis

$$E_i = np_i$$

where E_i is the expected frequency count for the *i*th level of the categorical variable, n is the total sample size, and p_i is the hypothesized proportion of observations in level *i*.

- Test statistic. The test statistic is a chi-square random variable (X^2) defined by the following equation.

$$X^2 = \Sigma [(O_i - E_i)^2 / E_i]$$

where O_i is the observed frequency count for the *i*th level of the categorical variable, and E_i is the expected frequency count for the *i*th level of the categorical variable.

- P-value. The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a chi-square, use the Chi-Square Distribution Calculator to assess the probability associated with the test statistic. Use the degrees of freedom computed above.

Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Test Your Understanding

Problem

Acme Toy Company prints baseball cards. The company claims that 30% of the cards are rookies, 60% veterans, and 10% are All-Stars.

Suppose a random sample of 100 cards has 50 rookies, 45 veterans, and 5 All-Stars. Is this consistent with Acme's claim? Use a 0.05 level of significance.

Solution

The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.
  - Null hypothesis: The proportion of rookies, veterans, and All-Stars is 30%, 60% and 10%, respectively.
  - Alternative hypothesis: At least one of the proportions in the null hypothesis is false.

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- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. Using sample data, we will conduct a chi-square goodness of fit test of the null hypothesis.
- **Analyze sample data**. Applying the chi-square goodness of fit test to sample data, we compute the degrees of freedom, the expected frequency counts, and the chi-square test statistic. Based on the chi-square statistic and the degrees of freedom, we determine the P-value.

$$DF = k - 1 = 3 - 1 = 2$$

$$(E_i) = n * p_i$$
$$(E_1) = 100 * 0.30 = 30$$
$$(E_2) = 100 * 0.60 = 60$$
$$(E_3) = 100 * 0.10 = 10$$

$$X^2 = \Sigma [(O_i - E_i)^2 / E_i]$$
$$X^2 = [(50 - 30)^2 / 30] + [(45 - 60)^2 / 60] + [(5 - 10)^2 / 10]$$
$$X^2 = (400 / 30) + (225 / 60) + (25 / 10) = 13.33 + 3.75 + 2.50 = 19.58$$

where DF is the degrees of freedom, k is the number of levels of the categorical variable, n is the number of observations in the sample, E_i is the expected frequency count for level i, O_i is the observed frequency count for level i, and X^2 is the chi-square test statistic.

The P-value is the probability that a chi-square statistic having 2 degrees of freedom is more extreme than 19.58.

We use the Chi-Square Distribution Calculator to find $P(X^2 > 19.58) = 0.0001$.

- **Interpret results**. Since the P-value (0.0001) is less than the significance level (0.05), we cannot accept the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the variable under study was categorical, and each level of the categorical variable had an expected frequency count of at least 5.

HYPOTHESIS TEST ON A MEAN

This lesson explains how to conduct a hypothesis test of a mean, when the following conditions are met:

- The sampling method is simple random sampling.
- The sampling distribution is normal or nearly normal.

Generally, the sampling distribution will be approximately normally distributed if any of the following conditions apply.

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- The population distribution is normal.
- The population distribution is symmetric, unimodal, without outliers, and the sample size is 15 or less.
- The population distribution is moderately skewed, unimodal, without outliers, and the sample size is between 16 and 40.
- The sample size is greater than 40, without outliers.

This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false; and vice versa.

The table below shows three sets of hypotheses. Each makes a statement about how the population mean μ is related to a specified value *M*. (In the table, the symbol ≠ means " not equal to ".)

Set	Null hypothesis	Alternative hypothesis	Number of tails
1	$\mu = M$	$\mu \neq M$	2
2	$\mu > M$	$\mu < M$	1
3	$\mu < M$	$\mu > M$	1

The first set of hypotheses (Set 1) is an example of a two-tailed test, since an extreme value on either side of the sampling distribution would cause a researcher to reject the null hypothesis. The other two sets of hypotheses (Sets 2 and 3) are one-tailed tests, since an extreme value on only one side of the sampling distribution would cause a researcher to reject the null hypothesis

Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. It should specify the following elements.

- Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.
- Test method. Use the one-sample t-test to determine whether the hypothesized mean differs significantly from the observed sample mean.

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Analyze Sample Data

Using sample data, conduct a one-sample t-test. This involves finding the standard error, degrees of freedom, test statistic, and the P-value associated with the test statistic.

- Standard error. Compute the standard error (SE) of the sampling distribution.

$$SE = s * sqrt\{ (1/n) * [(N - n) / (N - 1)] \}$$

where *s* is the standard deviation of the sample, N is the population size, and *n* is the sample size. When the population size is much larger (at least 20 times larger) than the sample size, the standard error can be approximated by:

$$SE = s / sqrt(n)$$

- Degrees of freedom. The degrees of freedom (DF) is equal to the sample size (n) minus one. Thus, DF = n - 1.
- Test statistic. The test statistic is a t statistic (t) defined by the following equation.

$$t = (x - \mu) / SE$$

where x is the sample mean, μ is the hypothesized population mean in the null hypothesis, and SE is the standard error.

- P-value. The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a t statistic, use the t Distribution Calculator to assess the probability associated with the t statistic, given the degrees of freedom computed above. (See sample problems at the end of this lesson for examples of how this is done.)

Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Problem 1: Two-Tailed Test

An inventor has developed a new, energy-efficient lawn mower engine. He claims that the engine will run continuously for 5 hours (300 minutes) on a single gallon of regular gasoline. From his stock of 2000 engines, the inventor selects a simple random sample of 50 engines for testing. The engines run for an average of 295 minutes, with a standard deviation of 20 minutes. Test the null hypothesis that the mean run time is 300 minutes against the alternative hypothesis that the mean run time is not 300 minutes. Use a 0.05 level of significance. (Assume that run times for the population of engines are normally distributed.)

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Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

$$\text{Null hypothesis: } \mu = 300$$
$$\text{Alternative hypothesis: } \mu \neq 300$$

Note that these hypotheses constitute a two-tailed test. The null hypothesis will be rejected if the sample mean is too big or if it is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. The test method is a one-sample t-test.
- **Analyze sample data**. Using sample data, we compute the standard error (SE), degrees of freedom (DF), and the t statistic test statistic (t).

$$SE = s / \text{sqrt}(n)$$
$$SE = 20 / \text{sqrt}(50) = 20/7.07 = 2.83$$

$$DF = n - 1 = 50 - 1 = 49$$

$$t = (x - \mu) / SE = (295 - 300)/2.83 = -1.77$$

where s is the standard deviation of the sample, x is the sample mean, μ is the hypothesized population mean, and n is the sample size.

Since we have a two-tailed test, the P-value is the probability that the t statistic having 49 degrees of freedom is less than -1.77 or greater than 1.77.

We use the t Distribution Calculator to find P(t < -1.77) = 0.04, and P(t > 1.77) = 0.04. Thus, the P-value = 0.04 + 0.04 = 0.08.

- **Interpret results**. Since the P-value (0.08) is greater than the significance level (0.05), we cannot reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the population was normally distributed, and the sample size was small relative to the population size (less than 5%).

Problem 2: One-Tailed Test

Bon Air Elementary School has 1000 students. The principal of the school thinks that the average IQ of students at Bon Air is at least 110. To prove her point, she administers an IQ test to 20 randomly selected students. Among the sampled students, the average IQ is 108 with a standard deviation of 10. Based on these results, should the principal accept or reject her original hypothesis? Assume a significance level of 0.01. (Assume that test scores in the population of engines are normally distributed.)

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Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

Null hypothesis: μ >= 110
Alternative hypothesis: μ < 110

Note that these hypotheses constitute a one-tailed test. The null hypothesis will be rejected if the sample mean is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.01. The test method is a one-sample t-test.
- **Analyze sample data**. Using sample data, we compute the standard error (SE), degrees of freedom (DF), and the t statistic test statistic (t).

SE = s / sqrt(n)
SE = 10 / sqrt(20) = 10/4.472 = 2.236

DF = n - 1 = 20 - 1 = 19

t = (x - μ) / SE = (108 - 110)/2.236 = -0.894

where s is the standard deviation of the sample, x is the sample mean, μ is the hypothesized population mean, and n is the sample size.

Here is the logic of the analysis: Given the alternative hypothesis ($\mu < 110$), we want to know whether the observed sample mean is small enough to cause us to reject the null hypothesis.

The observed sample mean produced a t statistic test statistic of -0.894. We use the t Distribution Calculator to find $P(t < -0.894) = 0.19$. This means we would expect to find a sample mean of 108 or smaller in 19 percent of our samples, if the true population IQ were 110. Thus the P-value in this analysis is 0.19.

- **Interpret results**. Since the P-value (0.19) is greater than the significance level (0.01), we cannot reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the population was normally distributed, and the sample size was small relative to the population size (less than 5%

HYPOTHESIS TEST ON PROPORTIONS

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This lesson explains how to conduct a hypothesis test of a proportion, when the following conditions are met:

- The sampling method is simple random sampling.
- Each sample point can result in just two possible outcomes. We call one of these outcomes a success and the other, a failure.
- The sample includes at least 10 successes and 10 failures.
- The population size is at least 20 times as big as the sample size.

This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false; and vice versa.

Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. It should specify the following elements.

- Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.

- Test method. Use the one-sample z-test to determine whether the hypothesized population proportion differs significantly from the observed sample proportion.

Analyze Sample Data

Using sample data, find the test statistic and its associated P-Value.

- Standard deviation. Compute the standard deviation (σ) of the sampling distribution.

$$\sigma = \text{sqrt}[P * (1 - P) / n]$$

Where P is the hypothesized value of population proportion in the null hypothesis, and n is the sample size.

- Test statistic. The test statistic is a z-score (z) defined by the following equation.

$$z = (p - P) / \sigma$$

where P is the hypothesized value of population proportion in the null hypothesis, p is the sample proportion, and σ is the standard deviation of the sampling distribution.

- P-value. The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a z-score, use the Normal Distribution Calculator to assess the probability associated with the z-score. (See sample problems at the end of this lesson for examples of how this is done.)

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Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Test Your Understanding

In this section, two hypothesis testing examples illustrate how to conduct a hypothesis test of a proportion. The first problem involves a a two-tailed test; the second problem, a one-tailed test.

Problem 1: Two-Tailed Test

The CEO of a large electric utility claims that 80 percent of his 1,000,000 customers are very satisfied with the service they receive. To test this claim, the local newspaper surveyed 100 customers, using simple random sampling. Among the sampled customers, 73 percent say they are very satisified. Based on these findings, can we reject the CEO's hypothesis that 80% of the customers are very satisfied? Use a 0.05 level of significance.

Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

Null hypothesis: P = 0.80
Alternative hypothesis: P ≠ 0.80

Note that these hypotheses constitute a two-tailed test. The null hypothesis will be rejected if the sample proportion is too big or if it is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. The test method, shown in the next section, is a one-sample z-test.
- **Analyze sample data**. Using sample data, we calculate the standard deviation (σ) and compute the z-score test statistic (z).

$$\sigma = \text{sqrt}[P * (1 - P) / n]$$
$$\sigma = \text{sqrt} [(0.8 * 0.2) / 100]$$
$$\sigma = \text{sqrt}(0.0016) = 0.04$$

$$z = (p - P) / \sigma = (.73 - .80)/0.04 = -1.75$$

where P is the hypothesized value of population proportion in the null hypothesis, p is the sample proportion, and n is the sample size.

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Since we have a two-tailed test, the P-value is the probability that the z-score is less than -1.75 or greater than 1.75.

We use the Normal Distribution Calculator to find P(z < -1.75) = 0.04, and P(z > 1.75) = 0.04. Thus, the P-value = 0.04 + 0.04 = 0.08.

- **Interpret results**. Since the P-value (0.08) is greater than the significance level (0.05), we cannot reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the sample included at least 10 successes and 10 failures, and the population size was at least 10 times the sample size.

Problem 2: One-Tailed Test

Suppose the previous example is stated a little bit differently. Suppose the CEO claims that *at least* 80 percent of the company's 1,000,000 customers are very satisfied. Again, 100 customers are surveyed using simple random sampling. The result: 73 percent are very satisfied. Based on these results, should we accept or reject the CEO's hypothesis? Assume a significance level of 0.05.

Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

Null hypothesis: P >= 0.80
Alternative hypothesis: P < 0.80

Note that these hypotheses constitute a one-tailed test. The null hypothesis will be rejected only if the sample proportion is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. The test method, shown in the next section, is a one-sample z-test.
- **Analyze sample data**. Using sample data, we calculate the standard deviation (σ) and compute the z-score test statistic (z).

$$\sigma = \sqrt{P * (1 - P) / n}$$
$$\sigma = \sqrt{(0.8 * 0.2) / 100}$$
$$\sigma = \sqrt{0.0016} = 0.04$$

$$z = (p - P) / \sigma = (.73 - .80)/0.04 = -1.75$$

where P is the hypothesized value of population proportion in the null hypothesis, p is the sample proportion, and n is the sample size.

Since we have a one-tailed test, the P-value is the probability that the z-score is less than -1.75. We use the Normal Distribution Calculator to find P(z < -1.75) = 0.04. Thus, the P-value = 0.04.

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- **Interpret results**. Since the P-value (0.04) is less than the significance level (0.05), we cannot accept the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the sample included at least 10 successes and 10 failures, and the population size was at least 10 times the sample size

HYPOTHESIS TEST ON THE DIFFERENCE BETWEEN MEANS

This lesson explains how to conduct a hypothesis test for the difference between two means.

The test procedure, called the **two-sample t-test**, is appropriate when the following conditions are met:

- The sampling method for each sample is simple random sampling.
- The samples are independent.
- Each population is at least 20 times larger than its respective sample.
- The sampling distribution is approximately normal, which is generally the case if any of the following conditions apply.
    - The population distribution is normal.
    - The population data are symmetric, unimodal, without outliers, and the sample size is 15 or less.

- The population data are slightly skewed, unimodal, without outliers, and the sample size is 16 to 40.
- The sample size is greater than 40, without outliers.

This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false; and vice versa.

The table below shows three sets of null and alternative hypotheses. Each makes a statement about the difference *d* between the mean of one population μ_1 and the mean of another population μ_2. (In the table, the symbol ≠ means " not equal to ".)

Set	Null hypothesis	Alternative hypothesis	Number of tails
1	$\mu_1 - \mu_2 = d$	$\mu_1 - \mu_2 \neq d$	2
2	$\mu_1 - \mu_2 > d$	$\mu_1 - \mu_2 < d$	1
3	$\mu_1 - \mu_2 < d$	$\mu_1 - \mu_2 > d$	1

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The first set of hypotheses (Set 1) is an example of a two-tailed test, since an extreme value on either side of the sampling distribution would cause a researcher to reject the null hypothesis. The other two sets of hypotheses (Sets 2 and 3) are one-tailed tests, since an extreme value on only one side of the sampling distribution would cause a researcher to reject the null hypothesis.

When the null hypothesis states that there is no difference between the two population means (i.e., d = 0), the null and alternative hypothesis are often stated in the following form.

$$H_0: \mu_1 = \mu_2$$
$$H_a: \mu_1 \neq \mu_2$$

Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. It should specify the following elements.

- Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.
- Test method. Use the two-sample t-test to determine whether the difference between means found in the sample is significantly different from the hypothesized difference between means.

Analyze Sample Data

Using sample data, find the standard error, degrees of freedom, test statistic, and the P-value associated with the test statistic.

- Standard error. Compute the standard error (SE) of the sampling distribution.

$$SE = sqrt[(s_1^2/n_1) + (s_2^2/n_2)]$$

where s_1 is the standard deviation of sample 1, s_2 is the standard deviation of sample 2, n_1 is the size of sample 1, and n_2 is the size of sample 2.

- Degrees of freedom. The degrees of freedom (DF) is:

$$DF = (s_1^2/n_1 + s_2^2/n_2)^2 / \{ [(s_1^2 / n_1)^2 / (n_1 - 1)] + [(s_2^2 / n_2)^2 / (n_2 - 1)] \}$$

If DF does not compute to an integer, round it off to the nearest whole number. Some texts suggest that the degrees of freedom can be approximated by the smaller of $n_1 - 1$ and $n_2 - 1$; but the above formula gives better results.

- Test statistic. The test statistic is a t statistic (t) defined by the following equation.

$$t = [(x_1 - x_2) - d] / SE$$

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where x_1 is the mean of sample 1, x_2 is the mean of sample 2, d is the hypothesized difference between population means, and SE is the standard error.

- P-value. The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a t statistic, use the t Distribution Calculator to assess the probability associated with the t statistic, having the degrees of freedom computed above. (See sample problems at the end of this lesson for examples of how this is done.)

Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Test Your Understanding

In this section, two sample problems illustrate how to conduct a hypothesis test of a difference between mean scores. The first problem involves a two-tailed test; the second problem, a one-tailed test.

Problem 1: Two-Tailed Test

Within a school district, students were randomly assigned to one of two Math teachers - Mrs. Smith and Mrs. Jones. After the assignment, Mrs. Smith had 30 students, and Mrs. Jones had 25 students.

At the end of the year, each class took the same standardized test. Mrs. Smith's students had an average test score of 78, with a standard deviation of 10; and Mrs. Jones' students had an average test score of 85, with a standard deviation of 15.

Test the hypothesis that Mrs. Smith and Mrs. Jones are equally effective teachers. Use a 0.10 level of significance. (Assume that student performance is approximately normal.)

Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

$$\text{Null hypothesis: } \mu_1 - \mu_2 = 0$$
$$\text{Alternative hypothesis: } \mu_1 - \mu_2 \neq 0$$

Note that these hypotheses constitute a two-tailed test. The null hypothesis will be rejected if the difference between sample means is too big or if it is too small.

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- **Formulate an analysis plan**. For this analysis, the significance level is 0.10. Using sample data, we will conduct a two-sample t-test of the null hypothesis.
- **Analyze sample data**. Using sample data, we compute the standard error (SE), degrees of freedom (DF), and the t statistic test statistic (t).

$$SE = \text{sqrt}[(s_1^2/n_1) + (s_2^2/n_2)]$$
$$SE = \text{sqrt}[(10^2/30) + (15^2/25] = \text{sqrt}(3.33 + 9)$$
$$SE = \text{sqrt}(12.33) = 3.51$$

$$DF = (s_1^2/n_1 + s_2^2/n_2)^2 / \{ [(s_1^2 / n_1)^2 / (n_1 - 1)] + [(s_2^2 / n_2)^2 / (n_2 - 1)] \}$$
$$DF = (10^2/30 + 15^2/25)^2 / \{ [(10^2 / 30)^2 / (29)] + [(15^2 / 25)^2 / (24)] \}$$
$$DF = (3.33 + 9)^2 / \{ [(3.33)^2 / (29)] + [(9)^2 / (24)] \} = 152.03 / (0.382 + 3.375) = 152.03/3.757 = 40.47$$

$$t = [(x_1 - x_2) - d] / SE = [(78 - 85) - 0] / 3.51 = -7/3.51 = -1.99$$

where s_1 is the standard deviation of sample 1, s_2 is the standard deviation of sample 2, n_1 is the size of sample 1, n_2 is the size of sample 2, x_1 is the mean of sample 1, x_2 is the mean of sample 2, d is the hypothesized difference between the population means, and SE is the standard error.

Since we have a two-tailed test, the P-value is the probability that a t statistic having 40 degrees of freedom is more extreme than -1.99; that is, less than -1.99 or greater than 1.99.

We use the t Distribution Calculator to find $P(t < -1.99) = 0.027$, and $P(t > 1.99) = 0.027$. Thus, the P-value = 0.027 + 0.027 = 0.054.

- **Interpret results**. Since the P-value (0.054) is less than the significance level (0.10), we cannot accept the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the samples were independent, the sample size was much smaller than the population size, and the samples were drawn from a normal population.

Problem 2: One-Tailed Test

The Acme Company has developed a new battery. The engineer in charge claims that the new battery will operate continuously for *at least* 7 minutes longer than the old battery.

To test the claim, the company selects a simple random sample of 100 new batteries and 100 old batteries. The old batteries run continuously for 190 minutes with a standard deviation of 20 minutes; the new batteries, 200 minutes with a standard deviation of 40 minutes.

Test the engineer's claim that the new batteries run at least 7 minutes longer than the old. Use a 0.05 level of significance. (Assume that there are no outliers in either sample.)

Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

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- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

$$\text{Null hypothesis: } \mu_1 - \mu_2 >= 7$$
$$\text{Alternative hypothesis: } \mu_1 - \mu_2 < 7$$

Note that these hypotheses constitute a one-tailed test. The null hypothesis will be rejected if the mean difference between sample means is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. Using sample data, we will conduct a two-sample t-test of the null hypothesis.
- **Analyze sample data**. Using sample data, we compute the standard error (SE), degrees of freedom (DF), and the t statistic test statistic (t).

$$SE = \text{sqrt}[(s_1^2/n_1) + (s_2^2/n_2)]$$
$$SE = \text{sqrt}[(40^2/100) + (20^2/100]$$
$$SE = \text{sqrt}(16 + 4) = 4.472$$

$$DF = (s_1^2/n_1 + s_2^2/n_2)^2 / \{ [(s_1^2 / n_1)^2 / (n_1 - 1)] + [(s_2^2 / n_2)^2 / (n_2 - 1)] \}$$
$$DF = (40^2/100 + 20^2/100)^2 / \{ [(40^2 / 100)^2 / (99)] + [(20^2 / 100)^2 / (99)] \}$$
$$DF = (20)^2 / \{ [(16)^2 / (99)] + [(2)^2 / (99)] \} = 400 / (2.586 + 0.162) = 145.56$$

$$t = [(x_1 - x_2) - d] / SE = [(200 - 190) - 7] / 4.472 = 3/4.472 = 0.67$$

where s_1 is the standard deviation of sample 1, s_2 is the standard deviation of sample 2, n_1 is the size of sample 1, n_2 is the size of sample 2, x_1 is the mean of sample 1, x_2 is the mean of sample 2, d is the hypothesized difference between population means, and SE is the standard error.

Here is the logic of the analysis: Given the alternative hypothesis ($\mu_1 - \mu_2 < 7$), we want to know whether the observed difference in sample means is small enough (i.e., sufficiently less than 7) to cause us to reject the null hypothesis.

The observed difference in sample means (10) produced a t statistic of 0.67. We use the t Distribution Calculator to find $P(t < 0.67) = 0.75$.

This means we would expect to find an observed difference in sample means of 10 or less in 75% of our samples, if the true difference were actually 7. Therefore, the P-value in this analysis is 0.75.

- **Interpret results**. Since the P-value (0.75) is greater than the significance level (0.05), we cannot reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the samples were independent, the sample size was much smaller than the population size, and the sample size was large without outliers.

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Hypothesis Test: Difference Between Paired Means

This lesson explains how to conduct a hypothesis test for the difference between paired means.

The test procedure, called the **matched-pairs t-test**, is appropriate when the following conditions are met:

- The sampling method for each sample is simple random sampling.
- The test is conducted on paired data. (As a result, the data sets are *not* independent.)
- The sampling distribution is approximately normal, which is generally true if any of the following conditions apply.
    - The population distribution is normal.
    - The population data are symmetric, unimodal, without outliers, and the sample size is 15 or less.
    - The population data are slightly skewed, unimodal, without outliers, and the sample size is 16 to 40.
    - The sample size is greater than 40, without outliers.

This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false; and vice versa.

The hypotheses concern a new variable d, which is based on the difference between paired values from two data sets.

$$d = x_1 - x_2$$

where x_1 is the value of variable x in the first data set, and x_2 is the value of the variable from the second data set that is paired with x_1.

The table below shows three sets of null and alternative hypotheses. Each makes a statement about how the true difference in population values μ_d is related to some hypothesized value D. (In the table, the symbol ≠ means " not equal to ".)

Set	Null hypothesis	Alternative hypothesis	Number of tails
1	$\mu_d = D$	$\mu_d \neq D$	2
2	$\mu_d > D$	$\mu_d < D$	1
3	$\mu_d < D$	$\mu_d > D$	1

The first set of hypotheses (Set 1) is an example of a two-tailed test, since an extreme value on either side of the sampling distribution would cause a researcher to reject the null hypothesis. The other two sets of hypotheses (Sets 2 and 3) are one-tailed tests, since an extreme value on only one side of the sampling distribution would cause a researcher to reject the null hypothesis.

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Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. It should specify the following elements.

- Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.
- Test method. Use the matched-pairs t-test to determine whether the difference between sample means for paired data is significantly different from the hypothesized difference between population means.

Analyze Sample Data

Using sample data, find the standard deviation, standard error, degrees of freedom, test statistic, and the P-value associated with the test statistic.

- Standard deviation. Compute the standard deviation (s_d) of the differences computed from n matched pairs.

$$s_d = \text{sqrt}\ [\ (\Sigma(d_i - d)^2 / (n - 1)\ ]$$

where d_i is the difference for pair *i*, d is the sample mean of the differences, and n is the number of paired values.

- Standard error. Compute the standard error (SE) of the sampling distribution of d.

$$SE = s_d * sqrt\{ (1/n) * [(N - n) / (N - 1)] \}$$

where s_d is the standard deviation of the sample difference, *N* is the number of matched pairs in the population, and *n* is the number of matched pairs in the sample. When the population size is much larger (at least 20 times larger) than the sample size, the standard error can be approximated by:

$$SE = s_d / sqrt(n)$$

- Degrees of freedom. The degrees of freedom (DF) is: DF = n - 1 .
- Test statistic. The test statistic is a t statistic (t) defined by the following equation.

$$t = [(x_1 - x_2) - D] / SE = (d - D) / SE$$

where x_1 is the mean of sample 1, x_2 is the mean of sample 2, d is the mean difference between paired values in the sample, D is the hypothesized difference between population means, and SE is the standard error.

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- P-value. The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a t statistic, use the t Distribution Calculator to assess the probability associated with the t statistic, having the degrees of freedom computed above. (See the sample problem at the end of this lesson for guidance on how this is done.)

Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Test Your Understanding

Problem

Forty-four sixth graders were randomly selected from a school district. Then, they were divided into 22 matched pairs, each pair having equal IQ's. One member of each pair was randomly selected to receive special training. Then, all of the students were given an IQ test. Test results are summarized below.

Pair	Training	No training	Difference, d	$(d - d)^2$
1	95	90	5	16
2	89	85	4	9
3	76	73	3	4
4	92	90	2	1

5	91	90	1	0
6	53	53	0	1
7	67	68	-1	4
8	88	90	-2	9
9	75	78	-3	16
10	85	89	-4	25
11	90	95	-5	36

Pair	Training	No training	Difference, d	$(d - d)^2$
12	85	83	2	1
13	87	83	4	9
14	85	83	2	1
15	85	82	3	4
16	68	65	3	4
17	81	79	2	1
18	84	83	1	0
19	71	60	11	100
20	46	47	-1	4
21	75	77	-2	9
22	80	83	-3	16

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$$\Sigma(d - d)^2 = 270$$
$$d = 1$$

Do these results provide evidence that the special training helped or hurt student performance? Use an 0.05 level of significance. Assume that the mean differences are approximately normally distributed.

Solution

The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

$$\text{Null hypothesis: } \mu_d = 0$$
$$\text{Alternative hypothesis: } \mu_d \neq 0$$

Note that these hypotheses constitute a two-tailed test. The null hypothesis will be rejected if the difference between sample means is too big or if it is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. Using sample data, we will conduct a matched-pairs t-test of the null hypothesis.
- **Analyze sample data**. Using sample data, we compute the standard deviation of the differences (s), the standard error (SE) of the mean difference, the degrees of freedom (DF), and the t statistic test statistic (t).

$$s = \text{sqrt} [(\Sigma(d_i - d)^2 / (n - 1)]$$
$$s = \text{sqrt}[270/(22-1)]$$
$$s = \text{sqrt}(12.857) = 3.586$$

$$SE = s / \text{sqrt}(n) = 3.586 / [\text{sqrt}(22)]$$
$$SE = 3.586/4.69 = 0.765$$

$$DF = n - 1 = 22 - 1 = 21$$

$$t = [(x_1 - x_2) - D] / SE$$
$$t = (d - D)/ SE = (1 - 0)/0.765 = 1.307$$

where d_i is the observed difference for pair *i*, d is mean difference between sample pairs, D is the hypothesized mean difference between population pairs, and n is the number of pairs.

Since we have a two-tailed test, the P-value is the probability that a t statistic having 21 degrees of freedom is more extreme than 1.307; that is, less than -1.307 or greater than 1.307.

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We use the t Distribution Calculator to find P(t < -1.307) = 0.103, and P(t > 1.307) = 0.103. Thus, the P-value = 0.103 + 0.103 = 0.206.

- **Interpret results**. Since the P-value (0.206) is greater than the significance level (0.05), we cannot reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the samples consisted of paired data, and the mean differences were normally distributed. In addition, we used the approximation formula to compute the standard error, since the sample size was small relative to the population size.

HYPOTHESIS TEST ON THE DIFFERENCE BETWEEN PAIRED MEANS

This lesson explains how to conduct a hypothesis test for the difference between paired means.

The test procedure, called the **matched-pairs t-test**, is appropriate when the following conditions are met:

- The sampling method for each sample is simple random sampling.
- The test is conducted on paired data. (As a result, the data sets are *not* independent.)
- The sampling distribution is approximately normal, which is generally true if any of the following conditions apply.

- The population distribution is normal.
- The population data are symmetric, unimodal, without outliers, and the sample size is 15 or less.
- The population data are slightly skewed, unimodal, without outliers, and the sample size is 16 to 40.
- The sample size is greater than 40, without outliers.

This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false; and vice versa.

The hypotheses concern a new variable d, which is based on the difference between paired values from two data sets.

$$d = x_1 - x_2$$

where x_1 is the value of variable x in the first data set, and x_2 is the value of the variable from the second data set that is paired with x_1.

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The table below shows three sets of null and alternative hypotheses. Each makes a statement about how the true difference in population values μ_d is related to some hypothesized value D. (In the table, the symbol ≠ means " not equal to ".)

Set	Null hypothesis	Alternative hypothesis	Number of tails
1	$\mu_d = D$	$\mu_d \neq D$	2
2	$\mu_d > D$	$\mu_d < D$	1
3	$\mu_d < D$	$\mu_d > D$	1

The first set of hypotheses (Set 1) is an example of a two-tailed test, since an extreme value on either side of the sampling distribution would cause a researcher to reject the null hypothesis. The other two sets of hypotheses (Sets 2 and 3) are one-tailed tests, since an extreme value on only one side of the sampling distribution would cause a researcher to reject the null hypothesis.

Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. It should specify the following elements.

- Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.
- Test method. Use the matched-pairs t-test to determine whether the difference between sample means for paired data is significantly different from the hypothesized difference between population means.

Analyze Sample Data

Using sample data, find the standard deviation, standard error, degrees of freedom, test statistic, and the P-value associated with the test statistic.

- Standard deviation. Compute the standard deviation (s_d) of the differences computed from n matched pairs.

$$s_d = \text{sqrt} [(\Sigma(d_i - d)^2 / (n - 1)]$$

Where d_i is the difference for pair *i*, d is the sample mean of the differences, and n is the number of paired values.

- Standard error. Compute the standard error (SE) of the sampling distribution of d.

$$SE = s_d * \text{sqrt}\{ (1/n) * [(N - n) / (N - 1)] \}$$

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Where s_d is the standard deviation of the sample difference, *N* is the number of matched pairs in the population, and *n* is the number of matched pairs in the sample. When the population size is much larger (at least 20 times larger) than the sample size, the standard error can be approximated by:

$$SE = s_d / \text{sqrt}(n)$$

- Degrees of freedom. The degrees of freedom (DF) is: DF = n - 1 .
- Test statistic. The test statistic is a t statistic (t) defined by the following equation.

$$t = [(x_1 - x_2) - D] / SE = (d - D) / SE$$

where x_1 is the mean of sample 1, x_2 is the mean of sample 2, d is the mean difference between paired values in the sample, D is the hypothesized difference between population means, and SE is the standard error.

- P-value. The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a t statistic, use the t Distribution Calculator to assess the probability associated with the t statistic, having the degrees of freedom computed above. (See the sample problem at the end of this lesson for guidance on how this is done.)

Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Test Your Understanding

Problem

Forty-four sixth graders were randomly selected from a school district. Then, they were divided into 22 matched pairs, each pair having equal IQ's. One member of each pair was randomly selected to receive special training. Then, all of the students were given an IQ test. Test results are summarized below.

Pair	Training	No training	Difference, d	$(d - d)^2$
1	95	90	5	16
2	89	85	4	9
3	76	73	3	4
4	92	90	2	1
5	91	90	1	0
6	53	53	0	1
7	67	68	-1	4
8	88	90	-2	9
9	75	78	-3	16
10	85	89	-4	25
11	90	95	-5	36

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Pair	Training	No training	Difference, d	$(d - d)^2$
12	85	83	2	1
13	87	83	4	9
14	85	83	2	1
15	85	82	3	4
16	68	65	3	4
17	81	79	2	1
18	84	83	1	0
19	71	60	11	100
20	46	47	-1	4
21	75	77	-2	9
22	80	83	-3	16

$$\Sigma(d - d)^2 = 270$$
$$d = 1$$

Do these results provide evidence that the special training helped or hurt student performance? Use an 0.05 level of significance. Assume that the mean differences are approximately normally distributed.

Solution

The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

$$\text{Null hypothesis: } \mu_d = 0$$
$$\text{Alternative hypothesis: } \mu_d \neq 0$$

Note that these hypotheses constitute a two-tailed test. The null hypothesis will be rejected if the difference between sample means is too big or if it is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. Using sample data, we will conduct a matched-pairs t-test of the null hypothesis.
- **Analyze sample data**. Using sample data, we compute the standard deviation of the differences (s), the standard error (SE) of the mean difference, the degrees of freedom (DF), and the t statistic test statistic (t).

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$$s = \text{sqrt}\ [\ (\Sigma(d_i - d)^2 / (n - 1)\ ]$$
$$s = \text{sqrt}[\ 270/(22\text{-}1)\ ]$$
$$s = \text{sqrt}(12.857) = 3.586$$

$$SE = s / \text{sqrt}(n) = 3.586 / [\ \text{sqrt}(22)\ ]$$
$$SE = 3.586/4.69 = 0.765$$

$$DF = n - 1 = 22 - 1 = 21$$

$$t = [\ (x_1 - x_2) - D\ ] / SE$$
$$t = (d - D)/\ SE = (1 - 0)/0.765 = 1.307$$

where d_i is the observed difference for pair *i*, d is mean difference between sample pairs, D is the hypothesized mean difference between population pairs, and n is the number of pairs.

Since we have a two-tailed test, the P-value is the probability that a t statistic having 21 degrees of freedom is more extreme than 1.307; that is, less than -1.307 or greater than 1.307.

We use the t Distribution Calculator to find $P(t < -1.307) = 0.103$, and $P(t > 1.307) = 0.103$. Thus, the P-value = 0.103 + 0.103 = 0.206.

- **Interpret results**. Since the P-value (0.206) is greater than the significance level (0.05), we cannot reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the samples consisted of paired data, and the mean differences were normally distributed. In addition, we used the approximation formula to compute the standard error, since the sample size was small relative to the population size.

HYPOTHESIS TEST ON THE DIFFERENCE BETWEEN PROPORTIONS

This lesson explains how to conduct a hypothesis test to determine whether the difference between two proportions is significant.

The test procedure, called the two-proportion z-test, is appropriate when the following conditions are met:

- The sampling method for each population is simple random sampling.
- The samples are independent.
- Each sample includes at least 10 successes and 10 failures.
- Each population is at least 20 times as big as its sample.

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This approach consists of four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results.

State the Hypotheses

Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. The table below shows three sets of hypotheses. Each makes a statement about the difference *d* between two population proportions, P_1 and P_2. (In the table, the symbol ≠ means " not equal to ".)

Set	Null hypothesis	Alternative hypothesis	Number of tails
1	$P_1 - P_2 = 0$	$P_1 - P_2 \neq 0$	2
2	$P_1 - P_2 \geq 0$	$P_1 - P_2 < 0$	1
3	$P_1 - P_2 \leq 0$	$P_1 - P_2 > 0$	1

The first set of hypotheses (Set 1) is an example of a two-tailed test, since an extreme value on either side of the sampling distribution would cause a researcher to reject the null hypothesis. The other two sets of hypotheses (Sets 2 and 3) are one-tailed tests, since an extreme value on only one side of the sampling distribution would cause a researcher to reject the null hypothesis.

When the null hypothesis states that there is no difference between the two population proportions (i.e., d = 0), the null and alternative hypothesis for a two-tailed test are often stated in the following form.

$$H_0: P_1 = P_2$$
$$H_a: P_1 \neq P_2$$

Formulate an Analysis Plan

The analysis plan describes how to use sample data to accept or reject the null hypothesis. It should specify the following elements.

- Significance level. Often, researchers choose significance levels equal to 0.01, 0.05, or 0.10; but any value between 0 and 1 can be used.
- Test method. Use the two-proportion z-test (described in the next section) to determine whether the hypothesized difference between population proportions differs significantly from the observed sample difference.

Analyze Sample Data

Using sample data, complete the following computations to find the test statistic and its associated P-Value.

- **Pooled sample proportion.** Since the null hypothesis states that $P_1=P_2$, we use a pooled sample proportion (p) to compute the standard error of the sampling distribution.

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$$p = (p_1 * n_1 + p_2 * n_2) / (n_1 + n_2)$$

where p_1 is the sample proportion from population 1, p_2 is the sample proportion from population 2, n_1 is the size of sample 1, and n_2 is the size of sample 2.

- **Standard error.** Compute the standard error (SE) of the sampling distribution difference between two proportions.

$$SE = sqrt\{ p * (1 - p) * [(1/n_1) + (1/n_2)] \}$$

where p is the pooled sample proportion, n_1 is the size of sample 1, and n_2 is the size of sample 2.

- **Test statistic.** The test statistic is a z-score (z) defined by the following equation.

$$z = (p_1 - p_2) / SE$$

where p_1 is the proportion from sample 1, p_2 is the proportion from sample 2, and SE is the standard error of the sampling distribution.

- **P-value.** The P-value is the probability of observing a sample statistic as extreme as the test statistic. Since the test statistic is a z-score, use the Normal Distribution Calculator to assess the probability associated with the z-score. (See sample problems at the end of this lesson for examples of how this is done.)

The analysis described above is a two-proportion z-test.

Interpret Results

If the sample findings are unlikely, given the null hypothesis, the researcher rejects the null hypothesis. Typically, this involves comparing the P-value to the significance level, and rejecting the null hypothesis when the P-value is less than the significance level.

Test Your Understanding

In this section, two sample problems illustrate how to conduct a hypothesis test for the difference between two proportions. The first problem involves a a two-tailed test; the second problem, a one-tailed test.

Problem 1: Two-Tailed Test

Suppose the Acme Drug Company develops a new drug, designed to prevent colds. The company states that the drug is equally effective for men and women. To test this claim, they choose a a simple random sample of 100 women and 200 men from a population of 100,000 volunteers.

At the end of the study, 38% of the women caught a cold; and 51% of the men caught a cold. Based on these findings, can we reject the company's claim that the drug is equally effective for men and women? Use a 0.05 level of significance.

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Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

$$\text{Null hypothesis: } P_1 = P_2$$
$$\text{Alternative hypothesis: } P_1 \neq P_2$$

Note that these hypotheses constitute a two-tailed test. The null hypothesis will be rejected if the proportion from population 1 is too big or if it is too small.

- **Formulate an analysis plan**. For this analysis, the significance level is 0.05. The test method is a two-proportion z-test.
- **Analyze sample data**. Using sample data, we calculate the pooled sample proportion (p) and the standard error (SE). Using those measures, we compute the z-score test statistic (z).

$$p = (p_1 * n_1 + p_2 * n_2) / (n_1 + n_2)$$
$$p = [(0.38 * 100) + (0.51 * 200)] / (100 + 200)$$
$$p = 140/300 = 0.467$$

$$SE = \text{sqrt}\{ p * (1 - p) * [(1/n_1) + (1/n_2)] \}$$
$$SE = \text{sqrt} [0.467 * 0.533 * (1/100 + 1/200)]$$
$$SE = \text{sqrt} [0.003733] = 0.061$$

$$z = (p_1 - p_2) / SE = (0.38 - 0.51)/0.061 = -2.13$$

where p_1 is the sample proportion in sample 1, where p_2 is the sample proportion in sample 2, n_1 is the size of sample 1, and n_2 is the size of sample 2.

Since we have a two-tailed test, the P-value is the probability that the z-score is less than -2.13 or greater than 2.13.

We use the Normal Distribution Calculator to find $P(z < -2.13) = 0.017$, and $P(z > 2.13) = 0.017$. Thus, the P-value = 0.017 + 0.017 = 0.034.

- **Interpret results**. Since the P-value (0.034) is less than the significance level (0.05), we cannot accept the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the samples were independent, each population was at least 10 times larger than its sample, and each sample included at least 10 successes and 10 failures.

Problem 2: One-Tailed Test

Suppose the previous example is stated a little bit differently. Suppose the Acme Drug Company develops a new drug, designed to prevent colds. The company states that the drug is more effective for women than for men. To test this claim, they choose a a simple random sample of 100 women and 200 men from a population of 100,000 volunteers.

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At the end of the study, 38% of the women caught a cold; and 51% of the men caught a cold. Based on these findings, can we conclude that the drug is more effective for women than for men? Use a 0.01 level of significance.

Solution: The solution to this problem takes four steps: (1) state the hypotheses, (2) formulate an analysis plan, (3) analyze sample data, and (4) interpret results. We work through those steps below:

- **State the hypotheses.** The first step is to state the null hypothesis and an alternative hypothesis.

$$\text{Null hypothesis: } P_1 >= P_2$$
$$\text{Alternative hypothesis: } P_1 < P_2$$

Note that these hypotheses constitute a one-tailed test. The null hypothesis will be rejected if the proportion of women catching cold (p_1) is sufficiently smaller than the proportion of men catching cold (p_2).

- **Formulate an analysis plan**. For this analysis, the significance level is 0.01. The test method is a two-proportion z-test.
- **Analyze sample data**. Using sample data, we calculate the pooled sample proportion (p) and the standard error (SE). Using those measures, we compute the z-score test statistic (z).

$$p = (p_1 * n_1 + p_2 * n_2) / (n_1 + n_2)$$
$$p = [(0.38 * 100) + (0.51 * 200)] / (100 + 200)$$
$$p = 140/300 = 0.467$$

$$SE = sqrt\{ p * (1 - p) * [(1/n_1) + (1/n_2)] \}$$
$$SE = sqrt [0.467 * 0.533 * (1/100 + 1/200)]$$
$$SE = sqrt [0.003733] = 0.061$$

$$z = (p_1 - p_2) / SE = (0.38 - 0.51)/0.061 = -2.13$$

where p_1 is the sample proportion in sample 1, where p_2 is the sample proportion in sample 2, n_1 is the size of sample 1, and n_2 is the size of sample 2.

Since we have a one-tailed test, the P-value is the probability that the z-score is less than -2.13. We use the Normal Distribution Calculator to find $P(z < -2.13) = 0.017$. Thus, the P-value = 0.017.

- **Interpret results**. Since the P-value (0.017) is greater than the significance level (0.01), we cannot reject the null hypothesis.

Note: If you use this approach on an exam, you may also want to mention why this approach is appropriate. Specifically, the approach is appropriate because the sampling method was simple random sampling, the samples were independent, each population was at least 10 times larger than its sample, and each sample included at least 10 successes and 10 failures.

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POWER OF A HYPOTHESIS TEST

The probability of *not* committing a Type II error is called the **power** of a hypothesis test.

Effect Size

To compute the power of the test, one offers an alternative view about the "true" value of the population parameter, assuming that the null hypothesis is false. The **effect size** is the difference between the true value and the value specified in the null hypothesis.

Effect size = True value - Hypothesized value

For example, suppose the null hypothesis states that a population mean is equal to 100. A researcher might ask: What is the probability of rejecting the null hypothesis if the true population mean is equal to 90? In this example, the effect size would be 90 - 100, which equals -10.

Factors That Affect Power

The power of a hypothesis test is affected by three factors.

- Sample size (n). Other things being equal, the greater the sample size, the greater the power of the test.

- Significance level (α). The higher the significance level, the higher the power of the test. If you increase the significance level, you reduce the region of acceptance. As a result, you are more likely to reject the null hypothesis. This means you are less likely to accept the null

hypothesis when it is false; i.e., less likely to make a Type II error. Hence, the power of the test is increased.

- The "true" value of the parameter being tested. The greater the difference between the "true" value of a parameter and the value specified in the null hypothesis, the greater the power of the test. That is, the greater the effect size, the greater the power of the test.

Test Your Understanding

Problem 1

Other things being equal, which of the following actions will reduce the power of a hypothesis test?

I. Increasing sample size.
II. Increasing significance level.
III. Increasing beta, the probability of a Type II error.

(A) I only
(B) II only
(C) III only
(D) All of the above
(E) None of the above

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Solution

The correct answer is (C). Increasing sample size makes the hypothesis test more sensitive - more likely to reject the null hypothesis when it is, in fact, false. Increasing the significance level reduces the region of acceptance, which makes the hypothesis test more likely to reject the null hypothesis, thus increasing the power of the test. Since, by definition, power is equal to one minus beta, the power of a test will get smaller as beta gets bigger.

Problem 2

Suppose a researcher conducts an experiment to test a hypothesis. If she doubles her sample size, which of the following will increase?

I. The power of the hypothesis test.
II. The effect size of the hypothesis test.
III. The probability of making a Type II error.

(A) I only
(B) II only
(C) III only
(D) All of the above
(E) None of the above

Solution

The correct answer is (A). Increasing sample size makes the hypothesis test more sensitive - more likely to reject the null hypothesis when it is, in fact, false. Thus, it increases the power of the test. The effect size is not affected by sample size. And the probability of making a Type II error gets smaller, not bigger, as sample size increases.

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TOPIC 4

CORRELATION AND REGRESSION ANALYSIS

CORRELATION

This is an important statistical concept which refers to interrelationship or association between variables.

The purpose of studying correlation is for one to be able to establish a relationship, plan and control the inputs (independent variables) and the output (dependent variables)

In business one may be interested to establish whether there exists a relationship between the

i) Amount of fertilizer applied on a given farm and the resulting harvest
ii) Amount of experience one has and the corresponding performance
iii) Amount of money spent on advertisement and the expected incomes after sale of the goods/service

There are two methods that measure the degree of correlation between two variables these are denoted by R and r.

(a) **Coefficient of correlation** denoted by r, this provides a measure of the strength of association between two variables one the dependent variable the other the independent variable. r can range between +1 and – 1 for perfect positive correlation and perfect negative correlation respectively with zero indicating no relation i.e. for perfect positive correlation y increase linearly with x increament.

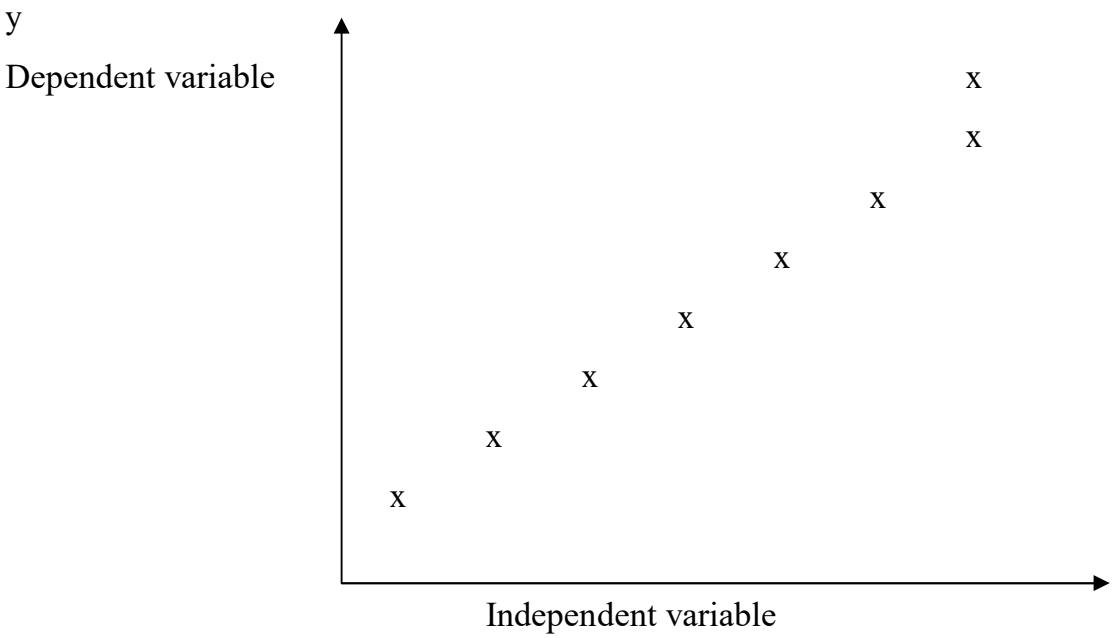
(b) Rank correlation coefficient denoted by R is used to measure association between two sets of ranked or ordered data. R can also vary from +1, perfect positive rank correlation to -1 perfect negative rank correlation where O or any number near zero representing no correlation.

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SCATTER GRAPHS

- A scatter graph is a graph which comprises of points which have been plotted but are not joined by line segments
- The pattern of the points will definitely reveal the types of relationship existing between variables
- The following sketch graphs will greatly assist in the interpretation of scatter graphs.

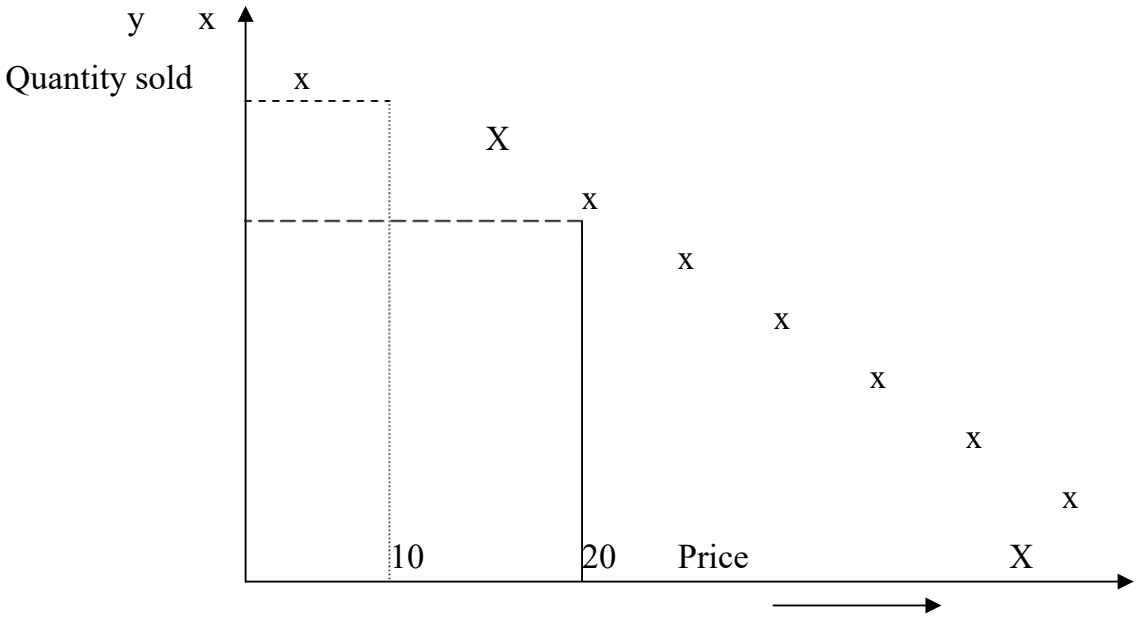
Perfect positive correlation



NB: For the above pattern, it is referred to as perfect because the points may easily be represented by a single line graph e.g. when measuring relationship between volumes of sales and profits in a company, the more the company sales the higher the profits.

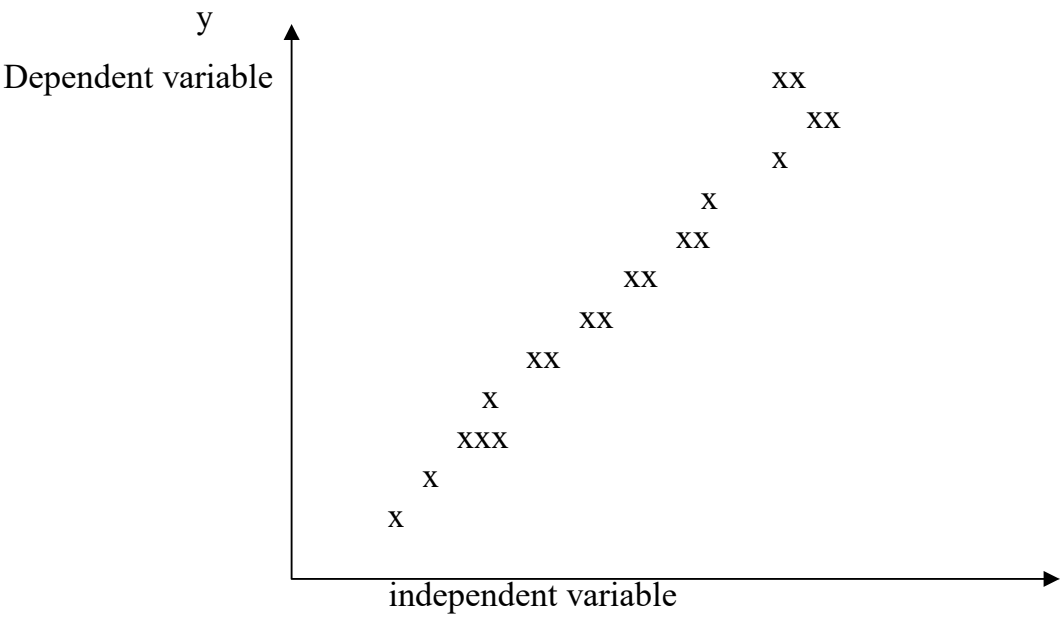
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Perfect negative correlation

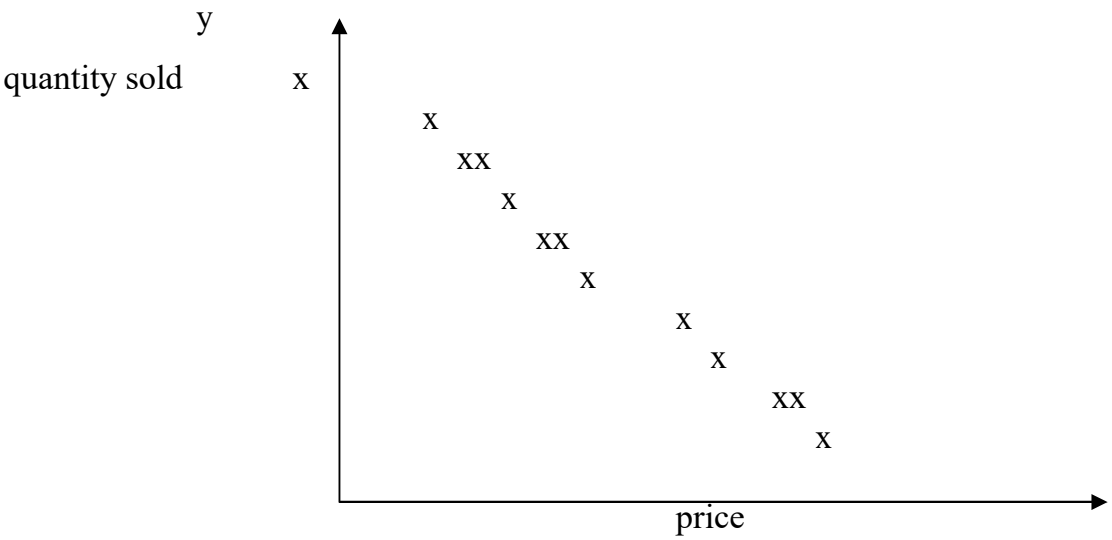


This example considers volume of sale in relation to the price, the cheaper the goods the bigger the sale.

High positive correlation

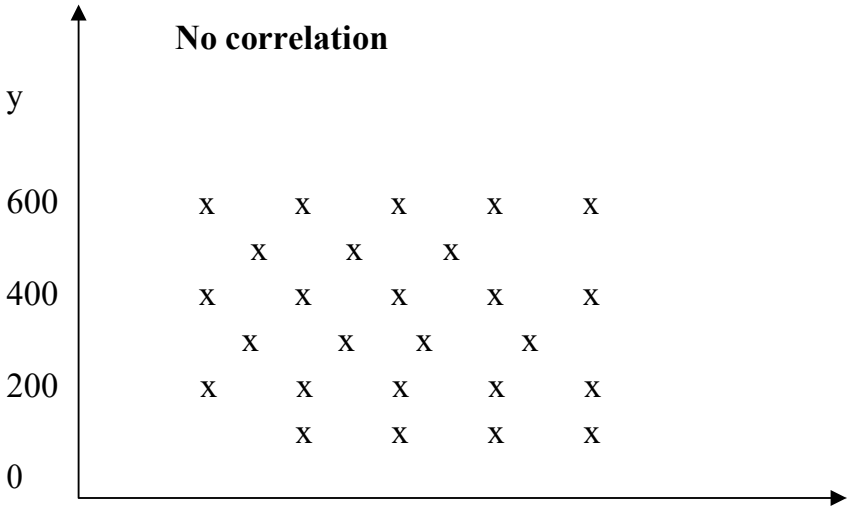


High negative correlation



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No correlation



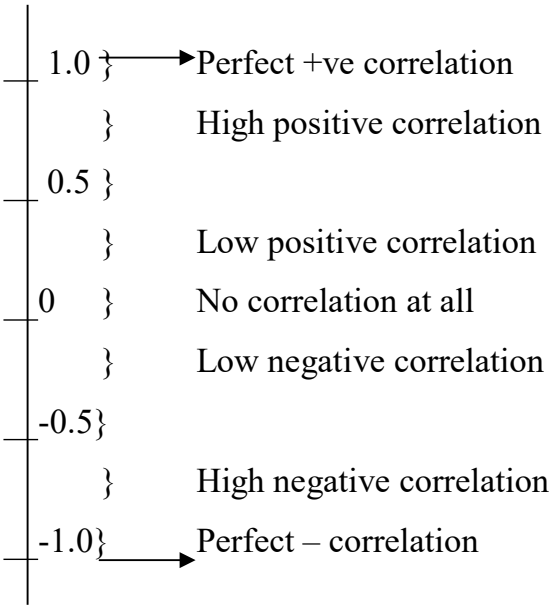
10 20 30 40 50 x

h) Spurious Correlations

- In some rare situations when plotting the data for x and y we may have a group showing either positive correlation or –ve correlation but when you analyze the data for x and y in normal life there may be no convincing evidence that there is such a relationship. This implies therefore that the relationship only exists in theory and hence it is referred to as spurious or non sense e.g. when high passrates of student show high relation with increased accidents.

CORRELATION COEFFICIENT

- These are numerical measures of the correlations existing between the dependent and the independent variables
- These are better measures of correlation than scatter groups
- The range for correlation coefficients lies between +ve 1 and –ve 1. A correlation coefficient of +1 implies that there is perfect positive correlation. A value of –ve shows that there is perfect negative correlation. A value of 0 implies no correlation at all
- The following chart will be found useful in interpreting correlation coefficients



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There are usually two types of correlation coefficients normally used namely;-

Product Moment Coefficient (r)

It gives an indication of the strength of the linear relationship between two variables.

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - \left(\sum x\right)^2} \times \sqrt{n\sum y^2 - \left(\sum y\right)^2}}$$

note that this formula can be rearranged to have different outlooks but the result is always the same.

Example

The following data was observed and it is required to establish if there exists a relationship between the two.

X	15	24	25	30	35	40	45	65	70	75
Y	60	45	50	35	42	46	28	20	22	15

SOLUTION

Compute the product moment coefficient of correlation (r)

X	Y	X^2	Y^2	XY
15	60	225	3,600	900
24	45	576	2,025	1,080
25	50	625	2,500	1,250
30	35	900	1,225	1,050
35	42	1,225	1,764	1,470
40	46	1,600	2,116	1,840
45	28	2,025	784	1,260
65	20	4,225	400	1,300
70	22	4,900	484	1,540
75	15	5,625	225	1,125
$\sum X = 424$	$\sum Y = 363$	$\sum X^2 = 21,926$	$\sum Y^2 = 15,123$	$\sum XY = 12,815$

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$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - \left(\sum x\right)^2} \times \sqrt{n\sum y^2 - \left(\sum y\right)^2}}$$

$$r = \frac{10 \times 12,815 - 424 \times 363}{\sqrt{\left(10 \times 21,926 - 424^2\right)} \times \sqrt{\left(10 \times 15,123 - 363^2\right)}}$$

$$= \frac{-25,762}{\sqrt{(39,484)} \times \sqrt{(19,461)}} = -0.93$$

The correlation coefficient thus indicates a strong negative linear association between the two variables.

Interpretation of r – Problems in interpreting r values

NOTE:

- A high value of r (+0.9 or – 0.9) only shows a strong association between the two variables but doesn't imply that there is a causal relationship i.e. change in one variable causes change in the other it is possible to find two variables which produce a high calculated r yet they don't have a causal relationship. This is known as spurious or nonsense correlation e.g. high pass rates in QT in Kenya and increased inflation in Asian countries.
- Also note that a low correlation coefficient doesn't imply lack of relation between variable but lack of linear relationship between the variables i.e. there could exist a curvilinear relation.
- A further problem in interpretation arises from the fact that the r value here measures the relationship between a single independent variable and dependent variable, where as a particular variable may be dependent on several independent variables (e.g. crop yield may be dependent on fertilizer used, soil exhaustion, soil acidity level, season of the year, type of seed etc.) in which case multiple correlation should be used instead.

THE RANK CORRELATION COEFFICIENT (R)

Also known as the spearman rank correlation coefficient, its purpose is to establish whether there is any form of association between two variables where the variables and arranged in a ranked form.

$$R = 1 - \frac{6\sum d^2}{n\left(n^2 - 1\right)}$$

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Where d = difference between the pairs of ranked values.

n = numbers of pairs of rankings

Example

A group of 8 accountancy students are tested in Quantitative Analysis and Law II. Their rankings in the two tests were.

Student	Q. A. ranking	Law ranking	d	d^2
A	2	3	-1	1
B	7	6	1	1
C	6	4	2	4
D	1	2	-1	1
E	4	5	-1	1
F	3	1	2	4
G	5	8	-3	9
H	8	7	1	1

$\sum d^2 = 22$

d = Q. T. ranking – Law II ranking

$$R = 1 - \frac{6\sum d^2}{n\left(n^2 - 1\right)} = 1 - \frac{6 \times 22}{8\left(8^2 - 1\right)}$$

$= 0.74$

Thus we conclude that there is a reasonable agreement between student's performances in the two types of tests.

NOTE: in this example, if we are given the actual marks then we find r. R varies between +1 and -1.

Tied Rankings

A slight adjustment to the formula is made if some students tie and have the same ranking the adjustment is

$\frac{t^3 - t}{12}$ where t = number of tied rankings the adjusted formula becomes

$$R = 1 - \frac{6\left(\sum d^2 + \frac{t^3 - t}{12}\right)}{n\left(n^2 - 1\right)}$$

Example

Assume that in our previous example student E& F achieved equal marks in Quantitative Analysis. and were given joint 3rd place.

Solution

Student	**Q. T. ranking**	**Law II ranking**	**d**	**d^2**
A	2	3	-1	1
B	7	6	1	1
C	6	4	2	4
D	1	2	-1	1
E	3 ½	5	-1 ½	2 ¼
F	3 ½	1	2 ½	6 ¼
G	5	8	- 3	9
H	8	7	1	1
				$\sum d^2 = 26\frac{1}{2}$

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$$R = 1 - \frac{6\left(\sum d^2 + \frac{t^3 - t}{12}\right)}{n - \left(n^2 - 1\right)} = 1 - \frac{6\left(26\frac{1}{2} + \frac{2^3 - 2}{12}\right)}{8\left(8^2 - 1\right)}$$

$= 0.68$

NOTE: It is conventional to show the shared rankings as above, i.e. E, & F take up the 3rd and 4th rank which are shared between the two as 3½ each.

COEFFICIENT OF DETERMINATION

This refers to the ratio of the explained variation to the total variation and is used to measure the strength of the linear relationship. The stronger the linear relationship the closer the ratio will be to one.

Coefficient determination = Explained variation

Total variation

Example (Rank Correlation Coefficient)

In a beauty competition 2 assessors were asked to rank the 10 contestants using the professional assessment skills. The results obtained were given as shown in the table below

Contestants	**1st assessor**	**2nd assessor**
A	6	5
B	1	3
C	3	4
D	7	6
E	8	7
F	2	1
G	4	8
H	5	2
J	10	9
K	9	10

Required

Calculate the rank correlation coefficient and hence comment briefly on the value obtained

			d	d2
A	6	5	1	1
B	1	3	-2	4
C	3	4	-1	1
D	7	6	1	1
E	8	7	1	1
F	2	1	1	1
G	4	8	-4	16
H	5	2	3	9
J	10	9	+1	1
K	9	10	-1	1

$\Sigma d^2 = 36$

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∴ The rank correlation coefficient R

$$R = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 36}{10(10^2 - 1)}$$

$$= 1 - \frac{216}{990}$$

$$= 1 - 0.22$$

$$= 0.78$$

Comment: since the correlation is 0.78 it implies that there is high positive correlation between the ranks awarded to the contestants. $0.78 > 0$ and $0.78 > 0.5$

Contestant	1st assessor	2nd assessor	d	d^2
A	1	2	-1	1
B	5 (5.5)	3	2.5	6.25
C	3	4	-1	1
D	2	1	1	1
E	4	5	-1	1
F	5 (5.5)	6.5	-1	1
G	7	6.5	-0.5	0.25
H	8	8	0	0

$\Sigma d^2 = 11.25$

Required: Complete the rank correlation coefficient

$$\therefore R = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 11.25}{8(63)}$$

$$= 1 - \frac{67.5}{504}$$

$$= 1 - 0.13$$

$$= 0.87$$

This implies high positive correlation

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Example (Rank Correlation Coefficient)

Sometimes numerical data which refers to the quantifiable variables may be given after which a rank correlation coefficient may be worked out.

Is such a situation, the rank correlation coefficient will be determined after the given variables have been converted into ranks. See the following example;

Candidates	Math	r	Accounts	r	d	d2
P	92	1	67	5	-4	16
Q	82	3	88	1	2	4
R	60	5(5.5)	58	7(7.5)	-2	4
S	87	2	80	2	0	0
T	72	4	69	4	0	0
U	60	5(5.5)	77	3	-2.50	6.25
V	52	8	58	7(7.5)	0.5	0.25
W	50	9	60	6	3	9
X	47	10	32	10	0	0
Y	59	7	54	9	-2	4

$\Sigma d^2 = 43.5$

$$\therefore \text{Rank correlation } r = 1 - \frac{6\sum d^2}{n\left(n^2-1\right)}$$

$$= 1 - \frac{6 \times 43.5}{10\left(10^2-1\right)}$$

$$= 1 - \frac{261}{990}$$

$$= 1 - 0.26$$

= 0.74 (High positive correlation between Mathematics marks and Accounts)

Example
(Product moment correlation)
The following data was obtained during a social survey conducted in a given urban area regarding the annual income of given families and the corresponding expenditures.

Family	(x)Annual income £ 000	(y)Annual expenditure £ 000	xy	x^2	Y^2
A	420	360	151200	176400	129600
B	380	390	148200	144400	152100
C	520	510	265200	270400	260100
D	610	500	305000	372100	250000
E	400	360	144000	160000	129600
F	320	290	92800	102400	84100
G	280	250	70000	78400	62500
H	410	380	155800	168100	144400
J	380	240	91200	144400	57600
K	300	270	81000	90000	72900
Total	4020	3550	1504400	1706600	1342900

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Required
Calculate the product moment correlation coefficient briefly comment on the value obtained
The produce moment correlation

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - \left(\sum x\right)^2} \times \sqrt{n\sum y^2 - \left(\sum y\right)^2}}$$

Workings:

$$\overline{X} = \frac{4020}{10} = 402 \qquad \overline{Y} = \frac{3550}{10} = 355$$

$$r = \frac{10(1,504,400) - (4020)(3550)}{\sqrt{10(1,706,600) - 4020^2} \times \sqrt{10(1,342,900) - (3550)^2}}$$

= 0.89

Comment: The value obtained 0.89 suggests that the correlation between annual income and annual expenditure is high and positive. This implies that the more one earns the more one spends.

REGRESSION ANALYSIS

In statistics, **regression analysis** is a statistical technique for estimating the relationships among variables. It includes many techniques for modeling and analyzing several variables, when the focus is on the relationship between a dependent variable and one or more independent variables.

More specifically, regression analysis helps one understand how the typical value of the dependent variable changes when any one of the independent variables is varied, while the other independent variables are held fixed. Most commonly, regression analysis estimates the conditional expectation of the dependent variable given the independent variables — that is, the average value of the dependent variable when the independent variables are fixed. Less commonly, the focus is on a quantile, or other location parameter of the conditional distribution of the dependent variable given the independent variables. In all cases, the estimation target is a function of the independent variables called the **regression function**. In regression analysis, it is also of interest to characterize the variation of the dependent variable around the regression function, which can be described by a probability distribution.

Regression analysis is widely used for prediction and forecasting, where its use has substantial overlap with the field of machine learning. Regression analysis is also used to understand which among the independent variables are related to the dependent variable, and to explore the forms of these relationships. In restricted circumstances, regression analysis can be used to infer causal relationships between the independent and dependent variables. However this can lead to illusions or false relationships, so caution is advisable. A large body of techniques for carrying out regression analysis has been developed. Familiar methods such as linear regression and ordinary least squares regression are parametric, in that the regression function is defined in terms of a finite number of unknown parameters that are estimated from the data. Nonparametric regression refers to techniques that allow the regression function to lie in a specified set of functions, which may be infinite-dimensional.

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The performance of regression analysis methods in practice depends on the form of the data generating process, and how it relates to the regression approach being used. Since the true form of the data-generating process is generally not known, regression analysis often depends to some extent on making assumptions about this process. These assumptions are sometimes testable if many data are available. Regression models for prediction are often useful even when the assumptions are moderately violated, although they may not perform optimally. However, in many applications, especially with small effects or questions of causality based on observational data, regression methods can give misleading results.

- The general equation used in simple regression analysis is as follows

$$y = a + bx$$

Where y = Dependant variable

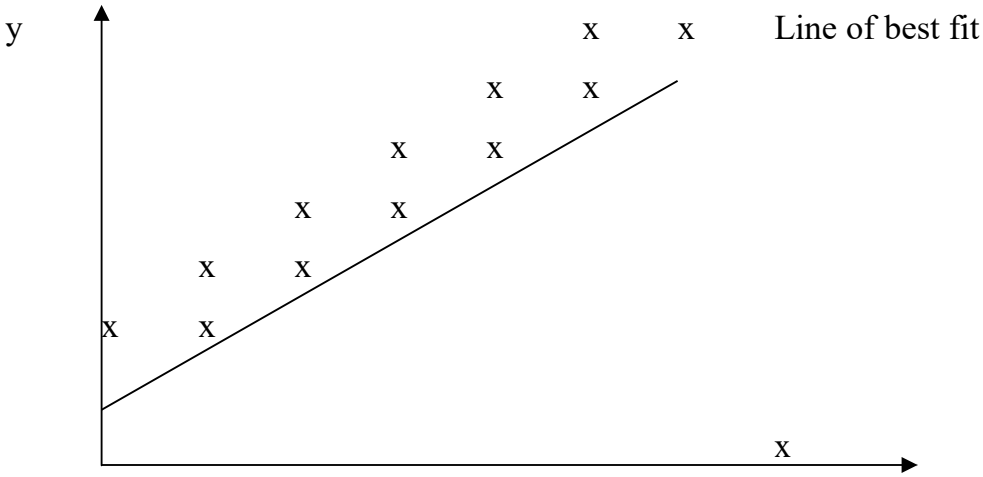
a= Interception y axis (constant)

b = Slope on the y axis

x = Independent variable

The determination of the regression equation such as given above is normally done by using a technique known as "the method of least squares'.

Regression equation of y on x i.e. y = a + bx



The following sets of equations normally known as normal equation are used to determine the equation of the above regression line when given a set of data.

$\Sigma y = an + b\Sigma x$

$\Sigma xy = a\Sigma x + b\Sigma x^2$

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Where Σy = Sum of y values

Σxy = sum of the product of x and y

Σx = sum of x values

Σx^2= sum of the squares of the x values

a = The intercept on the y axis

b = Slope gradient line of y on x

NB: The above regression line is normally used in one wayonly i.e. it is used to estimate the y values when the x values are given.

Regression line of x on y i.e. x = a + by

- The fact that regression lines can only be used in one way leads to what is known as a regression paradox
- This means that the regression lines are not ordinary mathematical line graphs which may be used to estimate the x and y simultaneously
- Therefore one has to be careful when using regression lines as it becomes necessary to develop an equation for x and y before doing the estimation.

The following example will illustrate how regression lines are used

Example

An investment company advertised the sale of pieces of land at different prices. The following table shows the pieces of land their acreage and costs

Piece of land	(x)Acreage Hectares	(y) Cost £ 000	xy	x^2
A	2.3	230	529	5.29
B	1.7	150	255	2.89
C	4.2	450	1890	17.64
D	3.3	310	1023	10.89
E	5.2	550	2860	27.04
F	6.0	590	3540	36
G	7.3	740	5402	53.29
H	8.4	850	7140	70.56
J	5.6	530	2969	31.36
	$\underline{\Sigma x}$ =44.0	Σy =4400	Σxy= 25607	Σx^2 = 254.96

Required

Determine the regression equations of

i) y on x and hence estimate the cost of a piece of land with 4.5 hectares

ii) Estimate the expected average if the piece of land costs £ 900,000

$\Sigma y = an + b\Sigma xy$

$\Sigma xy = a\sum x + b\Sigma x^2$

$900 = -13.59 + 102.78x$

$x = (\frac{900+13.59}{102.78})$

≈ 8.889

By substituting of the appropriate values in the above equations we have

$4400 = 9a + 44b$ ..... (i)

$25607 = 44a + 254.96b$..... (ii)

By multiplying equation …. (i) by 44 and equation …… (ii) by 9 we have

$193600 = 396a + 1936b$..... (iii)

$230463 = 396a + 2294.64b$..... (iv)

By subtraction of equation …. (iii) from equation …… (iv) we have

$36863 = 358.64b$

$102.78 = b$

by substituting for b in .....(i)

$4400 = 9a + 44(102.78)$

$4400 - 4522.32 = 9a$

$-122.32 = 9a$

$-13.59 = a$

Therefore the equation of the regression line of y on x is

$Y = 13.59 + 102.78x$

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When the acreage (hectares) is 4.5 then the cost

(y) = -13.59 + (102.78 x 4.5)

= 448.92

= £ 448, 920

Note that

Where the regression equation is given by

y= a + bx

Where a is the intercept on the y axis and

b is the slope of the line or regression coefficient

n is the sample size then,

Intercept a = $\frac{\sum y - b\sum x}{n}$

Slope b = $\frac{n\sum xy - \sum x\sum y}{n\sum x^2 - \left(\sum x\right)^2}$

Example

The calculations for our sample size n = 10 are given below. The linear regression model is y = a + bx

Table:

Distance x miles	**Time y mins**	**xy**	**x^2**	**y^2**
3.5	16	56.0	12.25	256
2.4	13	31.0	5.76	169
4.9	19	93.1	24.01	361
4.2	18	75.6	17.64	324
3.0	12	36.0	9.0	144
1.3	11	14.3	1.69	121
1.0	8	8.0	1.0	64
3.0	14	42.0	9.0	196
1.5	9	13.5	2.25	81
4.1	16	65.6	16.81	256
Total 28.9	136	435.3	99.41	1972
Σx	Σy	Σxy	Σx^2	Σy^2

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The Slope b = $\frac{10\times 435.3 - 28.9\times 136}{10\times 99.41 - 28.9^2} = \frac{422.6}{158.9} = 2.66$

and the intercept a = $\frac{136 - (2.66\times 28.9)}{10} = 5.91$

We now insert these values in the linear model giving

y = 5.91 + 2.66x

or

Delivery time (mins) = 5.91 + 2.66 (delivery distance in miles)

The slope of the regression line is the estimated number of minutes per mile needed for a delivery. The intercept is the estimated time to prepare for the journey and to deliver the goods, that is the time needed for each journey other than the actual traveling time.

PREDICTION WITHIN THE RANGE OF SAMPLE DATA

We can use the linear regression model to predict the mean of dependant variable for any given value of independent variable

For example if the sample model is given by

Time (min) = 5.91 + 2.66 (distance in miles)

Then the distance if 4.0 miles then our estimated mean time is

$\acute{Y} = 5.91 + 2.66 \times 4.0 = 16.6$ minutes

NB: The regression line of y on x can be used in extrapolation too.

MULTIPLE LINEAR REGRESSION MODELS

There are situations in which there is more than one factor which influence the dependent variable

Example

Cost of production per week in a large department

Factors

i) Total numbers of hours worked
ii) Raw material used during the week
iii) Total number of items produced during the week
iv) Number of hours spent on repair and maintenance

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It is sensible to use all the identified factors to predict department costs

Scatter diagram will not give the relationship between the various factors and total costs

The linear model for multiple linear regression if of the type; (which is the line of best fit).

$y = \alpha + b_1x_1 + b_2x_2 + \ldots\ldots\ldots\ldots + b_nx_n$

We assume that errors or residuals are negligible.

In order to choose between the models we examine the values of the multiple correlation coefficient r and the standard deviation of the residuals α.

A model which describes well the relationship between y and x's has multiple correlation coefficient r close to $\pm$ and the value of α which is small.

ILLUSTRATION

Odino chemicals limited are aware that its power costs are semi variable cost and over the last six months these costs have shown the following relationship with a standard measure of output.

Month	Output (standard units)	Total power costs £ 000
1	12	6.2
2	18	8.0
3	19	8.6
4	20	10.4
5	24	10.2
6	30	12.4

Required

i) Using the method of least squares, determine on appropriate linear relationship between total power costs and output

ii) If total power costs are related to both output and time (as measured by the number of the month) the following least squares regression equation is obtained

Power costs = 4.42 + (0.82) output + (0.10) month

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Where the regression coefficients (i.e. 0.82 and 0.10) have t values 2.64 and 0.60 respectively and coefficient of multiple correlation amounts to 0.976

Compare the relative merits of this fitted relationship with one you determine in (a). Explain (without doing any further analysis) how you might use the data to forecast total power costs in seven months.

SOLUTION

a)

Output (x)	Power costs (y)	x^2	y^2	xy
12	6.2	144	38.44	74.40
18	8.0	324	64.00	144.00
19	8.6	361	73.96	163.40
20	10.4	400	108.16	208.00
24	10.2	576	104.04	244.80
30	12.4	900	153.76	372.00
$\Sigma x = 123$	$\Sigma y = 55.8$	$\Sigma x^2 = 2705$	$\Sigma y^2 = 542.36$	$\Sigma xy = 1,206.60$

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}$$

$$= \frac{6\times1206.6-123\times55.8}{6\times2705-(123)^2}$$

$$= \frac{376.2}{1101}$$

$$= 0.342$$

$$a = \frac{1}{n}(\Sigma y - b\Sigma y)$$

$$= \frac{1}{6} \text{ x } (55.8 - 0.342) \text{ x } 123$$

$$= 2.29$$

$\therefore$ (Power costs) = 2.29 + 0.342

b. For linear regression calculated above, the coefficient of correlation r is

$$r = \frac{(6\times1206.6)-(123\times55.8)}{\sqrt{6\times2705-123\times123}\sqrt{6\times542.36-55.8\times55.8}}$$

$$= \frac{376.2}{\sqrt{1101\times140.52}}$$

$$= 0.96$$

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This show a strong correlation between power cost and output. The multiple correlation when both output and time are considered at the same time is 0.976.

We observe that there has been very little increase in r which means that inclusion of time variable does not improve the correlation significantly

The value for time variable is only 0.60 which is insignificant as compared with a t value of 2.64 for the output variable

In fact, if we work out correlation between output and time, there will be a high correlation. Hence there is no necessity of taking both the variables. Inclusion of time does improve the correlation coefficient but by a very small amount.

If we use the linear regression analysis and attempt to find the linear relationship between output and time i.e.

Month	Output
1	12
2	18
3	19
4	20
5	24
6	30

The value of b and a will turn out to be 3.11 and 9.6 i.e. relationship will be of the form

Output = 9.6 + 3.11 × month

For this equation forecast for 7th month will be

Output = 9.6 + 3.11 × 7

= 9.6 + 21.77

= 31.37 units

Using the equation , Power costs = 2.29 + 0.34 × output

= 2.29 + 0.34 × 31.37

= 2.29 + 10.67

= 12.96 i.e. £ 12,960

Non Linear Relationships

In the scatter diagram and the correlation coefficient do not indicate linear relationship, then the relationship may be non – linear

Two such relationships are of peculiar interest

$y = ab^x$

Both of these can be reduced to linear model. Simple or multiple linear regression methods are then used to determine the values of the coefficients

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Exponential model

$y = ab^x$

Take log of both sides

$\log y = \log a + \log b^x$

$\log x = \log a + x\log b$

Let log y = Y and log a = A and log b = B

Then Y = A + Bx. This is a linear regression model

i. Geometric model

$y = ax^b$

using the same technique as above

$\log y = \log a + b\log x$

$Y = A + bX$

Where Y = log

A = log a

X = log x

Using linear regression technique (the method of least squares), it is possible to calculate the value of a and b

COMPUTER OUTPUT OF LINEAR REGRESSION ANALYSIS

The analysis in multiple regression can be done or achieved using statistical computer software's like SPSS,STAT,MINITAB,SAS etc.
The results of the software are normally presented in a special table known as the analysis of variance (anova) table
The ANOVA table is presented as follows:

Source of variation	**Degrees of freedom**	**Sum of squares**	**Mean squares**	**F - Ratio**
Model	K – 1	SSR	MSR = $\frac{SSR}{k-1}$	$\frac{MSR}{MSE}$
			MSE = $\frac{SSE}{n-k}$	
Error	n – K	SSE		
Total	n – 1	TSS		

Where;

K – Total number of variation (both independent + dependent)
N – Number of observation / pairs of data / sample size
SSR – Sum of square regression
SSE – Sum of square errors
MSR – Mean square regression
MSE – Mean square error

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NB $\{R^2 = \frac{SSR}{TSS}\}$ the coefficient of determination

SSR = n $[(b_o \sum y + b_1 \sum x_1 y + b_2 \sum x_2 y) \quad (\sum y)^2]$

F – Ratio – is a measure of accuracy or how good or adequate the regression model is for prediction.
NB: The F – Ratio is compared to a tabulated value. If F-Ratio is 0 ≥ F tabulated then the regression model is adequate for prediction.
F – Ratio ≥ F tabulated – adequate

T-RATIOS AND CONFIDENCE INTERVAL FOR THE COEFFICIENTS

Test for the significance

The test helps in determining the predictor variables which are crucial in affecting the response variable.

In order to determine the significance, the procedure below is adopted:

Step 1:

Compute the t – calculated value for each predictor variable where

$$T_{calc} = \frac{Estimated\ /\ slope}{standarderror\ (Se)}$$

Step 2:

Determine the t – critical value from the student t-table where

$$t\ critical = t\ n - k; 1 - \frac{\propto}{2}$$

$\propto$ - Significance level = probability of rejecting the predictor variable (S)

$(1 - \frac{\propto}{2})$ = Probability of accepting the variable (S)

Step 3:

If $|t_{cal} \geq t\ critical$, the predictor variable significant in affecting the response variable.

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ILLUSTRATION

An economic working for RTC Limited suspects that the annual demand of the company's sole product depends on the disposable income of consumers and the unit price of the product.

A regression analysis of the annual demand against the disposable income and the unit price of the product has been undertaken using the information available on the product over the last 10 years. A section of the results obtained using statistical software is given below:

Analysis of variance

Source	Degree of freedom	Sum of squares
Model	2	93.5176
Error	7	1.8823
Total	9	95.4

Parameter estimates

Variable	Estimate	Standard error
Constant	0.700	0.9106
Income	2.467	0.1374
Price	-0.659	0.0967

Required:

(a) The estimated regression equation.
(b) Interpret the meaning of each of the above parameter estimates.
(c) The coefficient of determination and interpret your result.
(d) Test the adequacy of the model for prediction (F-table value = 9.55).
(e) Test the significance of each predictor variable in explaining the annual demand of the product (Use a significance level of 1%).

SOLUTION

(a) $\hat{Y} = 0.700 + 2.467$ (income) – 0.659 (price).

- The constant = 0.700, gives the demand realized in absence of the two predictor variables.
- The coefficient for income shows that a shilling increase in disposable income leads to 2.467 units increase in demand.
- A unit increase in the unit price leads to 0.659 units decrease in the annual demand.

(b) $R^2 = \frac{93.5176}{95.4} \times 100 = 98.0\%$

The model helps to explain 98% of the annual demand.

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(c) **Mean sum of squares** **F – value** $= \frac{93.5176 \div 2}{46.7588 \qquad 173.9 \qquad 1.8823 \div 7}$

Table value = $F_{0.01}$ (2, 7) = 9.55
Since 173.9> 9.55, we conclude that the available evidence shows the model is adequate.

(d) Income $\frac{t - value}{17.95}$
Price -6.81
Table value = $t_{0.01}$ (7) = 3.50
Since t – value > 3.50, then both predictors are significant prediction of demand.

Statistical inference

It is the process of drawing conclusions about attributes of a population based upon information contained in a sample taken from the population.

It is divided into estimation of parameters and testing of hypothesis. Symbols for statistic of population parameters are as follows.

	Sample Statistic	**Population Parameter**
Arithmetic mean	$\bar{x}$	μ
Standard deviation	s	σ
Number of items	n	N

Statistical estimation

It is the procedure of using statistic to estimate a population parameter

It is divided into point estimation (where an estimate of a population parameter is given by a single number) and interval estimation (where an estimate of a population is given by a range in which the parameter may be considered to lie) e.g. a bus meant to take a class of 100 students (population N) for trip has a limit to the maximum weight of 600kg of which it can carry, the teacher realizes he has to find out the weight of the class but without enough time to weigh everyone he picks 25 students selected at random (sample n = 25). These students are weighed and their average weight recorded as 64kg ($\bar{X}$ - mean of a sample) with a standard deviation (s), now using this the teacher intends to estimate the average weight of the whole class (μ – population mean) by using the statistical parameters standard deviation (s), and mean of the sample ($\bar{x}$).

Characteristic of a good estimator

(i) Unbiased: where the expected value of the statistic is equal to the population parameter e.g. if the expected mean of a sample is equal to the population mean

(ii) Consistency: where an estimator yields values more closely approaching the population parameter as the sample increases

(iii) Efficiency: where the estimator has smaller variance on repeated sampling.

(iv) Sufficiency: where an estimator uses all the information available in the data concerning a parameter

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Confidence Interval

The interval estimate or a 'confidence interval' consists of a range (an upper confidence limit and lower confidence limit) within which we are confident that a population parameter lies and we assign a probability that this interval contains the true population value

The confidence limits are the outer limits to a confidence interval. Confidence interval is the interval between the confidence limits. The higher the confidence level the greater the confidence interval. For example

A normal distribution has the following characteristic

i. Sample mean ± 1.960 σ includes 95% of the population

ii. Sample mean ± 2.575 σ includes 99% of the population

1. LARGE SAMPLES

These are samples that contain a sample size greater than 30(i.e. n>30)

(a) Estimation of population mean

Here we assume that if we take a large sample from a population then the mean of the population is very close to the mean of the sample

Steps to follow to estimate the population mean includes

i. Take a random sample of n items where (n>30)

ii. Compute sample mean ($\bar{X}$) and standard deviation (S)

iii. Compute the standard error of the mean by using the following formular

$$S_{\bar{x}} = \frac{s}{\sqrt{n}}$$

where $S_{\bar{x}}$ = Standard error of mean

S = standard deviation of the sample

n = sample size

iv. Choose a confidence level e.g. 95% or 99%

v. Estimate the population mean as below:

Population mean $\mu = \bar{X} \pm$ (appropriate number) $\times S_x$ –

'Appropriate number' means confidence level e.g. at 95% confidence level is 1.96 this number is usually denoted by Z and is obtained from the normal tables.

Example

The quality department of a wire manufacturing company periodically selects a sample of wire specimens in order to test for breaking strength. Past experience has shown that the breaking strengths of a certain type of wire are normally distributed with standard deviation of 200 kg. A random sample of 64 specimens gave a mean of 6200 kgs. Find out the population mean at 95% level of confidence

Solution

Population mean = $\bar{X}$ 1.96 S_x –

Note that sample size is large i.e. n > 30 whereas s and $\bar{x}$ are given thus step i), ii) and iv) are provided.

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Here: $\bar{X}$ = 6200 kgs

$$S_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{200}{\sqrt{64}} = 25$$

Population mean = 6200 ± 1.96(25)

= 6200 ± 49

= 6151 to 6249

At 95% level of confidence, population mean will be in between 6151 and 6249

FINITE POPULATION CORRECTION FACTOR (FPCF)

If a given population is relatively of small size and sample size is more than 5% of the population then the standard error should be adjusted by multiplying it by the finite population correction factor

FPCF is given by $= \sqrt{\frac{N-n}{n-1}}$

where N = population size

n = sample size

Example

A manager wants an estimate of sales of salesmen in his company. A random sample 100 out of 500 salesmen is selected and average sales are found to be Shs. 75,000. if a sample standard deviation is Shs. 15,000 then find out the population mean at 99% level of confidence

Solution

Here N = 500, n = 100, $\bar{X}$ = 75000 and S = 15000

Now

Standard error of mean

$$= S_{\bar{x}} = \frac{s}{\sqrt{n}} \times \sqrt{\frac{N-n}{n-1}}$$

$$= \frac{15000}{\sqrt{100}} \times \sqrt{\frac{(500-100)}{(500-1)}}$$

$$= \frac{15000}{10} \times \sqrt{\frac{400}{499}}$$

$$= \frac{15000}{10}(0.895)$$

$S_{\bar{x}}$ = 1342.50 at 99% level of confidence

Population mean $= \bar{X} \pm 2.58\ S_{\bar{x}}$

=shs 75000 ± 2.58(1342.50)

=shs 75000 ± 3464

= Shs 71536 to 78464

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b) Estimation of difference between two means

We know that the standard error of a sample is given by the value of the standard deviation (σ)divided by the square root of the number of items in the sample ($\sqrt{n}$).

But, when given two samples, the standard errors is given by

$$S_{(\bar{x}A-\bar{x}B)} = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}}$$

Also note that we do estimate the interval not from the mean but from the difference between the two sample means i.e. $(\bar{X}_A - \bar{X}_B)$.

The appropriate number of confidence level does not change

Thus the confidence interval is given by;

$(\bar{X}_A - \bar{X}_B) \pm$ Confidence level $S_{(\bar{x}A-\bar{x}B)}$

$= (\bar{X}_A - \bar{X}_B) \pm Z\ S_{(\bar{x}A-\bar{x}B)}$

ILLUSTRATION

Given two samples A and B of 100 and 400 items respectively, they have the means $\overline{X_1} = 7$ ad $\overline{X_2} = 10$ and standard deviations of 2 and 3 respectively. Construct confidence interval at 70% confidence level?

SOLUTION

Sample	A	B
	$\overline{X_1} = 7$	$\overline{X_2} = 10$
	$n_1 = 100$	$n_2 = 400$
	$S_1 = 2$	$S_2 = 3$

The standard error of the samples A and B is given by

$$S_{(\bar{x}A - \bar{x}B)} = \sqrt{\frac{4}{100} + \frac{9}{400}}$$

$$= \sqrt{\frac{25}{400}} = \frac{5}{20}$$

$$= ¼ = 0.25$$

At 70% confidence level, then appropriate number is equal to 1.04 (as read from the normal tables)

$\overline{X}_1 - \overline{X}_2 = 7 - 10 = -3 = 3$

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We take the absolute value of the difference between the means e.g. the value of $|X|$ = absolute value of X i.e. a positive value of X.

Confidence interval is therefore given by

= 3± 1.04 (0.25) From the normal tables a z value of 1.04 gives a value of 0.7.

= 3± 0.26

= 3.26 and 2.974

Thus $2.974 \leq X \leq 3.26$

Example

A comparison of the wearing out quality of two types of tyres was obtained by road testing. Samples of 100 tyres were collected. The miles traveled until wear out were recorded and the results given were as follows.

Tyres	T1	T2
Mean	$\overline{X}_1 = 26400$ miles	$\overline{X}_2 = 25000$ miles
Variance	$S^2_1 = 1440000$ miles	$S^2_2 = 1960000$ miles

Find a confidence interval at the confidence level of 70%

Solution

$\overline{X}_1 = 26400$

$\overline{X}_2 = 25000$

Difference between the two means

$$\left(\overline{X}_1 - \overline{X}_2\right) = (26400 - 25000)$$
$$= 1{,}400$$

Again we take the absolute value of the difference between the two means
We calculate the standard error as follows

$$S_{(\overline{x}A - \overline{x}B)} = \sqrt{\frac{S^2}{n_1} + \frac{S^2}{n_2}}$$

$$= \sqrt{\frac{1{,}440{,}000}{100} + \frac{1{,}960{,}000}{100}}$$

$$= 184.4$$

Confidence level at 70% is read from the normal tables as 1.04 (Z = 1.04).
Thus the confidence interval is calculated as follows

$= 1400 \pm (1.04)\ (184.4)$

$= 1400 \pm 191.77$

or $(1400 - 191.77)$ to $(1400 + 191.77)$

$1{,}208.23 \leq X \leq 1591.77$

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a) Estimation of population proportions

This type of estimation applies at the times when information cannot be given as a mean or as a measure but only as a fraction or percentage

The sampling theory stipulates that if repeated large random samples are taken from a population, the sample proportion "p' will be normally distributed with mean equal to the population proportion and standard error equal to

$S_p = \sqrt{\frac{Pq}{n}}$ = Standard error for sampling of population proportions

Where n is the sample size and q = 1 – p.

The procedure for estimating a proportion is similar to that for estimating a mean, we only have a different formula for calculating standard.

ILLUSTRATION

In a sample of 800 candidates, 560 were male. Estimate the population proportion at 95% confidence level.

SOLUTION

Here

Sample proportion (P) = $\frac{560}{800} = 0.70$

$q = 1 - p = 1 - 0.70 = 0.30$

$n = 800$

$$\sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.70)(0.30)}{800}}$$

$S_p = 0.016$

population proportion

$= P \pm 1.96\ S_p$ where $1.96 = Z$.

$= 0.70 \pm 1.96\ (0.016)$

$= 0.70 \pm 0.03$

$= 0.67$ to 0.73

= between 67% to 73%

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ILLUSTRATION

A sample of 600 accounts was taken to test the accuracy of posting and balancing of accounts where in 45 mistakes were found. Find out the population proportion. Use 99% level of confidence

SOLUTION

Here

$n = 600;\ p = \frac{45}{600} = 0.075$

$q = 1 - 0.075 = 0.925$

$$S_p = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.075)(0.925)}{600}}$$

$= 0.011$

Population proportion

$= P \pm 2.58\ (S_p)$

$= 0.075 \pm 2.58\ (0.011)$

$= 0.075 \pm 0.028$

$= 0.047$ to 0.10

= between 4.7% to 10%

b) Estimation of difference between population proportions

Let the two proportions be given by P_1 and P_2, respectively

Then the difference (absolute) between the two proportions is given by $(P_1 - P_2)$

The standard error is given by

$$S_{(P1-P2)} = \sqrt{\frac{pq}{n_1} + \frac{pq}{n_2}} \text{ where } p = \frac{p_1 n_1 + p_2 n_2}{n_1 + n_2} \text{ and } q = 1 - p$$

Then given the confidence level, the confidence interval between the two population proportions is given by

$$(P_1 - P_2) \pm \text{Confidence level } S_{(P1-P2)}$$

$$= (P_1 - P_2) \pm Z \sqrt{\frac{pq}{n_1} + \frac{pq}{n_2}}$$

Where $P = \frac{p_1 n_1 + p_2 n_2}{n_1 + n_2}$ always remember to convert P_1 & P_2 to P.

2. SMALL SAMPLES

(a) Estimation of population mean

If the sample size is small (n<30) the arithmetic mean of small samples are not normally distributed. In such circumstances, students t distribution must be used to estimate the population mean.

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In this case

Population mean $\mu = \overline{X} \pm ts_{\bar{x}}$

$\overline{X}$ = Sample mean

$S_{\bar{x}} = \frac{s}{\sqrt{n}}$

S = standard deviation of samples = $\sqrt{\frac{\sum(x - \bar{x})^2}{n-1}}$ for small samples.

n = sample size

v = n – 1 degrees of freedom.

The value of t is obtained from students t distribution tables for the required confidence level

Example

A random sample of 12 items is taken and is found to have a mean weight of 50 grams and a standard deviation of 9 grams

What is the mean weight of population

a) with 95% confidence
b) with 99% confidence

Solution

$\overline{X} = 50$; S = 9; v = n – 1 = 12 – 1 = 11; $S_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{9}{\sqrt{12}}$

$\mu = x' \pm ts_{\bar{x}}$

At 95% confidence level

$$\mu = 50 \pm 2.262\left(\frac{9}{\sqrt{12}}\right)$$

$$= 50 \pm 5.72 \text{ grams}$$

Therefore we can state with 95% confidence that the population mean is between 44.28 and 55.72 grams

At 99% confidence level

$$\mu = 50 \pm 3.25\left(\frac{-9}{\sqrt{12}}\right)$$

$$= 50 \pm 8.07 \text{ grams}$$

Therefore we can state with 99% confidence that the population mean is between 41.93 and 58.07 grams

Note: To use the t distribution tables it is important to find the degrees of freedom (v = n – 1). In the example above v = 12 – 1 = 11

From the tables we find that at 95% confidence level against 11 and under 0.05, the value of t = 2.201

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PRACTICE EXERCISES

QUESTION 1

Unlisted plc hopes to achieve a Stock Market quotation for its shares. A profit forecast is necessary and, in order to achieve such a forecast, the company has experimented with a number of approaches.

The following are details from a linear regression on the last 11 years' profit figures:

x = years (expressed 1to 11)

y = annual profit figures

$\sum x = 66$

$\sum y = 212.10$

$\sum x^2 = 506$

$\sum xy = 1{,}406.70$

$\sum y^2 = 4{,}254.08$

$\sum (y - \hat{y})^2 = 0.916$ where $\hat{y}$ represents profit values estimated by the regression line.

The following formulae are given:

$$\text{Standard error of the regression line } \sigma_R = \sqrt{\frac{\sum (y - \hat{y})^2}{df}}$$

$$\text{Coefficient of correlation (r)} = \sqrt{\frac{\textit{Explained variation}}{\textit{Total variation}}}$$

You are required:

a) To obtain the simple least squares regression line of Y on X;
b) To use the line to estimate profit in each of the next two years;
c) To calculate the coefficient of determination for the line and to explain its meaning;
d) To calculate the standard error of the regression line and to use this to obtain the 95% confidence interval for the line;
e) On the basis of the information given on your answer (a) to (d) to determine whether it is likely that the regression line will be a good estimator of profit.

Solution:

a) $y = a + bx$

Where a and b are determined as follows

$$a = \frac{\sum y}{n} - b\frac{\sum x}{n}$$

$$b = \frac{n\sum xy - \sum x\sum y}{n\sum x^2 - \left(\sum x\right)^2}$$

So given that $\sum x = 66$, $\sum y = 212.1$, $\sum x^2 = 506$, $\sum xy = 1{,}406.7$, $\sum y^2 = 4{,}254.08$

x = number of years, y = annual profit

Then $b = \frac{11\times 1406.7 - 66\times 212.1}{11\times 506 - (66)^2} = 1.219$

And $a = \frac{212.1}{11} - 1.219\times\frac{66}{11} = 11.967$

So $y = 11.967 + 1.219\times x$

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b) 12th year profit $y_{12} = 11.967 + 1.219\times 12 = 26.595$

13th year profit $y_{13} = 11.967 + 1.219\times 13 = 27.814$

c) $$r^2 = \frac{\left(n\sum xy - \sum x\sum y\right)^2}{\left(n\sum x^2 - \left(\sum x\right)^2\right) - \left(n\sum y^2 - \left(\sum y\right)^2\right)}$$

$$r^2 = \frac{\left(11\times 1406.7 - 66\times 212.1\right)^2}{\left(11\times 506 - 66^2\right) - \left(11\times 4251.08 - 212.1^2\right)}$$

$r^2 = 0.9944$

99.44% of the variation in annual profit can be predicted by change in actual values of numbers of years.

d) $S_e = \sqrt{\frac{\sum y - a\sum y - b\sum xy}{n-2}} = \sqrt{\frac{(y-\hat{y})}{n-1}} = \sqrt{\frac{0.916}{9}} = 0.319$

Given 95% confidence interval for the line, at 9 degrees of freedom the t value is $t_{95\%,9} = 2.2622$ The confidence interval for the regression line is:

$$y \pm t_{95\%} \times S_e \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{\sum x^2 - \frac{(\sum x)^2}{n}}} \text{ and given } \bar{x} = \frac{\sum x}{n} = \frac{66}{11} = 6$$

$$y \pm 2.2622 \times 0.319 \sqrt{\frac{1}{11} + \frac{(x-6)^2}{506 - \frac{(66)^2}{11}}}$$

$$y \pm 0.722 \sqrt{\frac{1}{11} + \frac{(x-6)^2}{110}}$$

e) The regression line will be a good estimator of profit because r^2 was high (meaning that variation in profit can be highly explained by actual number of years). The standard error of regression line was also very small.

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QUESTION 2

The following regression equation was calculated for class of 24 CPA II students. -

$\hat{y} = 3.1 + 0.021x_1 + 0.075x_2 + 0.043x_3$

Standard error (0.0190) (0.034) (0.018)

Where y=students score on a theory examination

x_1 = Students rank (from the bottom) in high school

x_2 = Students verbal aptitude score

x_3 =A measure of students character

Required:

a) Calculate the t ratio and the 95% confidence interval for each regression coefficient.

b) What assumptions did you make in (a) above? How reasonable are they?

c) Which regressor gives the strongest evidence of being statistically discernible?

d) In writing up a final regression equation, should one keep the first regressor in the equation, or drop if? Why?

Solution:

a) $t = \frac{b_i - 0}{S_{b_i}} = \frac{\text{slope}_i}{\text{standard error of slope}}$

Confidence interval = $b_i \pm t_{0.975\%,n-1-3} S_{b_i}$

$$t_{0.975,24-1-3} = 2.09$$

	Calculated t	Confidence interval
For X_1:	$t = \frac{0.021}{0.019} = 1.11$	$0.021 \pm 2.09 \times 0.019 = 0.021 \pm 0.04$
X_2:	$t = \frac{0.075}{0.034} = 2.206$	0.075 ± 0.071
X_3:	$t = \frac{0.043}{0.018} = 2.389$	0.043 ± 0.038

b) The assumptions include:

- Error or residuals are independent and normally distributed for a given value of x.
- Expected value of error is equal to zero
- Variance of errors is the same for all x's.

These assumptions are set up to enable one to come up with a projection of the population from the sample. So they are reasonable.

c) X_1 gives the strongest evidence of being statistically discernible because the t statistic calculated is within the required range.

d) The decision to keep or drop the first regressor will be based not only on t-test, but also looking at the r^2 and standard error of the regression in general. The main objective is to include the regressor that reduces standard error of regression and r^2 value is large. Other than just having the t test alone. In this case since t calculated is within the required range and standard error of regression is low, then it will be appropriate to include the first regressor x_1 in the final regression

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QUESTION 3

a) Does finding a no linear relationship between two variables mean no relationship?

b) Does a high correlation mean that one variable causes another variable to vary?

Solution:

a) Not finding a linear relationship does not necessarily mean that a relationship does not exist. Other relationships may exist that are non-linear. May be logarithmic, exponential or quadratic. Linear relationship is of the form $y = a + bx_1 + cx_2$ for a 2 variable for example.

b) Correlation measures the direction and extent one variable (dependent) is affected by another variable (independent). So high correlation means the independent variable causes the dependent variable to vary.

c) Given that x=30 then:

i) $\hat{y} = 27.32 + 1.3 \times 30 = 66.32$

ii) The relationship is linear with a given value of 27.32 even without exposure to insecticides. This value of y increases for any hour of exposure to insecticide by a factor of 1.3.

iii) Coefficient of determination r^2=0.86

H_o: r = 0 A relationship exists

H_1: r ≠ 0 A relationship does not exist.

$$t = \sqrt{\frac{r^2(n-2)}{(1-r^2)}} = \sqrt{\frac{0.86(16-2)}{1-0.86}} = 9.27 > t_{0.975\%,14} = 2.14$$

So we reject H_0 and accept H_1 that a relationship actually exists

iv) Assumptions include:

- Relationship is linear
- Independent variable x is known, so used to predict y
- Errors are normally distributed with expected value of zero for any value of x
- Variance of errors is a constant
- Errors are independent

Note: Test statistic is distributed as student's t with n – 2 degrees of freedom and is given by:

$t = {}_{r;\, n-2;} = 2.14$ from t-tables

QUESTION 4

Kenya Graduate School (KGS) offers a variety of graduate courses. However, its main emphasis has been on information science (IS) courses. Due to the laboratory equipment requirements for IS courses, KGS has to estimate in advance the expected students enrolments. Over the last 5 years, the students enrolments, by quarter, has been:

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Years

Quarter	**1991**	**1992**	**1993**	**1994**	**1995**
First	30	32	41	45	73
Second	42	107	93	101	181
Third	100	71	139	151	227
Fourth	66	47	62	67	109

Required:

a) Determine the estimates, by quarter, for year 1996. Justify the method you use.

b) If linear multiple regression were to be used in order to determine the predicting equation, what other variables would be included?

c) How would the expected enrolments be compared to the actual enrolments?

Note:

$\sum x = 210$ $\quad\quad$ $\sum y = 1{,}784$

$\sum x^2 = 2{,}870$ $\quad\quad$ $\sum xy = 22{,}253$

Solution:

a) Let y be enrolment and x be quarter of a year. Then $y = a + bx$ where

$$a = \frac{\sum y}{n} - b\frac{\sum x}{n}$$

$$b = \frac{n\sum xy - \sum x\sum y}{n\sum x^2 - \left(\sum x\right)^2}. \text{ So } b = \frac{20\times 22253 - 210\times 1784}{20\times 2870 - (210)^2} = 5.29$$

And $a = \frac{1784}{20} - 5.29\times\frac{210}{20} = 33.61$ giving the expression for y as follows:

$$y = 33.61 + 5.29\times x$$

1996	**Quarter x**	$y = 33.61 + 5.29\times x$
First	21	144.7
Second	22	150.0
Third	23	155.3
Fourth	24	160.6

There is an overall trend of increased enrolment with time. Other than the seasonal variation, the relationship can be seen to be linear. So the regression equation is appropriate.

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b) The other factors to be included are income, level of education and population growth.

c) The expected enrolment will be followed as a general trend with seasonal variations.

Justification of calculation.

$$r^2 = \frac{\left(n\sum xy - \sum x\sum y\right)^2}{\left(n\sum x^2 - \left(\sum x\right)^2\right)\left(n\sum y^2 - \left(\sum y\right)^2\right)}$$

$$r^2 = \frac{20\times 22253 - 210\times 1784}{20\times 2870 - (210)^2 - \left(20\times 211254 - 1784^2\right)}$$

$$r^2 = 0.77$$

And r=0.6 meaning there is a positive correlation and 77% of the variation is explained by the quarters.

Calculation of $\sum y^2$

Quarter	Enrolment y	y^2
1	30	900
2	42	1764
3	100	10000
4	66	4356
5	32	1024
6	107	11449
7	71	5041
8	47	2209
9	41	1681
10	93	8649
11	139	19321
12	62	3844
13	45	2025
14	101	10201
15	151	22801
16	67	4489
17	73	5329
18	181	32761
19	227	51529
20	109	11881
Sum		211254

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QUESTION 5

a) Define the goodness of fit test. How is it applied in accounting?

b) A research studying the role of stress and its implication on personal life in respect of job change over by low cadre staff, came up with the following data. It relates to 30 firms over 3-year period

No. of people changing jobs in a year	0	1	2	3	4	5	6	7	8	9
Observed frequency	8	18	19	20	16	12	8	4	3	2

By fitting a Poisson distribution to get expected frequency, test its goodness of fit.

Solution:

i) Goodness of fit test is a test on how well empirical distribution(obtained from sample data) can fit theoretical distribution (like normal, Poisson or binomial distributions) using the χ^2 test. Accountants can use it to determine whether a given age-debtors distribution can be approximated by a given function. Also while forecasting past data or surveyed data can be compared with assumed distribution to come up with a conclusion that the distribution function represents the forecast

Accountants can also come up with appropriate wage/salary given that a certain distribution exists between staff turnover and salary/wages

ii) A table to aid in calculation of distribution and x^2 is as follows:

No.of people changing	Observed values			Poison distribution	$(f_0 - f_e)^2$
x	O	f_0	f_0x	f_e	f_e
0	8	0.073	0.000	0.039	0.030
1	18	0.164	0.164	0.126	0.012
2	19	0.173	0.345	0.204	0.005
3	20	0.182	0.545	0.222	0.007
4	16	0.145	0.582	0.180	0.007
5	12	0.109	0.545	0.117	0.001
6	8	0.073	0.436	0.064	0.001
7	4	0.036	0.255	0.030	0.002
8	3	0.027	0.218	0.012	0.019
9	2	0.018	0.164	0.004	0.044
Total	**110**	**1**	**3.255**		**0.127**

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Poisson distribution $f_e = \frac{e^{-\lambda}\lambda^x}{x!}$ and $f_o = \frac{\text{observed value}}{\text{total value}}$

Mean $\lambda = \frac{\sum f_0 x}{\sum f_0} = 3.255$

$\chi^2 = \sum \frac{(f_0 - f_e)^2}{f_e} = 0.127 < \chi^2_{0.05,8\,df} = 15.5$ so the Poisson distribution fits well for the data.

QUESTION 6

With reference to linear regression define the following terms:

i) Scatter diagram.
ii) Bivariate distribution.
iii) Positive correlation.
iv) Confidence interval.
v) Auto correlation.

Solution:

i) Scatter diagram is a plot of a distribution in its ungrouped form on a graph
ii) Bivariate distribution is a distribution of two variables
iii) Positive correlation occurs when movement of one variable in one direction causes the other variable to move in the same direction
iv) Confidence interval is the limit at which a parameter or the linear regression itself is taken to represent a given distribution
v) Autocorrelation occurs when a series' errors or disturbance covariance is not equal to zero so the least squares estimated are not the best linear unbiased estimates.

TOPIC 5

TIME SERIES

Definition

This is a sequence of a variable values that change over a uniform set of time. The variable values represent statistical data while time can be in seconds, hours. days, weeks etc. Many business and economic studies are based on time series data.

Examples

1. Monthly production level for a company over several years
2. Weekly sales for a chain of supermarkets over a couple of months etc.

Time series components

All-time series contain at least one of the following four components:

1. Secular trend
2. Seasonal variations
3. Cyclical variations
4. Random/ irregular erratic variations

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1. **Secular trend (T)**

This is the general underlying tendency of the time series data to increase, decrease or remain constant for a long period of time.

The importance of the trend includes the following:

- It permits to project past patterns or trend into the future.
- It is used to describe a historical pattern in the given data. This may be used to evaluate the success or failure of a given action.
- Identifying the secular trend enables its elimination in the trend component and thus makes it easier to study other components of the time series.

2. seasonal variations/variations (S)

Are periodic movements of the data where the duration is less than a year. The factors that mainly cause these variations are: -

a) climatic changes
b) the customs and habits that people follow at different times

The main objective of measuring the seasonal variations is to isolate them so that their effect can be understood and used for future extrapolation.

3. Cyclical variations/ fluctuations (C)

Are periodic movements within the time series data where the duration is more than a year. They are not as regular as the seasonal variations but their sequence of change is the same. The causes of the

cyclical variations are the four phases of an economic cycle which include: the boom/peak, decline/downturn, depression/trough and recovery/upswing.

4. random/residual/irregular erratic occurrences (R)

These are completely unpredictable variations within the data caused by unpredictable events like sickness, machine breakdown, weather conditions, strikes etc. They are non-recurring influences which cannot be mathematically captured yet they have profound consequences on a time series.

Time series (decomposition)

This analysis provides techniques that may be used to isolate the four components of a time series. Decomposition may be used to measure the degree of impact each component has on the direction of time series itself i.e the influence each component has on the movement of the time series. In this analysis a standard line diagram representing the time series data is also plotted. The diagram is known as histogram or a time series plot. This is a plot of the variable values on the y axis against time points on the x axis

ILLUSTRATION

The data below represent the daily sales (sh000) for business is a week's period.

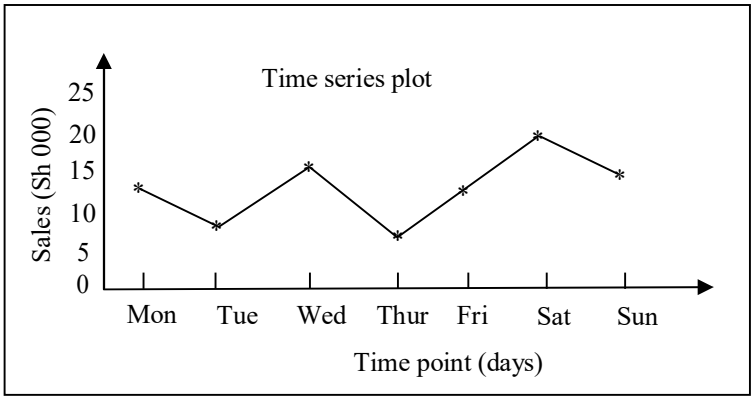
Mon	Tue	Wed	Thur	Friday	Sat	Sun
12	9	11	14	13	10	15

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Required

Plot a historigram of the above data.

SOLUTION



THE TREND ANALYSIS

This is the process of fining/superimposing a trend line on a time series plot. There are four method of doing as described below:

a) freehand/eye projection method
b) semi averages method
c) moving averages method
d) least square method

a. freehand/eye projection method

In this method the trend line is fitted on the time series plot using a free hand. However, the following points need to be considered:

i) The trend line should be a smooth one
ii) The line should bisect the fluctuations of the time series plot

Advantages of the method

- The method is the simplest
- It's flexible in that it can be used for both straight and curved trend lines.

Disadvantages

- The method is very subjective
- Because of its subjectivity, it doesn't have much value in forecasting

b. semi averages method

This is the easiest objective method that involves the calculation of two separate averages from a set of data that has been divided into two groups:

Procedure

i) Split the data into two halves namely lower and upper half
ii) Compute the arithmetic mean for each half
iii) Plot each mean against an appropriate time point which is the median of each set of data points
iv) Join the two points with a straight line to form the required trend line.

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Advantages

- Method is simple to understand
- It is an objective method

Disadvantages

- Method assumes a straight line trend which may not be always the case.
- Only two points are considered and hence the method is not a representative of all the data values

ILLUSTRATION

The data below relates to quarterly sales or a company over a period or 3yrs

Quarters (qrt) sales (sh million)

Years	1	2	3	4
2006	12	9	11	14
2007	12	10	17	20
2008	15	12	21	22

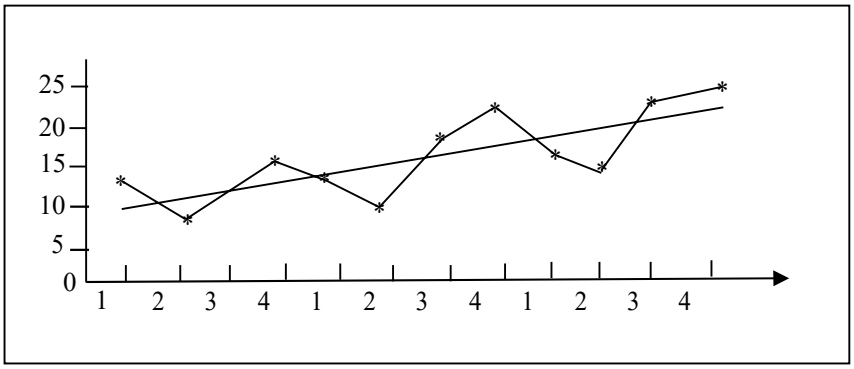
Required

A time series plot and the trend line using the moving averages method

SOLUTION

Lower half values	Upper half values
12,9,11,14,13,10	17,20,15,12,21,22
$X_1 = 11.5$	$X_2 = 17.83$
Time point: between quarters 3 and 4 (2006)	Time point: between quarters 1 and 2 (2008)

Plot



c) Moving averages (M.A) method

These are successive and overlapping arithmetic means for a set of data grouped into equal number of values known as the order or period. The moving averages represent the trend line values.

NB: each moving average value must correspond with an appropriate time point which is the median of the time points for the odd set of values being averaged.

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ILLUSTRATION

The data below shows the monthly sales (sh million) made by Excel ltd. for the year 2008.

Month	Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec
Sales (Sh 000)	190	180	204	272	255	196	212	238	245	264	280	270

Required

The moving averages of order 3

Solution

Month	Sales	M.A (order 3) (represent trend values)
J	190	-
F	180	(190 + 180 + 204)/3 = 191.33
M	204	(180 + 204 + 272)/3 = 218.67
A	272	(204 + 272 + 255)/3 = 243.67
M	255	(272 + 255 + 196)/3 = 241
J	196	(255 + 196 + 212)/3 = 221
J	212	(196 + 212 + 238)/3 = 215.33
A	238	(212 + 238 + 245)/3 = 231.67
S	245	(238 + 245 + 264)/3 = 249
O	264	(245 + 264 + 280)/3 = 263
N	280	(264 + 280 + 270)/3 = 271.33
D	270	-

CENTERED MOVING AVERAGES

When the order of the moving averages consists of even set or values, the calculated moving averages do not have corresponding time point as was the case for odd period. In this case a process known as centering is used where we deliberately force the precompiled moving averages to have their corresponding time points.

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The centering process involves computing moving averages of order 2 based on the previously computed moving averages. The resultant moving averages have corresponding time points and they represent the trend values.

ILLUSTRATION

The data below relates to the number of beds occupied in a hotel

Bed occupancy

Quarters (Q)

Years	1	2	3	4
2006	60	88	100	76
2007	67	99	110	92
2008	79	105	118	98

Required:

Centered moving averages of order 4.

SOLUTION

Yr	Q	Y	M. A (order 4)	Centered M.A (order 2)	
2006	1	60		-	
	2	88	81	-	
	3	100	82.75	(81 + 82.75)/2 = 81.875	
	4	76	85.5	(82.75 + 85.5)/2 = 84.125	
2007	1	67	88	86.75	
	2	99	92	90	trend
	3	110	95	values	
	4	92	96.5	93.5	
2008	1	79	98.5	95.75	
	2	105	100	97.5	
	3	118		99.25	
	4	98		-	
				-	

WEIGHTED MOVING AVERAGES

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These are moving averages where each value per order/period is assigned its respective weight. In this case, each moving average is computed as follows:

Weighted moving average $= \frac{\sum[(datavalue S respective weight)}{\sum weights}$

Advantages of moving averages

- They show the true nature of the trend line whether it is linear or a curve.
- They normally smoothen the peaks and troughs of the original data.
- The method is simpler compared to the least square method.
- The method is representative as it takes into account all the data values

Disadvantages of moving averages

- There are some missing values at the start and at the end.
- There are no standard rules for determining the order of the moving averages.
- Since the moving averages cannot be expressed in form of a standard equation, they cannot be used on their own to make an objective forecast.

d) Least square method

This is the most popular method of fitting a linear trend on a set of plotted data. This method normally uses the equation of a straight line: y = a + bx.

Once the values of coefficients a and b have been computed, the above equation is transformed to a least square equation: t = a + bx where t represent the trend value for each value of x.

ILLUSTRATION

The data below represent the profit (sh. millions) made by a company over a period 3yrs.

Profit (Sh million)

Quarters (Q)

Years	**1**	**2**	**3**	**4**
2006	2.2	5.0	7.9	3.2
2007	2.9	5.2	8.2	3.8
2008	3.2	5.8	9.1	4.1

Required

Trend values using the least square regression method.

Solution

Yrs	**Q**	**Codes representing**				**Trend values (t) using the least square eqn**
	Quarters	**(x)**	**y**	**x^2**	**xy**	
2006	1	1	2.2	1	2.2	3.938 + 0.171 (1) = 4.11
	2	2	5.0	4	10	3.938 + 0.171 (2) = 4.282
	3	3	7.9	9	23.7	= 4.453
	4	4	3.2	16	12.8	= 4.624
2007	1	5	2.9	25	14.5	= 4.795
	2	6	5.2	36	31.2	= 4.966
	3	7	8.2	49	57.4	= 5.137
	4	8	3.8	64	30.4	= 5.308
2008	1	9	3.2	81	28.8	= 5.479
	2	10	5.8	100	58	= 5.65
	3	11	9.1	121	100.1	= 5.821
	4	12	4.1	144	49.2	= 5.992
		$\sum x = 78$	$\sum y = 60.6$	$\sum x^2 = 650$	$\sum xy = 418.3$	

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NB: The quarters have been assigned new codes (x) to represent continuity of the time series.

In least square method, y = a + bx

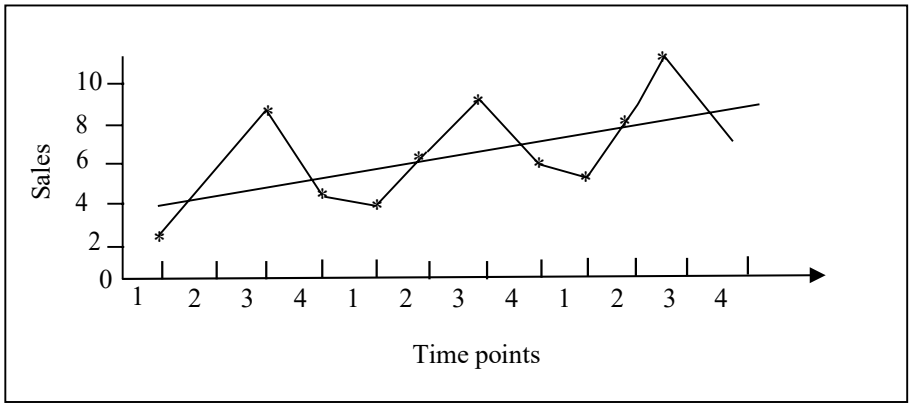
Where

$$b = \frac{n\sum Sy - \sum S\sum y}{n\sum S^2 - (\sum S)^2} = \frac{(12x418.3) - (78\ 60.6)}{(12\ 650) - (78)^2}$$

$$a = \frac{\Sigma}{n} - \frac{b\sum S}{n} = \frac{60.6}{12} \qquad 0.1718x\frac{78}{12} = 3.938$$

Therefore t = 3.938 + 0.171 x (least square regression equation)

Time series plot and the trend line.



Advantages

- There is no room for subjectivity
- It gives the best trend line which is the line of best fit
- It takes into account all the data values.
- There are no missing trend values as was the case for moving averages

Disadvantages

- Method applicable for only linear trend.

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TIME SERIES MODELS

Are expressions that indicate the relationship between the various components that form an individual time series value. The two main models that are frequently used are:

i) The additive model
ii) The multiplicative model

The additive model is expressed as: Y = T + S+ C+R where T is the trend value, S-seasonal value, C- cyclical value and R- random value. This model assumes that components a re independent of each other. This is not realistic since components in a time series relate. This is a demerit of this model.

The multiplicative model is expressed as: Y = T * S* C*R. this model is preferred over the additive model since it assumes that the components interact with each other.

NB: The models are used in the computation of the seasonal component.

SEASONAL ANALYSIS

This analysis isolates the seasonal component of a time series. The computation of the seasonal values is the most important aspect in seasonal analysis because the values are used in extrapolating the time series. There are two types of seasonal values namely:

i) Specific seasonal values
ii) Typical seasonal values

The computation of the seasonal values depends on the stated model.

The specific seasonal values measure the short term effect of the seasons on the time series data while the typical measure the long term effect.

When the additive model is applicable, the specific and typical values are called **factors** (expressed as deviations). In the use of multiplicative model, the seasonal values are referred as **indices** (expressed as percentages).

Isolation of specific and typical seasonal factors using additive model

Given the time series values denoted by (y) and trend values denoted by (t), the following procedure is followed:

i) Compute the specific seasonal factors (y - t) where both values exist
ii) Find the arithmetic mean for the specific factors in each season
iii) Add the arithmetic means
iv) If the sum is not equal to zero adjust them with an adjustment factor/normalization ratio until a sum of zero is obtained.

Normalization ratio = $\frac{sumofthemeans}{numberofseasons/yr}$

v) The adjusted means with a sum of zero are the required typical seasonal factors.

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ILLUSTRATION

Years	Q	No. of beds (y)	t	y-t
2006	1	60	-	-
	2	88	-	-
	3	100	82	+18
	4	76	84	-8
2007	1	67	87	- 20
	2	99	90	+9
	3	110	94	+16
	4	92	96	-4
2008	1	79	98	-19
	2	105	99	+6
	3	118	-	-
	4	98	-	-

Seasonal arithmetic means

Quarters

Years	1	2	3	4	
2006	-	-	+18	-8	
2007	-20	+ 9	+16	-4	
2008	-19	+ 6	-	-	
Mean	-19.5	+ 7.5	+17	-6	Sum = -1
Adjusted means	-19.5 (-0.25) = -19.25	+7.5- (-0.25) = 7.75	+17-(-0.25) = -17.75	-6-(-0.25) = -5.75	Sum = 0

Normalization ratio = $\frac{-1}{4}$ = -0.25

Therefore the typical seasonal factors are:

Q1 = -19.25 Q2 = 7.75 Q3 = 17.25 Q4 = -5.75

Interpretation

Q1 and Q4 indicate that the long term effect of quarter 1 and 4 is to reduce the number of beds occupied by approximately 19 and 6 respectively.

Q2 and Q3 indicate that the long term effect of quarters 2 and 3 is to increase the number of beds occupied by approximately 8 and 7 respectively.

Isolation of specific and typical seasonal factors using multiplicative model

Given the y and t values, the following procedure is used in the computation of the seasonal indices:

1. Compute the specific seasonal indices y/t
2. Find the arithmetic mean of the specific seasonal indices for each season.
3. Find the sum of the seasonal means.
4. If the sum is not equivalent to the number of seasons per year, adjust them using a normalization ratio.

Normalization ratio = $\frac{numberofseasons/yr}{sumofthemeans}$

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The adjusted seasonal means are the required typical seasonal

ILLUSTRATION

Yrs	Q	y (profit)	t	y/t
2006	1	2.2	4.11	0.5353
	2	5.0	4.28	1.1682
	3	7.9	4.45	1.7753
	4	3.2	4.62	0.6926
2007	1	2.9	4.80	0.6042
	2	5.2	4.97	1.0463
	3	8.2	5.14	1.5953
	4	3.8	5.31	0.7156
2008	1	3.2	5.48	0.5839
	2	5.8	5.65	1.0265
	3	9.1	5.82	1.5636
	4	4.1	5.99	0.6845

Seasonal arithmetic means

Quarters

Yrs	**1**	**2**	**3**	**4**	
2006	0.5353	1.1682	1.7753	0.6926	
2007	0.6042	1.0463	1.5953	0.7156	
2008	0.5839	1.0265	1.5636	0.6845	
Mean	0.5745	1.0803	1.6447	0.6976	
Adjusted means	0.5745 x 1.00073 = 0.5749	1.0803 x 1.00073 = 1.0811	1.644 x 1.00073 =1.6459	0.6976 x 1.00073 = 0.6981	Sum = 3.9971 Sum = 4

Normalization ratio = $\frac{4}{3.9971}$ = 1.00073

Therefore the typical seasonal factors are:
Q1 = 0.5749 = 57.49% Q2 = 1.0811 = 108.11% Q3 = 1.6459 = 164.59% Q4 = 0.6981 = 69.81%

Interpretation
Q1 and Q4 indicate that, the long term effect of quarters I and 4 is to reduce profit by 42.51% and 30.19% respectively.
Q2 and Q3 indicate that, the long term effect of quarters 2 and 3 is to increase profit by 8.1 I % and 64.59% respectively.

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Deseasonalized values
These are time series values where the effects of the seasons have been removed. This is normally done using the typical seasonal values as exemplifies below:

1. Deseasonalizing using the additive model
In this case the typical seasonal factors are subtracted from their respective data values.

ILLUSTRATION

Deseasonalise the following time series using the following seasonal factors:

Q1 = -19.25 Q2 = 7.75 Q3 = 17.25 Q4 = -5.75

Years	Q	No. of beds (y)	Deseasonalised values
2006	1	60	60-(19.25) = 79.25
	2	88	88-(7.75) = 80.25
	3	100	100-(17.25) = 82.75
	4	76	76 -(-5.75) = 81.75
2007	1	67	67-(-19.25) = 86.25
	2	99	99-((7.75) = 91.25
	3	110	110-(17.25) = 92.75
	4	92	92-(-5.75) = 97.75
2008	1	79	79-(-19.25) = 98.25
	2	105	105-(7.75) = 97.25
	3	118	118-(17.25) = 100.75
	4	98	98-(-5.75) = 103.75

2. Deseasonalizing using the multiplicative model

In this case the time series values are divided by their respective typical seasonal indices,

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ILLUSTRATION

Given the following time series, deseasonalize it using the following seasonal indices.

Q1 = 0.5749 = 57.49% Q2 = 1.0811 = 108.11% Q3 = 1.6459 = 164.59% Q4 = 0.6981 = 69.81%

Yrs	Q	y (profit)	t
2006	1	2.2	2.2÷0.5749 = 3.83
	2	5.0	5.0÷1.0811 = 4.62
	3	7.9	7.9÷1.6459 = 4.80
	4	3.2	3.2÷0.6981 = 4.58
2007	1	2.9	2.9÷0.5749 = 5.04
	2	5.2	5.2÷1.0811 = 4.81
	3	8.2	8.2÷1.6459 = 4.98
	4	3.8	3.8÷0.6981 = 5.44
2008	1	3.2	3.2÷0.5749 = 5.57
	2	5.8	5.8÷1.0811 = 5.36
	3	9.1	9.1÷1.6459 = 5.53
	4	4.1	4.1÷0.6981 = 5.87

CYCLICAL VARIATION ANALYSIS

This involves the isolation of the cyclical component.

Procedure

- Obtain the trend and seasonal components
- Obtain a product of trend (T) and seasonal(S) components (assuming a multiplicative model).this product is called statistical norm.
  i.e. statistical norm = T*S
- Obtain the cyclical and irregular variations by dividing the data by the statistical norm.
  $\frac{y}{T\ S} = \frac{T\ S\ C\ R}{T\ S}$ = C*R
- Multiply the results by 100 to express the answer as a percentage.
- Eliminate the random/irregular variations by taking a four period centered moving average. This leaves only the cyclical variations.

ILLUSTRATION

Given the following time series work out the Cyclical variation analysis

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Yrs	Q	y (profit)
2006	1	2.2
	2	5.0
	3	7.9
	4	3.2
2007	1	2.9
	2	5.2
	3	8.2
	4	3.8
2008	1	3.2
	2	5.8
	3	9.1
	4	4.1

SOLUTION

Yrs	Q	1 y (profit)	2 T	3 S	4 TS	Cyclical /Random (1/4)	Cyclical component
2006	1	2.2	4.11	0.5749	2.362839	0.931083	-
	2	5.0	4.28	1.0811	4.627108	1.080589	-
	3	7.9	4.45	1.6459	7.324255	1.078608	103.559
	4	3.2	4.62	0.6981	3.225222	0.99218	103.647
2007	1	2.9	4.80	0.5749	2.75952	1.050907	100.871
	2	5.2	4.97	1.0811	5.373067	0.96779	99.916
	3	8.2	5.14	1.6459	8.459926	0.969276	99.887
	4	3.8	5.31	0.6981	3.706911	1.025112	99.220
2008	1	3.2	5.48	0.5749	3.150452	1.015727	98.750
	2	5.8	5.65	1.0811	6.108215	0.949541	97.951
	3	9.1	5.82	1.6459	9.579138	0.949981	-
	4	4.1	5.99	0.6981	4.181619	0.980481	-

Typical seasonal indices (s) are:-
Q1 = 0.5749 = 57.49% Q2 = 1.0811 = 108.11% Q3 = 1.6459 = 164.59% Q4 = 0.6981 = 69.81%

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Isolation of the irregular values

$$R = \frac{y}{TSC}$$

TIME SERIES FORECASTING/EXTRAPOLATION

Extrapolation refers to the process of predicting the time series to the most likely future value. Virtually every form of decision making and planning activity in business involves forecasting. Typical applications include: planning, inventory control, investment cash flow, cost projection demand forecasts, advertising planning, corporate planning budgeting etc.
Forecasting can be done using either quantitative or qualitative methods.

Quantitative methods

These are methods based on computations and are divided into two categories namely:

a. Simplistic methods
b. Composite methods

a. Simplistic methods

Are forecasting methods used where data values do not exhibit any trend. Data fluctuate or suddenly change from one time point to another. The methods are:-

i. the Naive method
ii. moving averages (smoothing method) first order exponential smoothing method

The Naive method

Simply estimate the value in the next time period to be equal to that of the last time period. i.e

$y_{t+1} = y_t$

Where y_{t+1} is the estimate of the value of the time series in the next time period.

y_t is the actual value in the current time period.

i. Moving averages

In this case, last computed moving average is considered to be the future forecast.

ILLUSTRATION

Yr	Investment (Sh 000)	MA of order 3	MA of order 4	Centered MA order 2
1997	73.2			
1998	68.1	71.37	72.50	
1999	72.8	72.27	72.15	72.33
2000	75.9	73.50	72.45	72.30
2001	71.8	72.33	71.25	71.85
2002	69.3	69.70	69.15	70.20
2003	68	68.27	68.68	68.91
2004	67.5	68.47	69.65	69.16
2005	69.9	70.20	71.48	70.56
2006	73.2	72.80	72.83	72.15
2007	75.3	73.80		
2008	72.9			

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Forecast: using the MA of order 3, 73.8 is the forecast for any future time period. The centered MA produces a forecast of 72 .15.

ii. First order exponential smoothing method

The method is a form of moving averages that makes use of a smoothing constant (c) which is a value between 0 and 1. The method involves the automatic weighting of past data such that the most current value receives the greatest weighting and the older observations receive a decreasing weighting. The method involves little record keeping of past data. The basic exponential smoothing formula takes the following form;

New forecast = last period's forecast + α (last period's actual value – last period's forecast)

Can also be written mathematically as:

$F_t = F_{t+1} + \alpha\ (A_{t-1} - F_{t+1})$

Alternative method.

$F_t = \alpha\ (A_{t-1}) + (1 - \alpha)\ (F_{t1})$

ILLUSTRATION

The following data shows weekly sales of a company:

Week	1	2	3	4	5	6	7	8
Sales (sh 000)	452	385	401	298	500	480	358	468

Required:

Given exponential smoothing constants (α) of 0.1 and 0.5, forecast the sales of the 9th week.

SOLUTION

Week	Sales	Forecast 1 (α = 0.1)	Forecast 2 (α = 0.5)
1	452	452 (assumed)	423 (assumed)
2	385	452 + 0.1 (452 – 452) = 452	437.5
3	401	452 + 0.1 (385 – 452) = 445.3	411.3
4	298	421.8 + 0.1 (401 – 421.8) = 419.7	406.2
5	500	419.7 + 0.1 (298 – 419.7) = 407.6	352.1
6	480	407.6 + 0.1 (500 – 407.6) = 416.8	426.1
7	358	416.8 + 0.1(480 – 416.8) = 423.1	453.1
8	468	423.1 + 0.1 (358 – 423.1) = 416.6	405.6
		416.6 + 0.1 (468 – 416.6) = 421.7	436.8

Sales for week 9 are sh 421700 using a smoothing constant of 0.1 and sh 436800 using α = 0.5

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ACCURACY OF FORECAST VALUES

The more accurate, reliable forecast would be one producing the smaller value of mean square error (MSE)

$$\text{MSE} = \frac{\sum(Ft - Æt)^2}{n-1}$$

Where Ft – forecast value for time period t

At – actual / observed value for time period t

(a) For α = 0.5

$\sum(Ft \quad At)^2$ = (423-452)^2 + (425.9-385)^2 +(421.8 -401)^2 + (419.7-298)^2 + (407.6 – 500)^2 + (416.8 -480)^2 + (423.1-358)^2+(416.6-468)^2 = 37169.31

$$\text{MSE} = \frac{37169.31}{8-1} = 5309.9$$

(b) For α = 0.5

$\sum(Ft \quad At)^2$ = (423-452)^2 + (425.9-385)^2 +(421.8 -401)^2 + (419.7-298)^2 + (407.6 – 500)^2 + (416.8 -480)^2 + (453.1-358)^2+(405.6-468)^2 = 53127.97

$$\text{MSE} = \frac{53127.97}{8-1} = 7589.\ 71$$

The forecast with a lower value of MSE is considered to be more accurate. Therefore, the forecast using $\alpha = 0.1$ appear to be more accurate.

COMPOSITE METHODS OF FORECASTING

These methods are used to forecast future values where a time series exhibit some trend. Data values reflect a tendency of either moving upward of downwards. The most common method is the least square regression method

NB: the predicted values are adjusted using the typical seasonal values

ILLUSTRATION

Forecasting using the least square method

Years	Q	1 y(profit)
2006	1	2.2
	2	5.0
	3	7.9
	4	3.2
2007	1	2.9
	2	5.2
	3	8.2
	4	3.8
2008	1	3.2
	2	5.8
	3	9.1
	4	4.1

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Least square regression equation: t = 3.938 + 0.171x

Typical seasonal indices are: =

Q1 = 0.5749 = 57.49% Q2 = 1.0811 = 108.11% Q3 = 1.6459 = 164.59% Q4 = 0.6981 = 69.81%

Required;-

Extrapolate the profit for the four quarters of year 2009.

SOLUTION

Forecasts (sh million)

2009 Q1 = (3.938 + 0.171x13) = 6.161

Q2 = (3.938 + 0.171x14) = 6.332

Q3 = (3.938 + 0.171x15) = 6.503

Q4 = (3.938 + 0.171x16) = 6.674

Adjusted forecasts

6.161x0.5749 = 3.54 6.332x1.0811 = 6.85 6.503x1.6459 = 10.7 6.674x0.6981=4.66

Qualitative methods-

These are methods which do not require any past records to make a forecast. Some of the most common methods include the following:

a) Delphi method

This method incorporates both judgmental and subjective factors. It is an iterative process that allows experts to make an objective forecast. There are 3 groups of participants involved namely:

- Decision makers
- Staff personnel
- Respondents

The decision making group usually consists of 5 - 10 experts who will be making the actual forecast. The staff personnel assist the decision makers by preparing, distributing, collecting and summarizing a series of questionnaires and survey results. The respondents are a group of people whose views and judgment are valued and are being sought. This group provides input to the decision makers before the forecast is made.

In this method, it is crucial to select participants from different functional fields due to the following reasons:

- To get diverse opinions
- To have diversity of ideas and experience
- To reduce prediction error
- To improve .on quality of final results

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b) Consumer market survey

This method solicits input from consumers or potential consumers regarding their future purchasing plans. The views are considered to be the forecast.

c)The jury of executive opinion

This method takes the opinion of a small group of high level manager only. Their suggested result on a particular aspect is considered to be the forecast.

PRACTICE EXERCISES

QUESTION 1

Find the moving average of the time series of quarterly production (in tons) of coffee in an Indian State as given below. After that, come up with a trend line to approximate the production in future.

Production (in Tons)

Year	Quarter I	Quarter II	Quarter III	Quarter IV
1983	-	-	12	16
1984	5	1	10	17
1985	7	1	10	16
1986	9	3	8	18
1987	5	2	15	5

Solution:

		x	A=y	Quarterly moving average	Centred moving average T	x^2	xy	A / T	Deseasonalised values A / S
1983	3	1	12			1	12		11.06
	4	2	16			4	32		8.122
				8.5					
1984	1	3	5		8.25	9	15	0.6	6.748
				8.0					
	2	4	1		8.125	16	4	0.123	4.902
				8.25					
	3	5	10		8.5	25	50	1.176	9.217
				8.75					
	4	6	17		8.75	36	102	1.943	8.629
				8.75					
1985	1	7	7		8.75	49	49	0.800	9.447
				8.75					
	2	8	1		8.625	64	8	0.116	4.902
				8.5					
	3	9	10		8.75	81	90	1.143	9.217
				9.0					
	4	10	16		9.25	100	160	1.730	8.122
				9.5					
1986	1	11	9		9.25	121	99	0.973	12.146
				9					
	2	12	3		9.25	144	36	0.324	14.706
				9.5			0		
	3	13	8		9	169	104	0.889	7.373
				8.5					
	4	14	18		8.375	196	252	2.149	9.137
				8.25					
1987	1	15	5		9.125	225	75	0.548	6.748
				10					
	2	16	2		8.375	256	32	0.239	9.804
				6.75					
	3	17	15			289	255		13.825
	4	18	5			324	90		2.538
Total		171	160			2109	1465		

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Approximating the trend to be linear, then

Trend line - T = a + b ×Quarter number.

$$a = \frac{\sum y}{n} - b\frac{\sum x}{n}$$

$$b = \frac{(n\sum xy - \sum x \times \sum y)}{n\sum x^2 - (\sum x)^2}$$

given that

$\sum x = 171$
$\sum x^2 = 2109$
$\sum y = 160$
$\sum xy = 1465$
$n = 18$

$$b = \frac{(18\times 1465\times 171 - 171\times 160)}{18\times 2109 - 171^2} = -0.1135$$

$$a = \frac{\sum y}{n} - b\times\frac{\sum x}{n} = \frac{160}{18} - (-0.1135)\times\frac{171}{18} = 9.9673$$

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So, T = 9.9673-0.1135 × Quarter number

Notes:
Any number of years moving average can be used. Quarterly moving average has been chosen in this case. Since it is not centred, centering is done as shown.
The trend can also be obtained from the time series as required here.
The summation for quarter numbers and the actual production are obtained. The additional values of summation of x^2 (quarter number squared) and summation of xy (production × quarter number) are obtained from the additional columns indicated.
The values of a and b of the trend line equation can then be obtained as shown.Though not required. It was possible to obtain deseasonalised data before obtaining the trend line. This means a better forecasting equation is obtained (moving average and trend equation would have been used)
Seasonal factor S is obtained by averaging the error variation A/T for each quarter as per the second table. Since the summation of the average is not equal to 4 (seasonal aspect) it has to be corrected by the factor 4/3.941.
The deseasonalised data is then obtained. Notice the way here a multiplicative model was chosen because of the way the seasonal aspect keeps on changing.

Determination of S

	1	2	3	4	
1983	-	-	-	-	
1984	0.6	0.123	1.176	1.943	
1985	0.8	0.116	1.143	1.730	
1986	0.973	0.324	0.889	2.149	
1987	0.548	0.239	-	-	Total
Average	0.730	0.201	1.069	1.941	3.941
Corrected	0.741	0.204	1.085	1.970	

So the summations will change to be as follows

$\sum x = 171$

$\sum x^2 = 2109$

$\sum y = 156.643$

$\sum xy = 1507.171$

$n = 18$

$$b = \frac{(18 \times 1507 \times 171 - 171 \times 156.643)}{18 \times 2109 - 271^2} = 0.0393$$

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$$a = \frac{\sum y}{n} - b \times \frac{\sum x}{n} = \frac{156.643}{18} - 0.0393 \times \frac{171}{18} = 5.1556$$

So, T = 5.1556 + 0.0393 × Quarter number

QUESTION 2

(a) Differentiate between the additive model and the multiplicative model as used in time series analysis.

(b) The sales data of XYZ Ltd. (in million of shillings) for the years 2001 and 2004 inclusive are as given below:

	Quarter			
Year	**1**	**2**	**3**	**4**
2001	40	64	124	58
2002	42	84	150	62
2003	46	78	154	96
2004	54	78	184	106

Required:

(i) The trend in the data using the least squares method.

(ii) The estimated sales for each quarter of the year 2004.

(iii) The percentage variation of each quarter's actual sales for the year 2004.

Solution:

a) - In an additive model seasonal variation, cyclical variation and random variation are expressed as absolute values

It is best applied where components are independent of each other e.g. where cyclical variations are not affected by value of trend.

Additive model is expressed as

$O = T + C + S + I$

Where O = Observed value of time series
T = Trend Component
C = Cyclical Component
S = Seasonal Component
I = Irregular Component

In Multiplicative model seasonal variation, cyclical variation and random variation are expressed as absolute values

It is best applied when components are interdependent e.g. when seasonal variation is affected by trend values

Multiplicative Model is expressed as $O = T \times C \times S \times I$

Where O = Observed value of time series
T = Trend Component
C = Cyclical Component
S = Seasonal Component
I = Irregular Component

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b)

	X	Y	XY	X^2
2001 Q1	1	40	4	1
Q2	2	64	128	4
Q3	3	124	372	9
Q4	4	58	232	16
2004 Q1	5	42	210	25
Q2	6	84	504	36
Q3	7	150	1050	49
Q4	8	62	496	64
2003 Q1	9	46	414	81
Q2	10	78	780	100
Q3	11	104	1694	121
Q4	12	96	1152	144
2004 Q1	13	54	702	169
Q2	14	78	1092	196
Q3	15	184	2760	225
Q4	16	106	1696	256
	136	1420	13,322	1,496

Let the regression line y on x be in the form
y= a + bx

$$b = \frac{n\sum XY - \sum X\sum Y}{n\sum X^2 - (\sum X)^2} = \frac{16 \times 13322 - 136 \times 1420}{16(1496) - (136)^2}$$

$$= \frac{20,032}{5,440}$$

$$= 3.68$$

$$A = \bar{Y} - b\bar{x}$$

$$= \bar{Y} = \frac{\sum y}{N} = \frac{1420}{16} = 88.75$$

$$\bar{X} = \frac{\sum x}{n} = \frac{136}{16} = 8.5$$

Therefore a = 88.75 – 3.68(8.5)
= 57.47

Trend line = y = a + bx
Y = 57.47 + 3.68x

ii) Estimated sales for each quarter of year 2004

	Sh 'M'
1st Quarter = 57.47 +	105.31
3.68(13) =	108.99
2nd Quarter = 57.47 +	112.67
3.68(14) =	116.35
3rd Quarter = 57.47 +	
3.68(15) =	
4th Quarter = 57.47 + 3.68	
(16) =	

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iii) Percentage variation = $\frac{\text{Actual Sales}}{\text{Estimated Sales}} \times 100\%$

1st Quarter $= \frac{54}{105.31} \times 100 = 51.2\%$

2nd Quarter $= \frac{78}{108.99} \times 100 = 71.57\%$

3rd Quarter $= \frac{184}{112.6} \times 100 = 163\%$

4th Quarter $= \frac{106}{116.35} \times 100 = 91.01\%$

QUESTION3

(a) State the principal components of a time series.

(b) (i) Explain the difference between multiplicative and additive models as used in time series.

(ii) State the conditions under which each model is used.

(c) The table below shows the sales of new cars by quarters during a period of three years:

Year	Quarter 1 Sh. "million"	Quarter 2 Sh. "million"	Quarter 3 Sh. "million"	Quarter 4 Sh. "million"
2001	55.0	76.5	61.2	77.8
2002	54.4	65.9	52.7	81.4
2003	59.3	83.2	78.5	93.0

Required:

(i) Explain the purpose of the seasonal index

(ii) The seasonal index for each quarter assuming an additive model.

Solution:

(a) Principal components of a time series are:

- Secular trend (T)
- Seasonal variation (S)
- Cyclic variation (C)
- Random variation (R)

(b) (i) Difference between multiplicative and additive models:

- Multiplicative model expresses the time series model as a product of the four principle components.
  That is Y = TSCR
- Additive model expresses the time series model as a sum of the four principle components.
  That is Y = T + C + R + S

(ii) Conditions under which each model is used;

- Multiplicative model is used if the four principle components are not independent.
- Additive model is used when the four principle components are independent.

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QUESTION 4

a) write short notes on mean absolute deviation

b) The following date relates to the sales datea of an engineering firm;

Profits (Sh. "Million")

Quarter

Year	1	2	3	4
2003	5.5	5.4	7.2	6.0
2004	4.8	5.6	6.3	5.6
2005	4.0	6.3	7.0	6.5
2006	5.2	6.5	7.5	7.2
2007	6.0	7.0	8.4	7.7

Required:

i) The deseasonalised sales of the engineering firm

ii) Trend line using the least square method

Solution:

a) Mean Absolute Deviation

it is a measure of the overall error of the forecasts made. It is calculated by dividing the summation of the forecast errors by the number of time periods.

$$MAD = \frac{\Sigma / \text{forecast error} /}{N}$$

$$= \frac{\Sigma / Y_t - F_t /}{n}$$

Calculation of MAD uses absolute value i. e the signs are disregarded.

(b) (i)

Year	Quarter	Actual Sales	Centered 4 quarter moving total	Centered 4 quarter moving average	Ratio to moving average In %
2003	I	5.5			
	II	5.4			
	III	7.2	23.8	6.0	120.0
	IV	6.0	23.5	5.9	101.7
2004	1	4.8	23.5	5.9	101.7
	II	5.6	22.5	5.6	100.0
	III	6.3	21.9	5.5	114.5
	IV	5.6	21.9	5.6	101.8
2005	I	4.0	22.6	5.7	70.2
	II	6.3	23.4	5.9	106.8
	III	7.0	24.4	6.1	114.8
	IV	6.5	25.1	6.3	103.2
2006	I	5.2	25.5	6.4	81.3
	II	6.5	26.1	6.5	100.0
`	III	7.5	26.8	6.7	111.9
	IV	7.2	27.5	6.9	104.3
2007	I	6.0	28.2	7.1	84.5
	II	7.0	28.9	7.2	97.2
	III	8.4			
	IV	7.7			

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SEASONAL INDICES

QUARTERS

YEAR	I	II	III	IV
2003	-	-	120.0	101.7
2004	82.8	100.0	114.5	101.8
2005	70.2	106.8	114.8	103.2
2006	81.3	100.0	111.9	104.3
2007	84.5	97.2	-	-
Mean	79.7	101.0	115.3	102.8
Seasonal index	79.9	101.3	115.6	103.2

Deseasonalised Sales

YEAR	QUARTER	ACTUAL SALES	SEASONAL INDEX	DESEASONALISED SALES
2003	**I**	5.5	0.799	6.9
	II	5.4	1.013	5.3
	III	7.2	1.156	6.2
	IV	6.0	1.032	5.8
2004	**I**	4.8	0.799	6.0
	II	5.6	1.013	5.5
	III	6.3	1.156	5.4
	IV	5.6	1.032	5.4
2005	**I**	4.0	0.799	5.0
	II	6.3	1.013	6.0
	III	7.0	1.156	6.2
	IV	6.5	1.032	6.0
2006	**I**	5.2	0.799	6.3
	II	6.5	1.013	6.5
	III	7.5	1.156	6.4
	IV	7.2	1.013	6.5
2007	**I**	6.0	0.799	7.5
	II	7.0	1.013	6.9
	III	8.4	1.156	7.3
	IV	7.7	1.032	7.5

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(II)

YEAR	Quarter	Deseasonalised Sales	X*	X^2	XY
2003	I	6.9	-19	361	-131.1
	II	5.3	-17	289	-90.1
	III	6.2	-15	225	-93.0
	IV	5.8	-13	169	-75.4
2004	I	6.0	-11	121	-66.0
	II	5.5	-9	81	-49.5
	III	5.4	-7	49	-37.8
	IV	5.4	-5	25	-27.0
2005	I	5.0	-3	9	-15.0
	II	6.2	-1	1	-6.2
	III	6.0	1	1	6.0
	IV	6.3	3	9	18.9
2006	I	6.5	5	25	32.5
	II	6.4	7	49	44.8
	III	6.5	9	81	58.5
	IV	7.0	11	121	77.0
2007	I	7.5	13	169	97.5
	II	6.9	15	225	103.5
	III	7.3	17	289	124.1
	IV	7.5	19	361	142.5
		125.6		2660	114.2

$$b = \frac{\Sigma XY}{\Sigma X^2} = \frac{114.2}{2660} = 0.04$$

$$A = \frac{\Sigma Y}{n} = \frac{125.6}{20} = 6.3$$

$Y = 6.3 + 0.04X$

A= 5.173
B =0.106
Y = 5.173 + 0.106x

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TOPIC 6

LINEAR PROGRAMMING

INTRODUCTION

Business organizations have various objectives which they have to meet using a certain available resources that are usually in scarce supply, for instance:

i) A manufacturing company deems to provide quality products and make profit through utilization of the limited resources like personnel, material, machine, lime, market etc.

ii) A hospital has the main objective of maintaining and restoring good health to its patients at an affordable cost to the patients. Resources include medical personnel, number of beds, pharmacies and laboratories.

In such examples, mathematical programming(MP)provides a technique that may be used to make decision on the best way to allocate the limited resources in order to 235inimize profit or minimize cost.

Programming refers to a mathematical technique which is iterative. Iteration is a technique which converges towards an optimal solution using the same basic steps in a repetitive manner. The solution keeps improving until it can improve no more i.e. until the best solution is obtained given that circumstance.

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Mathematical Programming therefore is a mathematical decision tool that aids managers in seeking either the maximization n of profit, minimization of cost or both within an environment of scarce/limited resources. Such scarce resources are called constraints e.g. raw materials labour supply, market etc. The maximization of profit and in minimization of cost are known as objectives. The decision problems can be formulated and solved as mathematical programming problems. Mathematical programming involves optimization of a certain function called the objective function subject to certain constraints.

The mathematical programming techniques can be divided into 7 categories namely:

1. linear programming
2. non-liner programming
3. integer programming
4. dynamic programming
5. stochastic programming
6. parametric programming
7. goal programming

1. Linear programming (LP) method

This method is a technique for choosing the best alternative from aset of feasible alternatives whereby the objective function and constraints are expressed as linear mathematical functions. In order to apply linear programming(LP), the following requirements should be met:

i) There should be a clearly identifiable objective which is measured quantitatively.
ii) The activities to be included should be distinctly identifiable and measurable in quantitative terms.
iii) The resources of the system should be identifiable and measurable quantitatively and also in limited supply.
iv) The relationships representing objective function and the constraints equations or inequalities must be linear in nature.
v) There should be a series of feasible alternative courses of action available to the decision maker, which are determined by the resource constraints.

Business application of linear programming

a) Determination or optimal product mix in industries.
b) Determination of optimal machine and labour contribution
c) Determination of optimal use of storage and shipping facilities
d) Determining the best route in transport industry.
e) Todetermine investment plans.
f) To find the appropriate number of financial auditors
g) Assigning advertising expenditures to different media plans.
h) Determining theamount of fertilizer to apply per acre in the agricultural sector.
i) Determiningcampaign strategies in politics.
j) Determining the best marketing strategies.

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Basic assumptions of linear programming (LP)

i. **Certainty**– values (numbers) in the objective and constraint are known with certainty and do not change during the period being studied.
ii. **Proportionality/linearity**– a basic assumption of linear programming(LP) is that proportionality exists in the objective function and the constraints inequalities- e.g. if a production of 1unit of a product uses 3 hours of a particular scarce resource, then making 10 units use 30 hours of the resource.
iii. **Additivity**– the total of all the activities is given by the sum total of each activity conducted separately. For instance, the total profit in the objective function is determined by the sum of the profit contributed by each of the products separately.
iv. **Divisibility/continuity**– solutions need not be in whole numbers (integers) Instead, they are divisible and may take any fractional value.
v. **Non negativity/finite choice**– negative values of physical quantities are impossible, you simply cannot produce negative number of chairs, shirts, lamps or computers.
vi. Time factors are ignored. All production are assumed to be instantaneous.

vii. Costs and benefits which cannot be quantified easily like goodwill, liquidity and labour stability are ignored.
viii. Interdependence between demand products is ignored, products may be complementary or a substitute for one another.

Advantages of linear programming (LP)

i) Improves the quality of decisions.
ii) Helps in attaining the optimum use of production factors.
iii) It highlights the bottlenecks in the production process
iv) It gives insight and perspective into problem situations,
v) Improves the knowledge and skills of tomorrow's executives,
vi) Enable one to consider all possible solutions to problems.
vii) Enables one to come up with better and more successful decisions
viii) It is a better tool for adjusting to meet changing conditions.

Disadvantages of Linear programming

i) It treats all relationships as linear.
ii) It is assumed that any activity is infinitely divisible.
iii) It takes into account single objective only i.e. profit maximization or cost minimization
iv) It can be adopted only under the condition of certainty i.e. recourses, per unit contribution, costs etc. are known with certainty. This does not hold in real situations

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Mathematical formulation of linear programming problems

Formulating a linear program involves developing a mathematical model to represent the managerial problem. The step in formulating a linear program follows:

a) Completely understand the managerial problem being faced
b) Identify the objective and the constraints.
c) Define the decision variables.
d) Use the decision variables to write mathematical expression for the objective function and the constraints.

ILLUSTRATION

Maximization case

A company produces inexpensive tables and chairs. The production process for each is similar in that both require a certain number of hours of carpentry work and a certain number of labour hours in the painting department. Each table takes 4 hours of carpentry and 2 hours in the painting shop. Each chair requires 3 hours of carpentry and 1 hour in painting. During the current production period, 240 hours of carpentry time are available and 100 hours in painting time are available. Each table sold yield a profit of $7 and each chair produced is sold for a $5 profit.

Formulate this problem as a linear programming problem to determine as to how many tables and chairs should be produced so that the firm can maximize the profit. Assume that there are no marketing constraints so that all that is produced can be sold.

SOLUTION

The objective function:

The goal of the firm is the maximization of profit, which would be obtained by producing and selling the tables and chairs.

It we let x_1 be the number of tables, x_2 be the number of chairs and Z be the total profit.

Then $Z = 7x_1 + 5x_2$ (this is the objective function which is linear in nature)

NB: since the problem calls for a decision about the optimal (best possible) values of x_1 and x_2, these are known as the **decision variables.**

Constraint

These are the resources which must be in limited supply. The mathematical relationship which it used to explain this limitation is inequality (a mathematical relationship involving $\leq$ or $\geq$ sign).

Each table requires 4 hours of carpentry while a chair requires 3hours. Hence the total consumption of carpentry hours would be $4x_1 + 3x_2$, which cannot exceed the total availability of 240 hours. This constraint can be expressed as an inequality of the form. $4x_1 + 3x_2 \leq 240$. Similarly, a table requires 2 hours of painting while a chair requires 1 hour, With the availability of 100 hours, we have $2x_1 + x_2 \leq 100$ as the painting constraint.

Non-negativity condition:

Obviously x_1 and x_2 being the number of units produced cannot have negative values. Symbolically, $x_1 \geq 0$ and $x_2 \geq 0$ (this is the non-negativity condition)

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Hence the above linear programming problem can be summarized as follows:

Maximize $Z = 7x_1 + 5x_2$		(profit)	this formulation is called
Subject to:	$4x_1 + 3x_1 \leq 240$	(carpentry hours constraint)	either the LPP model
	$2x_1 + x_2 \leq 100$	(painting hours constraint)	or Primal LP model
	$x_1 \geq 0, x_2 \geq 0$	(non-negativity restriction)	

ILLUSTRATION

Minimization case

The Star hotel was burned down in a fire and the manager decided to accommodate the guests in 4 – person and 8-person tents. The tents were to be hired at a cost of $15 and $ 45 per night respectively, the space available could accommodate at most 13 tents and the manager had to cope with at least 64 guests. Formulate this as a linear programming model that could be used to determine the number of tents of each type that could pull up in order to minimize the overall cost.

SOLUTION

Let x_1 be the number of 4-person tents to be pitched

x_1 be the number of 8-person tents to be pitched

Objective function:

Minimize cost, $C = 15x_1 + 45x_1$

Subject to:

$4x_1 + 8x_1 \geq 64$

$x_1 + x_1 \leq 13$

$x_1, x_2 \geq 0$

Generalized formulation of LPP

If there are n decision variables and m constraints in the problem, the mathematical formulation of the LP is:-

Optimization (Max) $Z = C_1x_1 + C_2x_1 + \ldots\ldots\ldots\ldots C_ax_a$

Subject to the constraints:

$a_{11}x_1 + a_{12}x_1 + \ldots\ldots\ldots\ldots + a_{1a}x_n \leq b_1$

$a_{21}x_1 + a_{22}x_2 + \ldots\ldots\ldots\ldots + a_{2n}x_n \leq b_2$

$\vdots \quad \vdots$

$a_{m1}x_1 + a_{m2}x_2 + \ldots\ldots\ldots\ldots + a_{mn}x_n \leq b_m$

$x_1, x_2 \ldots\ldots\ldots\ldots\ldots_n \geq 0$

Where

x_2– decision variable

c_j– constant presenting per unit contribution of the objective function of the jth decision variable a_{ij}– constant representing, exchange coefficient of the jth decision variable in the ith constant

b, - constant representing the ith constraint requirement of availability

In shorter form, the problem can be written us:

Maximise = $Z = \sum_{j-i}^{n} c_j x_j$

Subject to

$\sum_{j-i}^{n} c_j x_j \leq b_1$ For i= 1,2 ..... m

$x_j \leq b_1$ For i= 1,2 ..... n

In Matrix notation, an LPP can be expressed as follows:

Minimization problem		Minimization problem	
Maximize	$Z = Cx$	Minimize	$Z = Cx$
Subject to:	$AX \leq B$	Subject to:	$AX \geq B$
	$X \geq 0$		$X \geq 0$

Where

C= row matrix containing the coefficients m the objective function

X = column matrix containing decision variables

A = matrix containing the coefficients in the constraints

B = column matrix containing the RHS values of the constraints

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NB:
Generally, the constraints in the maximization problems are of the ≤ type, and the ≥ type in minimization problems. But a given problem may contain a mix of the constraints, involving the signs ≤,≥ and/ or =.
Usually, the decision variables are non-negative. However, this may not be always the case.
For instant, if an investor is dealing in shares, he can decide to buy more, sell or retain what he has. Therefore if x represents the number of shares, then x= 0 (indicates no new investment x>0(indicates new investment) and x< 0 (indicates selling of the available shares). Hence x shall be unrestricted in sign or it is a free variable.

SOLUTION TO LINEAR PROGRAMMING PROBLEMS

The linear programming(LP) problems can be solved by the help of the following methods:
1. Graphical method
2. Simplex method (algebraic method)

The purpose of the graphical method is to provide a grasp of the basic concepts that are used it simplex method. The simplex method is the major method of solving Linear programming models.
Graphical method can be used for problems with two decision variables only.
Problems with more than two variables must be solved by the simplex method

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1) Graphical solution method

Graphical method is the simplest to use and should be used whenever possible. In order to solve a LP problem graphically, the following procedure is adopted:

i) Formulate the appropriate Linear programming problem
ii) Graph the constraint inequalities as follows: treat each inequality as though it were equality and for each equation, arbitrarily select two sets of coordinate points. Plot of equality and connect them with appropriate lines.
iii) Identify the solution space or the feasible region which satisfies all the constraints simultaneously. For ≤ constraints, this region is below the lines and for ≥ constraints region is above the lines. (Shade the unwanted region).
iv) Locate the solution points of the feasible region. These points always occur at the corner points of the feasible region.
v) Evaluate the objective function at each of the corner points (this method is called vertex/corner point method).

NB: The optimal production mix can also be obtained using isoprofit method (for maximization problem) or Isocost method (for minimization problem). The isoprofit is a straight line representing all combinations of $x_1$and x_2 for a particular profit level.

Procedure for corner point method

- Graph all constraints and find the feasible region (this is the area which does not contravene any of the restrictions and is therefore the area that contains all possible solutions)
- Find the corner points of the feasible region

- Compute the profit (or cost) at each of the feasible comer points
- Select the corner point with the best value of the objective function in step 3. This is the optimal solution/production mix.

Procedure for Isoprofit or Isocost method

1. Consider the objective function and equate it to an arbitrary profit (or cost) value.
2. Plot the objective function (on the graph) which yield; the first isoprofit (or Isocost) that must go through the feasible region.
3. Plot a few more isoprofit (or Isocost) lines to the right (or left) which are parallel to the first one.
4. The isoprofit (Isocost) line that touches the furthest (closest) point of the feasible region yields the optimal production mix.
5. Identify the optimum value of the objective function i.e. the optimal solution
6. Interpret the results.

ILLUSTRATION

Plot the following on a graph

$4x + 2y \leq 100$
$4x + 6y \leq 180$
$x + y \leq 40$
$x \leq 20$
$y \geq 10$

$x \geq 0$ – non – negativity constraint.

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Plotting

Let 4x + 2y = 100

Coordinates

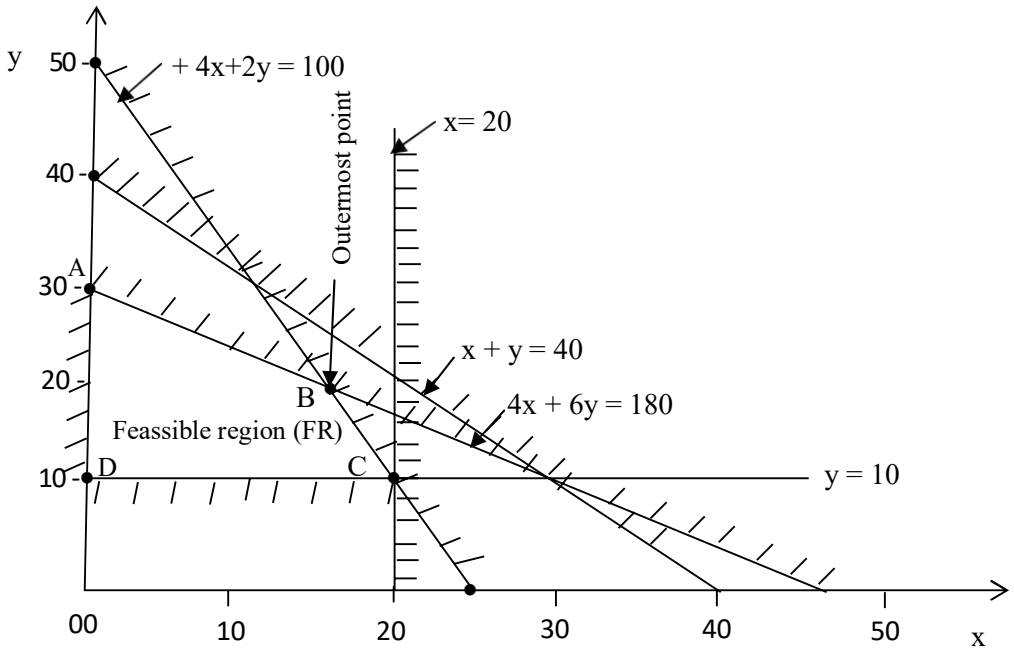
x	y
25	0

Let 4x + 6y = 180

x	y
45	0

Let x + y = 40; x = 20; y = 10

x	y
40	0



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NB: The optimal solution as found at one of the corners of the feasible region (corner method)

Vertex	X	Y	Z
A	0	30	3(0) + 4 (30) = 120
B	15	20	3(15) + 4 (20) = 125
C	20	10	3(20) + 4 (10) = 100
D	0	10	= 40

To find B

$$\begin{array}{r} 4x + 6y = 180 \\ \underline{4x + 2y = 100} \\ 4y = 80 \end{array}$$

Therefore y = 20 x = 15

Optimal solution / product mix

Unit of A = 15

Unit of B = 20

Maximum profit = $125

ILLUSTRATION

5x + 7 y = Z

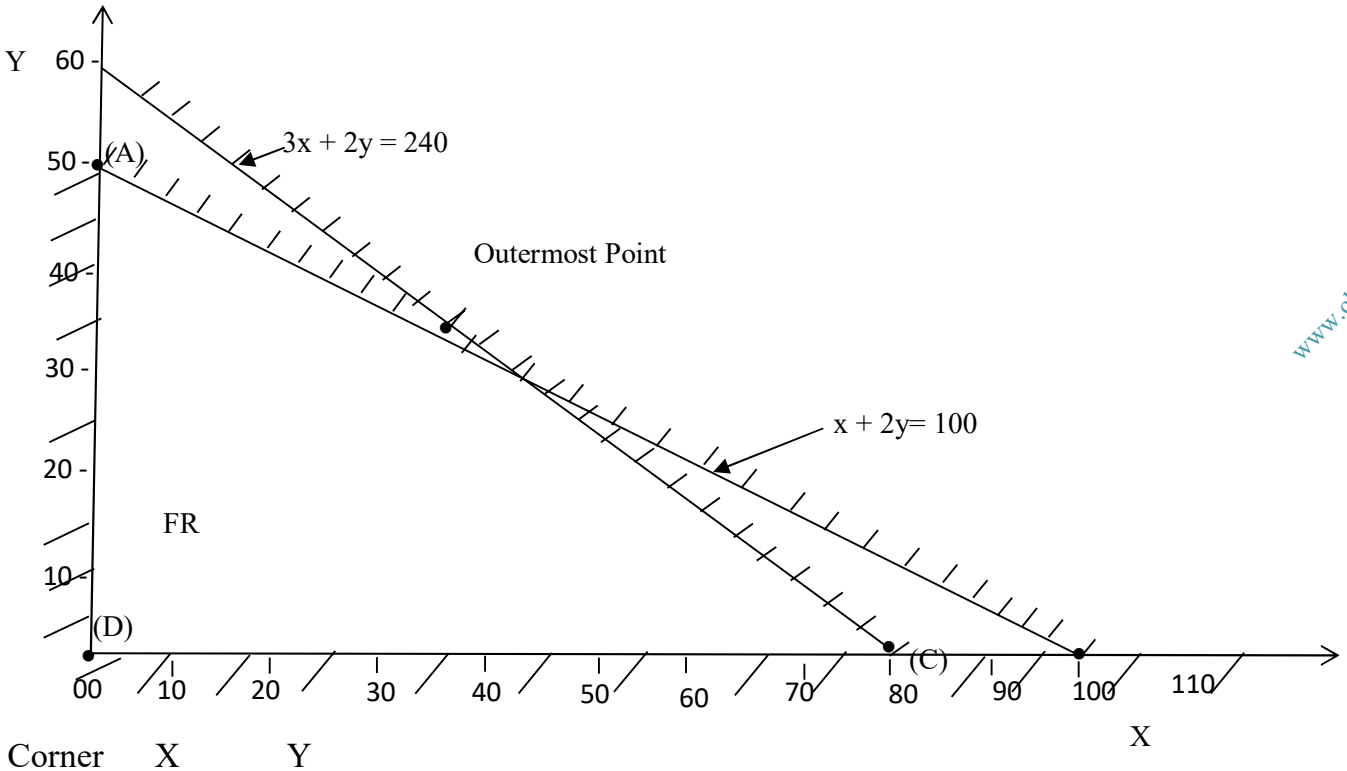
Subject to;

$3x + 4y \leq 240$

$x + 2y \leq 100$

$x \geq 0, y \geq 0$

3x + 4y = 240		x + 2y = 100	
x	y	x	y
80	0	100	0
0	60	0	50



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Corner	X	Y
A	0	50
B	40	30
C	80	0
D	0	0

To solve for B

1(3x + 4y = 240)

3(x + 2y = 100)

3x + 4y = 240

3x + 6y = 300

$\frac{-2y}{-2} = \frac{-60}{-2}$

∴ y = 30 ; x = 40

	Z
A	5(0) + 7(50) = 350
B	5(40) + 7(30) = 410
C	5(80) + 7(0) = 400
D	5(0) + 7(0) = 0

∴ Optimal Solution
Unit x = 40 units
y = 30 units
Maximum profit, Z = 410

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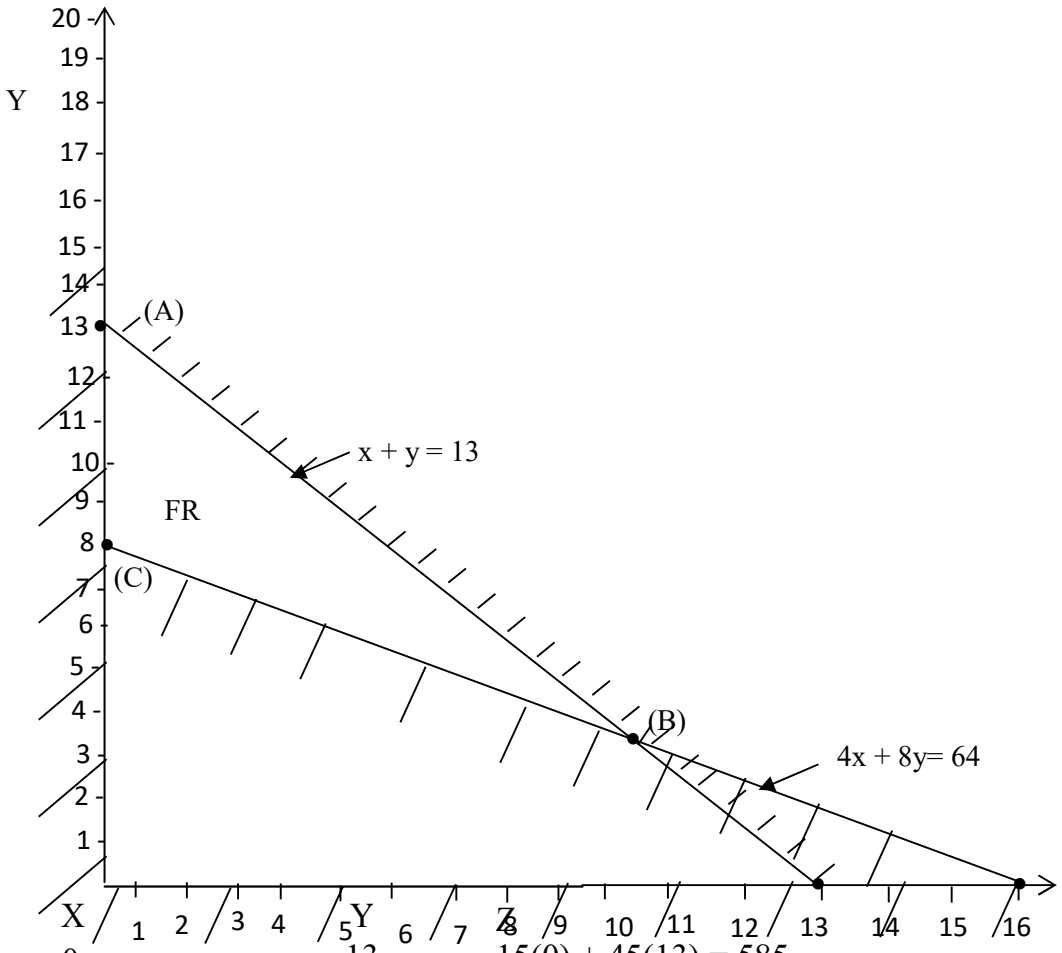
ILLUSTRATION

To solve for

Min C = 15x + 45y
Subject to;
x + y ≤ 13
4x + 8y ≥ 64
x, y≥ 0

x + y = 13	
x	y
0	13
13	0

4x + 8y = 64	
x	y
0	8
16	0



A	0	13	15(0) + 45(13) = 585
B	10	3	15(10) + 45(3) = 285
C	0	8	15(0) + 45(8) = 360

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∴ Optimal solution

x = 10

y = 3

To solve for B

1(4x + 8y = 64)

4(x + y = 18)

4x + 8y = 64

4x + 4y = 52

4y = 12

y = 3; x = 10

Binding and non-binding constraints

Once the optimal solution is obtained, the constraints can be classified as either binding or non-binding. A constraint is binding if the left and right hand sides of its inequality function are equal when the optimal values are substituted. If the substitution does not lead to equality, then the constraint is non-binding.

SIMPLEX METHOD (gauss Jordan method)

This is an iterative method of solving LPP, It's appropriate where the graphical method is not applicable. Note that the graphical method is limited to two variables. Therefore the simplex method comes in handy for more variables (though can be used for two variables). The method considers only those feasible solutions which are provided by the corner points and indicate whether a given solution is optimal or not. Each iteration in the simplex method produces a feasible solution and an answer better than the previous one i.e. either greater contribution in maximizing problems, or less cost in minimizing problems.This method yields not only the optimal solution to the Xi variables and the maximum (or minimum cost) but valuable economic information as well.

NB: the non-negativity assumption is ignored since it is automatically taken care of by the simplex.

The method involves a number of tableaus.

Formulating the simplex model

- State the objective function and the various inequalities representing the constraints (primal LP model).
- Convert each inequality to an equation by adding an extra variable called a slack (s) variable, The slack represents an unused resource. This is analogous to starting the evaluation process in the graphical approach at the point of origin where both x_1 and x_2 are equal to zero. This yields the canonical form of the primal model.

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ILLUSTRATION

Primal model

Minimize $Z = 12x_1 + 20x_2$

Subject to:

(Constraint 1) $3x_1 + 4x_2 \geq 96$

(Constraint 2) $6x_1 + 6x_2 \leq 168$

(Constraint 3): $x_1 \geq 18$

$x_1, x_2 \geq 0$

Canonical form

Let $S_1 S_2$ and S_3 be the slack variables for constraints 1, 2 and 3 respectively.

Hence: Maximise $z = 12x_1 + 20x_2 + Os_1 + Os_2 + Os_3$

Subject to:

(1) $3x_1 + 4x_2 + 1s_1 + Os_2 + Os_3 = 96$ (S_1, is the slack for constraint 1)

(2) $6x_1 + 6x_2 + Os_1 + 1s_2 + Os_3 = 168$ (S_2, is the slack for constraint 2)

(3) $Ox_1 + x_2 + Os_1 + Os_2 + 1s_3 = 18$ (S_2, is the slack for constraint 3)

Ignore non-negativity

Solving the above problem using the simplex algorithm

TABLEAU 1

Place all of the coefficient (in the canonical form) into a tabular form:

Profit / unit	**Production mix / solution variables**	**Real variables**	**Slack variables**	**Constant**	
Cj	Solution mix Basis	12 20 X_1 X_2	0 0 0 S_1 S_2 S_3	RHS Quantity	Profit per unit row
0 0 0	S_1 S_2 S_3	3 6 6 6 0 1	1 0 0 0 1 0 0 0 1	96 168 18	Constraint equation rows.
	Zj Cj – Zj	0 0 12 20	0 0 0 0 0 0	0 0	Gross profit net profit row

Leaving variable

Entering pivot column

SOLUTION TABLEAU I

The tableau shows a feasible solution with nil production, nil contribution and maximum unused capacity as represented by the values of the slack variables:

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Hence, Decision variables	**Slack variables**
$X_1 = 0$	$S_1 = 0$
$X_2 = 0$	$S_2 = 0$
$Z = 0$	$S_3 = 0$

Notes

i. If a variable is not in the basis of solution mix, it is said to be non-basic and it has a solution of zero e.g. $X_1 = 0$ and $X_2 = 0$ in tableau I.

ii. If a variable is in the basis, then it is a basis variable and its value is non-zero. It takes the value on the right hand side(RHS) of the tableau,

iii. For any basis variable, there shall be a unique" 1" in the column and the rest of the values in that column will be zero,

iv. For optimality assessment (maximization problem), the solution is not optimal as long as there is a positive value in the Cj-Zj row (hence tableau l is not optimal)

v. If solution is not optimal identify 2 variables:
   - Entering variable – given by the largest positive net contribution in the Cj – Zj row
   - Leaving variable – given by the smallest ratio between the right hand side (RHS) and pivot column (illustrated in tableau 2 below)

TABLEAU 2

Improve the initial solution by going through the following steps:

a) Select the highest contribution in Cj – Zj row i.e 20 under X_2 column.

b) Divide the right hand side(RHS) quantity values with the numbers in the X_2 column to get the ratio:

96/4=24
68/6= 28
18/1=18
0/0 = 0 (ignored)

c) Select the row that gives the lowest non-zero ratio i.e. 18. The intersection between column identified in step (a) and row (c) gives an element known as **pivot element (1)**

d) Divide all the elements in the identified row (S_3) by the pivot element (1) and change the solution variable the heading of the identified column (X2)

Profit / unit	**Production mix / solution variables**	**Real variables**	**Slack variables**	**Constant**
Cj	Solution mix basis	12 20 X_2 X_2	0 0 0 S_1 S_2 S_3	RHS Quantity
0 0 20	S_1 S_2 S_3	3 4 6 6 0 1	1 0 0 0 1 0 0 0 1	96 168 18
	Zj Cj – Zj	0 0 12 20	0 0 0 0 0 0	0 0

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NB : row X_2 indicate that 18 units of X_2 are produced with profit per unit being sh.20.

e) Carry out repetitive row by row operations using row 3 (X_2) which makes all the other elements in the pivot elements column into zeros as shown below.

Row operations

	3	4	1	0	0	96
$S_1 - 4X_2$ =	- 0	4	0	0	4	72
	3	0	1	0	-4	24 (replace row S_1 by this row
$S_1 - 6X_2$ =	6	6	0	1	0	168
	- 0	6	0	0	6	108
	6	0	0	1	-6	60 (replace row $S_1$by this row

f) Compute the Zj and Cj-Zj values

Zj = ∑ (profit per unit x variable value)

The above information is presented in tableau 2 below

Profit / unit	**Production mix / solution variables**	**Real variables**	**Slack variables**	**Constant**
Cj	**Solution mix basis**	**12 20** **X_1 X_2**	**0 0 0** **$S_1S_2S_2$**	**RHS Quantity**
0 (new row) 0 (new row) 20	S_1 S_2 S_3	3 4 6 6 0 1	1 0 -4 0 1 -6 0 0 1	24 60 18
	Zj Cj – Zj	0 20 12 0	0 0 20 0 0 -20	360 0

SOLUTION TABLEAU 2

Solution mix **Slack variables**

X_1=O S_1=24

X_2= 18 S_2 =60

Z= 360 S_3=0

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Tableau 2 remarks

- Solution is not optimal since there is a positive element in the $C_j - Z_j$ row
- Entering variable is X, (it is the only positive value)

NB: Ignore a negative or undefined ratios

TABLEAU 3

Using table 2, repeat steps (a) to (f)

- Highest contribution =12
- Right hand side(RHS) column ratios; 24/3=8: (\0/6=10: 18/0(.ignore)
- Lowest right hand side(RHS) ratio=8 thus pivot element is 3
- Divide all elements in row S_1 (tableau 2) with the pivot element und replace S_1 with X_1
- Row operation on S_2

$S_1 - 4x_3$ =	6 - 6	0 0	0 2	1 0	-6 -8	60 48
	0	0	-2	1	2	12 (used to replace row S_1

Compute the Zj and Cj – Zj values.

Above information is presented in tableau 3 below.

Profit / unit	Production mix / solution variables	Real variables	Slack variables	Constant
Cj	Solution mix basis	12 20 X_1 X_2	0 0 0 S_1 S_2 S_2	RHS Quantity
12 (new row) 0 (new row) 20	X_1 S_2 X_2	1 0 0 0 0 1	$^1/_3$0 $^{-4}/_3$ -2 1 2 0 0 1	8 12 18
	Zj Cj – Zj	12 20 0 0	4 0 4 -4 0 -4	456

Solution is now optimal since there is no positive element in the Cj-Zj row

Optimal Solution

Decision variables	Slack variables
$X_1 = 8$	$S_1 = 0$ (scarce)
$X_2 = 18$	$S_2 = 12$ (abundant)
Profit = 456	$S_3 = 0$ (scarce)

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Notes

- If a resource is fully utilized (scarce), then its shadow price is non-zero but if it is abundant the shadow price is zero. The shadow prices are found in the Cj-Zj row.
- Constraint 1 is a scarce resource with slack of zero and shadow price of sh4. This implies that the resource is fully utilized and therefore a unit increase of constraint 1 leads to an increase in profit by sh4.
- Constraint 3 is a scarce resource with a slack of zero and shadow price of sh4
- Constraint 2 is abundant with an excess of 12 units and hence its availability is not scarce. Thus, is shadow price is zero.

ILLUSTRATION

Suppose the rum gets an offer of additional unit of material at $2.1 per unit; is it worth it.

Solution

a) Material is a scarce resource and hence worthwhile to acquire more provided acquisition cost less than $4 (shadow price).

b) Since acquisition cost of $2.1 is less than $4 it is profitable to acquire more material.For every additional unit of material acquired, profit will increase by $(4-2.1) $1.9

Interpretation of S_1 and S_3 column in the optimal solution (Tableau 3)

Bringing 1 unit of S_1 into the basis has the following consequences:

- 1/3 units of X_1 will get out of the basis. This is because one unit of X_1 requires 3 units of constraint 1.

- In giving up 1/3 of X_1, 1/3x6 =2 Units of constraint 2 will be released.
- Bringing in 1 unit of S_1 units, the basis has no effect on X_2 (zero coefficient)

If one unit of S_3 goes into the basis the following are the consequences:

- One unit of X_2will not be produced.
- Resources which would have been used to produce one unit of X_2 will be available to produce units of X_1 and so 1 unit of S_3 into the basis means we can produce an additional 4/3 units of X_1.This is because 4 units of constraint will be available, but a unit of X_1 requires 3 units
- Production of A 4/3 units of X_1 will require 4/3*6=8 units of constraint 2. However non production of a unit of X_2 will free 6 units of constraint 2 so that the net number of constraint 2 units which will get out of the basis is 8-6=2 units of constraint 2.

If one unit of S_3 gets into the basis, 1 unit or' X_2 will get out and vice versa

SURPLUS AND ARTIFICIAL VARIABLES IN LINEAR PROGRAMMING PROBLEMS (LPP)

SURPLUS VARIABLE I

This is a variable used to make a (≥ or =) constraint into an equality constraint when formulating an LPP into standard (canonical) form. Failure to do so, the simplex technique is unable to set up an initial solution in the first tableau. The surplus represents the amount of resources used above the allocated.

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ILLUSTRATION

Consider the constraint $5x_1 + 10x_2+8X_3 \geq 210$ (there will be excess resource)
In order to use the simplex procedure, we standardize this constraint by subtracting a variable in the left hand side (LHS) to change the inequality into equality. Thus: $5x_1 + 10x_2+8x_3 - R_1 = 210$
Where R_1, is a surplus variable and is the amount by which the solution exceeds the constraint resource. Because of its analogy to a slack variable, surplus is sometimes called negative slack.

If for example, a solution to an LPP involving the above constraint is $x_1 + 20x_2=8_1x_3 = 5$, then the amount of surplus or unused resource could be computed as follows:

$5(20) + 10(8) +8 (5) - R_1 = 210$
$- R_1 = 210 - 220$
$R_1 = 10$ surplus units of first resource

ARTIFICIAL VARIABLE (A)

This is a variable which is used in conjunction with (≥) and (=) constraints in order to get an initial feasible (realistic) solution.

≥ Constraint

For any problem being solved using simplex method, the decision variables must equal to zero for the initial solution. For the preceding constraint, if the decision variables $X_1=X_2=X_3=0$; - R_1 =200 (violates non negativity)

NB. In LP no variable of whatever type is allowed to be negative.

Remedy

Introduce a fictitious (artificial) variable A_1,then we can write the standardized constraint as follows:

$5X_1+ 10X_2+ 8X_3- R_1 + A_1 = 90$

Then in the initial simplex solution, we render $X_1=X_2=X_3=R_1$ =0 (which doesn't violate non negativity)

= Constraint

Consider the constraint $25x_1+30x_2 = 90$

For initial solution $X_1=X_2= 0$

Substituting in the constraint: 0=90 [incorrect]

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Remedy

Introduce an artificial variable to the LHS, call it A_2

Thus $25x_1+ x_2+R_1+ A_2=90$

Hence for initial solution: $x_1=x_2=0$ so that A_1 =90 (correct)

NB: artificial variables have no physical meaning and drop out of the solution mix before the final tableau

SURPLUS AND ARTIFICIAL VARIABLES IN THE OBJECTIVE FUNCTION

Whenever an artificial or surplus variable is added to one of the constraints, it must also be included in the other equations and in the problem's objective function, just us was done for slack variables. Since artificial variables must be forced out of the solution, we can assign a very high cost (M) to each. This method is called the **penalty or the Big M method** of getting rid of an artificial variable. In minimization problems, variables with high cost leave the solution quickly, or never enter it at all. Surplus variables, like slack variables, carry a zero cost.

If a problem had an objective function that reads;.

Minimize cost $C= 5x_1+ 9x_2+ 7x_3$

And constraints such as the two mentioned previously, the completed objective function and constraints would appear as follows.

Minimize cost = $5x_1 + 9x_2 + 7x_3 + 0R_1 + MA_1 + MA_2$

Subject to: $5x_1 + 10x_2 + 8x_3 + R_1 + 1A_1 + 0A_1 = 210$

$5x_1 + 30x_2 + 0x_3 + 0R_1 + 0A_1 + 1A_2 = 90$

SIMPLEX SOLUTION TO MINIMIZATION PROBLEM

Procedure:

i. Choose the variable with a 'negative Cj-Zj that indicates tile largest decrease in cost to enter the solution. The corresponding column is the pivot column.
ii. Determine the row to be replaced by selecting the one with the smallest (nonnegative) quantity –to- pivot column substitution rate ratio. This is the: pivot row.
iii. Calculate new values for the pivot row.
iv. Calculate new values for the other rows.
v. Calculate the Z_j and C_j-Z_j values for this tableau. If there are any Cj-Zj numbers less than 0, return to step '

ILLUSTRATION

Minimize $C = 5P_1 + 8P_2$

Subject to: (1) $P_1 + P_1 \geq 500$

(2) $P_1 \leq 400$

(3) $P_2 \geq 200$

(4) $P_1, P_2 \geq 0$

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SOLUTION

Standard (canonical) form

Minimize cost $= 5P_1 + 8P_2 + 0S_1 + MA_1 + MA_2$

Subject to: (1) $p_1 + p_2 + 1A_1 = 500$

(2) $P_1 + S_1 = 400$

(3) $P_2 - R_2 + A_1 = 200$

NOTE: The simplex iterations for such solution of minimization problem are identical to those of maximization problem, except that, for optimality in minimization will have been achieved when all Cj-Zj values are zero or positive- just the opposite from the maximization case.

Thus the entering variable will be indicated by Cj-Zj value which is the largest negative.

Tableau 1

Cj →		**5**	**8**	**0**	**0**	**0**	**M**	**M**		
↓	**Basis**	**p_1**	**p_2**	**R_1**	**S_1**	**R_2**	**A_1**	**A_2**	**RHS**	**Ratio**
M	A_1	1	1	-1	0	0	1	0	500	500/1=500
0	S_2	1	0	0	1	0	0	0	400	400/0
M	A_2	0	1	0	0	-1	0	1	200	(ignore)
Zj		M	2M	-M	0	-M	M	M		200/1 = 200
Cj-Zi		5-M	8-2M	+M	0	-M	0	0		

The number of basis variables in the initial tableau must be equal to the number of inequalities developed.

NB: The numbers in the Zj row = $\sum(Cj$ column x corresponding numbers in each other column)
e.g. Zj (for P_1 column)= M(l)+0(1)+M(0)=M

Tableau I Remarks

- It is not optimal since we have negative values in the Cj-Zj row.
- Entering variable is P_1 (with largest negative value)
- Leaving variable is A_2 (with smallest ratio)

$A_1 - A_2$	1	1	-1	0	0	1	0	500
=	- 0	1	0	0	-1	0	1	200
New A_2	1	0	-1	0	1	1	-1	300

Tableau 2

Cj →		**5**	**8**	**0**	**0**	**0**	**M**	**M**		
↓	**Basis**	**P_1**	**P_2**	**R_1**	**S_1**	**R_2**	**A_1**	**A_2**	**RHS**	**Ratio**
M	A_1	1	0	-1	0	1	1	-1	300	300/1=300
0	S_1	1	0	0	1	0	0	0	400	400/1 = 400
8	P_2	0	1	0	0	-1	0	1	200	200/1 = 200
Zj		M	8	-M	0	M -8	M	-M+8	300 M+	
Cj-Zi		5-M	0	+M	0	-M-8	0	2M-8	1600	

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Remarks

- The solution is not optimal since we have negative values ,in Cj-Zj row
- Entering variable is P_I and leaving variable is A_1

Row operation

$S_1 - A_1$=	1	0	1	0	0	0	400
	- 1	0	0	1	1	-1	300
New S_2	0	0	1	-1	-1	1	100 (shown in tableau 3 below)

Tableau 3

Cj →		**5**	**8**	**0**	**0**	**0**	**M**	**M**	
↓	**Basis**	**p_1**	**p_2**	**R_1**	**S_1**	**R_2**	**A_1**	**A_2**	**RHS**
5	P_1	1	0	-1	0	1	1	-1	300
0	S_1	0	0	1	1	-1	-1	0	400
8	P_2	0	1	0	0	-1	0	1	200
Zj		5	8	-5	0	-3	5	3	3100
Cj-Zi		0	0	5	0	3	M-5	M-3	

Tableau 3 Remarks
Tableau 3 is optimal since there is no negative value in the Cj – Zj row.

Solution values of the variables

Decision variables	**Slack variables**	**Artificial variables**	**Surplus variables**
$P_1 = 300$	$S_1 = 100$	$A_1 = 0$	$R_1 = 0$
$P_2 = 200$		$A_2 = 0$	$R_2 = 0$

Minimized cost = $3100

SPECIAL CASES OF SIMPLEX SOLUTIONS

Infeasibility

This is a condition that arises when there is no solution to a LPP that satisfies all the constraints given. Graphically, it means that no feasible solution region exists. This condition might occur if the problem was formulated with conflicting constraints. The cause of infeasibility is resource availability which may not be enough to meet the obligation.

An infeasible solution is indicated by looking at the final tableau. In it, till Cj-Zj elements will have the proper sign to imply optimality, but an artificial variable will still be among the basis to imply feasibility.

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ILLUSTRATION

Cj →		**5**	**8**	**0**	**0**	**M**	**M**	
↓	**Basis**	**x_1**	**x_2**	**S_1**	**S_2**	**A_1**	**A_2**	**RHS**
5	x_1	1	0	-2	3	-1	0	200
0	x_2	0	1	1	2	-2	0	100
M	A_2	0	0	0	-1	-1	1	20
Zj		5	8	-2	31-M	-21-M	M	1800 +
Cj-Zi		0	0	2	-31+M	21+2M	0	20M

Decision variables

$X_1 = 200$ $A_2 = 20$ $X_2 = 100$

Since A_2 is basic i.e. has non zero value in the optimal solution, then this is an infeasible solution. The cause of infeasibility is either conflicting constraints of improper formulation of LPP.

Unboundedness

A linear programming problem (LPP) is said to be unbounded if the objective function values can be Improved without limitation. For maximization problem the values can increase indefinitely while in minimization the values can decrease to zero.
Will be recognized in simplex iteration before an optimal solution is reached if all ratios are either negative or undefined (∞).

Degeneracy/Redundancy

This concept is applied with regard to constraints. A constraint which does not form part of the boundary marking the feasible region when plotted is said to be redundant. The inclusion or exclusion of such a redundant constraint does not affect the optimal solution to the problem.

NB: a redundant constraint is not necessarily a non-binding constraint

.

ILLUSTRATION

Cj → / ↓		5	8	0	0	0			Ratio
	Basis	x_1	x_2	x_2	S_2	S_2	S_3	**RHS**	
8	x_1	¼	1	1	2	0	0	10	10/0.25=40
0	S_2	4	0	1/3	-1	1	0	20	20/4=5 two smallest
0	s_2	2	0	2	2/5	0	0	10	ratio
Zj		2	8	-2	16	0	0	80	imply
Cj-Zi		3	0	-6	-16	0	0		degeneracy

Theoretically, degeneracy could lead to a situation known as cyclical in which the simplest algorithm alternate back and forth between two same non optimal solutions.

Multiple/alternate optimal solutions

Multiple or alternate optimal solutions can be spotted by examining the final tableau. If the Cj-Zj value is equal to zero for a variable that is not in the solution mix basis, then, more than one optimal solution exist.

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ILLUSTRATION

Cj → / ↓		3	2	0	0		
	Basis	x_1	x_2	x_2	S_2	**RHS**	**Ratio**
2	x_2	3/2	1	1	0	6	4
0	S_2	1	0	½	-1	3	3
Zj		3	2		0	12	
Cj-Zi		0	0		0		

Remarks

i) Solution is optimal since there is no positive element in the Cj-Zj row

ii) Although decision variable X_1 is not in the basis, its coefficient shadow price in the Cj-Z) row is zero: meaning that if it goes to the basis, the objective value (Z) will not change. Thus, it means that there is more than one optimal solution.

DUALITY THEOREM IN LPP

Associated with any LPP is another mirror image-like problem. If the original problem is called the primal, the image is known us the dual.

The dual and the primal are so much related such that all the information required to formulate one of them, will also be required to formulate the other.

Furthermore, the solution to one of them can be used to obtain the solution to the other,

If the decision variables for the primal problem are production mix (number of product mix combination), then the decision variables of the dual will be the opportunity cost or shadow prices of the resources,

The optimal solutions for the primal and the dual are equivalent but they are derived through alternative procedures. The dual contains economic information useful to management and it may also be easier to solve in terms of less computations than the primal problem.

Steps to form a dual from a primal

a) If the primal is maximization, the dual is a minimization and vice versa,
b) The RHS values of the primal constraints become the dual's objective function coefficients,
c) The primal objective function coefficients become the RHS values of the dual constraints,
d) The transpose of the primal constraint coefficients become the dual constraint coefficients.
e) Constraints inequality signs are reversed.
f) If the constraints are mixed in terms of inequality signs ensure they all face the same direction so as to correctly formulate the dual from the primal. NB: multiplying the inequality by (-1) reverses its direction.

ILLUSTRATION

Formulate the duality of the following primal LP

Maximize $Z = 12x_1 + 20x_2$

Subject to:

$3x_1 + 4x_2 \leq 96$ (material)

$6x_1 + 6x_2 \leq 168$ (labour hours)

$x_1 \leq 18$ (chair demand)

$x_1, x_2 \leq 0$ (non negativity)

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SOLUTION

Let y_i, be shadow price or the opportunity cost of constraints of resource I = 1,2,3

C = total opportunity cost (to be minimized)

Hence, Min $C = 96y_1 + 168y_2 + 18y_3$

Subject to (1) $3y_1 + 6y_2 \leq 12$

(2) $4y_1 + 6y_2 + y_3 \leq 20$

$y_1, y_2, y_3 \leq 0$

EXERCISE

Formulate the duals of the following

Primal problems	**Dual formulation solutions**
1. Max $Z = 25x_1 + 12x_2 + 15x_3$	Max $C = 20y_1 + 55y_2$
Subject to: $x_1 + x_2 \leq 20$	Subject to: $y_1 + 3x_2 \geq 25$
$3x_1 + 5x_2 + 3x_3 \leq 55$	$y_1 + 5y_2 \geq 12$
$X_1, x_2, x \geq 0$	$3y_3 \geq 15$
	$y_1, y_2, y_3 \geq 0$

Primal

2. Max $C = 20x_1 + 15x_2 + 17x_3$

Subject to: $2x_1 + 3x_2 + 4x_3 \leq 15$

$x_1 \geq 15$

$x_1 + x_3 \geq 100$

$x_1, x_2, x_3 \leq 0$

Dual

Multiply constraints (1) by – 1

To get $-2x_1 - 3x_2 - 4x_3 \geq -15$

Thus, Max $Z = -15y_1 + 5y_2 + 100y_3$

Subject to: $-2y_1 + y_2 \leq 20$

$-3y_1 + y_2 \leq 15$

$-4y_2 + y_3 \leq 17$

$y_1, y_2, y_3 \geq 0$

SHADOW PRICE

This is the increase in the objective function value that results from a one unit increase in the right-hand side of that constraint. It gives the contribution of one additional unit of a scarce resource. Graphically, a shadow price is determined by adding 1 unit to the right hand side value in question and then resolving for the optimal solution in terms of the same two binding constraints. The shadow price is equal to the difference in the values of the objective function between the new and original problems.

The shadow price for a binding constraint is non-zero while that of non-binding constraint is zero. The shadow prices can be obtained using arithmetic method or through dual formulation. NB: the solutions to a dual formulation are the shadow prices, Hence shadow prices are also culled dual prices.

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ILLUSTRATION 1

Determine the shadow prices for the following LPP

Maximize $Z = 12x_1 + 20x_2$

Subject to:

$3x_1 + 4x_2 \leq 96$ (material)

$6x_1 + 6x_2 \leq 168$ (labour hours)

$x_1 \leq 18$ (chair demand)

$x_1, x_2 \leq 0$ (non negativity)

SOLUTION

The optimal solution to the problem using the graphical method is:

$X_1 = 8$ and $X_2 = 18$, giving a contribution of \$456.

Binding constraints are material and chair demand

Shadow price for material (using arithmetic method)

Increase material by 1 unit. Hence, $3x_1 + 4x_2 \leq 97$. Graphically, this inequality shifts outwards slightly thereby expanding the feasible region.

New optimal point will have x2 = 18 substitute 18 into the new material constraints

$= 3X_1 + 4(18) = 97$

$X_1 = (97 - 72)/3 = 25/3$

New value of $Z = 12\ (25/3) + 20(18) = \$460$

Old Z = \$456

Shadow price \$4

ILLUSTRATION 2

Determine the shadow price of chair demand

Increase the chair demand by1. Hence $X_2 = 19$ to solve for X_1, substitute the value X_2 into the material constraint: $3X_1 + 4(19) = 96$

$X_1 = \frac{96-76}{3} = \frac{20}{3}$

Substitute the values of X_1, and X_2 into the objective function.

New contribution $Z = 12(20/3) + 20\ (19) = 460$

Therefore shadow price = 460 – 456 = $4

SENSITIVITY ANALYSIS

In the optimal solution to LPP we assume complete certainty in the data and relationships of a problem i.e. prices are fixed, resources are known, time' needed to produce a unit exactly set and production is instantaneous. This scenario is called deterministic assumptions. However, in the real world, conditions are dynamic and changing. To accommodate this, we relax the assumptions in LP and investigate the consequences of changes in the following:

i) The contribution rates for each variable.
ii) Technological coefficients (the numbers in the constraints equations)
iii) Available resources (the RHS quantities in each constraint).

This investigation is the one called sensitivity analysis, parametric programming, optimality analysis, post-optimality analysis or 'what if' analysis.

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If a minor change in a factor causes a relatively large change in the optimal solution, we say that the LPP is sensitive to the factor, otherwise it is insensitive or robust meaning it is tolerant to the factor. An important function of sensitivity analysis is to allow managers to experiment with values of the input parameters.

Types of sensitivity analysis

1. Changes in the RHS values of constraints. (Also called RHS ranging).
2. Changes in the objective function coefficients (profit or loss per unit), (also called coefficient ranging).
3. Changes in LHS coefficients of constraints e.g. resource input per unit of a decision variable such as materials labour hours etc,
4. the additional or removal of a constraint
5. Additional or removal of a decision variable.

The most commonly carried out sensitivity analysis are (1) and (2) due to the following reasons:

Number 3 are technical inputs which are usually dependent on technological progress hence long-term in nature. NB, LP is a short term planning.

Numbers 4 and 5 consist of changes which are so fundamental and long-term in nature, hence requiring formulation of a totally new problem.

TRANSPORTATION AND ASSIGNMENT PROBLEMS

Distribution Problems

Distribution problems are special types of mathematical problems which deals with assigning tasks or transporting items, with the objective of either 260inimize260g the gain or 260inimize260g the losses on cost.

There are two types of distribution problems namely:

1) Assignment problems
2) Transportation problems

ASSIGNMENT PROBLEMS

Assignment Models

The following example will be used as a basis of the step-by-step explanation.

ILLUSTRATION 1

A company employs services engineers based at various locations throughout the country to service and repair their equipment installed in customer's premises. Four requests for services have been received and the company finds that four engineers are available. The distances each of the engineers is from the various customers in the following table and the company wishes to assign engineers to customers to minimize the total distances to be travelled.

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Customers

	W	X	Y	Z
Alf	25	18	23	14
Bill	38	15	53	23
Charlie	15	17	41	30
Dave	26	28	36	29

Step 1. Reduce each column by the smallest figure in that column. The smallest figures are 15, 15, 23 and 14 and deducting these values from each element in the columns produces the following table.

Table 2

	W	X	Y	Z
A	10	3	0	0
B	23	0	30	9
C	0	2	18	16
D	11	13	13	15

Step 2 Reduce each row by the smallest figure in that row.

The smallest figures are 0, 0, 0 and 11 and deducting these values gives the following table.

Table 3

	W	X	Y	Z
A	10	3	0	0
B	23	0	30	9
C	0	2	18	16
D	0	2	2	4

Note: Where the smallest value in a row is zero (i.e. as in rows A, B and C above) the rows is, of course, unchanged.

Step 3 Cover all the zero in the table 3 by the *minimum possible* number of lines. The lines may be horizontal or vertical.

Table 4

	W	X	Y	Z
A	10	3	0	0
B	23	0	30	9
C	0	2	18	16
D	0	2	2	4

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Note: Line 3́, covering Row B, could equally well have been drawn covering column X.

Step 4.Compare the number of lines with the number of assignments to be made (in this example there are 3 lines and 4 assignments).If the number of line *equals* the number of assignments to be made go to step 6.

If the number of lines is less than the number of assignments to be made (i.e. as in this example which has three lines and four assignments) then

a) Find the smallest uncovered element from step 3, called **X** (in Table 4 this value is 2).
b) Subtract X to every element in the matrix.
c) Add back to every element covered by a line. If an element is covered by two lines, for example, cell **A: W** in Table 4, **X** is added twice.

Note: The effect of these steps is that **X** is subtracted from all uncovered by one line remain unchanged, and elements covered by two lines are increased by **X**.

Note: The effect of these steps is that X is subtracted from all uncovered elements, elements covered but one line remains unchanged, and elements covered by two lines are increased by X.

Carrying out this procedure on Table 4 produces the following results:

In Table 4 the smallest elements is 2. New table is

Table 5

	W	X	Y	Z
A	12	3	0	0
B	25	0	30	9
C	0	0	16	14
D	0	0	0	2

Note: It will be seen that cells A: W and B: W have been increased by 2; cells A : X, A : Y,A :Z, B :X,B:Y, B:Z, C:W and D:W are unchanged, and all other cells have been reduced by 2.

Step 5. Repeat steps 3 and step 4 until the number of lines covering the zero equals the number of assignments without any further repetition, thus:

Table 6

	W	X	Y	Z	
A	~~12~~	~~3~~	~~0~~	~~0~~	Line 1
B	~~25~~	~~0~~	~~30~~	~~9~~	Line 2
C	~~0~~	~~0~~	~~16~~	~~14~~	Line 3
D	~~0~~	~~0~~	~~0~~	~~2~~	Line 4

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Step 6 when the number of lines equals the number of assignments to be made using the following rules:

a) Assign to any zero which is unique to both a column and a row.
b) Assign to any zero which is unique to a column or a row.
c) Ignoring assignments already made repeat rule (b) until all assignments are made.

Carrying out this procedure for our example results in the following:

a) (Zero unique to both a column and a row). None in this example.
b) (Zero unique column or row). Assign B to X and A to Z. The position is now as follows.

Table 7

	W	X	Y	Z
A	Row	Satisfied		Column satisfied
B	Row	Satisfied		Column satisfied
C	0	Column Satisfied	16	Column Satisfied
D	0	Column Satisfied	0	Column Satisfied

c) Repeating rule (b) results in assigning D to Y and C to W.

Notes:

a) Should the final assignment not be to a zero, then more lines than necessary were used in step 3.
b) If a block of 4 or more zero's is left for the final assignment, then a choice of assignment exits with the same mileage.

Step 7 Calculate the total mileage of the final assignment.

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A to Z	Mileage	14
B to X		15
C to W		15
D toY		36
		80 Miles

The assignment technique for minimizing

A minimizing assignment problem typically involves making assignments so as to minimize contribution. To minimize only one step 1 from above differs-the columns are reduced by the largest number in each column. From then on the same rules apply that are used for minimization.

Maximising example

ILLUSTRATION 2

The previous example No.1 will be used with the changed assumptions that the figures relate to contribution and not mileage and that it is required to 263inimize contribution .The solution would be reached as follows.(In each case the step number corresponds to the solution given for Example No 1.)

Original data

Table 8

	W	X	Y	Z	
A	25	18	23	14	Contributions
B	38	15	53	23	to be gained
C	15	17	41	30	
D	26	28	36	29	

Step 1: Reduce each column by the largest figure in that column and ignore the resulting signs

Table 9

	W	X	Y	Z
A	13	10	30	16
B	0	13	0	7
C	23	11	12	0
D	12	0	17	1

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Step 2. Reduce each row by smallest figures in that row.

Table 10

	W	X	Y	Z
A	3	0	20	6
B	0	13	0	7
C	23	11	12	0
D	12	0	17	1

Step 3.Cover zeros by minimum possible number of lines.

Table 11

	W	X	Y	Z
A	3	0	20	6
B	0	13	0	7
C	23	11	12	0
D	12	0	17	1

Step 4. If a number of lines equals the number of assignments to be made go to step 6.If less, (as in this example), carry out the 'uncovered element' procedure previously described. This results in the following table:

Table 12

	W	X	Y	Z
A	0	0	17	6
B	0	16	0	10
C	20	11	9	0
D	9	0	14	1

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Step 5 Repeat steps 3 and step 4 until the number of lines covering the zero equals the number of assignments without any further repetition, thus:

Table 13

	W	X	Y	Z
A	0	0	17	6
B	0	16	0	10
C	20	11	9	0
D	9	0	14	1

Step 6. Make assignment in accordance with the rules previously described which result in the following assignment:

C to Z

D to X

A to W

B to Y

Step 7.Calculate contribution to gained from the assignments.

C to Z	30
D to X	28
A to W	25
B to Y	53
	136

Notes:

a) It will be apparent that minimizes assignment problems can be solved in virtually the same manner as minimizes problems.
b) The solution methods given are suitable for any size of matrix. If a problem is as small as the illustration used in this chapter, it can probably be solved merely by inspection.

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Unequal sources and destinations

To solve assignments problems in the manner described the matrix must be square, i.e. the supply must equal the requirements. Where the supply and requirements are not equal, an artificial source or destinations must be created to square the matrix. The cost/mileage/contributions etc for the fictitious column or row be zero throughout.

Solution method

Having made the sources equal the destinations, the solutions method will be as normal, treating the fictitious elements as though they were real. The solution method will automatically assign a source or destination to the fictitious row or column and the resulting assignment will incur zero cost or gain zero contribution.

NB

a) The assignment technique can be used for repairing type of problems, e.g. taxis to customers, jobs to personnel.
b) Most practical problems of size illustrated could be solved fairly readily using nothing more than commonsense. However, the technique illustrated can be used to solve much larger problems.

TRANSPORTATION PROBLEM

A transportation problem is a special type of mathematical programming that deals with the shipment or transportation of items from several sources to several destinations.
Transportation deals number of sources of supply (e.g a manufacturing company, warehouse) and a number of destinations (e,g shops, houses) so as to minimize transportation costs of supplying items from a set of source points to a set of destinations.
The usual objective is minimizes the total payoff. (Profit or cost)

A major characteristic of this problem is the linearity requirement, i.e. transport cost fom one point to another must be clearly defined, if it will cost sh.50 to transport a bag from a warehouse to shop A then it will cost sh.250 to transport 5 bags.

Transportation problems can be formulated using either a transportation table or formulating it as a inear programming format (L.P.P).

Assumptions

1) The number of sources and destinations must be known.
2) There must be a balance between supply and demand
3) The unit shipping payoff must be known
4) The units requirement at the destination and the units at the source must also be known.
5) If no balance between supply and demand a dummy source or destination has to be introduced.
6) It is not a must that there should be a balance between the number of sources and the number of destination.
7) The units can be shipped from any source to any destination.

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ILLUSTRATION 1

A computer support firm has three branches at different parts of the city, it receives orders for a total of 15 desktop computers from four customers. In total in the three branches there are 15 machines available. The management wish to minimize delivery costs by dispatching the computers from the appropriate branch for each customer.
Details of the availabilities, ‘requirements, and transport costs per computer are given in the following table.

Table 1

Cost in Ksh			Customer A	Customer B	Customer C	Customer D	Total
		Computers	3	3	4	5	15
	Branch X.	2	13	11	15	20	transportation
Available	Branch Y	6	17	14	12	13	cost
	Branch Z	7	18	18	15	12	per unit
	Total	15					

Solution

Step 1 Make an initial feasible allocation of deliveries by selecting the cheapest route first, and allocate as many as possible then the next cheapest and so on. The result of such an allocation is as follows.

Table 2

			Requirement			
			A	**B**	**C**	**D**
		Computers	3	3	4	5
	X	2 Units		2 (1)		
Available	Y	6 Units	1 (4)	1 (3)	4 (2)	
	Z	7 Units	2 (5)			5 (2)

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Note: the number in the table represent deliveries of computers and the number in the brackets (1), (2), etc represent the sequence in which they are inserted, lowest cost first i.e.

		Sh.
1. 2 units X → B sh.11/unit	Total cost	22
2. 4 units Y → C sh.12/unit	Total cost	48
5 units Z → D sh.12/unit	Totals cost	60
3. The next lowest cost move which is feasible i.e. doesn't exceed row or column totals is 1 unit Y → B sh.14/unit		14
4. similarly the next lowest feasible allocation 1 unit Y→ A sh.17/unit		17
5. finally to fulfill the row /column totals 2 units Z → A sh.18/unit		36
		197

Step 2 Check solution obtained to see if it represents the minimum cost possible. This is done by calculating 'shadow costs' (i.e. an imputed cost of not using a particular route) and comparing these with the real transport costs to see whether a change of allocation is desirable.

This is done as follows:

Calculate a nominal 'dispatch' and 'reception' cost for each occupied cell by making an assumption that the transport cost per unit is capable of being split between dispatch and reception costs thus:

D(X) + R(B) = 11
D(Y) + R(A) = 17
D(y) + R(B) = 14
D(Y) + R© = 12
D(Z) + *R(A)* = 18
D(Z) + R(D) = 12

Where D(X), D(Y) and D(Z) represent Dispatch cost from depots X, Y and Z, and R(A) R(B), R(C) and R(D) represent Reception costs at customers A, B, C, D.

By convention the first depot is assigned the value of zero i.e. D(X) = 0 and this value is substituted in the first equation and then all the other values can be obtained thus

R(A)	= 14	D(X)	= 0
R(B)	= 11	D(Y)	= 3
R(C)	= 9	D(Z)	= 4
R(D)	= 8		

Using these values the shadow costs of the unoccupied cells can be calculated. The unoccupied cells are X : A, X : C, X : D, Y : D, Z : B, Z : C.

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					Shadow costs
D(X) + R(A)	= 0	+	14	=	14
D(X) + RI	= 0	+	9	=	9
D(X) + R(D) =	0	+	8	=	8
	3	+	8	=	11
D(Z) + R(B)	= 4	+	11	=	15
D(Z) + RI	= 4	+	9	=	13

These computed 'shadow costs' are compared with the actual transport costs (from *Tab*- I), Where the *actual* costs are *less than shadow costs,* overall costs can be reduced by allocating units into that cell.

	Actual cost	-	Shadow cost	+ Cost increase - Cost reduction
CellX:A	13	-	14	= -1
X : C	15	-	9	= +6
X : D	20	-	8	= + 12
Y: D	13	-	11	= +2
Z : B	18	-	15	= +3
Z : C	15	-	13	= +2

The meaning of this is that total costs could be reduced by sh.1 for every unit that can be transferred into cell X : A. As there is a cost reduction that can be made the solution , Table 2 is not optimum.

Step 3: Make the maximum possible allocation of deliveries into the cell where actual costs are less than shadow costs using occupied cells i.e.

Cell X : A from Step 2, The number that can be allocated is governed by the need to keep within the row and column totals. This is done as follows:

Table 3

			Requirement			
			A	B	C	D
			3	3	4	5
	X	2 Units	+	2 -		
Available	Y	6 Units	1 -	1 +	4	
	Z	7 Units	2			5

Table 3 is a reproduction of Table 2 with a number of + and – inserted. These were inserted for the following reasons.

Cell X : A + indicates a transfer *in* as indicated in Step 2

Cell X : B – indicates a transfer *out* to maintain Row X total.

Cell Y : B + indicates a transfer *in* to maintain Column B total

Cell Y : A – indicates a transfer *out* to maintain Row Y and Column A totals.

The maximum number than can be transferred into Cell X : A is the lowest number in the Minus cells i.e. cells Y : *A*, and X : B which is 1 unit.

Therefore 1 unit is transferred in the + and – sequence described above resulting in the following table

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Table 4

			Requirement			
			A	B	C	D
			3	3	4	5
	X	2 Units	1	1		
Available	Y	6 Units		2	4	
	Z	7 Units	2			5

The total cost of this solution is

		Sh.
Cell X:A	1 unit @ sh.13	= 13
Cell X:B	1 Unit @ sh.11	= 11
Cell Y:B	2 Units @ sh.14	= 28
Cell Y:C	4 Units @ sh.12	= 48
Cell Z:A	2 Units @ sh.18	= 36
Cell Z:D	5 Units @ sh.12	= 60
		196

The new total cost is sh.1 less than the total cost established in Step 1. This is the result expected because it was calculated in Step 2 that sh.1 would be saved for every unit we were able to transfer to Cell X : A and we were able to" transfer 1 unit only.

Notes: Always commence the + and – sequence with a + in the cell indicated by the (actual cost – shadow cost) calculation. Then put a – in the occupied cell in the same row which has an occupied

cell in its column. Proceed until a – appears in the same column as the original +.

Step 4 Repeat Step 2 i.e. check that solution represents minimum cost. Each of the processes in Step 2 are repeated using the latest solution (Table 4) as a basis, thus: Nominal dispatch and reception costs for each occupied cell.

D(X) + R(A) = 13
D(X) + R(B) = 11
D(y) + R(B) = 14
D(Y) + R(C) = 12
DZ) + *R(A)* = 18
D(Z) + R(D) = 12

On setting D(X) to be 0, the rest of the values are found to be

R(A) = 13 D(X) = 0
R(B) = 11 D(Y) = 3
R(C) = 9 D(Z) = 5
R(D) = 7

Using these values the shadow costs of the unoccupied cells are calculated. The unoccupied cells are X:C , X:D, Y:A, Y:D, Z:B, and Z:C

Therefore;

D(X) + RI = 9
D(X) + R(D) = 7
D(Y) + R(A) = 16
D(Y) + R(D) = 10
D(Z) + R(B) = 16
D(Z) + RI = 14

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The computed shadow costs are compared with actual costs to see if any reduction in cost is possible.

	Actual cost	-	Shadow cost	+ Cost increase - Cost reduction
Cell X :C	15	-	9 =	+6
X:D	20	-	7=	+13
Y:A	17	-	16 =	+1
Y:D	13	-	10 =	+3
Z:B	18	-	16 =	+2
Z:C	15	-	14 =	+1

It will be seen that all the answers are positive, therefore no further cost reduction is possible and optimum solution has been reached.
Thus the optimal solution is represented by *table 4*

UNEQUAL SUPPLY AND DEMAND QUANTITIES

ILLUSTRATION 2

Wanjiru books supplies in a firm dealing with import of books and it has three stores strategically situated around the country. Yesterday the company received orders to supply 100 books from 4 schools, of the books ordered the firm has 110 books in stock. The firm wishes to minimize cost and its seeking your advice, advise the firm.

Below is a table of availability and requirement;

			Required				
			Sch. A	Sch. B	Sch. C	Sch. D	Total
		Books	25	25	42	8	100
	Store I	40	Sh.3	16	9		transport
	Store II	20	Sh.1	9	3	8	costs per
Available	Store III	50	Sh.4	5	2	5	Book
	Total	110					

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Solution

Step 1*:* add a dummy destination to table 5 with zero transport costs and requirements equal to the surplus availability.

			Required					
			Sch. A	Sch. B	Sch. C	Sch. D	Dummy	Total
		Books	25	25	42	8	10	100
	Store I	40	Sh.3	16	9		0	transport
	Store II	20	Sh.1	9	3	8	0	costs per
Available	Store III	50	Sh.4	5	2	5	0	Book
	Total	110						

***Step* 2** Now that the quantity available equals the quantity required (because of insertion of the dummy) the solution can proceed in exactly the same manner described in the first example. First set up an initial feasible solution

			Requirement				
			A	B	C	D	Dummy
			25	25	42	8	10
	I	40	5 [4]	17 [6]		8 [3]	10 [7]
Available	II	20	20 [1]				
	III	50		8 [5]	42 [2]		

The numbers in the table represent the allocations made and the numbers in brackets represent the sequence they were inserted based on lowest cost and the necessity to maintain row/column totals. The residue of 10 was allocated to the dummy. The cost of this allocation is

	Sh.	**Sh.**
I→A	5 units @ 3	15
I→B	17 units @ 16	272
I→D	8units @ 2	16
I→Dummy	10 units @ zero cost	
II→A	20 units @ 1	20
III→B	8 units @ 5	40
III→C	42 units @ 2	84
		447

Step 3. Check solution to see if it represents the minimum cost possible in the same manner as previously described i.e.

Dispatch & Reception Costs of used routes:

D(I) + R(A)	= 3
D(I) + R(B)	= 16
D(I) + R(D)	= 2
D(I) + R(Dummy)	= 12
D(II) + *R(A)*	= 1
D(III) + R(B)	= 5
D(III) + RI	= 2

Setting D(I) at zero the following values are be obtained

R(A)	=3	D(I)	=0
R(B)	=16	D(I)	=-2
R(C)	=13	D(III)	=-11
R(D)	=2		
R(Dummy)	=0		

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Using these values the shadow costs of the unused routes can be calculated .The unused routes are I:C,II:B,II:C,II:D,II:Dummy,III:D,and Dummy

ShadowCosts

			£	
D (I) +	RI	=	0+13	=13
D (II). +	R (B)	=	-2+16	=14
D (II). +	RI	=	-2+13	=11
D (II) +	R (D)	=	-2+ 2	=0
D (II) +	R (Dummy)	=	-2+0	=-2
D (III) +	R (A)	=	-11+3	=-8
D (III) +	R (D)	=	-11+2	=-9
D (III) +	R (Dummy)	=	-11+0	=-11

The shadow costs are then deducted from actual costs

It will be seen that total cost can be reduced by £8 per unit for every unit that can be transferred into Cell II:C

Step4.Make the maximum possible allocation of deliveries into Cell II:C.This is done by inserting a sequence of +and -,maintaining row and column totals.

			Requirements				
			A	B	C	D	Dummy
			25	25-	42	8	10
	I	40	5+	17-		8	10
Available	II	20	20-				
	III	50		8+	42-		

The maximum transferable number is the lowest number in the minus cell, i.e. 17. After the transfer is made we get;

			A	B	C	D	Dummy
			25	25-	42	8	10
	I	40	22	0		8	10
Available	II	20	3		17		
	III	50		25	25		

Step 3 is repeated again to check if the cost is minimum after setting D(I) = 0.
In our case after deducting shadow costs from actual costs we find that there are no more negative numbers thus we deduce from the last table that the minimum transportation cost is,

$(22\times3) + (8\times2) + (10\times0) + (3\times1) + (17\times3) + (25\times5) + (25\times2) = Sh.311$

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Maximization using Transportation

Transportation problems are usually minimizing problems, on occasions problems are framed so that the objective is to make the allocations from sources to destinations in a manner which maximizes contribution or profit. These problems are dealt with similar to minimizing problems but the reverse of it. i.e.

a) Make initial feasible allocation on basis of maximum contribution first, then next highest and so on.
b) For optimum, the differences between actual and shadow contributions for the unused routes should be *all negative.* If not, make allocation into cell with the *largest positive* difference.
c) In case there are more items available than are required, a dummy destination with zero contribution should be introduced and the maximizing procedure in ***a).*** followed

PRACTICE EXERCISES

QUESTION 1

A company wishes to purchase additional machinery in a capital expansion program. Three types of machines are to be purchased: A, B, and C. Machine A costs $25,000 and requires 200 square feet of floor space for its operation. Machine B costs $30,000 and requires 250 square feet of floor space. Machine C costs $22,000 and requires 175 square feet of floor space. The total budget for this expansion program is $350,000. The maximum available floor space for the new machines is 4,000 square feet. The company also wishes to purchase at least one of each machine.

Given that machines A, B, and C can produce 250, 260, and 225 pieces per day, the company wants to determine how many machines of each type it should purchase so as to maximize daily output (in units) from the new machines.

a) Explicitly define your decision variables and formulate the LP model.
b) Assess the validity of the four underlying LP assumptions for this problem.
c) Solve and analyse the problem using a computer package

Solution:

a) Let a, b, and c, be number of machines A, B, and C. These are the decision variables.

Formulation of LP model

Maximise

Output $U = 250a + 260b + 225c$

Subject to the constraints.

Capital budget $25a + 30b + 22c \leq 350$ '000'

Floor space $200a + 250b + 175c \leq 4000$ Square feet

$a, b, c \geq 1$

b) Linear / Proportion – the number of units with capital budget and floor space are linearly related.

Deterministic – the coefficients for the variables and constraints are known with certainty.

Additive – Buying one more of a given machine gives more production or additional production. Effect is additive.

Divisible – This requires that the machines and given constraints to be divisible. In this case the assumption does not hold. Here we have to take a machine as a whole and not ½ or ¼ or fraction of the machine.

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c) Computer solution and analysis.

Target Cell (Max)

Name	**Original Value**	**Final Value**
zfunc sol	0	3527.045455

Adjustable Cells

Name	**Original Value**	**Final Value**
sol a	0	1
sol b	0	1
sol c	0	13.40909091

Constraints

Name	**Cell Value**	**Status**	**Slack**
capbudget '000' sol	350	Binding	0
flospace (sq ft) sol	2796.590909	Not Binding	1203.409
sol a	1	Binding	0

sol b	1	Binding	0
sol c	13.40909091	Not Binding	12.40909

Adjustable Cells

Name	Final Value	Reduced Gradient
sol a	1	-5.681818182
sol b	1	-46.81818182
sol c	13.40909091	0

Constraints

Name	Final Value	Dual Price
capbudget '000' sol	350	10.22727273
flospace (sq ft) sol	2796.590909	0

Target Name	Value
zfunc sol	3527.045455

Adjustable Name	Value	Lower Limit	Target Result	Upper Limit	Target Result
sol a	1	1	3527.04	1	3527.04
sol b	1	1	3527.045	1	3527.04
sol c	13.40909091	1	735	13.40909091	3527.04

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The solution of the problem is as follows

The number of machines to buy is:

A=1

B=1

C=13

The maximum output is 3527 pieces per day.

The capital budget will be used up completely and the floor space will be having a slack of 1203 square feet. So the dual price of the capital budget is $10.22.

Note: In exams, the solution from a computer package will be given and the student will be required to interpret the solution

QUESTION 2

A pension fund wishes to invest in one or more of six possible investments. Financial analysts have estimated the present value of effective annual estimate. The data in this table indicate that the present value of investing $10,000 in alternative 1 is the sum of $1,200 (0.12×$10,000) for year 1, $1,000 (0.10×10,000) for year 2, and $800 (0.08×$10,0000 for year 3, for a total present value of $3,000.

Effective Annual Rate of return

Investment	Year 1	Year 2	Year 3
1	0.12	0.10	0.08
2	0.14	0.10	0.10
3	0.15	0.12	0.08
4	0.10	0.12	0.15
5	0.08	0.12	0.18
6	0.25	0.15	0.05

Management has decided that $300,000 will be invested. At least $50,000 is to be invested in alternative 2 and no more than $40,000 in alternative 5. Total investment in alternative 4 and 6 should not exceed $75,000, as these are risky investments.

If the objective is to maximize the present value of total dollar return for the three-year period, formulate the LP model for how much capital to invest in each alternative.

Can you comment on how relevant each underlying LP assumption is to this problem?

Solution:

Let x_1,x_2,x_3,x_4,x_5, andx_6 be the amount invested for the 3-year period for alternatives 1, 2, 3, 4, 5, and 6. Then the objective function will be:

z – Total dollar return.

Maximize $z = 0.3x_1 + 0.34x_2 + 0.35x_3 + 0.37x_4 + 0.38x_5 + 0.45x_6$

Subject to the constraints:

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 300{,}000$ $ Capital budget.

$x_2 \geq 50{,}000$ $ min limit of alternative 2

$x_5 \leq 40{,}000$ $ max limit of alternative 5

$x_4 + x_6 \leq 75{,}000$ $ max limit of total of 4 and 6

$x_1,x_2,x_3,x_4,x_5,x_6 \geq 0$

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Effective Annual rate of return

Investment	Year 1	Year 2	Year 3	Total
1	0.12	0.10	0.08	0.3
2	0.14	0.10	0.10	0.34
3	0.15	0.12	0.08	0.35
4	0.10	0.12	0.15	0.37
5	0.08	0.12	0.18	0.38
6	0.25	0.15	0.05	0.45

Computer solution and analysis.

Target Cell (Max)

Name	Original Value	Final Value
return sol	0	113200

Adjustable Cells

Name	Original Value	Final Value
x1	0	0
x2	0	50000
x3	0	135000
x4	0	0
x5	0	40000
x6	0	75000

Constraints

Name	Cell Value	Status	Slack
capbudget	300000	Binding	0
limitalt2	50000	Binding	0
limitalt5	40000	Binding	0
limitalt4&6	75000	Binding	0
x1	0	Binding	0
x2	50000	Not Binding	50000
x3	135000	Not Binding	135000
x4	0	Binding	0
x5	40000	Not Binding	40000
x6	75000	Not Binding	75000

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Name	Final Value	Reduced Gradient
x1	0	-0.0499
x2	50000	0
x3	135000	0
x4	0	-0.0801
x5	40000	0
x6	75000	0

Name	Final Value	Dual Price
capbudget sol	300000	0.35

limitalt2	50000	-0.01
limitalt5	40000	0.03
limitalt4&6	75000	0.1

Target

Name	**Value**
Return sol	113200

Adjustable Name	**Value**	**Lower Limit**	**Target Result**	**Upper Limit**	**Target Result**
sol x1	0	0	113200	0	113200
sol x2	50000	50000	113200	50000	113200
sol x3	135000	135000	113200	135000	113200
sol x4	0	0	113200	0	113200
sol x5	40000	40000	113200	40000	113200
sol x6	75000	75000	113200	75000	113200

The present amount value to be invested in each of the alternatives is:

Alternative 1- $0
Alternative 2- $50,000
Alternative 3- $135,000
Alternative 4- $0
Alternative 5- $40,000
Alternative 6- $7,5000

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There is no slack in any of the constraints. The dual prices are:

Capital budget $0.35
Limitation on alternative 2 $0.01
Limitation on alternative 5 $0.03
Limitation on alternative4 and 6 $0.1

Note: Even though solution was not requested, it has been presented to add on intepretation of LP computer analysis.

b) Assumptions:

- Proportional – Increase in more investment ⇒ more return.
- Deterministic – Estimates of present value for each alternative (though estimated).
- Additive – To make up total dollar return, the investment return are added together.
- Divisible – Fractional amounts of the investment options are possible.

QUESTION 3

a) A small company will be introducing a new line of lightweight bicycle frames to be made from special aluminium alloy and steel alloy. The frames will be produced in two models, deluxe and professional. The anticipated unit profits are currently Sh.1,000 for a deluxe frame and Sh.1,500 for a professional frame. The number of kilogrammes of each alloy needed per frame is

summarized in the table below. A supplier delivers 100 kilogrammes of the aluminium alloy and 80 kilogrammes of the steel alloy weekly.

	Aluminium alloy	Steel alloy
Deluxe	2	3
Professional	4	2

Required:

i) Determine the optimal weekly production schedule.

ii) Within what limits must the unit profits lie for each of the frames for this solution to remain optimal?

b) Explain the limitations of the technique you have used to solve part (a) above.

Solution:

a)

i) Simplex method will be appropriate.

Formulation of problem.

Objective function.

Let x_1 and x_2 be the number of Deluxe and Professional bicycle frames produced respectively per week.

$z = 1000x_1 + 1500x_2$ Profit sh.

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Constraints:

$2x_1 + 4x_2 \leq 100$ Aluminum alloy

$3x_1 + 2x_2 \leq 80$ Steel alloy

$x_1, x_2 \geq 0$

In standard form:

$0 = z - 1000x_1 - 1500x_2 + 0s_1 + 0s_2$

$100 = 2x_1 + 4x_2 + s_1 + 0s_2$

$80 = 3x_1 + 2x_2 + 0s_1 + s_2$

Table 1

	x_1	x_2	s_1	s_2	Solution	Ratio	
s_1	2	4	1	0	100	25	⇐
s_2	3	2	0	1	80	40	
z	-1000	-1500	0	0	0		
	⇑						

Table 2

x_2	1/4	1	1/4	0	25	50	
s_2	2	0	-1/2	1	30	15	⇐
z	-250	0	375	0	37,500		
	⇑						

Table 3

x_2	0	1	3/8	-1/4	17.5	
x_1	1	0	-1/4	1/2	15	
z	0	0	312.5	125	41,250	

Stop here

The optimal weekly production schedule is as follows:
Deluxe bicycle Frame = 17.5 ≈17
Professional bicycle Frame = 15

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ii) Let Δ_1 be the change in profit from Deluxe bicycle frame.
Δ_2 be the change in profit from Professional bicycle frame. So
$C_1 = 1000 + \Delta_1$ and $C_2 = 1500 + \Delta_2$ limit of profit.

From the final table:
To avoid entry of

s_1 $\Rightarrow 312.5 - 1/4\Delta_1 > 0$ $\Rightarrow$ $\Delta_1 < 1250$

s_2 $\Rightarrow 125 + 1/2\Delta_1 > 0$ $\Rightarrow$ $\Delta_1 > -250$

From the two conditions:
$-250 < {}_1 < 1250$ and
$750 < C_1 < 2250$

To avoid entry of

s_1 $\Rightarrow 312.5 + 3/8\Delta_2 > 0$ $\Rightarrow \Delta_2 > -833.33$

s_2 $\Rightarrow 125 - 1/4\Delta_2 > 0$ $\Rightarrow -\Delta_2 > -500 \Rightarrow \Delta_2 < 500$

So from the two conditions:
$-833.33 < {}_2 < 500$
And C_2 varies as follows

$666.7 < C_2 < 2000$

NOTE: This problem could be solved graphically with part (i) Easily determined. Part (ii) Limits will be determined from equating slopes of the objective function which has coefficients with constraints nearest to it.

For part (ii), accurate drawings will be required. Intuition will have to be followed and there will be an assumption that fractions are possible.

b) The technique is really involving.

Assumes fractions are possible, which is not really the case like here where we cannot make ½ a bicycle frame.

QUESTION 4

a) Define the following terms as used in linear programming:
   i) Feasible solution
   ii) Transportation problem
   iii) Assignment problem

b) The TamuTamu products company ltd is considering an expansion into five new sales districts. The company has been able to hire four new experienced salespersons. Upon analysing the new salesperson's past experience in combination with a personality test which was given to them, the company assigned a rating to each of the salespersons for each of the districts .These ratings are as follows:

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		Districts				
		1	**2**	**3**	**4**	**5**
Salespersons	**A**	92	90	94	91	83
	B	84	88	96	82	81
	C	90	90	93	86	93
	D	78	94	89	84	88

The company knows that with four salespersons, only four of the five potential districts can be covered.

Required:

i) The four districts that the salespersons should be assigned to in order to maximize the total of the ratings
ii) Maximum total rating.

Solution:

a)

i) A feasible solution is one that satisfies the objective function and given constraints
ii) Transportation problem is a special linear programming problem where there a number of sources and destinations and an optimum allocation plan is required. Total demand equal total supply

iii) Assignment problem is a special kind of transportation problem where the number of sources equals the number of destinations. That means for every demand there is one supply.

b) This is a case of assignment problem.

Assignment problems usually require that the number of sources equal the number of supply. Here there are 5 districts and only 4 salespersons. A dummy salesperson E is introduced with zero ratings.

		Districts				
		1	2	3	4	5
	A	92	90	94	91	83
Sales persons	B	84	88	96	82	81
	C	90	90	93	86	93
	D	78	94	89	84	88
	E	0	0	0	0	0

By following the Hungarian method:

Firstly:

For each row, the lowest rating is reduced from each rating in the particular row. This results to a row reduced rating table. Then all the zeroes are to be crossed by the least number of vertical and horizontal lines. If the number of lines equal the number of rows (or columns = 5 in this case) then the final assignment has been determined. Otherwise the following steps are followed.

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	1	2	3	4	5
A	9	7	11	8	0
B	3	7	15	1	0
~~C~~	~~4~~	~~4~~	~~7~~	~~0~~	~~7~~
~~D~~	~~0~~	~~16~~	~~11~~	~~6~~	~~10~~
~~E~~	~~0~~	~~0~~	~~0~~	~~0~~	~~0~~

Secondly, for each column, the lowest rating is reduced from every rating in the particular column. In this case the table will remain the same since the dummy salesperson has ratings of zero for every district.

Thirdly a revision of the opportunity-rating table is done.

The smallest rating in the table not covered by the lines is taken (in this case it is one). This is reduced from all the uncrossed ratings and added to the ratings at the intersection of the crossings. Then all the zeroes are to be crossed by the least number of vertical and horizontal lines. If the number of lines equal the number of rows (or columns = 5 in this case) then the final assignment has been determined.

Otherwise the following steps are followed.

	1	2	3	4	5
A	8	6	10	8	0
B	2	6	14	0	0
C	4	4	7	0	8
D	0	16	11	6	11
E	0	0	0	0	1

Third step is repeated as follows:

	1	2	3	4	5
A	6	4	8	8	0
B	0	4	12	0	0
C	2	2	5	0	8
D	0	16	11	8	13
E	0	0	0	2	3

Still the optimal solution has not been reached. Third step is again repeated to give the following table:

	1	2	3	4	5
A	6	2	6	8	0
B	0	2	10	0	0
C	2	0	3	0	8
D	0	14	9	8	13
E	0	0	0	4	5

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An optimal assignment can now be determined since the number of lines crossing the ratings is equal to 5.

Lastly, the assignment procedure is that a row or column with only one zero is identified and assigned. This row or column is now eliminated. The other zeroes are then assigned until the last zero is assigned. This step-by-step assignment is shown on the following table from the first one to the fifth one.

		District				
		1	**2**	**3**	**4**	**5**
	A	6	2	6	8	[0]^{4}
Sales person	**B**	0	2	10	[0]^{5}	0
	C	2	[0]^{3}	3	0	8
	D	[0]^{1}	14	9	8	13
	E	0	0	[0]^{2}	4	5

The assignment is as follows

Salesperson	District	Rating
A	5	83
B	4	82
C	2	90
D	1	78
Total rating		333

The total rating is 333.

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TOPIC 7

DECISION THEORY

INTRODUCTION

Decision theory is a body of knowledge and related analytical techniques of different degrees of formality designed to help a decision maker choose among a set of alternatives in light of their possible consequences. Decision theory can apply to conditions of **certainty, risk, or uncertainty**.
In
It helps operations mangers with decisions on process, capacity, location and inventory, because such decisions are about an uncertain future.

Types of decisions

There are many types of decision making

1. **Decision making under uncertainty**
   Decision under certainty means that each alternative leads to one and only one consequence and a choice among alternatives is equivalent to a choice among consequences.
2. **Decision making under certainty**
   Whenever there exists only one outcome for a decision we are dealing with this category e.g. linear programming, transportation assignment and sequencing e.t.c.
3. **Decision making using prior data**
   It occurs whenever it is possible to use past experience (prior data) to develop probabilities for the occurrence of each data
4. **Decision making without prior data**
   No past experience exists that can be used to derive outcome probabilities in this case the decision maker uses his/her subjective estimates of probabilities for various outcomes

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DECISION MAKING UNDER UNCERTAINTY

Several methods are used to make decision in circumstances where only the pay offs are known and the likelihood of each state of nature are known

a) MAXIMIN METHOD

This criteria is based on the 'conservative approach' to assume that the worst possible is going to happen. The decision maker considers each strategy and locates the minimum pay off for each and then selects that alternative which maximizes the minimum payoff

Illustration

Rank the products A B and C applying the Maximin rule using the following payoff table showing potential profits and losses which are expected to arise from launching these three products in three market conditions
(see table 1 below)

Pay off table in £ 000's				
	Boom condition	**Steady state**	**Recession**	**Mini profits row minima**
Product A	+8	1	-10	-10
Product B	-2	+6	+12	-2
Product C	+16	0	-26	-26

Table 1

Ranking the MAXIMIN rule = BAC

b) MAXIMAX METHOD

This method is based on 'extreme optimism' the decision maker selects that particular strategy which corresponds to the maximum of the maximum pay off for each strategy

ILLUSTRATION

Using the above example

	Max. profits row maxima
Product A	+8
Product B	+12
Product C	+16

Ranking using the MAXIMAX method = CBA

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c) MINIMAX REGRET METHOD

This method assumes that the decision maker will experience 'regret' after he has made the decision and the events have occurred. The decision maker selects the alternative which minimizes the maximum possible regret.

Illustration

Regret table in £ 000's				
	Boom condition	Steady state	Recession	Mini regret row maxima
Product A	8	5	22	22
Product B	18	0	0	18
Product C	0	6	38	38

A regret table (table 2) is constructed based on the pay off table. The regret is the 'opportunity loss' from taking one decision given that a certain contingency occurs in our example whether there is boom steady state or recession

The ranking using MINIMAX regret method = BAC

d) THE EXPECTED MONETARY VALUE METHOD

The expected pay off (profit) associated with a given combination of act and event is obtained by multiplying the pay off for that act and event combination by the probability of occurrence of the given event. The expected monetary value (EMV) of an act is the sum of all expected conditional profits associated with that act

Illustration

A manager has a choice between

i) A risky contract promising shs 7 million with probability 0.6 and shs 4 million with probability 0.4 and

ii) A diversified portfolio consisting of two contracts with independent outcomes each promising Shs 3.5 million with probability 0.6 and shs 2 million with probability 0.4

Can you arrive at the decision using EMV method?

Solution

The conditional payoff table for the problem may be constructed as below.

(Shillings in millions)

Event E1	Probability (E1)	Conditional pay offs decision		Expected pay off decision	
	(i)	Contract (ii)	Portfolio(iii)	Contract (i) x (ii)	Portfolio (i) x (iii)
E1	0.6	7	3.5	4.2	2.1
E2	0.4	4	2	1.6	0.8
			EMV	5.8	2.9

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Using the EMV method the manager must go in for the risky contract which will yield him a higher expected monetary value of shs 5.8 million

e) EXPECTED OPPORTUNITY LOSS (EOL) METHOD

This method is aimed at minimizing the expected opportunity loss (OEL). The decision maker chooses the strategy with the minimum expected opportunity loss

f) THE HURWIZ METHOD

This method was the concept of coefficient of optimism (or pessimism) introduced by L. Hurwicz. The decision maker takes into account both the maximum and minimum pay off for each alternative and assigns them weights according to his degree of optimism (or pessimism). The alternative which maximizes the sum of these weighted payoffs is then selected

g) THE LAPLACE METHOD

This method uses all the information by assigning equal probabilities to the possible payoffs for each action and then selecting that alternative which corresponds to the maximum expected pay off

ILLUSTRATION

A company is considering investing in one of three investment opportunities A, B and C under certain economic conditions. The payoff matrix for this situation is economic condition.

Investment opportunities	1 £	2 £	3 £
A	5,000	7,000	3,000
B	-2,000	10,000	6,000
C	4,000	4,000	4,000

Determine the best investment opportunity using the following criteria.

i) Maximin
ii) Maximax
iii) Minimax
iv) Hurwicz (Alpha = 0.3

SOLUTION

Economic condition

Investment opportunities	1 £	2 £	3 £	Minimum £	Maximum £
A	5000	7000	3000	3000	7000
B	-2000	10000	6000	-2000	10000
C	4000	4000	4000	4000	4000

i) Using the Maximin rule Highest minimum = £ 4000

Choose investment C

ii) Using the Maximax rule Highest maximum = £ 10000

Choose investment B

iii) Minimax Regret rule

	1	2	3	Maximum regret
A	0	3000	3000	3000
B	7000	0	0	7000
C	1000	6000	2000	6000

Choose the minimum of the maximum regret i.e. £3000

Choose investment A

iv) Hurwicz rule: expected values

For A (7000 x 0.3) + (3000 x 0.7) = 2100 + 2100 = £4200

For B (10000 x 0.3) + (-2000 x 0.7) = 3000- 1400 = £ 1600

For C (4000 x 0.3) + (4000 x 0.7) = 1200 + 2800 = £ 4000

Best outcome is £ 4200 choose investment A

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Value of perfect information

It relates to the amount that we would pay for an item of information that would enable us to forecast the exact conditions of the market and act accordingly. The Value of perfect information is the amount by which the expected payoff will improve if the manager knows which event will occur.

The expected value of perfect information EVPI is the expected outcome with perfect information minus the expected outcome without perfect information namely the maximum expected monetary value.

ILLUSTRATION

From table 1 above and given that the probabilities are Boom 0.6, steady state 0.3 and recession 0.1 then

When conditions of the market are; boom launch product C: profit = 16

When conditions of the market are; steady state launch product B: profit = 6

When conditions of the market are; recession launch product B: profit = 12

The expected profit with perfect information will be (16 x 0.6) + (6 x 0.3) + (12 x 0.1) = 12.6

our expected profit choosing product C is 7the maximum price that we would pay for perfect information is 12.6 – 7 = 5.6

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DECISION TREES AND SUB SEQUENTIAL DECISIONS

A decision tree is a graphic display of various decision alternatives and the sequence of events as if they were branches of a tree. It is the general approach to a wide range of decisions such as, product planning, process management, capacity, and location. It is particularly valuable for evaluating different capacity expansion alternatives when demand is uncertain and sequential decisions art involved. For example, a company may expand a facility in 1996 only to discover in 1998 that demand is much higher than forecasted. In that case, a second decision may be necessary to determine whether to expand once again or build a second facility.

A decision tree is a schematic model of alternatives available to the decision maker, along with their possible consequences. The name derives from the tree- like appearance of the model. It consists of a number of square nodes representing decision points that are left by branches (which should be read from left to right), representing the alternatives. Branches leaving circular, or chance, nodes represent the events. The probability of each chance event, P(E), is shown above each branch. The probabilities for all branches leaving a chance node must sum to 1.0. The conditional payoff, which is the payoff for each possible alternative event combination, is shown at the end of each combination. Pay offs are given only at the outset, before the analysis begins, for the end points of each alternative event combination.

Symbols

- The symbol ▭ and ◯ indicates the decision point and the situation of uncertainty or event respectively. The node depicted by a square is a **decision node** while **outcome nodes** are depicted by a circle.
- Decision nodes: points where choices exist between alternatives and managerial decisions is made based on estimates and calculations of the returns expected.
- Outcome nodes are points where the events depend on probabilities

ILLUSTRATION 1

A retailer must decide whether to build a small or a large facility at a new location. Demand at the location can be either small or large with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high the manager may choose not to expand (payoff = Sh223,000) or to expand (payoff = Sh270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is Sh200,000. If a large facility is built and demand proves to be low, the choice is to do nothing (Sh40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only Sh20,000; the payoff grows to Sh220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is Sh800,000. ***Draw a decision tree***. Then analyze it to determine the expected payoff for each decision and event node. Which alternatives building a small facility or building a large facility, the higher expected payoff?

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Solution The decision tree in Figure below shows the event probability and the pay-off for each of the seven alternative event combinations. The first decision is whether to build a small or a large facility. Its node is shown first, to the left because it is the decision the retailer must make now. The second decision node - whether to expand at a later date is reached only if a small facility is built and demand turns out to be high. Finally the third decision point - whether to advertise-is reached only if the retailer builds a large facility and demand turns out to be low.

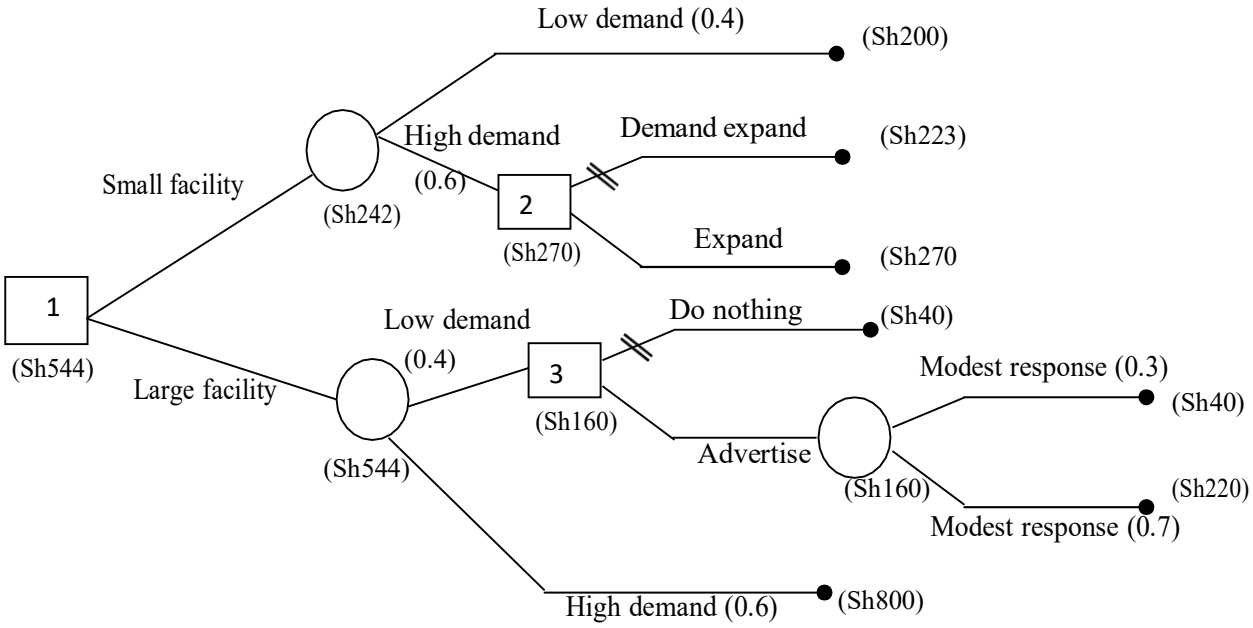


ILLUSTRATION 2

Kauzi Agro mills ltd (KAM) is considering whether to enter a very competitive market. In case KAM decided to enter this market it must either install a new forging process or pay overtime wages to the entire workers. In either case, the market entry could result in

i) high sales
ii) medium sales
iii) low sales
iv) no sales

a) Construct an appropriate tree diagram
b) Suppose the management of KAM has estimated that if they enter the market there is a 60% chance of their stakeholders approving the installation of the new forge. (this means that there is a 40% chance of using overtime) a random sample of the current market structure reveals that KAM has a 40% chance of achieving high sales, a 30% chance of achieving medium sales, a 20% chance of achieving low sales and a 10% chance of achieving no sales. Construct the appropriate probability tree diagram and determine the joint probabilities for various branches
c) Market analysts of KAM have indicated that a high level of sales will yield shs 1,000,000 profit; a medium level of sales will result in a shs 600000 profit a low level of sales will result in a shs 200000 profit and a no sales level will cause KAM a loss of shs 500000 apart from the cost of any equipment. Entering the market will require a cash outlay of either shs 300000 to purchase and install a forge or shs 10000 for overtime expenses should the second option be selected.

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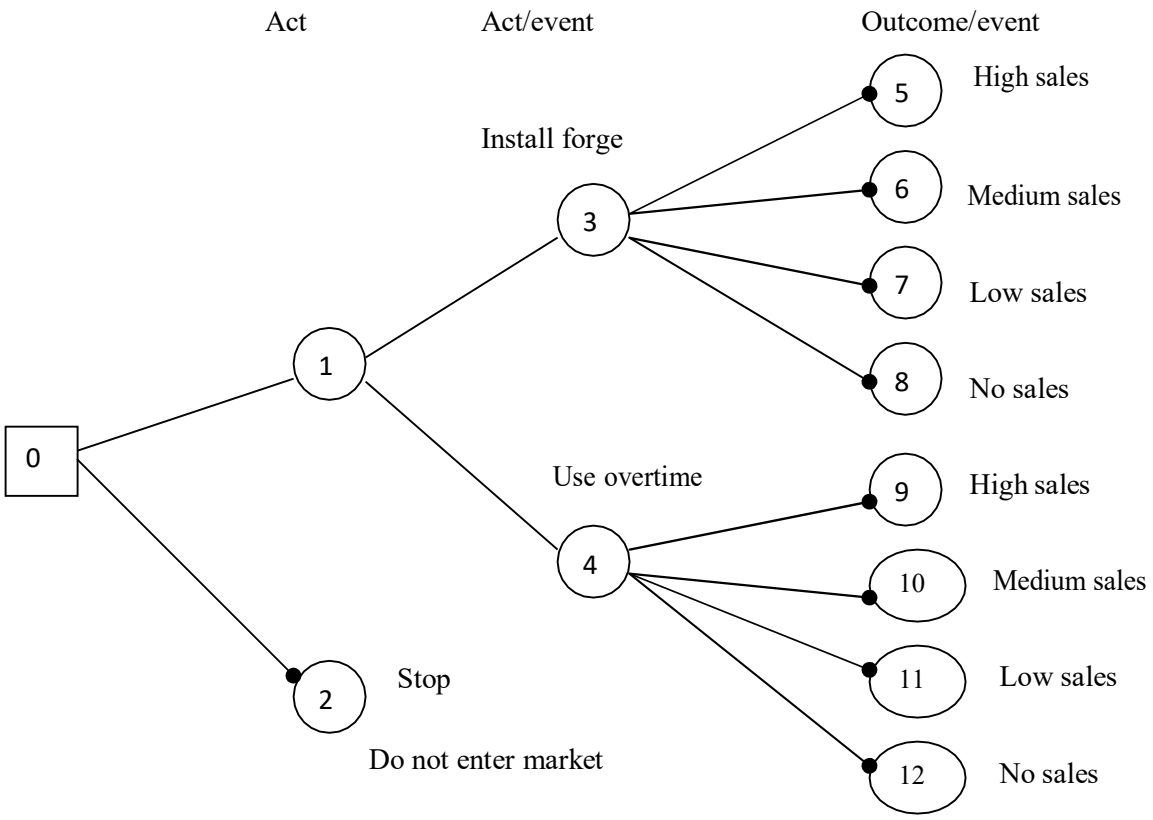
SOLUTION

a) The tree diagram for this problem is illustrated as follows:

The 1st stage of drawing a tree diagram is to show all decision points and outcome points done from left to right, concentrate first on the logic of the problem and on probabilities or values involved. This is called **forward pass**.

The resultant is the figure below:

Tree diagram

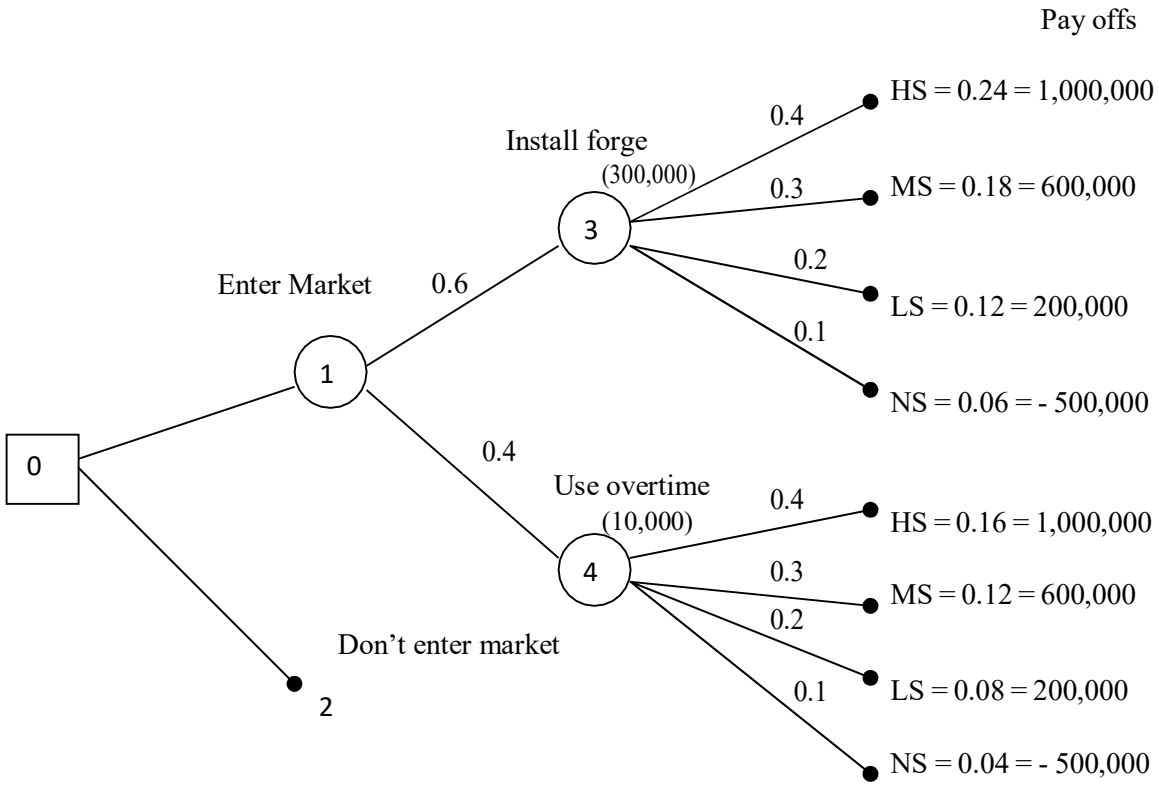


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The entire sample space of act event choices is available to KAM are summarized in the table shown below

Path	Summary of alternative Act event sequence
0 – 1 – 3 – 5	Enter market, install forge, high sales
0 – 1 – 3 – 6	Enter market, install forge, medium sales
0 – 1 – 3 – 7	Enter market, install forge, low sales
0 – 1 – 3 – 8	Enter market, install forge, no sales
0 – 1 – 4 – 9	Enter market, use overtime, high sales
0 – 1 – 4 –10	Enter market, use overtime, medium sales
0 – 1 – 4 – 11	Enter market, use overtime, low sales
0 – 1 – 4 – 12	Enter market, use overtime, no sales
0 – 2	Do not enter the market

b) The appropriate probability tree is shown in the figure below. The alternatives available to the management of KAM are identified. The joint probabilities are the result of the path sequence that is followed. For example, the sequence 'enter market install forge, low sales' yields (0.6) (0.2) = 0.12 = probability to install forge and get low sales.



(c) The overall decision is determined after analysis of the expected values at various points so the correct decision (with the highest expected value is made. The stage is worked from right to left and is known as the **backward pass.**

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- The expected value for a decision is the highest pay off value where as the E.V for an outcome is the summation of probability x pay off value of each branch. In both cases any expenditure incurred due to the selection of the said option is deducted.
- In our case

Node 3 = $[(0.4 \times 1{,}000{,}000) + (0.3 \times 600{,}000) + (0.2 \times 200{,}000) + (0.1 \times -50{,}000)]$

- 300,000

E.V. = 615,000 – 300,000 = 315,000

Node 4 = $[(0.4 \times 1{,}000{,}000) + (0.3 \times 600{,}000) + (0.2 \times 200{,}000) + (0.1 \times -50{,}000)]$

- 10,000

E.V. = 615,000 – 10,000 = 605,000

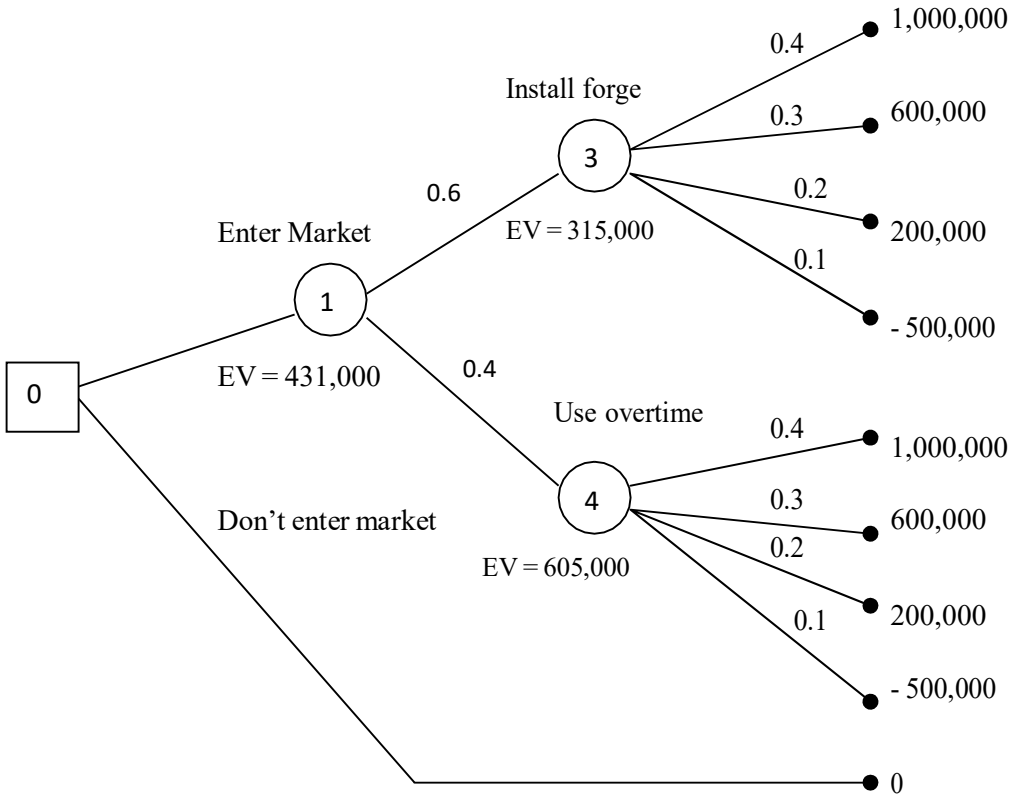
Node 1 = (0.6 × 315,000) + (0.4 × 605,000)

E.V. = 431,000

Node 0 = The highest of (0;431,000)

Since not entering the market has a 0 expected value = 431,000 = thus the decision should be to enter the market.

This is represented as below in a tree diagram.



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BAYES THEORY AND DECISION TREES

It makes an application of bayes' theorem to solve typical decision problems. This is examined a lot so it isimportant to clearly understand it.

ILLUSTRATION

Magana Creations is a company producing Ruy Lopez brand of cars. It is contemplating launching a new model, the Guioco. There are several possibilities that could be opted for.

- Continue producing Ruy Lopez which has profits declining at 10% per annum on a compounding zestimated profit of Shs. 30,000.
- Launch Guioco with prior market research costing Shs. 30,000 the market research will indicate whether future sales are likely to be 'good' or 'bad.' If the research indicates 'good' then the management will spend Shs. 35,000 more on capital equipment and this will increase annual profits to Shs. 100,000 if sales are actually high. If however sales are actually low, annual profits will drop to Shs. 25,000. Should market research indicate 'good' and management not spend more on promotion the profit levels will be as for 2nd scenario above.
- If the research indicate 'bad' then the management will scale down their expectations to give annual profit of Shs. 50,000 when sales are actually low, but because of capacity constraints if sales are high profit will be Shs. 70,000.

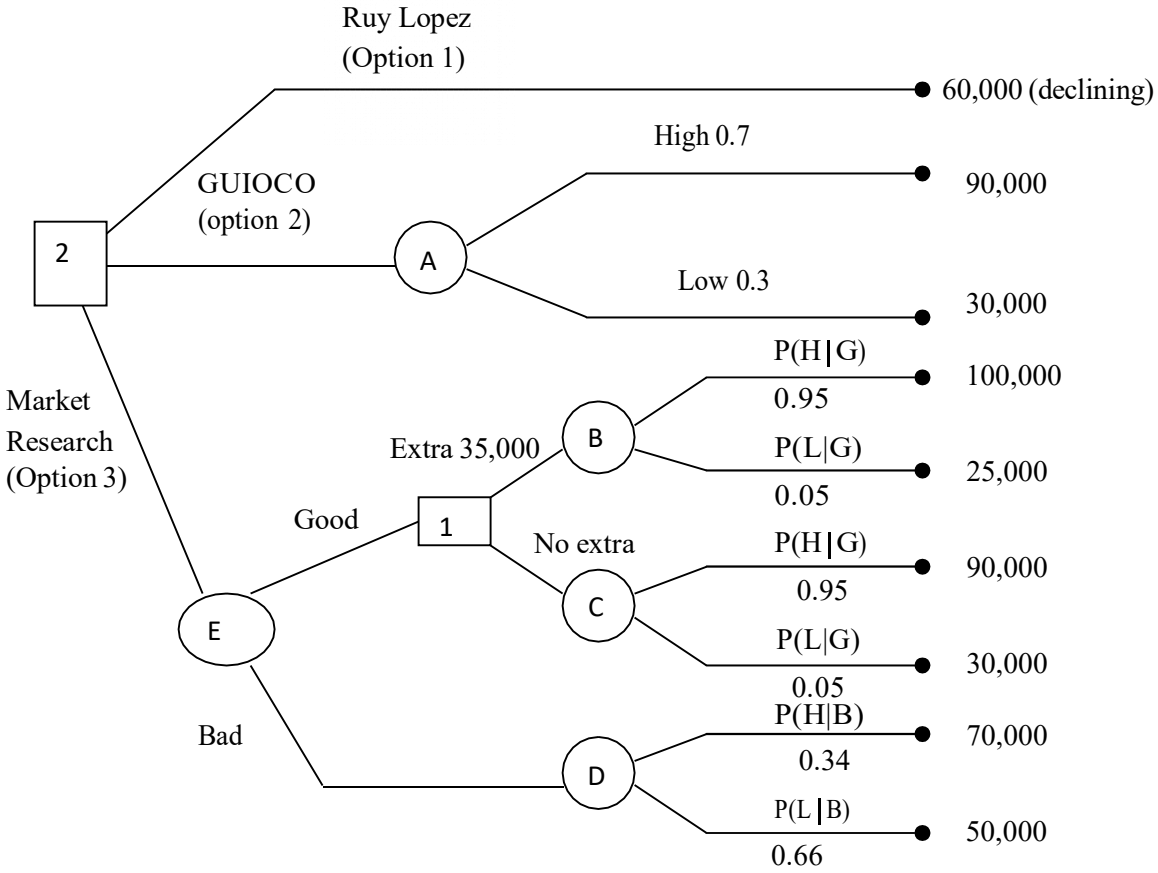
Past history of the market research company indicated the following results.

		Actual sales	
		High	**Low**
Predicted sales level	Good	0.8*	0.1
	Bad	0.2	0.9

When actual sales were high the market research company had predicted good sales level 80% of the time.

Required:
Use a time horizon of 6 years to indicate to the management of the company which option theory should adopt (Ignore the time value of money).

Solution
(a) First draw the decision tree diagram



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Computations; note how probability figures are arrived at.

- The decision tree dictates that the following probabilities need to be calculated.

P(G) For market research
P(B)

P(H|G)
P(L|G)
P(H|B) For sales outcome;
P(L|B)

P(G|H) = 0.8
P(B|H) = 0.2
P(G|L) = 0.1 Given
P(B|L) = 0.9
P(H) = 0.7
P(L) = 0.3

Good	P(G&H) = P(H) × P(G|H) 0.7 × 0.8 = 0.56	P(G&L) = P(L) × P(G|L) 0.3 × 0.1 = 0.03
Bad	B&H = P(H) × P(B|H) 0.7 × 0.2 = 0.14	P(B&L) = P(L) × P(B|L) 0.3 × 0.9 = 0.27
	High 0.7	Low 0.3

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P(G) = P(G and H) + P(G and L)
= 0.56 + 0.03 = 0.59

P(B) = P(B and H) + P(B and L)
= 0.14 + 0.27 = 0.41
Note that P(G) + P(B) = 0.59 + 0.41 = 1.00

From Bayes' rule;

$$P(H|G) = \frac{P(G|H) \times P(H)}{P(G)} = \frac{0.56}{0.59} = 0.95$$

$$P(L|G) = \frac{P(G|L) \times P(L)}{P(G)} = \frac{0.03}{0.59} = 0.05$$

$$P(H|B) = \frac{P(B|H) \times P(H)}{P(B)} = \frac{0.14}{0.41} = 0.34$$

$$P(L|B) = \frac{P(B|L) \times P(L)}{P(B)} = \frac{0.27}{0.41} = 0.66$$

Evaluating financial outcome:

Option 1:

Last year Shs. 60,000 profits

Year		Shs.
1 =	$60,000 \times 0.9$ =	54,000.0
2 =	$60,000 \times 0.9^2$ =	48,000.0
3 =	$60,000 \times 0.9^3$ =	43,740.0
4 =	$60,000 \times 0.9^4$ =	39,366.0
5 =	$60,000 \times 0.9_5$ =	35,429.5
6 =	$60,000 \times 0.9^6$ =	31,886.5
		253,022.0

Option 2

Expected value of Giuoco

Node (A): $0.7(90,000 \times 6) + 0.3(30,000 \times 6)$

= 378,000 + 54,000 = Shs. 432,000

Note that the figures a multiplied by 6 to account for the 6 years.

Option 3

Expected value of market research

Node (B): $0.95(100,000 \times 6) + 0.05(25,000 \times 6)$

= 570,000 + 7,500 = Shs. 577,500

Deduct Shs. 35,000 for extensions

= 542,500.

Node (C): $0.95(90,000 \times 6) + 0.05(30,000 \times 6)$

= 513,000 + 9,000 = Shs. 522,000

Node 1: Compare B and C

B is higher, thus = 542,000.

Node (D): $0.34(70,000 \times 6) + 0.66(50,000 \times 6)$

142,800 + 198,000 = Shs. 340,800

Node 2: Shs. 340,800 or 0 – no launch

Node (E): $0.59 \times 542,500 + 0.41 \times 340,800$

320,075 + 139,728 = Shs. 459,803

Less market research expenditure

459,803 – 30,000 = Shs. 429,803

Node 2: Final decision summary

Option 1 EMV = 253,022

Option 2 EMV = 432,000

Option 3 EMV = 429,803

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Therefore we chose option 2 since it has the highest EMV.

Advantages of decision trees

1. it clearly brings out implicit assumptions and calculations for all to see question and revise
2. it is easy to understand

Disadvantages

1. it assumes that the utility of money is linear with money
2. it is complicated by introduction of more variables and decision alternatives
3. it is complicated by presence of interdependent alternatives and dependent variables

PRACTICE EXERCISES

QUESTION 1

The following is a payoff table for a particular venture.

		States of nature				
		θ_1	θ_2	θ_3	θ_4	θ_5
	D_1	150	225	180	210	250
Decision	D_2	180	140	200	160	225
Alternatives	D_3	220	185	195	190	180
	D_4	190	210	230	200	160

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Determine the optimal decision using:

a) Max-min criterion.
b) Max-max criterion.
c) Min-max regret criterion.
d) Maximum expected payoff (assuming equal likelihood of states of nature).

Solution:

Optimal decision using:

a) Max-min criterion – Choose decision that maximizes the minimum profit.
   Min-max –choose decision that minimizes the maximum loss.

		Worst outcome	
Decision alternatives	D1	150	
	D2	140	
	D3	180	⇐Decision taken
	D4	160	

b) Max-max criterion – Choose decision that maximizes the maximum profit.
Min-min –choose decision that minimizes the minimum loss.

		Best outcome	
Decision alternatives	D1	250	⇐Decision taken
	D2	225	
	D3	220	
	D4	230	

c) Min-max regret criterion –from regret table, choose the decision that minimizes the maximum regret.
Regret = maximum payoff for a state of nature less the payoff of a given state in a decision alternative. E.g. regret for: $D_1\theta_1 = 220 - 150 = 70$
$D_3\theta_1 = 210 - 190 = 20$

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Regret table:

		States of Nature						
		θ_1	θ_2	θ_3	θ_4	θ_5	Max	Either
	D1	70	0	50	0	0	70	⇐Decision
Decision	D2	40	85	30	50	25	85	
alternative	D3	0	40	35	20	70	70	⇐Or this
	D4	30	15	0	10	90	90	

d) Maximum expected payoff –assuming equal likelihood of states of nature, decision that maximizes the expected payoff determined is taken.

For example:
Expected payoff for D_2 = Payoff $(D_2\theta_1 + D_2\theta_2 + D_2\theta_3 + D_2\theta_4 + D_2\theta_5)/5$
$= (180 + 140 + 200 + 160 + 225)/5 = 181$

	Expected Payoff	
Decision alternative	D1	203 ⇐Decision taken
	D2	181
	D3	194
	D4	198

QUESTION 2

Assume that Table question 1, is a loss table rather than a payoff table. Determine the optimal decision using:

a) The min-max criterion,
b) The min-min criterion,
c) The min-max regret criterion, and
d) The minimum expected loss criterion (again assuming equal likelihood of states of nature).

Solution:

a) Min-max

		Worst outcome	
Decision alternatives	D1	250	
	D2	225	
	D3	220	⇐Decision taken
	D4	230	

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b) Min-min

		Best outcome	
Decision alternatives	D1	180	
	D2	140	⇐Decision taken
	D3	180	
	D4	160	

c) Min-max regret

Regret = loss of a given state in a decision alternative less minimum loss for a given state of nature. E.g. regret for $D_3\theta_5 = 180 - 160 = 20$

Regret table:

		States of Nature						
		θ_1	θ_2	θ_3	θ_4	θ_5	Min	
Decision alternative	D1	0	85	0	50	90	90	
	D2	30	0	20	0	65	65	⇐Decision taken
	D3	70	45	15	30	20	70	
	D4	40	70	50	40	0	70	

d) Minimum expected loss

	Expected loss	
	D1	203
Decision alternative	D2	181 ⇐Decision taken
	D3	194
	D4	198

QUESTION 3

The following table is a payoff table for a particular venture.

		States of nature					
		θ_1	θ_2	θ_3	θ_4	θ_5	θ_6
Decision Alternative	D_1	280	300	260	360	400	450
	D_2	320	420	540	300	280	380
	D_3	200	360	400	440	250	320
	D_4	350	260	390	500	380	260

The relative likelihood's of occurrence for the states of nature are $f(\theta_1) = 0.18$, $f(\theta_2) = 0.10$, $f(\theta_3) = 0.16$, $f(\theta_4) = 0.24$, $f(\theta_5) = 0.20$, and $f(\theta_6) = 0.12$.

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Required:

a) Determine the decision alternative that maximizes expected payoff.
b) Determine expected value under certainty.
c) What is the expected value of perfect information?

Solution:

a) Expected payoff for a decision = Σ(Payoff; $\times$ $f(\theta)$ is
Where i = 1, 2, 3, 4 decision alternative
j = 1, 2, 3, 4, 5, 6 states of nature

	Expected payoff	
	D1	342.4
Decision alternatives	D2	359.6
	D3	330
	D4	378.6 ⇐Decision taken

b) Expected value under certainty: Under certainty given any state of nature a decision maker will choose the alternative with the highest payoff as follows:

	States of nature						
	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	
Certain payoff	350	420	540	500	400	450	
Probability	0.18	0.1	0.16	0.24	0.2	0.12	Total
Expected value	63	42	86.4	120	80	54	**445.4**

c) Expected value of perfect information is equal to expected value under certainty less the expected value under uncertainty

Value in (b) –Value in (a) = 445.4 – 378.6 = 66.8

QUESTION 4

An urban cable television company is investigating the installation of cable TV system in urban areas. The engineering department estimates the cost of the system (in present worth Sh.) to be Sh. 7 million. The sales department has investigated four pricing plans. For each pricing plan, the marketing division has estimated the revenue per household in present worth Sh. to be:

Plan	Revenue per household (Sh.)
I	150
II	180
III	200
IV	240

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The sales department estimates that the number of household subscribers would be approximately, either 10,000, 20,000, 30,000, 40,000, 50,000 or 60,000.

Required;-

a) Construct a payoff table for this problem.

b) What would be the company's optimal decision under the optimistic approach and the minimax regret approach.

c) Suppose that the sales department has determined the number of subscribers will be a function of the pricing plan.

The probability distributions for the pricing plans are given below.

	Probability under pricing plan			
Number of subscribers	**I**	**II**	**III**	**IV**
10,000	0	0.05	0.10	0.20
20,000	0.05	0.10	0.20	0.25
30,000	0.05	0.20	0.20	0.25
40,000	0.40	0.30	0.20	0.15
50,000	0.30	0.20	0.20	0.10
60,000	0.20	0.15	0.10	0.05

Which pricing plan is optimal?

d) Briefly explain the main difference between the approaches used in part (b) and (c) above.

Solution:

a) Payoff = (Revenue / Household × No. of households) – Initial cost

Payoffs in millions

Plan	Revenue	No. of households 10,000	20,000	30,000	40,000	50,000	60,000
I	150	-5.5	-4	-2.5	-1	0.5	2
II	180	-5.2	-3.4	-1.6	0.2	2	3.8
III	200	-5	-3	-1	1	3	5
IV	240	-4.6	-2.2	0.2	2.6	5	7.4

b) Optimistic approach means that the max-max criterion is used.

Plan	Max	
I	2	
II	3.8	
III	5	
IV	7.4	⇐Adopt Plan IV

Min-max regret means, from the opportunity loss table, the minimum of the maximum is actually chosen.

The opportunity loss table.

Opportunity loss or regret = max payoff for a given number of households less the payoff of a given number of household and given plan. E.g. Plan III for 40,000 household, = 0.26 - 1 = 1.6 million shillings

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	No. of households							
Plan	10,000	20,000	30,000	40,000	50,000	60,000	Max	
I	0.9	1.8	2.7	3.6	4.5	5.4	5.4	
II	0.6	1.2	1.8	2.4	3	3.6	3.6	
III	0.4	0.8	1.2	1.6	2	2.4	2.4	
IV	0	0	0	0	0	0	0	⇐Adopt

c) Given the probabilities, the payoff table will change to be as follows

Payoff = Payoff as determined from part of a) multiplied by the given probability under the respective pricing plans.

e.g. for Plan II for 3,000 household = -1.6 × 0.2

= -0.32

Expected payoff = sum of all the payoffs for a given plan.

	No. of households							
Plan	10,000	20,000	30,000	40,000	50,000	60,000	Expected	
I	0	-0.2	-0.125	-0.4	0.15	0.4	-0.175	
II	-0.26	-0.34	-0.32	0.06	0.4	0.57	0.11	⇐Adopt
III	-0.5	-0.6	-0.2	0.2	0.6	0.5	0	
IV	-0.92	-0.55	0.05	0.39	0.5	0.37	-0.16	

The pricing plan to follow is Plan II, which gives a higher expected payoff of sh110, 000.

d) The approach used in part (b) is that of decision making under uncertainty. Probabilities of occurrence as much as the outcomes are not known with certainty.
The approach in (c) on the other hand is decision making under risk. Probabilities of occurrence of an event is known with given amount. This gives expected payoff for any decision undertaken.

QUESTION 5

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Explain the following terms as used in decision analysis.

a) Decision making under risk versus uncertainty.
b) Decision trees versus probability trees.
c) Minimax versus maximax criterion.
d) Pure strategy versus mixed strategy games.
e) Games with more than two persons versus non zero-sum games

Solution:

a) Decision making under risk is when decisions are made using already known probabilities for states of nature or outcomes. The probabilities can come from previous data.
Decision making under certainty is when a decision is made where there is no prior probabilities for states of nature or outcomes.
b) Decision tree is a diagrammatic representation of decisions given different states of nature. Nodes and branches are used to represent the decisions and outcomes from given decisions.
Probability tree is a diagrammatic representation of the sequence of outcomes given certain probabilities.
c) Minimax criterion-involves choosing the alternative with minimum regret from choice of maximum regrets from given events.
Maximax criterion- involves choosing the alternative with maximum payoff from choice of maximum payoffs from given events.
d) Pure strategy in a game is where each player knows exactly what the other player is going to do. The same rule is still followed each time.

Mixed strategy is where there is a combination of the rules followed. Each player does not know what the other player is going to do. In this case probabilities are used to find what each player will do. The main aim is to maximize expected gains or to minimize losses.

e) Games represent a competitive situation where players aim to gain from each other.
Games with more than two persons represent real life situation where there are more than two persons as players. Each person seeks to gain from the others.
Non-zero sum games represent situation where it is not necessarily that what one losses is gained by another.

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TOPIC 8

GAME THEORY

Introduction

Game theory is used to determine the optimum strategy in a competitive situation

When two or more competitors are engaged in making decisions, it may involve conflict of interest. In such a case the outcome depends not only upon an individuals action but also upon the action of others. Both competing sides face a similar problem. Hence game theory is a science of conflict

Game theory does not concern itself with finding an optimum strategy but it helps to improve the decision process.

Game theory has been used in business and industry to develop bidding tactics, pricing policies, advertising strategies, timing of the introduction of new models in the market e.t.c.

RULES OF GAME THEORY

i) The number of competitors is finite
ii) There is conflict of interests between the participants
iii) Each of these participants has available to him a finite set of available courses of action i.e. choices
iv) The rules governing these choices are specified and known to all players
v) While playing each player chooses a course of action from a list of choices available to him
vi) the outcome of the game is affected by choices made by all of the players. The choices are to be made simultaneously so that no competitor knows his opponents choice until he is already committed to his own
vii) the outcome for all specific choices by all the players is known in advance and numerically defined
viii) When a competitive situation meets all these criteria above we call it a game

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NOTE: only in a few real life competitive situation can game theory be applied because all the rules are difficult to apply at the same time to a given situation.

ILLUSTRATION

Two players X and Y have two alternatives. They show their choices by pressing two types of buttons in front of them but they cannot see the opponents move. It is assumed that both players have equal intelligence and both intend to win the game.

This sort of simple game can be illustrated in tabular form as follows:

	Player Y		
		Button R	Button t
Player X	Button m	X wins 2 points	X wins 3 points
	Button n	Y wins 2 points	X wins 1 point

The game is biased against Y because if player X presses button m he will always win. Hence Y will be forced to press button r to cut down his losses

Alternative Illustration

	Player Y		
		Button R	**Button t**
Player X	Button m	X wins 3 points	Y wins 4 points
	Button n	Y wins 2 points	X wins 1 point

In this case X will not be able to press button m all the time in order to win(or button n). similarly Y will not be able to press button r or button t all the time in order to win. In such a situation each player will exercise his choice for part of the time based on the probability

Standard conventions in game theory

Consider the following table

	Y	
	3	-4
X	-2	1

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X plays row I, Y plays columns I, X wins 3 points
X plays row I, Y plays columns II, X looses 4 points
X plays row II, Y plays columns I, X looses 2 points
X plays row II, Y plays columns II, X wins 1 points

3, -4, -2, 1 are the known pay offs to X(X takes precedence over Y)
here the game has been represented in the form of a matrix. When the games are expressed in this fashion the resulting matrix is commonly known as PAYOFF MATRIX

STRATEGY

It refers to a total pattern of choices employed by any player. Strategy could be pure or a mixed one
In a pure strategy, player X will play one row all of the time or player Y will also play one of this columns all the time.
In a mixed strategy, player X will play each of his rows a certain portion of the time and player Y will play each of his columns a certain portion of the time.

VALUE OF THE GAME

The value of the game refers to the average pay off per play of the game over an extended period of time

ILLUSTRATION

$$\begin{array}{c} \\ Player\ X \end{array} \begin{array}{c} Player\ Y \\ \begin{bmatrix} 3 & 4 \\ - & \end{bmatrix} \end{array}$$

in this game player X will play his first row on each play of the game. Player y will have to play first column on each play of the game in order to minimize his looses
so this game is in favour of X and he wins 3 points on each play of the game.
This game is a game of pure strategy and the value of the game is 3 points in favour of X

ILLUSTRATION

Determine the optimum strategies for the two players X and Y and find the value of the game from the following pay off matrix

$$\begin{array}{c} \\ Player\ X \end{array} \begin{array}{c} Player\ Y \\ \begin{bmatrix} 3 & -1 & 4 & 2 \\ -1 & -3 & -7 & 0 \\ 4 & -7 & 3 & -9 \end{bmatrix} \end{array}$$

Strategy assume the worst and act accordingly
if X plays first
if X plays first with his row one then Y will play with his 2nd column to win 1 point similarly if X plays with his 2nd row then Y will play his 3rd column to win 7 points and if x plays with his 3rd row then Y will play his fourth column to win 9 points
In this game X cannot win so he should adopt first row strategy in order to minimize losses
This decision rule is known as 'maximum strategy' i.e. X chooses the highest of these minimum pay offs
Using the same reasoning from the point of view of y
If Y plays with his 1st column, then X will play his 3rd row to win 4 points
If Y plays with his 2nd column, then X will play his 1st row to lose 1 point
If Y plays with his 3rd column, then X will play his 1st row to win 4 points
If Y plays with his 4th column, then X will play his 1st row to win 2 points

Thus player Y will make the best of the situation by playing his 2nd column which is a 'Minimax strategy'
This game is also a game of pure strategy and the value of the game is –1(win of 1 point per game to y) using matrix notation, the solution is shown below

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$$\begin{array}{c} \\ \text{Player } X \end{array} \begin{array}{c} \text{Player } Y \\ \begin{bmatrix} 3 & -1 & 4 & 2 \\ -1 & -3 & -7 & 0 \\ 4 & -7 & 3 & -9 \end{bmatrix} \\ 4 \quad -1 \quad 4 \quad 2 \end{array} \begin{array}{c} \text{Row Minimum} \\ -1 \\ -7 \\ -9 \end{array}$$

column maximum

In this case value of the game is –1
Minimum of the column maximums is –1
Maximum of the row is also –1
i.e. X's strategy is maximim strategy
Y's strategy is Minimax strategy

SADDLE POINT

The saddle point in a pay off matrix is one which is the smallest value in its row and the largest value in its column. It is also known as equilibrium point in the theory of games.
Saddle point also gives the value of such a game. In a game having a saddle point, the optimum strategy for both players is to pay the row or column containing the saddle point.
Note: if in a game there is no saddle point the players will resort to what is known as mixed strategies.

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MIXED STRATEGIES

ILLUSTRATION

Find the optimum strategies and the value of the game from the following pay off matrix concerning two person game

$$\begin{array}{c} \\ \text{Player } X \end{array} \begin{array}{c} \text{Player } Y \\ \begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \end{array}$$

In this game there is no saddle point
Let Q be the proportion of time player X spends playing his 1st row and 1-Q be the proportion of time player X spends playing his 2nd row

Similarly:

Let R be the proportion of time player Y spends playing his 1st column and 1-R be the proportion of time player Y spends playing his second row

The following matrix shows this strategy

$$\begin{array}{cc} & \text{Player } Y \\ & R \quad 1-R \\ \text{Player } X \quad Q & \begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \\ 1-Q & \end{array}$$

X's strategy

X will like to divide his play between his rows in such a way that his expected winning or loses when Y plays the 1st column will be equal to his expected winning or losses when y plays the second column

Column 1

Points	Proportion played	Expected winnings
1	Q	Q
5	1-Q	5(1-Q)

Total = Q + 5(1 –Q)

Column 2

Points	Proportion played	Expected winnings
4	Q	4Q
3	1-Q	3(1-Q)

Total = 4Q + 3(1 –Q)

Therefore Q + 5(1-Q) = 4Q +3(1-Q)

Giving Q = $\frac{2}{5}$ and (1-Q) = $\frac{3}{5}$

This means that player X should play his first row $\frac{2}{5}^{th}$ of the time and his second row $\frac{3}{5}^{th}$ of the time

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Using the same reasoning

1×R + 4(1-R)= 5R +3(1-R)

Giving R = $\frac{1}{5}$ and (1-R) = $\frac{4}{5}$

This means that player Y should divide his time between his first column and second column in the ratio 1:4

$$\begin{array}{cc} & \text{Player } Y \\ & \frac{1}{5} \quad \frac{4}{5} \\ \text{Player } X \quad \frac{2}{5} & \begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \\ \frac{3}{5} & \end{array}$$

Short cut method of determining mixed matrices

$$\begin{array}{cc} & \text{Player } Y \\ \text{Player } X & \begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \end{array}$$

Step I

Subtract the smaller pay off in each row from the larger one and smaller pay off in each column from the larger one

$$\begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \quad \begin{matrix} 4-1=3 \\ 5-3=2 \end{matrix}$$

$5-1=4 \qquad 4-3=1$

Step II

Interchange each of these pairs of subtracted numbers found in step I

$$\begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \quad \begin{matrix} 2 \\ 3 \end{matrix}$$

$1 \quad 4$

Thus player X plays his two rows in the ratio 2: 3

And player Y plays his columns in the ratio 1:4

This is the same result as calculated before

To determine the value of the game in mixed strategies

In a simple 2 x 2 game without a saddle point, each players strategy consists of two probabilities denoting the portion of the time he spends on each of his rows or columns. Since each player plays a random pattern the probabilities are listed under

Pay off	Strategies which produce this pay off	Joint probability
1	Row I column I	$2/5 \times 1/5 = 2/25$
4	Row I column II	$2/5 \times 4/5 = 8/25$
5	Row II column I	$3/5 \times 1/5 = 3/25$
3	Row II column II	$3/5 \times 4/5 = 12/25$

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Expected value (or value of the game)

Pay off	Probability p(x)	Expected value x (p(x)
1	$2/25$	$2/25$
4	$8/25$	$32/25$
5	$3/25$	$15/25$
3	$12/25$	$36/25$

$\sum x\ p(x) = 85/25 \qquad = 17/5 = 3.4$

3.4 is the value of the game

DOMINANCE

Dominated strategy is useful for reducing the size of the payoff table

Rule of dominance

i) If all the elements in a column are greater than or equal to the corresponding elements in another column, then the column is dominated

ii) Similarly if all the elements in a row are less than or equal to the corresponding elements in another row, then the row is dominated

Dominated rows and columns may be deleted which reduces the size of the game

NB// always look for dominance and saddle points when solving a game

ILLUSTRATION

Determine the optimum strategies and the value of the game from the following 2xm pay off matrix game for X and Y

$$\begin{array}{c} \\ X \end{array} \begin{array}{c} \begin{array}{ccccc} & & & Y & \\ 6 & 3 & -1 & 0 & -3 \\ 3 & 2 & -4 & 2 & -1 \end{array} \end{array}$$

In this columns I, II, and IV are dominated by columns III and V hence Y will not play these columns

So the game is reduced to 2×2 matrix, hence this game can be solved using methods already discussed

$$\begin{array}{c} \\ X \end{array} \begin{array}{c} \begin{array}{cc} & Y \\ -1 & -3 \\ -4 & -1 \end{array} \end{array}$$

NON ZERO SUM GAMES

Until recently there was no satisfactory theory either to explain how people should play non zero games or to describe how they actually play such games

Nigel Howard (1966) developed a method which describes how most people play non zero sum games involving any number of persons

ILLUSTRATION

Each individual farmer can maximize his own income by maximizing the amount of crops that he produces. When all farmers follow this policy the supply exceeds demand and the prices fall. On the other hand they can agree to reduce the production and keep the prices high

This creates a dilemma to the farmer

This is an example of a non zero sum game

Similarly marketing problems are non zero sum games as elements of advertising come in. in such cases the market may be split in proportion to the money spent on advertising multiplied by an effectiveness factor

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PRISONERS DILEMMA

It is a type of non zero sum game and derives its name from the following story

The district attorney has two bank robbers in separate cells and offers each a chance of confession. If one confesses and the other does not then the confessor gets two years and the other one ten years. If both confess they will get eight years each. If both refuse to confess there is only evidence to ensure convictions on a lesser charge and each will receive 5 years

ILLUSTRATION

The table below is a pay off matrix for two large companies A and B. initially they both have the same prices. Each consider cutting their prices to gain market share and hence improve profit

Corporation B

Corporation A		**Maintain prices**	**Decrease prices**
	maintain prices	3,3 status quo	1 , 4 B gets market share and profit
	Decrease prices	4, 1, A gains market share and profit	(2,2) Both retain market share but lose profit

The entries in the pay off matrix indicate the order of preference of the players i.e. first A then B.

We may suppose that if both player study the situation, they will both decide to play row I column I(3,3).

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However

Suppose A's reasoning is as follows

- If B plays column I then I should play row 2 because I will increase my gain to 4
- In the same way B's reasoning may be as follows
- If A plays row I then I should play column 2 to get pay off 4 per play
- If both play 2(row 2 column 2) each two receives a pay off of 2 only
- In the long run pay off forms a new equilibrium point because if either party departs from it without the other doing so he will be worse off before he departed from it
- Game theory seems to indicate that they should play (2,2) because it is an equilibrium point but this is not intuitively satisfying. On the other hand (3,3) is satisfying but does not appear to provide stability. Hence the dilemma.

THEORY OF METAGAMES

This theory appears to describe how most people play non zero sum games involving a number of persons

Prisoners dilemma is an example of this. The aim is to identify points at which players actually tend to stabilize their play in non zero sum games.

This theory not only identifies equilibrium points missed by traditional game theory in games that have one or more such points but also does so in games in which traditional theory finds no such point

Its main aim is that each player is trying to maximize the minimum gain of his opponent

ADVANTAGES AND LIMITATIONS OF GAME THEORY

Advantage

Game theory helps us to learn how to approach and understand a conflict situation and to improve the decision making process

LIMITATIONS

1. Businessmen do not have all the knowledge required by the theory of games. Most often they do not know all the strategies available to them nor do they know all the strategies available to their rivals
2. there is a great deal of uncertainty. Hence we usually restrict ourselves to those games with known outcomes
3. The implications of the Minimax strategy is that the businessman minimizes the chance of maximum loss. For an ambitious business man, this strategy is very conservative
4. the techniques of solving games involving mixed strategies where pay off matrices are rather large is very complicated
5. in non zero sum games, mathematical solutions are not always possible. For example a reduction in the price of a commodity may increase overall demand. It is also not necessary that demand units will shift from one firm to another

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TOPIC 9

NETWORK PLANNING AND ANALYSIS

BASIC CONCEPTS

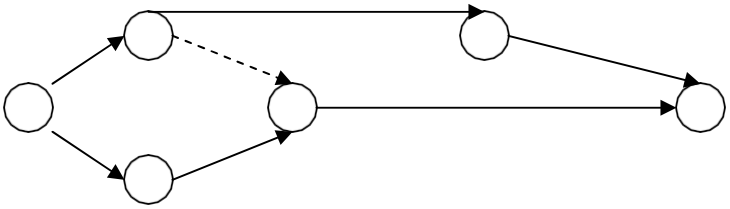
Network is a system of interrelationship between jobs and tasks for planning and control of resources of a project by identifying critical part of the project.

Activity: Task or job of work, which takes time and resource e.g building a bridge. Its represented by an arrow which indicates where the task begins and ends

Event (node): This is a point in time and it indicates the start or finish of an activity e.g in building a bridge, rails installed. Its represented by a circle.

Dummy activity: An activity that doesn't consume time or resources, its merely to show logical dependencies between activities so as abide by rules of drawing a network, its represented by dotted arrow

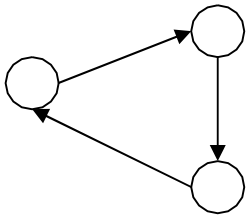
Network. This is a combination of activities and events (including dummy activities)



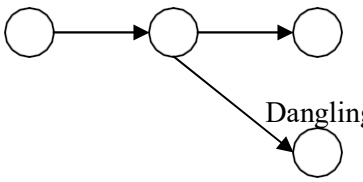
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Rules for Drawing a Network

a) A network should only have one start point and one finish point (start event and finish event)
b) All activities must have at least one preceding event (tail event)and at least one succeeding event (head event), but an activity may not share the same tail event and head event.
c) An activity can only start after its tail event has been reached
d) An event is only complete after all activities leading to it are complete.
e) Activities are identified by alphabetical or numeric codes i.e. A,B,C; 1,2,3 or identification by head or tail events 1-2, 2-4, 3-4,1-4…
f) Loops (a series of activities leading back to the same event) and danglers (activities which do not link to the overall project)are not allowed



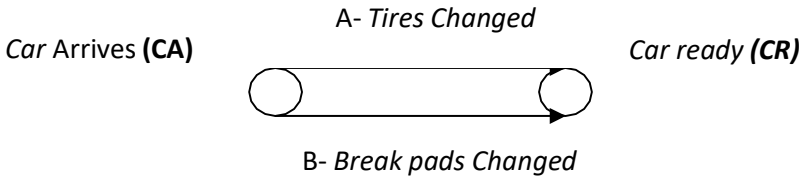
.1.1.1 Loop



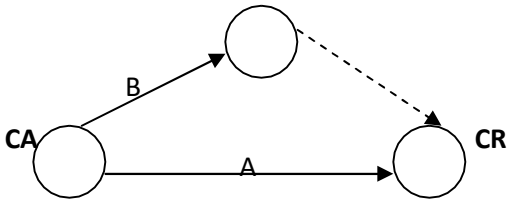
Dangling activity

Dummy Events

This is an event that does not consume time or resources, its represented by dotted arrow. Dummies are applied when two or more events occur concurrently and they share the same head and tail events e.g. when a car goes to a garage tires are changed and break pads as well, instead of representing this as;



These events are represented as;

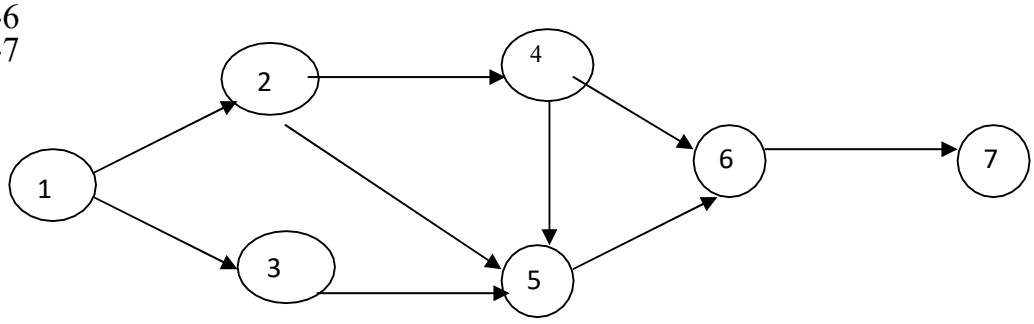


Example of a network

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Activities

1-2 - where 1 is the preceding event where as 2 is the succeeding event of the activity
1-3
2-4
2-5
3-5
4-5
4-6
5-6
6-7



TIME ANALYSIS

Assessing the time

After drawing the outline of the network time durations of the activities are then inserted.

a) Time estimates. The analysis of the projects time can be achieved by using :
   i. Single time estimates for each activity. These estimates would be based on the judgment of the individual responsible or by technical calculations using data from similar projects

ii. Multiple time estimates for each activity. the most usual multiple time estimates are three estimates for each activity , i.e. optimistic (O), Most Likely (ML), and Pessimistic (P). These three estimates are combined to give an expected time and the accepted formula is:

Expected time = $\frac{O + P + 4ML}{6}$

For example assume that the three estimates for an activity are

Optimistic 11 days
Most likely 15 days
Pessimistic 18 days

Expected time = $\frac{11 + 18 + 4(15)}{6}$

= **14.8 days**

b) Use of time estimates. as three time estimates are converted to a single time estimate there is no fundamental difference between the two methods as regards the basic time analysis of a network. However, on completion of the basic time analysis, projects with multiple time estimates can be further analyzed to give an estimate of the probability of completing the project by a scheduled date.

c) Time units. Time estimates may be given in any unit, i.e. minutes , hours, days depending on the project. All times estimates within a project must be in the same units otherwise confusion is bound to occur.

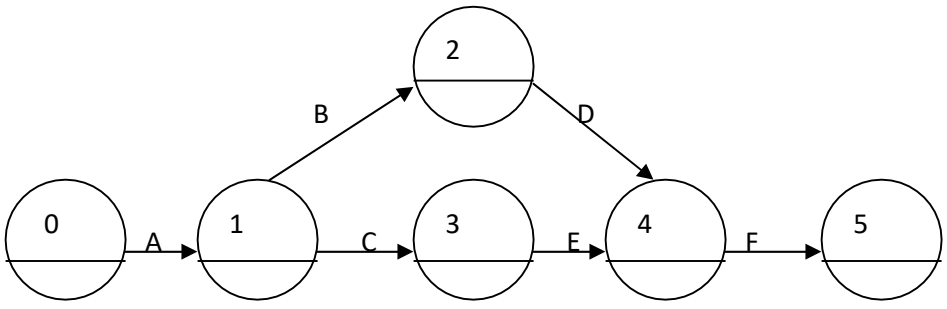
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Basic time analysis – critical path

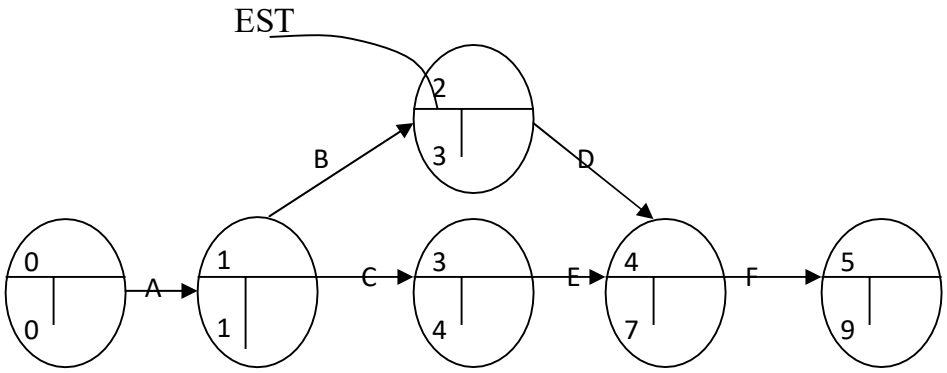
The critical path of a network gives the **shortest time in which the whole project can be completed.** It is the chain of activities with the longest duration times. There may be more than one critical path which may run through a dummy.

Earliest start times (EST) – Forward pass, Once the activities have been timed we can assess the total project time by calculating the ESTs for each activity. The EST is the earliest possible time at which a succeeding activity can start.

Assume the following network has been drawn and the activity times estimated in days.



The ESTs can be inserted as follows.



The method used to insert the ESTs is also known as the **forward pass,** this is obtained by;

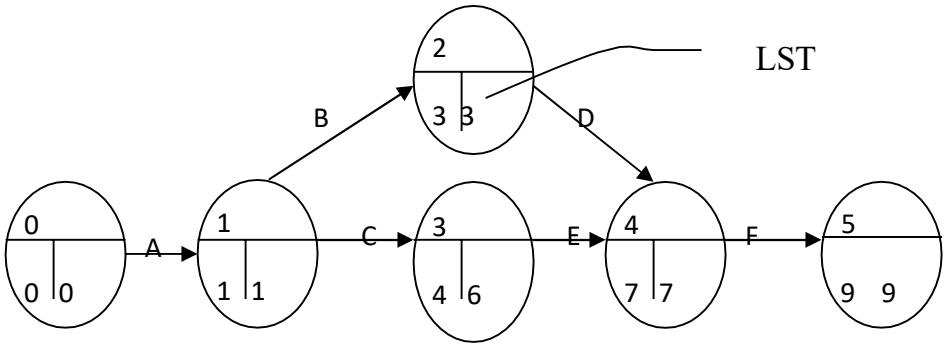
EST = The greater of [EST (tail event) + Activity duration]

a) Start from the start event giving it 0 values,
b) For the rest of the events EST is obtained by summing the EST of the tail event and the activity duration
c) Where two or more routes converge into an activity, calculate individual EST per route and the select the longest route (time)
d) The EST of the finish event is the shortest time the whole project can be completed.

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Latest Start Times (LST) – Backward pass. this is the latest possible time with which a preceding activity can finish without increasing the project duration. After this operation the critical path will be clearly defined.

From our example this is done as follows;



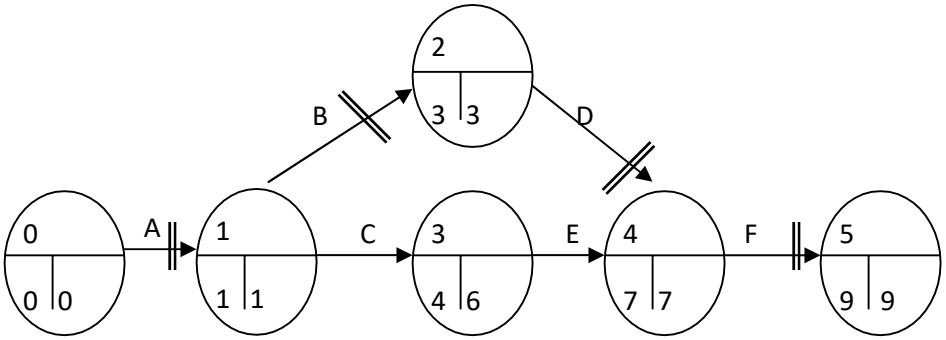
LST = Lowest of [LST (head event) – activity duration]

a) Starting at the finish event, insert the LST (i.e. 9 for our example) ,and work backwards through the network.
b) deduct each activity duration from the previously calculated LST (i.e. head LST).

c) Where the tails of activities join an event, the lowest number is taken as the LST for that event

Critical Path. . This is the chain of activities in a network with the longest duration Assessment of the resultant network shows that one path through the network (A, B, D, F) has EST's and LST's which are identical this is the critical path.

The critical path can be indicated on the network either by a different colour or by two small transverse lines across the arrows along the path thus in our example we have;



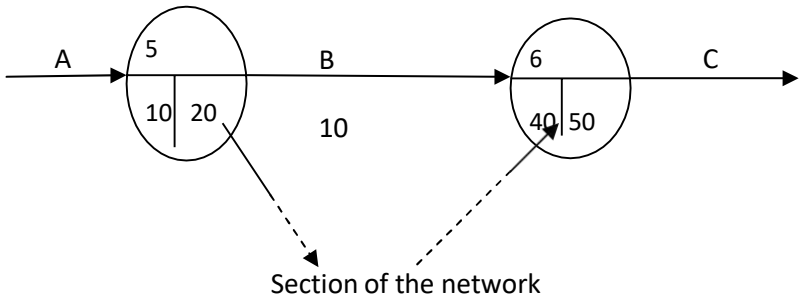
Activities along the critical path are vital activities which must be completed by their EST's/LST's otherwise the project will be delayed.

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Non critical activities (in the example above, C and E) have spare time or **float** available. C and/ or E could take up to an additional 2 days in total without delaying the project duration. If it is required to reduce the overall project duration then the time of one or more of the activities on the critical path must be reduced perhaps by using more labour, or better equipment to reducing job times.

FLOAT

Float or spare time can only be associated with activities which are non-critical. By definition, activities on the critical path cannot have float. There are three types of float, **Total Float, Free Float** and **Independent Float**. To illustrate these types of float we use the following example.



a) Total float. Amount of time by which a path of activities could be delayed without affecting the overall project duration. The path in this example consists of one activity only i.e. B

Total Float = Latest Finish time (LFT) - Earliest Start time(EST) time – Activity Duration

Total Float = 50 - 10 - 10
= 30 days

b) Free float Amount of time an activity can be delayed without affecting the commencement of a subsequent activity at its earliest start time, but may affect float of a previous activity.

Free Float = Earliest Finish Time(EFT) - EST - Activity Duration

Free Float= 40-10-10
= 20 days

c) Independent float. Amount of time an activity can be delayed when all preceding activities are completed as late as possible and all succeeding activities commenced as early a possible. Independent float therefore does not affect the float of either preceding or subsequent activities.

Independent float = EFT- Latest Start time (EST) - Activity Duration

Independent float= 40 - 20 - 10
= 10 days

Note:

- **for examination purposes, float always refers to total float**
- The total float can be calculated separately for each activity but it is often useful to find the total float over chains of non-critical activities between critical events

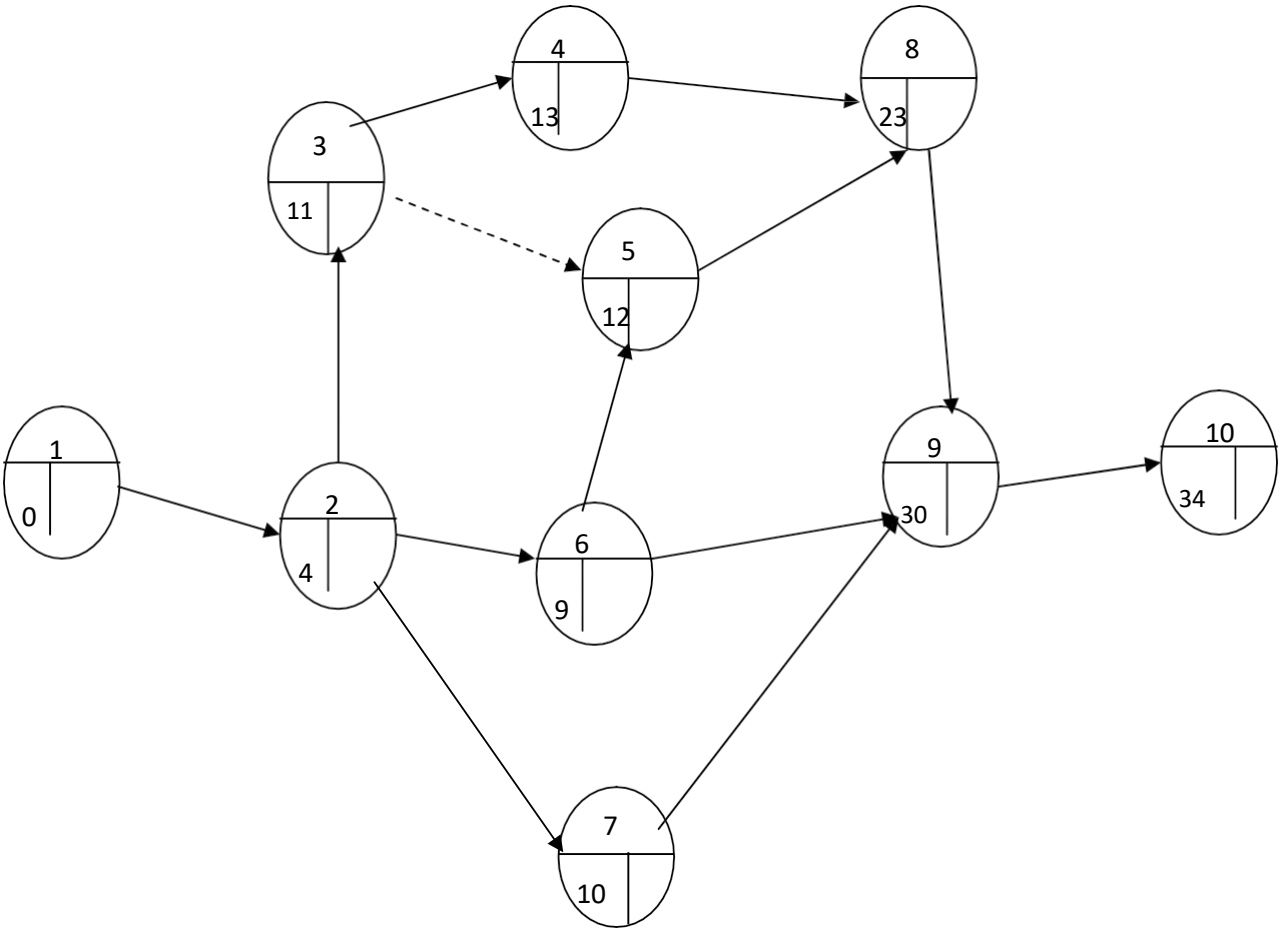
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Example.
The following represents activities of a network.

Activity	Preceding Activity	Duration Days
A	-	4
B	A	7
C	A	5
D	A	6
E	B	2
F	C	3
G	E	5
H	B,F	11
I	G,H	7
J	C	4
K	D	3
L	I,J,K	4

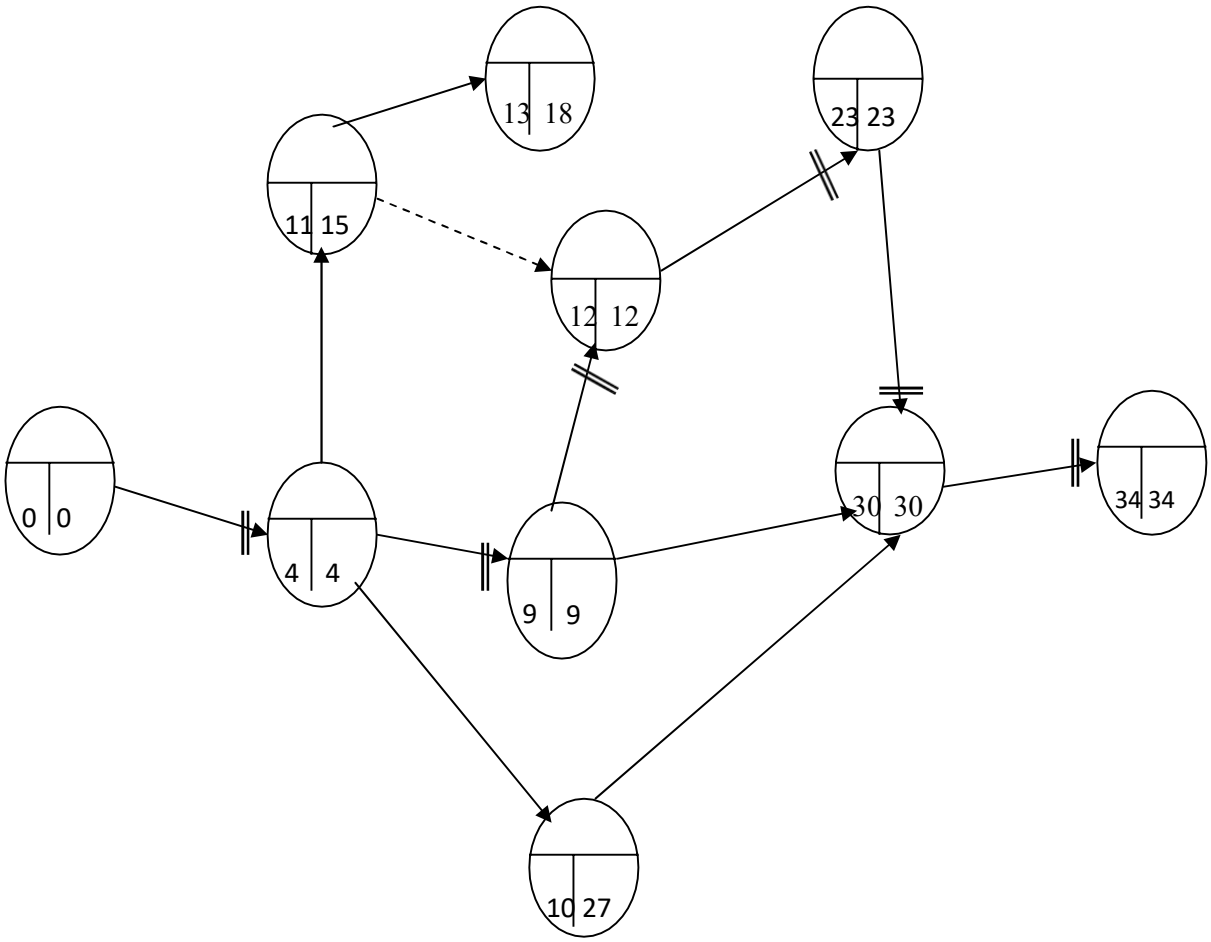
Required:

a) Draw the network diagram and find the critical path
b) Calculate the floats of the network in question



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- First we draw the network structure ensuring it fits the data above
- We then label all activities from 1 to 12 and indicate activity duration
- Conduct a forward pass operation (to obtain the diagram above)
- Operate backward pass to establish the critical path, thus we have…



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Therefore we get the critical path to be, A-C-F-H-I-L

b)

The floats of the network,

Activity	EST	LST	EFT	LFT	Activity Duration D	Total Float LFT -EST- D	Free Float EFT-EST-D	Independent Float EFT-LST-D
*A	0	0	4	4	4	-	-	-
B	4	4	11	15	7	4	-	-
*C	4	4	9	9	5	-	-	-
D	4	4	10	22	6	12	-	-
E	11	15	13	21	2	8	-	-
*F	9	9	15	15	3	-	-	-
G	13	21	23	23	5	5	5	-
*H	12	12	23	23	11	-	-	-
*I	23	23	30	30	7	-	-	-
J	9	9	30	30	4	17	17	17
K	10	22	30	30	3	17	17	5
*L	30	30	34	34	4	-	-	-

The total float on the non-critical chains are;

Non-critical chain	Time required (sum of duration)	Time available (LFT of last activity-EST of 1st activity)	Total Float over chain
B,E,G	14	19	5
B,Dummy	7	8	1
D,K	9	26	17
J	4	21	17

Slack

This is the difference between the EST and LST for each event. Strictly it does not apply to activities but on occasions the terms are confused in examination questions and unless the context makes it abundantly clear that event slack is required, it is likely that some form of activity float is required. Events on the critical path have zero slack.

Cost Scheduling

This is done by calculating the cost of various project durations, cost analysis seeks to find the cheapest way of reducing the overall cost duration of a project by increasing labour hours, equipment e.t.c.

Terminologies

Normal cost. The costs associated with a normal time estimate for an activity. Often the normal time estimate is set at the point where resources (labour, equipment, etc.) are used in the most efficient manner.

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Crash cost. The costs associated with the minimum possible time for an activity. Crash costs, because of extra wages, overtime premiums, extra facility costs are always higher than normal costs.

Crash time. The minimum possible time that an activity is planned to take. . The minimum time is invariably brought about by the application of extra resources, e.g. more labour or machinery.

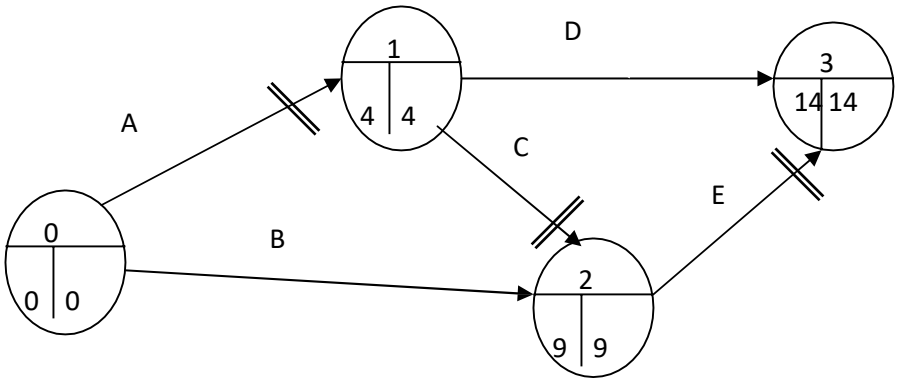
Cost slope. This is the average cost of shortening an activity by one time unit (day, week, month as appropriate). The cost slope is generally assumed to be linear and is calculated as follows:

$$\text{Cost slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash cost}}$$

Example

A project has the following activities and costs

Activity	Preceding Activity	Duration days	Crash time	Cost (Shs).	Crash cost	Cost slope
A	-	4	3	360	420	60
B	-	8	5	300	510	70
C	A	5	3	170	270	50
D	A	9	7	220	300	40
E	B,C	5	3	200	360	80



Project duration and costs

(a) Normal duration = 14 days
Critical path = A,C,E
Project cost (cost of all activities at normal time) = Shs. 1,250.

(b) Reduce by 1 day the activity on the critical path with the lowest cost slope. Thus we reduce C at extra cost of Shs. 50.

Now
Project duration = 13 days
Project cost = Shs. 1,300

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Note: that all activities are now critical.

(c) Further reducing the critical path by 1 day will require that more than one activity is affected because there exist several critical paths.

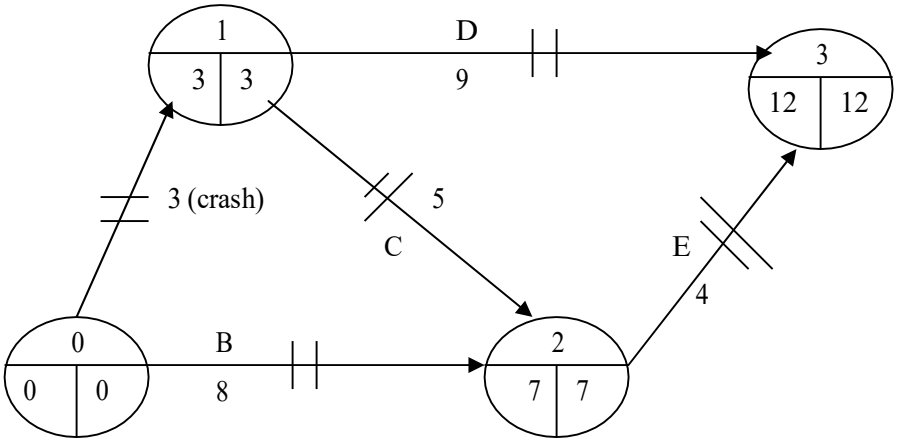
Reduce by 1 day	Extra cost	Activities critical
A and B	60 + 70 = 130	All
D and E	40 + 80 = 120	All
B, C and D	70 + 50 + 40 = 160	All
E and E	60 + 80 = 140	A, D, B, E

From this we realize that reducing D and E is the cheapest.

However closer examination of the fourth alternative reveals that C is now non-critical and has 1 day float. Since we earlier reduced C for Shs. 50, if we reduce A and E and increase C by a day which will save Shs. 50.

Then the net cost for 12 day duration = 1,300 + (140 – 50) = 1,390.

The network becomes;



(d) Next we reduce D & E

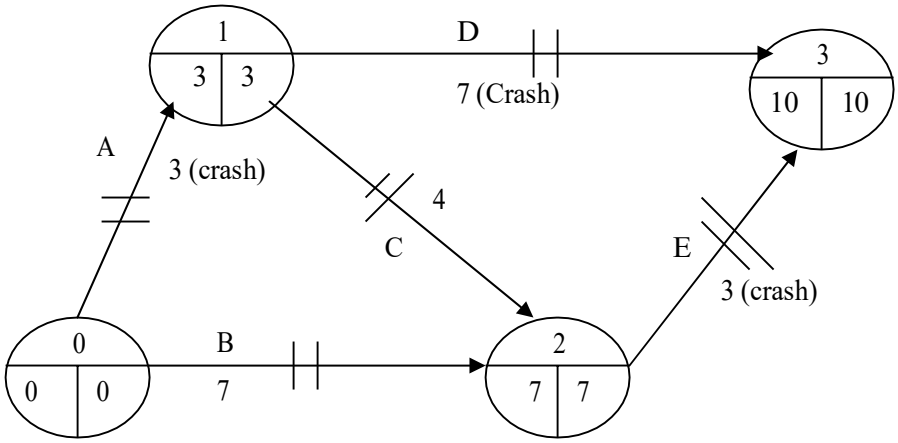
Project duration = 11 days

Project cost = 1,510

Critical activities = All

(e) Final reduction possible is by reducing B, C & D for Shs. 160 the network then becomes.

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Duration = 10 days

Cost = Shs. 1,670

Critical activities = All.

Note: only critical activities affect project duration.

: Always look for a possibility of increasing the duration of a previously crashed activity.

SCHEDULING RESOURCES AND GANTT CHART

Apart from time, cost network analysis also help in controlling and planning of resources.

Example
A project has the following activity durations and resource requirements.

Activity	Preceding activity	Duration (days)	Resource requirement (man power)
A	-	6	3
B	-	3	2
C	-	2	2
D	C	2	1
E	B	1	2
F	D	1	1

Required

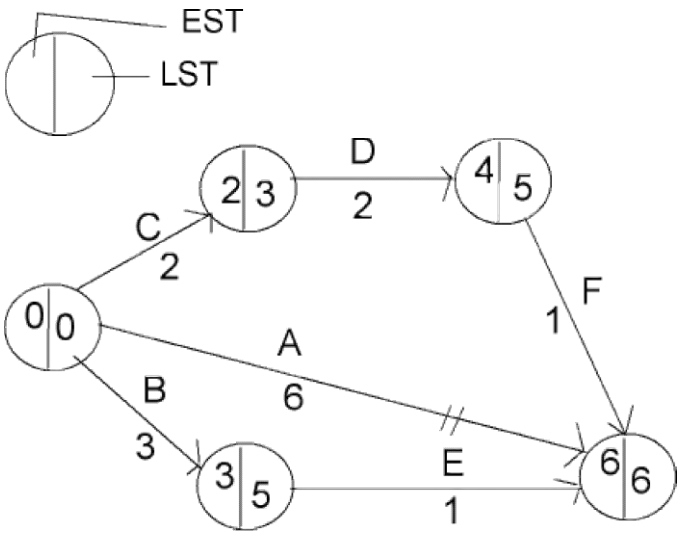
i) What is the networks critical path
ii) Draw a gantt chart diagram indicating activity times, using their estimate.
iii) Show resource requirement on a day to day basis assuming all events commence at their estimates.
iv) Assuming that only six employees are available, how will the activities be planned for?

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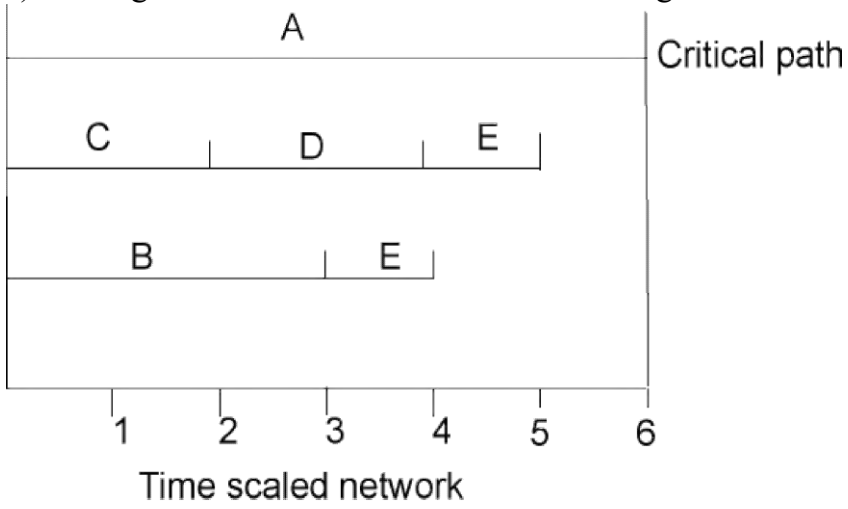
Solution

i)

Activities	Duration	EST	LST	Man power
A	6	0	0	3
B	3	0	0	2
C	2	0	0	2
D	2	2	3	1
E	1	3	5	2
F	1	4	5	1

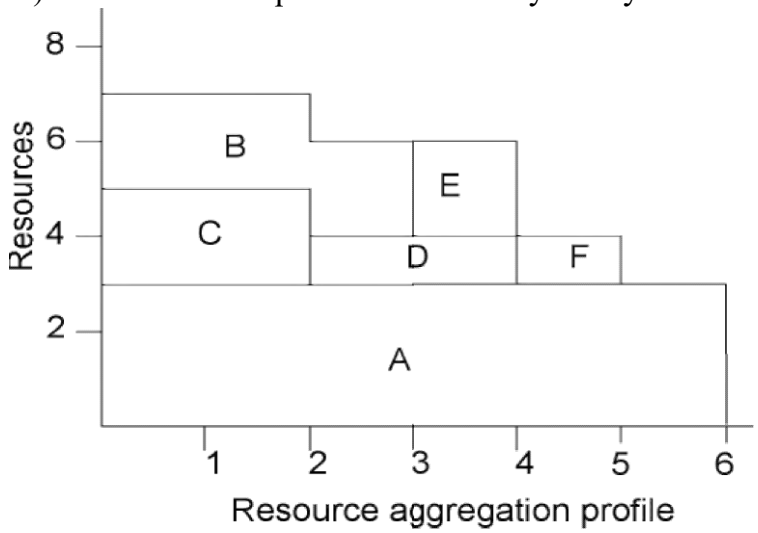


ii) A gantt chart or a bar chart. This is a diagram indicating a resource scaled network.

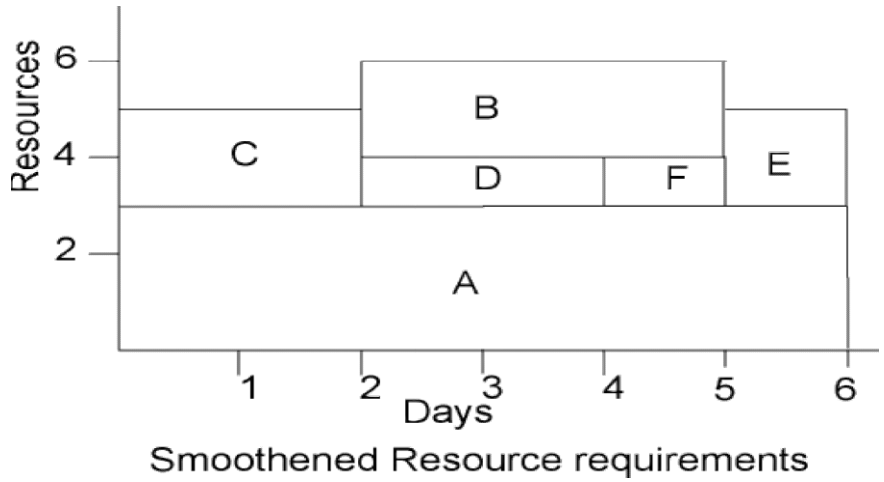


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iii) Resource requirements on a day to day basis.



iv) When on 6 manpower resources are available then we adjust the activities to accommodate this and still end at the given critical time duration i.e.



Smoothened Resource requirements

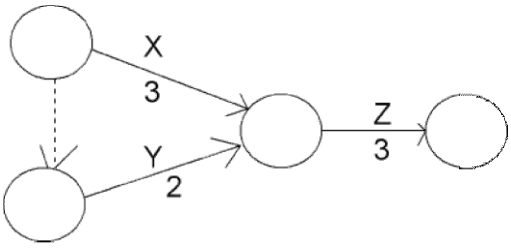
Node Networks

This network also known as a procedure diagram is represented with the same information as a network diagram.

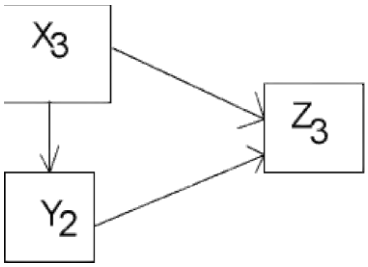
Its characteristics are;

i) Activities are shown in boxes instead of arrows
ii) Events are not represented.
iii) The arrows linking boxes indicate the sequence precedence of activities.
iv) Dummies aren't necessary.

E.g.

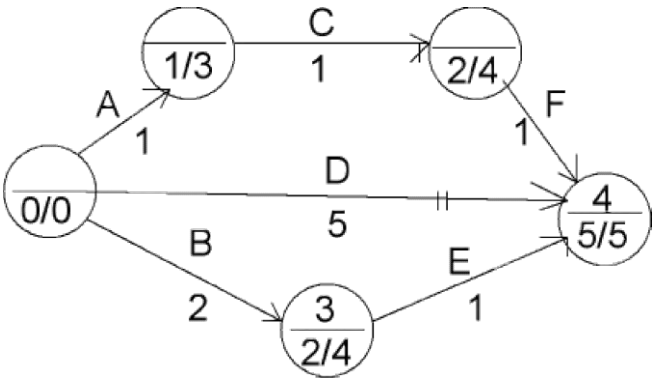


Would appear as

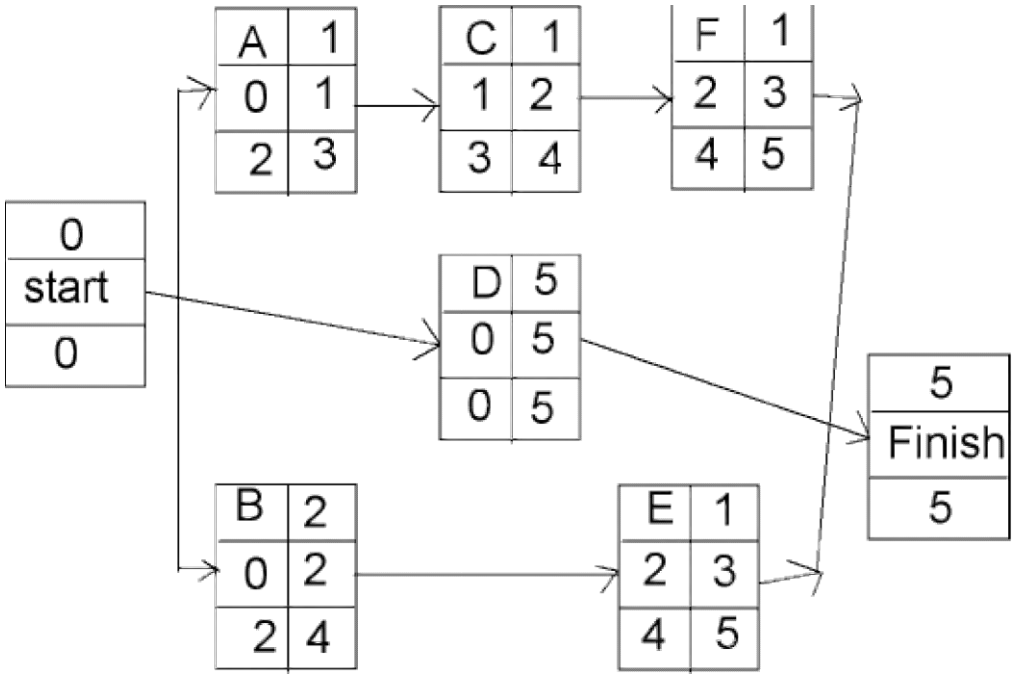


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A full activity node network is represented as;



This is represented as;



Note:

i) EST and LST are calculated by the same process we learnt earlier.

ii) EFT and LFT are calculated by adding the activity time duration to EST and LST respectively.

iii) Critical path is similarly identified by identifying equal EST and LST throughout the path.

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PRACTICE EXERCISES

QUESTION 1

a) For the product development project in question 1 consider the detailed time estimates given in the following table. Note that time estimates in the preceding exercise are equivalent to modal time estimates in this exercise.

Time Estimates (weeks)

Activity	Optimistic	Most likely	Pessimistic
A	1	3	4
B	1	1	2
C	4	5	9
D	1	1	1
E	4	6	12
F	1	1	2
G	1	2	3
H	6	8	10

Re-label your network in the question 1 to include expected duration $\overline{d}_{ij}$ (in place of activity duration d_{ij} and variances σ_{ij}.

Use equations below

$$d_{ij} = \frac{a_{ij} + 4m_{ij} + b_{ij}}{6} \quad \text{and} \quad \sigma_{ij}^2 = \left(\frac{b_{ij} - a_{ij}}{6}\right)^2$$

$$6\sqrt{\sigma_{ij}} = b_{ij} - a_{ij} \quad \text{or}$$

where: a_{ij} - optimistic time

b_{ij} - pessimistic time

b) Compare slacks to those in question 1.
c) Has the critical path changed?
d) Determine the following probabilities:
   i) That the project will be completed in 22 weeks or less.
   ii) That the project will be completed by its earliest expected completion date.
   iii) That the project takes more than 30 weeks to complete.

Solution:

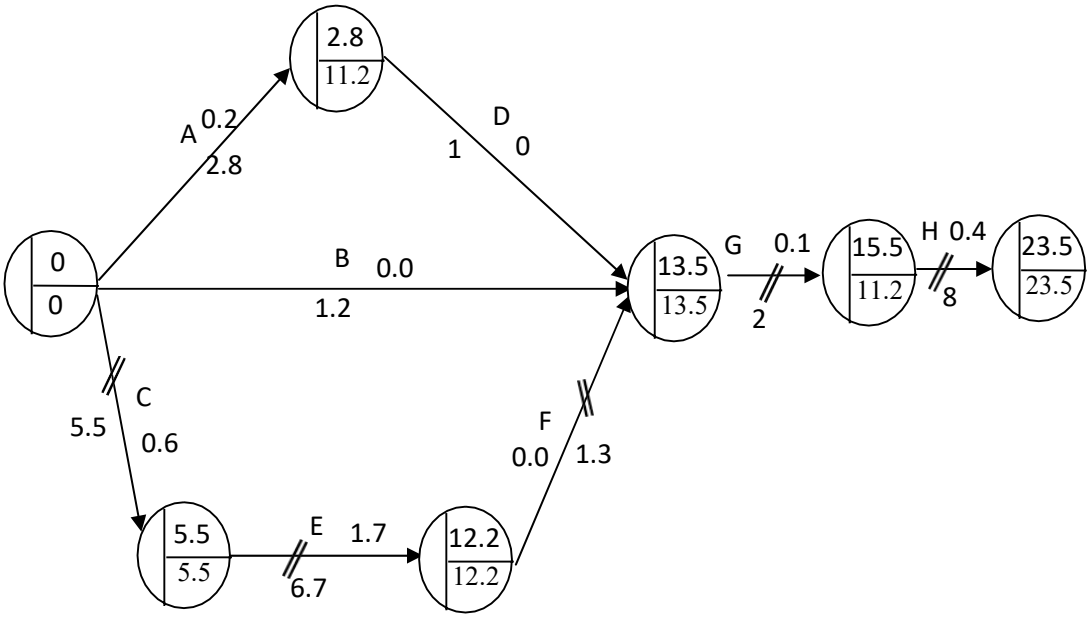
a) Calculation of estimated duration d_{ij} and standard deviation of duration σ_{ij} from the data of time estimates for the various activities is as follows:

$$d_{ij} = \left(\frac{a_{ij} + 4m_{ij} + b_{ij}}{6}\right) \text{ and } \sigma_{ij}^2 = \left(\frac{b_{ij} - a_{ij}}{6}\right)^2$$

Where: a_{ij}- optimistic time
b_{ij}- pessimistic time
m_{ij}- most likely time

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Activity	a_{ij}	m_{ij}	b_{ij}	d_{ij}	σ_{ij}^2	Slack	Comment
A	1	3	4	2.8	0.25	8.4	Not critical
B	1	1	2	1.2	0.03	12.3	Not critical
C	4	5	9	5.5	0.69	0	Critical
D	1	1	1	1.0	0.00	9.7	Not critical
E	4	6	12	6.7	1.78	0	Critical
F	1	1	2	1.2	0.03	0	Critical
G	1	2	3	2.0	0.11	0	Critical
H	6	8	10	8.0	0.44	0	Critical



b) The slacks in this situation are all more than in the situation where optimistic/pessimistic times are not included.

c) The critical path remained the same being C-E-F-G-H.

d)

i) The variance for the whole project is as follows

$\sigma^2=\sigma_A^2+\sigma_B^2+\sigma_C^2+\sigma_D^2+\sigma_E^2+\sigma_F^2+\sigma_G^2+\sigma_H^2$

$\sigma^2=0.25+0.03+0.69+0+1.78+0.03+0.11+0.44$

$\sigma^2=3.6$

The expected time of completion is T=23.5 weeks. The probability of completion of project within t=22 weeks is as follows:

$$P(t\leq T)=P\left(z\leq\frac{(t-T)}{\sigma}\right)$$

$$=P\left(z\leq\frac{(22-23.5)}{3.3}\right)$$

$$=P(z\leq -0.826)$$

From normal distribution table at z=-0.79, the required probability is (0.5-0.2967)=0.2033

So the probability of completing the project in 22 weeks is 0.2033.

ii) Expected time of completion is T = 23.5 weeks. So the probability of finishing the project within the earliest expected completion date is

$$P(t\leq 23.5)=P\left(z\leq\frac{23.5-23.5}{\sqrt{3.3}}\right)=P(z\leq 0)$$

From normal distribution tables at z=0 the probability =0.5. So the probability of finishing the project within the earliest expected completion date is 50%

iii) The probability of the project taking more than 30 days to complete

$$P(t\geq 30)=P\left(z\leq\frac{30-23.5}{\sqrt{3.3}}\right)=P(z\leq 3.26)$$

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From normal distribution tables at z=3.56 the probability =0. So the probability of the project being completed after 30 weeks =0.

QUESTION 2

a) A small construction project involves the following activities:

		Normal		Crash	
	Activity	**Time (days)**	**Cost (Sh.)**	**Time (days)**	**Cost (Sh.)**
1.2	Clear ground A	6	60,000	5	70,000
1.3	Lay foundation	5	30,000	3	50,000
2.4	Build walls	3	10,000	2	15,000
3.4	Roofing and pipingD	7	40,000	4	55,000
3.5	Painting	4	20,000	3	30,000
4.5	Landscaping	2	10,000	1	17,500

Required:

i) Determine the shortest time and associated cost to finish this project.

ii) If a penalty of Sh.4,500 must be charged for every day beyond 12 days, what is the most economical time for completing the project?

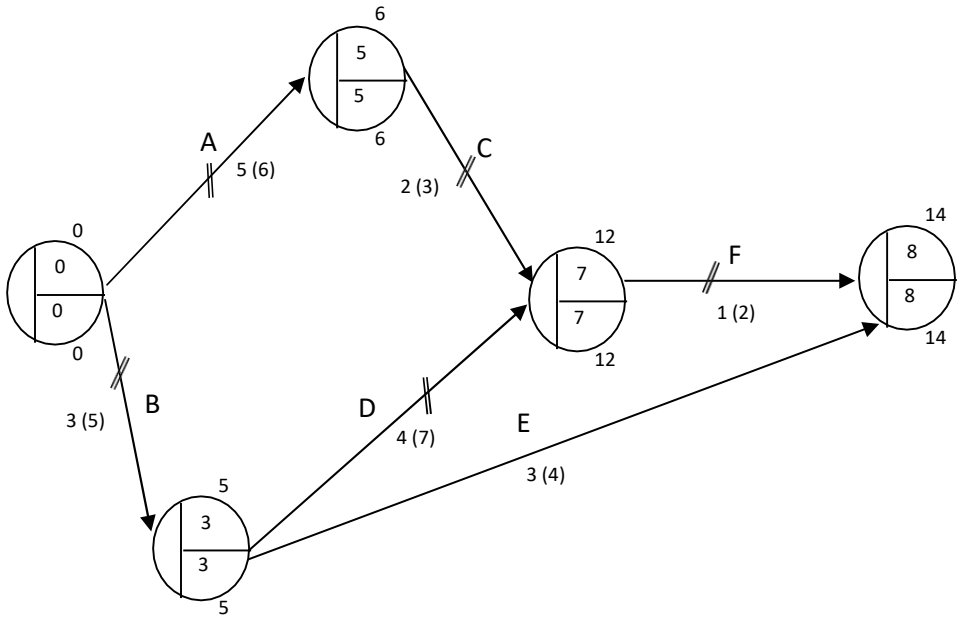
b) Explain the four different methods or approaches for organizing and displaying project information.

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Solution:

a)

i) The shortest time to finish the project is determined by crashing all the activities. The network diagram drawn using crash times is drawn as follows. (Note the normal duration for activities is in brackets and event times are above and below the events)



From the network diagram, the critical paths are A-C-F and B-D-F.
The project crash duration is 8 days.
The crash cost for the project is (70+50+15+55+30+17.5)×1,000=KSh. 237,500

ii) The cost schedule table is as follows.

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Normal	14	9	0	179
Compress D	13	4.5	5	179.5
Compress D	12	0	10	180
Crash D	11	0	15	185
Crash F (A-C-F & B-D-F critical)	10	0	22.5	192.5
Crash C	10	0	27.5	197.5
Compress B	9	0	37.5	207.5
Crash B	9	0	47.5	217.5
Crash A	8	0	57.5	227.5

The linear crashing cost per day $R=\left(\frac{\text{crash cost} - \text{normal cost}}{\text{normal time} - \text{crash time}}\right)$

Activity	**R**
A	10
B	10
C	5
D	5
E	10
F	7.5

It is economical to do the project within the normal duration of 14 days without crashing any activity.

Notes:
When normal activity durations are used, there is only one critical path B-D-F. So the activity with the lowest crash cost per day is D. It is compressed first by one day then again compressed and finally crashed. The opportunity cost decreased to zero while the additional crash cost increased progressively by Sh 5,000. At this point there are two critical paths A-C-F and B-D-F. Activity F is chosen to be crashed although it has a higher cost per day than activity C. This is because to crash C, activity B has to be compressed to be able to reduce the project duration by one day.
Activity C can then be crashed although this does not reduce the duration because of the other critical path B-D-F.

Compressing activity B after C is crashed results to reduction in project time of one day.
Crashing activity B will require that activity A is crashed too to come to the minimum project duration of 8 days.
At this point crashing terminates.

b) The different approaches to displaying project information are Gantt chart, project evaluation and review technique PERT, critical path analysis CPA and resource schedule charts.
Gantt chart involves displaying the activities on a graph against time. A line shows start, duration, end and float of activity.
CPA involves displaying project activities on network. The logical relationship between activities is shown together with activity durations. From this network, the critical activity can be determined.
PERT involves displaying project activities on a network like in CPA. The times for the duration used here are uncertain. So the expected time is used instead.
Resource schedule chart involves presenting project activity resources required and what is available on chart. For every resource, in a project, a resource chart is drawn.

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TOPIC 10

QUEUING THEORY

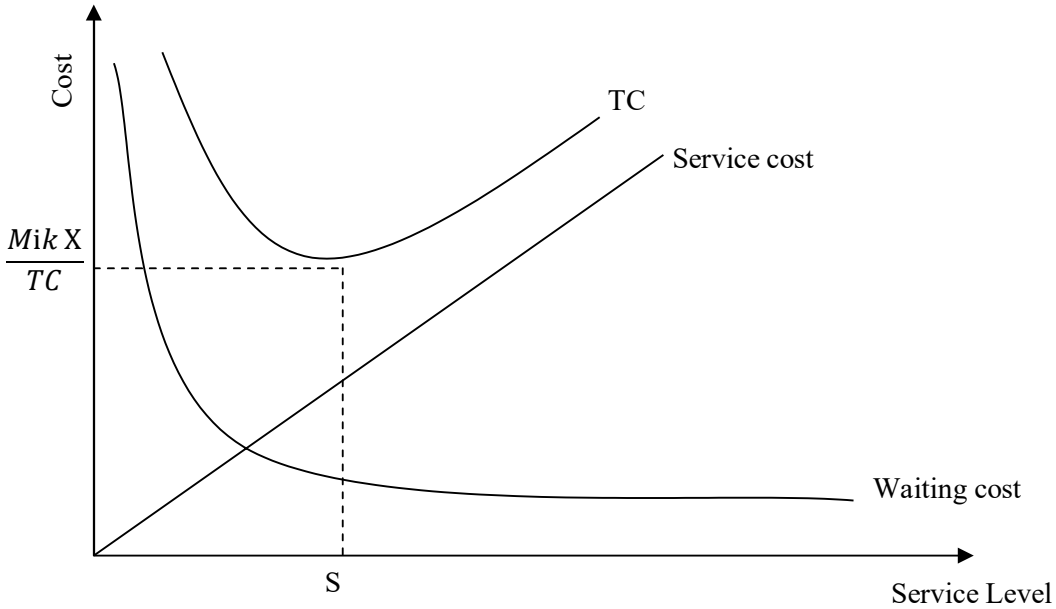
INTRODUCTION

Queuing Theory is the study of waiting line which consists of one or more customers waiting to be served. In queuing theory we analyze the following costs:

i) **Waiting costs:** These are the costs incurred by the customers waiting on the line. These costs decrease as the service level increases.
ii) **Service cost:** These are the costs incurred when the customer is being attended at the service facility.

The service costs increase as the service level increases. Therefore the total cost in queuing is the sum of the service costs and the waiting cost.
The main problem in queuing is to determine the optimal service level which minimizes the total cost.
Generally, the various costs in queuing can be summarized graphically as:

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Queuing theory has the following components:

1. Arrivals or calling population
2. Waiting line
3. The service channel or facility

OPERATING CHARACTERISTICS OF QUEUING SYSTEMS

Analysis of a queuing system involves a study in its different operating characteristics. Some of them are

1. Queue length (L_q)- The average number of customer in the queue waiting to get service . This excludeds the customer(s) being served
2. System length (L_s) - the average number of customers in the system including those waiting as well as those being served.
3. Waiting time in the queue (W_q) - the average time for which a customer has to wait in the queue to get service.
4. Total time in the system (W_S) - the average total time spent by a customer in the system from the moment he arrives till he leaves the system. It is taken to be the waiting time plus the service time.
5. Utilization factor (p) - It is the proportion of time a server actually spends with the customers. It is also called traffic Intensity.

WAITING TIME AND IDLE TIME COSTS

In order to solve a queuing problem, service facility must be manipulated so that an optimum balance is obtained between the cost of waiting time and the cost of idle time.

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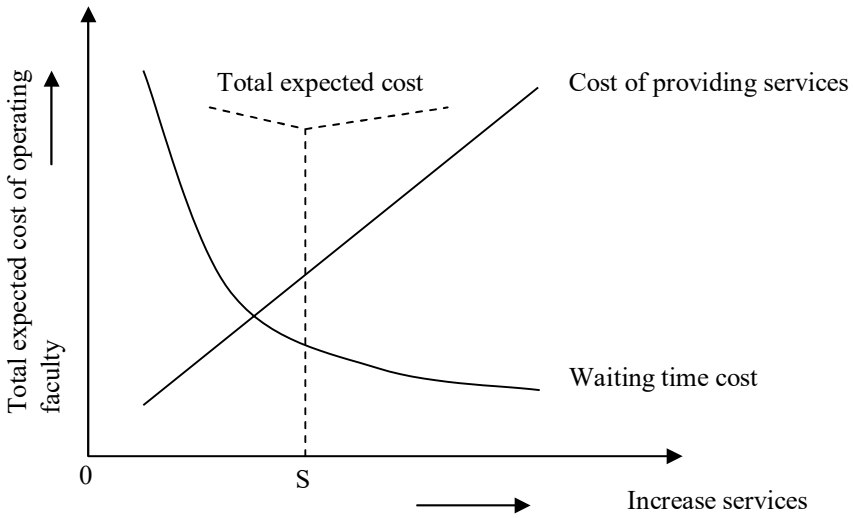
The cost of waiting customers generally includes either the indirect cost of lost business (because people go somewhere else, but less than they had intended to, or do not come again in future) or direct cost of idle equipment and persons; for example, cost of truck drivers and equipment waiting to be unloaded or cost of operating an airplane or ship waiting to land or dock.

The cost of lost business is not easy to assess, e.g., vehicle drivers wanting petrol will avoid pumps having long queues. To determine how much business is lost, some type of experimentation and data collection is required.

The cost of idle service facilities is the payment to be made to the servers engaged at the facilities for the period for which they remain idle.

The waiting time cost is added to the cost of providing service to establish a total expected cost.

The total expected cost is minimum at a service level denoted by point S. Thus the objective of the technique is really to determine that particular level of service which minimizes the total cost of providing service and waiting for that service.



Therefore the issue of concern to the management is to determine the optimal service rate, S, that will minimize the total cost associated with the waiting line

Let C_w = expected waiting cost / unit / unit time

L_s = expected (average) number of units in the system

C_f = cost of servicing one unit

Therefore expected waiting cost per time (period) = $C_w \times L_s = C_w \overline{\mu -}$

And expected service cost per unit time (period) = $Cf.\mu$

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Therefore total cost, $C = C_w \overline{\mu -} + \mu C_f$

This will be minimum if:- $\frac{d}{d\mu}(C) = 0$

$\frac{dS}{d\mu} = C_w \lambda (\mu \quad \lambda)^{-1+1} + C_f \mu = -C_w \overline{(\mu -)^2} + C_f = 0$ make μ the subject of formula

$(\mu \quad \lambda)^2 C_f = C_w \overline{(\mu -)^2} \times (\mu \quad \lambda)^2$

$(\mu \quad \lambda)^2 \frac{C_f}{C_f} = \frac{C_w -}{C_f}$

$\sqrt{(\mu \quad \lambda)^2} = \sqrt{\frac{\overline{C_w -}}{C_f}}$

$\mu \quad \lambda = \sqrt{\frac{\overline{C_w -}}{C_f}}$

$\mu = \sqrt{\frac{\overline{C_w}}{C_f}} \lambda + \lambda$

$$\mu = \lambda \pm \sqrt{\frac{C_w}{C_f}} \quad \lambda$$

NB; A plus and minus sign appear before the square root sign; A negative value of μ is not a possible answer in real life problems. μ given by the above equation is called minimum cost service rate,

ILLUSTRATION

Consider a situation in which the mean arrival rate is one customer every 4 minutes and the mean service time is 2/1/2 minutes. If the waiting cost is sh.5 per. unit per minute and the minimum cost of servicing one unit is sh. 4, find the minimum cost service rate.

SOLUTION

$\mu = \lambda \pm \sqrt{\frac{C_w}{C_f}} \quad \lambda$ But $C_f = 4$ (servicing rate)

$C_w = 5$ (Waiting)

$\lambda = ¼$

$$\mu = 0.25 \pm \sqrt{\frac{5 \times 0.25}{4}} = \sqrt{0.25 \pm 0.3125}$$

$$\mu = 0.25 \pm 0.56$$

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TRANSIENT AND STEADY STATES OF THE SYSTEM

Queuing theory analysis involves the study of system's behavior over time. If the operating characteristics (behavior of the system) vary with time, it is said to be in transient stage. Usually a system is transient during the early stages of its operation, when its behavior still depends upon the initial conditions. However, it is the 'long-run' behavior or the steady state condition of the system which is more important. A system is said to be in steady state condition if its behavior becomes independent of time.

As essential condition for reaching a steady-state is that the total elapsed time since the start of the operation must be sufficiently large (theoretically, it should tend to infinity), However, this is not the sufficient condition as the parameters of the system also affect its state, e.g. if the average arrival rate is less than average service rate and both are constant, the system eventually settles down to a steady state and the probability of finding a particular length of queue will be same at any time.If' the rates are not constant, the system will not reach a steady state, but it could remain stable. If the arrival rate is greater than service rate, the system cannot attain a steady state (regardless of the length of elapsed time): It is rather unstable, queue length increases steadily with time and theoretically, it could build up to infinity, Such state of the system is called explosive state.Evidently, imposing a limit on the maximum length of the queue (so the further arrivals are not accepted) automatically ensures stability, Queuing situations which are unstable for a limited time are common in practice, e,g, rush-hour traffic.

SINGLE CHANNEL SINGLE FACE SYSTEM (SIMPLE QUEUE)

This model has a poisson arrivals denoted by λ and exponential service denoted by μ. This model is also denoted as m|m|I (kendall notation) meaning single queue/single service.

Assumptions

1. Arrivals assumption

i) The size of the calling population is infinite
ii) Arrivals are random and they are specifically Poisson distributed
iii) The customers are patient (they wait in the same queue until they are served)

2. Waiting lines assumptions

i) The waiting line is unrestricted / unlimited in length (as long as it can be),
ii) The customers on service on FIRST COME FIRST SERVE (F.C.F.S)

3.Service facilities assumptions

i) The service time is random and specifically it's exponentially distributed.
ii) The queue design is a single phase, single channel design.

4. Other assumptions

i) Arrivals are random but the average or expected arrival rate is constant (λ)
ii) The service rate is also random but constant on average (μ)
iii) The service rate is higher than the arrival rate.

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QUEUING EQUATIONS (OPERATING CHARACTERISTICS)

There are three types of queuing equations - to do with numbers (length), times and probability

Numbers / Length

i. Length of the system (L_S) i.e. number of customers waiting on the queue plus those being served

$L_s = \frac{}{\mu -}$ where λ = arrival rate

μ = service rate

ii. Length of the queue (L_q) i.e. number of customers waiting in the queue

$= L_q = L_s \times \frac{}{\mu}$

$= \frac{}{P -} \times \frac{}{\mu}$

$= \frac{}{\mu(\mu -)}$

Times

i. Waiting time in the system i.e. average total time spent by a customer in the system from the moment he arrives till be leave the system. This is taken to be the waiting time plus the service time.

$W_s = {}^{I}/_{\mu} \quad \lambda$

ii. Waiting time in the queue i.e. average time for which a customer has to wait in a queue to get a service

$W_q = W_s \times {}^{\lambda}/_{\mu}$

$= \overline{\mu(\mu -)}$

Probabilities

i. Probability that the system is busy (also called utilization factor or traffic intensity).

P (row) = $\frac{}{\mu}$ or P = $\frac{}{\mu}$

ii. Probability the system is idle

$P_o = 1 - p$ (row)

$P_o = 1 - \frac{}{\mu}$

iii. The probability that the number of customers; n in the system is greater than k is given as follows:

$P (n > k) = [\frac{}{\mu}]^{k+1}$

iv. The probability that the number of customers n =k

$P (n = k) = [\frac{}{\mu}] \times 1 - [\frac{}{\mu}]$

v. The probability that the number of customer n ≥k

$P (n \geq k) = [\frac{}{\mu}]^{k}$

vi. Probability a customer spends time (T) greater than t the system (T> t)

$P (T > t) = e^{(- \mu)t}$

vii. The probability the service time T for a customer exceeds t

$P = e^{- \mu t}$

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ILLUSTRATION

The following information as regarding a particular garage

i) The arrival rate is 2 cars per hour
ii) Three cars are normally serviced per hour

Required:

a) Determine the service rate
b) Length of the queue
c) Length of the system
d) Determine the time a car takes being actually services
e) Determine the probability that the guard is busy,
f) Determine the probability that there are more than 5 cars in the garage

g) Determine the probability that a car will take less than I.5hrs in the garage

h) Determine the probability that a car will take more than 2S minutes being actually serviced

SOLUTION

i. Service rate p

1 car 20 min

μ = ?

$\mu = {}^{60}/_{20}$ = 3 cars per hour

ii. Length of the queue

$L_q = \frac{^{2}}{\mu(\mu-\ )}$ where λ is 2

$= \frac{2^2}{3(3-2)}$

= 1.333 cars

iii. Length of the system

$L_s = \frac{\ }{\mu-} = \frac{2}{(3-2)} = \frac{2}{1}$

= 2 cars

iv. The time a car takes being actually serviced

$t = W_s - W_q$

$W_s = \frac{\ }{\mu-}$

$= \frac{1}{3-2} = \frac{1}{1}$ = 1 min

$W_q = \frac{\ }{\mu(\mu-\ )} = \frac{2}{3(3-2)} = \frac{2}{3}$

$t = 1 - {}^{2}/_{3} = {}^{1}/_{3}$ hours or 20 min

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v. Probability that the garage is busy

$p = \frac{\ }{\mu} = \frac{2}{3}$ = 0.67 or 67%

vi. Probability that more than 5 cars in the garage

$P(n > k) = [\frac{\ }{\mu}]^{k+1} = [\frac{2}{3}]^{5+1}$

$= [\frac{2}{3}]^{6} = 0.088 = 8.8\%$

vii. Probability that cars will take less than 1.5 hours in the garage

$P(T > t) = e^{(\ -\mu)t}$

$(t < 1.5) = 1\text{-}p\ (t > 1.5)$

$P(T > 1.5) = e^{(3-2)1.5}$

$= e^{-1.5}$

e = 0.22 or 22%

P(T > 1.5) = 1-0.22

= 0.78 or 78%

viii. Probability that 25 minutes being serviced

P (T > t)

$P = e^{-\mu t}$

$t = {}^{25}/_{60} = 0.42$ hrs

$P = (t < 0.42) = e^{-3x\ 0.42}$

$P = e^{-1.26}$

$P = 0.283 \approx 28.3\%$

ILLUSTRATION

A change motor garage is able to install new car silencers at an average rate of three per hour or one every 2o minutes. Customers requiring this service arrive at the garage on an average of two customers per hour. The owner of the garage did a study on the queuing model relating to the garage services realized that they were single channel, single phase system.

Determine:

a) The average number of customers in the system -2
b) The average time spent by a customer in the system-1
c) The average number of customer in the queue-1.3
d) The utilization factor of the service facility-0.67

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SOLUTION

(a) The average number of customers in the system

$$\frac{}{\mu -} = \frac{2}{3-2}$$

$$= 2$$

(b) The average time spent by a customer

$$W = \frac{1}{\mu -} = \frac{1}{3-2} = 1$$

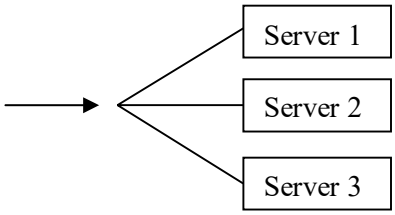
(c) The average number of customers in the que

$$L_q = \frac{2}{\mu(\mu -)} = \frac{2x2}{3(3-2)} = 1.333$$

(d) The utilization factor of the service facility

$$P_o = 1 - \frac{}{\mu} = 1 - 2/3 = 0.67$$

MULTI-CHANNEL SINGLE PHASE SYSTEM



The assumptions for the multichannel single phrase system are all the same as for those of simple queue.

Additional assumptions

1. All the service channels are identical in particular the service rate is equal for all the servers.
2. The combined service rate is greater than the arrival rate (i.e.) If the number of channels is C of M then $\mu > \lambda$ where μ = service rate
   $(\mu > \lambda)\lambda$ = arrival rate

In multiple channel system, it is assumed that the arrivals follow a Poisson probability distribution and the service times are exponentially distributed. The service is on the FCFS and all the servers are assumed to perform at the same rate.

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EQUATIONS FOR MULTI-CHANNEL QUEUING MODEL

If we let M to be equal to channels open then the following formulas may be used in waiting line analysis.

λ = average arrival rate

μ = average service rate at each channel

1. The probability that there are zero customers or units in the system

$$P_o = \frac{1}{\left[\sum_{n=0}^{n-1} \frac{1}{n!}\left[\frac{\lambda}{\mu}\right]\right] + \frac{1}{M!}\left[\frac{\lambda}{\mu}\right]^m \frac{M\mu}{M\mu-\lambda}}$$

2. Average number of customers or units in the system

$$L_s = \frac{h \times P\left(\frac{h}{P}\right)^m}{(M-1)|(MP-h)^2} \qquad P + \lambda/\mu$$

3. Average time unit spends in waiting time or being serviced

$$W_s = \frac{P\left(\frac{h}{P}\right)^m}{(M-1)|(MP-h)^2} \qquad P_o + 1/\mu = \frac{Ls}{h}$$

4. Average number of customers in line waiting for the service

   $L_q = L_o - = \frac{h}{P}$

5. Average time a customer or unit spends in the queue waiting for the service

   $W_q = W_s - = \frac{1}{P} = \frac{L_q}{h}$

6. Utilization rate e

   $P = \frac{h}{MP}$

Total cost in Queuing Model

The queuing model require a trade off or balance increased cost of providing better services and waiting costs associated with the queuing customers. The main costs in queuing are the waiting costs and the service cost therefore;

Total cost = waiting cost + service cost.

The total service cost = number of channels x cost per channel

Total waiting cost = (λw) CW

Where λw - Average number of arrivals x average wait per channel.

CW – Cost of waiting

If the waiting time cost is based on time in the channel then

Total cost = MCs + (λw)CW

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If the waiting cost is based on time in the queue then

Total cost = MCs + λwqCW

W – Average time in the channel / system

Wq – average time in the queue.

PRACTICE EXERCISES

QUESTION 1

In a three channel system, the rate of service at each channel is 5 customers per hour and customers arrive at the rate of 12 per hour. What is the probability that there are no customers in the system at a given point in time?

Solution:

C= 3, λ = 12, η = 5, ρ = 12/3 x 5 = 0.8

$$P_0 = \frac{3!\,(1 - 0.8)}{(0.8 \times 3)^3 + 3!\,(1 - 0.8\,(x)}$$

Where $x = {}^{c-1}\underline{1}\ (\rho c)^n$

$\Sigma\ n!$

$$= \frac{3 \times 2 \times 1\ (0.2)}{(2.4)^3 + 3 \times 2 \times 1)\ (0.2)\ (x)}$$

x is the sum of 3 figures giving $\frac{1}{0!}\ ((\rho c)^n$

where n = 0; $\frac{1}{0!}\ (0.8 \times 3)^o = 1.0$

n = 1 ; $\frac{1}{1!}\ (0.83 \times 3)^1 = 2.4$

n = 2 ; $\frac{1}{2!}\ (0.8 \times 3)^2 = ½\ (2.4)^2 = \underline{2.88}$

$\therefore X \quad \underline{6.28}$

$$P_o = \frac{1.2}{13.824 + 1.2\ (6.28)} = 0.056$$

C= 3, $\lambda = 12$, $\eta = 5$, $\rho = 12/3 \times 5 = 0.8$

$$P_0 = \frac{3!\ (1 - 0.8)}{(0.8 \times 3)^3 + 3!\ (1 - 0.8\ (x)}$$

Where $x = \frac{c-1}{\Sigma\ n!}\ \frac{1}{o}\ (\rho c)^n$

$$= \frac{3 \times 2 \times 1\ (0.2)}{(2.4)^3 + 3 \times 2 \times 1)\ (0.2)\ (x)}$$

x is the sum of 3 figures giving $\frac{1}{0!}\ ((\rho c)^n$

where n = 0; $\frac{1}{0!}\ (0.8 \times 3)^o = 1.0$

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$\therefore X \quad \underline{6.28}$

$$P_o = \frac{1.2}{13.824 + 1.2\ (6.28)} = 0.056$$

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QUESTION 2

A team of 15 men is employed to unload lorries at a terminal. The team works a 6 hour day during which 36 lorries arrive (i.e. 6 per hour) and it takes 7 ½ minutes to unload one lorry with the team acting as a single unit. Lorries are Served on a FIFO basis.

It has been estimated that the cost of keeping lorries waiting is Sh 6 per hour. Members of the team are each paid Sh 2.50 per hour. It is also estimated that if the size of the team increased to 20 men, the average service time would fall to 5 minutes.

Required;-

Calculate the cost of the present system and the cost of the proposed system, and determine whether an increase in the size of the team would be justified on grounds of cost.

Solution:

The cost of service with
15 man team = 15 x 2.50 x 6 = sh. 225 per day
20 man team = 20 x 2.50 x 6 = sh. 300 per day
The daily cost of lorry waiting time, at sh.6 per hour may be calculated in either of 2 ways. by calculating the average number of lorries in the system and multiplying this number by (sh 6 per hour x 6 hours per day) Sh. 36 per day or by calculating the average waiting time in the system, and multiplying this time by sh.6 per hour and by the number of lorries in a 6 hour day i.e. 36.

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Average number of customers in the system = $\frac{\lambda}{\mu - \lambda}$ or $\frac{P}{1 - P}$

15 man team
$\lambda = 6$ $\mu = 60/7.5 = 8$ $P = 0.75$

Average number in system = $\frac{6}{8 - 6} = 3$

Cost per day per customer = 36
Total daily cost of waiting time 3 x 36 = 108

20 man team
$\lambda = 6$ $\mu = 60/5 = 12$ $P = 0.5$

Average number in system = $\frac{6}{12 - 6} = 1$

Cost/per day per customer = 36
Total average daily cost of waiting time = 1x 36 = 36

Summary	**Sh**	**Summary**	**Sh**
Cost of service per day	225	Cost of service per day	300
Cost of waiting time per day	108	Cost of waiting time per day	36
Cost of system, per day	333	Cost of system per day	336

The 15 man team is marginally more economical by sh 3 per day.

QUESTION 3

Arrivals at a telephone booth are considered to be Poisson, with an average time of 10 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponentially with mean 3 minutes.

a) What is the probability that a person arriving at the booth will have to wait?
b) What is the average length of the queues that form from time to time?
c) The telephone department will install a second booth when convinced that an arrival would expect to have to wait at least three minutes for the phone. By how much must the flow of arrivals be increased in order to justify a second booth?

Solution:

Given $\lambda = 0.1$ arrival per minute

$\mu = 0.33$ service per minutes

a) Prob. (an arrival has to wait) $= 1 - P_0$

$$= \frac{\lambda}{\mu} = \frac{0.1}{0.33}$$

b) Average length of non-empty queues:

$$E(m/m>0) = \frac{\mu}{\mu - \lambda} = \frac{0.33}{0.33 - 0.1} = 1.43 \text{ persons}$$

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c) The average waiting time for an arrival before he gets service

$$E(w) = \frac{\lambda}{\mu(\mu - \lambda)}$$

If we fix $\mu = 0.33$, we want to find the new value of λ, say λ, for which $E(w) = 3$ minutes. Then, we have

$$3 = \frac{\lambda'}{0.33(0.33 - \lambda')}$$

$\lambda' = 0.16$ arrival per minute.

QUESTION 4

The repair of a Lathe requires four steps to be completed one after another in a certain order. The time taken to perform each step follows exponential distribution with a mean of 10 minutes and is independent of other steps. Machine breakdown follows Poisson process with mean rate of 3 breakdowns per hour. Answer the following:

i) What is the expected idle time of the machine, assuming there is only one repairman available in the workshop?
ii) What is the average waiting time of a breakdown machine in the queue?
iii) What is the expected number of broken down machines in the queue?

Solution:

Given $\lambda = 3$ per hour

$\mu = 6$ per hour

$s = 4$, since there are four steps to be completed one after another.

Using Erlangen model, we have

i) Average time an arrival spends in the system

$$= \frac{s+1}{2s} \frac{\lambda}{\mu(\mu-\lambda)} + \frac{1}{\mu}$$

$$= \frac{5}{2 \ x \ 4} \text{x} \frac{3}{6 \ x \ 3} + \frac{1}{6}$$

$$= \frac{5}{48} + \frac{1}{6} = \frac{13}{48}$$

$$= 16.25 \text{ minutes}$$

This would be, in other words, the expected, idle time of the machine.

ii) Average waiting time of the machine in the queue

$$= \frac{s+1}{2s} \frac{\lambda}{\mu(\mu-\lambda)}$$

$$= \frac{5}{2 \ x \ 4} \text{x} \frac{3}{6 \ x \ 3}$$

$$= \frac{5}{48} \text{hour} = 16.25 \text{ minutes}$$

iii) Average number of broken down machines in the queue

$$= \frac{s+1}{2s} \frac{\lambda^2}{\mu(\mu-\lambda)}$$

$$= \frac{5}{2 \ x \ 4} \text{x} \frac{3}{6 \ x \ 3}$$

$$= \frac{5}{16} = 0.3$$

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QUESTION 5

A tailoring shop with one man takes exactly one day to stitch a suit. Customer's arrival follow a Poisson pattern with mean rate of arrival of one in every two days. How long, on an average, customer is expected to wait in such a situation?

Solution:

Given

$$\lambda = \frac{1}{2} \text{ per day}$$

$$\mu = 1 \text{ per day}$$

$s = \infty$ since service time is constant

Thus, we have

$$E(w) = \frac{s+1}{2s} \frac{\lambda}{\mu\ (\mu - \lambda)}$$

$$= \frac{1}{2} \times \frac{0.5}{1\ (1.0 - 0.5)}$$

$$= \frac{1}{2}$$

Hence, a customer will have to wait for half a day.

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TOPIC 11

SIMULATION

INTRODUCTION

Simulation can be defined as a technique that imitates the operation as it evolves over time. It is basically a technique of conducting experiments on a model of a system. Simulation model usually takes the form of a set of assumptions about the operation of the system, expressed as mathematical or logical relations between the objects of interest in the system.

In order to study a system once it is defined, two alternatives are available:-

i) To study the actual system itself and the other
ii) To construct the model of the system and study the model

Generally the study of the actual system has the disadvantages of being time consuming, expensive and / or outright impossible (e.g. in a saw mill operation, it would be extremely time consuming and costly to try every possibility of cutting logs to maximize profit Likewise it would be impossible to study a proposed system without constructing some form of model.
Consequently models most existing or proposed systems are constructed and the models are analysed how the actual system will react to change. However, many realistic systems can't be modeled for solution by the standard operation research methods. Therefore some form of simulation must be used to provide the solution. Simulation is a general method which can-be used to solve problems in many areas of management such as

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i) Inventory management
ii) Queuing problems
iii) Capital budgeting
iv) Project management
v) Profit planning (CVP analysis etc.)

DEFINITION OF TERMS IN SIMULATION

a) A **System** - a system can be defined as a collection of entities that act & interact towards the accomplishment of some logical end. .
b) **State of a system**- This is the collection of the variables necessary' to describe the status of the system at any given time. Systems are usually classified as either discrete or continuous.
c) **A discrete system** is one which the state variable change only as discrete or countable points in time
d) **A continuous system**- is one in which the state variables change continuously over time
e) **Dynamic simulation**•-Representation a system as it evolves-overtime.
f) **Static simulation** model- Representation of a system at a particular point in time

g) **Model** –a model is a representation of the system and it usually takes the form of a set of assumption about the operation of the system

There are several types of simulation model namely:

1. Static simulation model
2. Dynamic simulation models
3. Deterministic simulation-models
4. Stochastic simulation models
5. Discrete simulation models
6. Continuous simulation models

Static simulation model

This is a representation of a system at a particular point in time.

Dynamic simulation model

This is a representation of a system as it evolves over time.

Deterministic simulation model

This is a model that contains No random variables.

Stochastic simulation model

This model contains one or more random variables.

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WHEN SIMULATION IS USED

i) When the assumptions made are unrealistic or unattainable.
ii) When the system takes too long to observe e.g. demographic / population issues(time compression advantage)
iii) When, the cost and the danger of experimenting with the real world situations is very high.
iv) Where there are difficulties in making observations e.g. space research and practice. Molecular research.

Variables in a simulation model

A business model usually consists of linked series of equations and formulae arranged so that they 'behave' in a similar manner to the real system being investigated. The formulae and equations use a number of factors or variables which can be classified into 4 groups.

(a) Input or exogenous variables
(b) Parameters
(c) Status variables
(d) Output or endogenous variables

These are described below.

a) Input variables

These variables are of two types - controlled and non-controlled.

Controlled variables: These are the variables that can be controlled by management. Changing the input values of the controlled values and noting the change in the output results is the prime activity of simulation. For example, typical controlled variables in an inventory simulation might be the re-order level and re-order quantity. These could be altered and the effect on the system outputs noted.
Non-controlled variables: These are Input variables which are not under management control. Typically these are probabilistic or stochastic variables i.e., they vary but in some uncontrollable probabilistic fashion.
For example, in a production simulation the number of breakdowns would be deemed to vary in accordance with a probability distribution derived from records of past breakdown frequencies.' In an inventory simulation demand and lead time would also be generally classified as non- controlled, probabilistic variables

b) Parameters

These are also input variables which, for a given simulation have a constant value. Parameters are factors which help to specify the relationships between other types of variables. For example in a production simulation a parameter (or constant) might be the time taken for routine maintenance, in an inventory simulation a parameter might be the cost of a stock-out.

c) Status variables

In some types of simulation the behavior of the system (rates, usages, speeds, demand and so on) varies not only according to individual characteristics but also according to the general state of the system at various times or seasons. As an example; in a simulation of supermarket demand and checkout queuing, demand will be probabilistic and variable on any given day but the general level of demand will be greatly influenced by the day of the week and the season of the year. Status variables would be required to specify the day(s) and season(s) to be used in a simulation.
Note: On occasions status variables and parameters would both be termed just parameters although strictly speaking there is a difference between the two concepts.

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d) Output variables

These are the results of the simulation. They arise from the calculations and tests performed in the model the input values of the controlled values. The values derived for me probabilistic elements and the specified parameters and status values. The output variables must be carefully chosen to reflect the factors which are critical to the really system being simulated and they related to the objectives of the really system. For example, output variables for an inventory simulation would typically include:

- Cost of stock holding
- Number of stock outs
- Number of unsatisfied orders
- Number of replenishment orders
- Cost of the re-ordering and so on

Constructing a simulation model

Some broad guidelines for constructing a simulation model are given below. These will be useful for dealing with examination questions but in this area especially, practice is vital.

Step 1: Identify the objective(s) of the simulation.

A detailed listing of the results expected from the simulation will help to clarify step 5 - the output variables.

Step 2 Identify the input variables. Distinguish between controlled and non-controlled variables

Step 3 Where necessary determine the probability distribution for the non-controlled variables

Step 4 Identify any parameters and status variables.

Step 5 Identify the output variables.

Step 6 Determine the logic of the model

This is the heart of the simulation construction. The key questions are: how are the input variables changed into output results? What formulae/decision rules are required? How will probabilistic elements be dealt with? How should the results be presented?

Some examples of the variables that may be used in simulation are:

i) A set of prices and cost and the standard relationships could be used to simulate profits

ii) The components of a queuing system such as the arrival rates and the service rates could be used to simulate a queuing system to generate such data as the waiting time, the length of a queue and the problems of a system being busy.

iii) In inventory management, variables such as demand and the lead time can be used in simulation to generate such cost data as the holding cost, shortage cost, ordering cost

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MONTE CARLO SIMULATION

This is a form of simulation that deals with the allocation of random numbers. When a system contains elements that exhibit chance in their behavior, the method of Monte Carlo sampling / simulation may be applied.

The basis of this method is experimentation on chance or probabilistic elements through sampling,

Steps in Monte Carlo Simulation

1. Set up probability distribution for the relevant random variables
2. Build up cumulative probability distributing for each of the variables in step 1.
3. Establish the intervals of the random numbers for each variable and allocate the random number ranges,
4. Obtain the random numbers -random numbers can be obtained from:
    a) Random number tables
    b) Calculators
    c) Computers

5 Run the simulation trials

THE ROLE OF COMPUTERS IN SIMULATION

1. It generates the random number
2. It simulates thousands of trials extremely fast, accurately and reliably. A computer can also stock large mass of data.
3. The computer simulates several combinations of the decision variables e.g. the re-order quantity and the re-order level in the inventory management or the service channel and the service time in queuing model in a matter of seconds.
4. It provides the management with printed reports which are very useful for decision making.

RANDOM NUMBER AND THEIR ALLOCATION

1. For uniform random number table, each unit has an equal chance of occurring at any point in the table i.e. a uniform distribution.
2. For uniform random number table, each unit has an equal chance of occurring at any point in the table i.e. a uniform distribution.
3. Each number is allocated once and only once.
4. The number allocated to a value of the random number is directly proportional to the problem of that value. i.e.

a) Single decimal probability distribution are allocated 10 digits i.e. from 0-9 or from 1 -0
b) Two decimal probability distribution are allocated 100 digits i.e. from 00-99 or 01 -DO
c) Three decimal probability distribution are located 1000 digits i.e., from 000-999 or 001 - 000 etc.

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Advantages of simulation

1. Simulation is well suited to problems which are difficult or impossible to solve analytically i.e. where main assumptions are unrealistic e.g., inventory management, queuing problems and capital budgeting. .
2. Simulation allows the analyst or the decision maker to experiment with the system behavior in a controlled environment instead of the real life setting which can be very costly or has inherent risks
3. It enables a decision maker to compress time in order to evaluate the long term effects of various alternatives.
4. Simulation can serve as a mode of training decision makers by enabling them to observe the behavior of a system under different circumstances without experimenting with the actual system e.g. military and business training/ gaming.
5. Simulation has the order of being relatively free from complicated mathematics thus very easy to understand for the operating personnel and for the non-technical managers.
6. Simulation models are comparatively flexible and can easily be modified to accommodate the changing environment e.g. a company manager can try several policy options in a matter of minutes

7. Simulation allows us to study the interactive effects of the individual components or variables to determine which ones are important.
8. Recent advancements in the software make some simulation models to be very easy to develop.

Disadvantages of simulation

1. Simulation is not precise i.e. it's not an approximation process and it does not necessarily yield an optimal answer but merely provides a set of system responses to the different operating conditions. In many cases, lacking precession is difficult to measure. However, as the number of simulation trials increases precision, increases provided that in the problem distribution the relevant variables do not change.
2. A good simulation model may be expensive in terms of design personnel (consultants, computing facilities software e.t.c.)
3. Simulation model is unique i.e. its solutions and inferences are not easily transferred to the other problems thus further increasing the cost of simulation.
4. Simulation can take time in terms of data collection and the designing of the model and this could delay the decision making which is costly in the long run.
5. In a number of situations it's not possible to quantify all the variables that affect the behaviour of a system.

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OTHER TYPES OF SIMULATION MODELS

Simulation models are often broken into three categories. The first, the Monte Carlo method just discussed, uses the concept probability distribution and random numbers to evaluate system responses to various policies. The two other categories are called operational gaming and system simulation. Although in theory the three models are distinctly different, the growth - of computerized simulation has tended to create a common basis in procedures and blur these differences.

a) Operational gaming

Operational gaming refers to simulation involving two or more competing players. The best examples are military games and business games. Both all participants to match their management arid decision-making skills in hypothetical situations of conflict

Military games are used world-wide to train a nation's top military officers, to test offensive and defensive strategies, and to examine the effectiveness of equipment and armies.

Business games, first developed by the firm Booz, Allen and Hamilton in the 1950s, are popular with both executives and business students. They provide an opportunity to test out business skills and decision-making ability in competitive environment. The person or team that performs best in the simulated environment is rewarded by knowing that his or' her company has been most successful in earning the largest profit, grabbing a high market share, or perhaps increasing the firm's trading value on the stock exchange.

b) Systems Simulation

Systems simulation is similar to business gaming in that it allows users to test various managerial policies and decisions to evaluate their effect on the operating environment. The variation of simulation models the dynamics of large systems. Such systems include corporate operations, the national economy, a hospital, or a city government system.

In a corporate operating system, sales, production levels, marketing policies, investments, union contracts, .utility rates, financing, and other factors are ail related in a series of mathematical' equations that are examined by simulation. In a simulation of an urban government, systems simulation may be employed to evaluate the impact of tax increase, capital expenditures for roads and buildings, housing availability, new garbage routs. in- migration and out-migration, locations of new schools or senior citizens centers, birth and death rates and many more vital issues. Simulation of economic systems, often called econometric models are used by government agencies, bankers, 'and large organizations to predict inflation rates, domestic and foreign money supplies, and unemployment levels.

ILLUSTRATION 1

XYZ Ltd is considering launching a new product which will require an investment of Shs. 5000. The product has a life of 1years. There are uncertain variables namely; selling price unity variable cost and demand as shown in probability distribution below:

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Selling price	Prob	Unit variable	Prob	Sales units demand	Prob
Sh 4	0.3	Sh 2	0.1	5000	0.2
Sh 5	0.5	Sh 3	0.6	4000	0.4
Sh 6	0.2	Sh 4	0.3	5000	0.4
	1.0		1.0		1.0

Using the following random numbers

806,043,632,140,360,589,161,573,996,497,726,953, 050,118,664886,924,077,008,401,658,401

Carry out 25 trial numbers

SOLUTION

1. Step one

Assign the random number ranges

i)

Selling price	Prob	Cum prob	RN ranges
Sh 4	0.3	0.3	0-2
Sh 5	0.5	0.8	3 - 7
Sh 6	0.2	1.0	8 - 9
	1.0		

ii)

Unit variable cost	**Prob**	**Cum prob**	**RN ranges**
Sh 2	0.1	0.1	0-0
Sh 3	0.6	0.7	1 - 6
Sh 4	0.3	1.0	7 - 9
	1.0		

iii)

Selling units /demand	**Prob**	**Cum prob**	**RN ranges**
Sh 3000	0.2	0.2	0- 1
Sh 4000	0.4	0.6	2 - 5
Sh 5000	0.4	1.0	6 - 9
	1.0		

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2. **Step 2**

Run the model

Average profit $= \frac{\text{Total profit}}{25 \text{ runs}}$

	RN	S.P (A)	R.N	Units variable cost (B)	(A-B=C) Unit contribution	R.N	D Quantity sold	CXD = E Total cost	F Fixed cost	E-F profit
1	8	6	8	4	2	8	5000	10000	5000	5000
2	0	4	3	3	1	6	5000	5000	5000	0
3	6	5	1	3	2	6	5000	10000	5000	5000
4	0	4	6	3	1	4	4000	4000	5000	(1000)
5	4	5	1	3	2	8	5000	10000	5000	5000
6	3	5	5	3	2	8	5000	10000	5000	5000
7	6	5	7	4	1	6	5000	5000	5000	0
8	3	5	3	3	2	9	5000	10000	5000	5000
9	2	4	9	4	0	2	4000	0	5000	(5000)
10	1	4	9	4	0	4	4000	0	5000	(5000)
11	4	4	4	3	1	0	3000	3000	5000	(2000)
12	0	4	4	3	1	7	5000	5000	5000	0
13	3	5	9	4	1	7	5000	5000	5000	0
14	6	5	7	4	1	0	3000	3000	5000	(2000)
15	0	4	7	4	0	0	3000	0	5000	(5000)
16	5	5	2	3	2	8	5000	10000	5000	5000
17	8	6	6	3	3	4	4000	12000	5000	7000
18	9	6	9	4	2	0	3000	6000	5000	1000
19	1	4	5	3	1	1	3000	3000	5000	(2000)
20	6	5	3	3	2	6	5000	10000	5000	5000
21	7	5	0	2	3	5	4000	12000	5000	7000
22	3	5	5	3	2	8	5000	10000	5000	5000
23	8	6	0	2	4	4	4000	16000	5000	11000
24	6	5	1	3	2	0	3000	6000	5000	1000
25	2	4	1	3	1	1	3000	3000	5000	(2000)
										46,000

Average profit $= \frac{46,000}{25} = 1,840$

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ILLUSTRATION 2

XYZ has a policy of ordering stock when level falls to 15 units the quantity ordered from the supply is always 20 units. The stock at beginning of 1st week is 20 units. The stock holding costs are sh. 10 per week / unit. The cost of placing one order = 25. The stock out cost are Sh100 per unit. The usage (demand) and lead time (time taken by supply to deliver stock) is uncertain as shown below.

NB: Ordering is done the following week upon discovery of the shortages.

Demand	Probability	Lead time	Probability
0	0.02	1	0.23
1	0.08	2	0.45
2	0.22	3	0.17
3	0.34	4	0.09
4	0.18	5	0.06
5	0.09		1.00
6	0.07		
	1.0		

Required;

Use the following random numbers

68 52 50 90 59 08 72 44 95 85 81 93 28 89 15 60 03

Use 14 trial numbers to find the cost

Demand	Probability	Cumm. prob	Random No. ranges
0	0.02	0.02	00-01
1	0.08	0.10	02-09
2	0.22	0.32	10-31
3	0.34	0.66	32-65
4	0.18	0.84	66-83
5	0.09	0.93	84-92
6	0.07	1.00	93-99

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Lead time	Probability	Cumm. prob	Random No. ranges
1	0.23	0.23	0-22
2	0.45	0.68	23-67
3	0.17	0.85	68-84
4	0.09	0.94	85-93
5	0.06	1.00	94-99

SOLUTION

Week /No.	Opening stock	Units ordered	Avail. for use	RN	Dem and	Closing stock	Order placed	RN	Lead time	Holding cost	Ordering cost	Total cost
1	20	-	20	68	4	16	No	15	1	160	-	160
2	16	-	16	52	3	13	No	60	2	130	-	130
3	13	-	13	50	3	10	Yes	03	1	100	25	125
4	10	20	30	90	5	25	No	68	3	250	-	250
5	25	-	25	59	3	22	No	52	2	220	-	220
6	22	-	22	08	1	21	No	50	2	210	-	210
7	21	-	21	75	4	17	No	90	4	170	-	170
8	17	-	17	44	3	14	No	59	2	140	-	140
9	14	-	14	95	6	8	Yes	08	1	80	25	105
10	8	20	28	85	5	23	No	72	3	230	-	230
11	23	-	23	91	-	19	No	44	2	190	-	190
12	19	-	19	93	6	13	No	95	5	130	-	130
13	13	-	13	28	2	11	Yes	85	4	110	25	135
14	11	-	11	89	5	6	No	81	3	60	-	60
												2,255

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Average cost = $\frac{\text{Total Cost}}{\text{14 Trial numbers}} = \frac{2255}{14}$ =161.07

PRACTICE EXERCISES

QUESTION 1

ABC Ltd. recently acquired a threshing machine with a useful life of 15 years. Over the useful life, the machine is likely to have periodic failures and breakdowns. Past data for similar machines indicate a probability distribution of failures as follows:

Number of failures	0	1	2	3
Probability	0.80	0.15	0.04	0.01

Required:

(i) Using the random numbers provided below, simulate the number of failures that will occur over the useful life of the machine.
Random numbers: 70,88,37,12,45,99,54,71,64,93,67,80,55,34,22

(ii) Determine the average annual failure rate.

Solution:

No of failures	Probability	Cumulative probability	R_N - Ranges
0	0.80	0.80	00 – 79
1	0.15	0.95	80 – 94
2	0.04	0.99	95 – 98
3	0.01	1.00	99 >

Simulation Worksheet

Years	Random numbers	No of failures
1	70	0
2	88	1
3	37	0
4	12	0
5	45	0
6	99	3
7	54	0
8	71	0
9	64	0
10	93	1
11	67	0
12	80	1
13	55	0
14	34	0
15	22	0
		6

Average annual failure rate $= \frac{6}{15} = 0.4$

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QUESTION 2

(a) Manukato Ltd. produces a designer perfume called "Hint of Elegance." Production of the perfume involves the use of two ingredients, X_1 and X_2 represented by the production function given below:

$$Y = \sqrt{X_1 X_2}$$

Where Y = Number of bottles of designer perfume produced.
X_1 = Units of ingredient 1.
X_2 = Units of ingredient 2.

Currently, the company is operating at a level where the daily usage of X_1 and X_2 is set at 250 units and 360 units respectively.

The price of the designer perfume and the cost of ingredients X_1 and X_2 are random variables. The data below relate to the three random variables.

Selling price of Y (per bottle) Shs.	**Probabilities**
4,000	0.15
4,500	0.35
5,000	0.20
5,500	0.30

Cost of ingredient X_1 Shs.	**Probabilities**
1,000	0.10
1,500	0.05
2,000	0.35
2,500	0.50

Cost of ingredient X2 Shs.	**Probabilities**
1,500	0.20
2,000	0.25
2,500	0.15
3,000	0.40

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Required:

(i) Calculate the daily expected profit of the company.

(ii) Simulate the company's profit for 10 days using the following random numbers:

58, 71, 96, 30, 24, 18, 46, 23, 34, 27, 85, 13, 99, 24, 44, 49,
18, 09, 79, 49, 74, 16, 32, 23, 02, 56, 88, 87, 59, 41, 06

(b) Nairobi Manufacturers Ltd. produces component X on machine Y at a rate of 4,000 units per month. Machine Z uses component X at the rate of 1,000 units per month, the remainder being put into stock. It costs Shs. 2,000 to set up machine Y while the stock holding cost is estimated at Shs. 2.50 per unit per annum plus a 20% opportunity cost of capital per annum. Each component costs Shs. 25 to produce.

Required:

(i) Compute the optimal batch size that should be produced using machine Y.

(ii) Assume that the actual set-up cost of machine Y is Shs. 1,000 instead of Shs. 2,000. Calculate the cost of prediction error.

Solution:

(a) (i) Let P = Selling price per bottle
C_1 = Cost of ingredient 1
C_2 = Cost of ingredient 2
Amount Produced daily = $\sqrt{250x360}$
= 300 units
$\therefore$ Profit = $300P - (250X_1 + 360X_2)$
Expected selling price
4000 x 0.15 + 4,500 x 0.35 + 5,000 x 0.20 + 5,500 x 0.3
= Shs. 4,825

Expected cost of ingredient 1
1000 x 0.1 + 1,500 x 0.05 + 2,000 x 0.35 +2,500 x 0.5
= Shs. 2,125

Expected cost of ingredient 2
1500 x 0.20 + 2,000 x 0.25 + 2,500 x 0.15 + 3,000 x 0.4
= Shs. 2,375
$\therefore$ Expected daily profit = (300 x 4825) – [(2,125 x 250) + (360 x 2,375)]
= 1,447,500 – 1,386,250
= Shs. 61,250

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(ii) Selling

Price Shs.	**Probs.**	**Cum Probs.**	**RN-Ranges**
4,000	0.15	0.15	01 – 15
4,500	0.35	0.50	16 – 50
5,000	0.20	0.70	51 – 70
5,500	0.30	1.00	71 - 00

Cost, ingredient 1

Sh.	**Probs.**	**Cum Probs.**	**RN-Ranges**
1,000	0.10	0.10	01 – 10
1,500	0.05	0.15	11 – 15
2,000	0.35	0.50	16 – 50
2,500	0.50	1.00	51 - 00

Cost, ingredient 2

Shs.	Probs.	Cum Probs.	RN-Ranges
1,500	0.20	0.20	01 – 20
2,000	0.25	0.45	21 – 45
2,500	0.15	0.60	46 – 60
3,000	0.40	1.00	61 - 00

Day	RN	Selling Price	Units	Total Revenue	RN	Cost X_1	Units	Total Cost X_1	RN	Cost X_2	Units	Total Cost $X_1 + X_2$	Total Cost $X_1 + X_2$	Daily Profit
		Shs.		Shs.000				Shs.000		Shs.		Shs. '000'	Shs. '000'	Shs.' 000'
300	58	5,000		1,500	71	2,500	250	625	96	3,000	360	1,080	1,705	(205)
2	30	4,500	300	1,350	24	2,000	250	500	18	1,500	360	540	1,040	310
3	46	4,500	300	1,350	23	2,000	250	500	34	2,000	360	720	1,220	130
4	27	4,500	300	1,350	85	2,500	250	625	13	1,500	360	540	1,165	185
5	99	5,500	300	1,650	24	2,000	250	500	44	2,000	360	720	1,220	430
6	49	4,500	300	1,350	18	2,000	250	500	09	1,500	360	540	1,040	310
7	79	5,500	300	1,650	49	2,000	250	500	74	3,000	360	1,080	1,580	70
8	16	4,500	300	1,350	32	2,000	250	500	23	2,000	360	720	1,220	130
9	02	4,000	300	1,200	56	2,500	250	625	88	3,000	360	1,080	1,705	(505)
10	87	5,500	300	1,650	59	2,500	250	625	41	2,000	360	720	1,345	305
														1,160

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Average Daily Profit = $\frac{1,160,000}{10}$

= Shs. 116,000

(b) EBQ = $\sqrt{\frac{2D_{Co.}}{C_h} x \frac{P}{P - D}}$

$\sqrt{\frac{2x12,000x2,000}{7.5} x \frac{48000}{48000 - 12000}}$

= 2921.19 or 2921 units

Where P is production rate
D is usage rate
$C_h = 2.50 + (20\% \times 25)$

(ii) Optimal EBQ = $\sqrt{\frac{2x12000x1000}{7.5}\left(\frac{48000}{48000-12000}\right)}$

= 2065.59 = 2066 units

TRC incurred= $\frac{12000(1000)}{2921}+\frac{1}{2}\frac{(2921)(48000-12000)}{48000}x7.5$

= Sh. 12323.49

TRC Optimal= $\frac{12000(1000)}{2066}+\frac{1}{2}\frac{(2066)(48000-12000)}{48000}x7.5$

= 11618.95

Cost of production error = 12323.49 – 11618.95

= Shs. 704.54

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